



COLLEGE OF COMPUTER AND INFORMATION SCIENCE

Academic Year 2024 – 2025

**SEIR-Viz: A Web-Based Epidemic Tool for Prediction and Visualization of Infectious
Disease Spread through SEIR Model Optimization**

Jeremy Aidan Lopez GENATO

Jon Peter Panganiban POMPERADA

John Bernard Gonzaga SAHAGUN

Thesis Adviser: Dennis A. Martillano

Submitted to the Faculty of Mapúa Malayan Colleges Laguna

In Partial Fulfillment of the Requirements for the degree of

Bachelor of Science in Computer Science

Table of Contents

Table of Contents.....	ii
List of Tables	viii
List of Figures	xiii
List of Appendices	xviii
Introduction.....	20
Background of the Study	20
Statement of the Problem.....	23
Rationale of the Study.....	26
Objectives	29
Scope and Delimitations of the Study.....	32
Conceptual Framework.....	37
Review of Literature	41
Epidemiology.....	41
Public Health Units	42

Epidemiological Model.....	44
Mathematical Modeling.....	45
Computational Modeling	46
Disease Simulation.....	47
Susceptible-Exposed-Infected-Recovered (SEIR) Model	48
Application of SEIR for Epidemiology	49
Parameter Modification, Estimation and Optimization	50
Simulated Annealing.....	50
Bayesian Inference.....	52
Math Model Validation.....	54
Epidemiologic Tools.....	55
Synthesis	57
Methodology	59
Research Design.....	59
Research Locale	59
Ethical Implication of the Research.....	61

Research Framework	63
Phase 1: Data Preparation	65
Phase 2: Modeling and Parameter Experimentation	82
Phase 3: Model Integration and Web Application Development	99
Phase 4: Testing and Analysis	108
Methodology Summary	121
Results and Discussion	125
Phase 1: Data Preparation	126
Data Cleaning.....	127
Data Transformation	131
Data Store.....	138
Phase 2: Modeling and Parameter Experimentation.....	140
SEIR Model Development and Optimization	141
SEIR Optimization Techniques	149
Model Selection	171
Model Confidence Testing.....	191

Phase 3: Model Integration and Web Application Development	200
SEIR Model Integration.....	201
Prototype Development	205
Deployment.....	219
Phase 4: Testing and Analysis	221
Functional Testing	221
Usability Testing.....	225
Conclusion	245
Recommendations.....	251
References.....	253
Appendices.....	269
Appendix A San Pedro ESU Work Flowchart.....	269
Appendix B SEIR Model of Different Studies	270
Appendix C San Pedro ESU Visualization and Posts.....	278
Appendix D Prototype Graphical User Interface of Web Application	280
Appendix E San Pedro ESU Achievement and Contributions	282

Appendix F Letter of Request to San Pedro Epidemiology and Surveillance Unit.....	284
Appendix G Interview with San Pedro ESU.....	286
Appendix H San Pedro Epidemiology and Surveillance Unit Organization Chart	287
Appendix I Sample DOH COVID-19 Data Drop Record	288
Appendix J Hyperparameter Estimation Techniques Used in Various Studies.....	290
Appendix K Usability Testing Questionnaire.....	291
Appendix L Transcript of Interview	296
Appendix M Transformed Dataset Data Dictionary	303
Appendix N Simulated Annealing Validation on Barangay Level.....	306
Appendix O Screenshot of SEIR-Viz Web Application.....	309
Appendix P Model Instance Generation Flow Chart.....	311
Appendix Q Functional Testing Activity.....	312
Appendix R Usability Testing Answered Forms.....	329
Appendix S ERD and Data Dictionary	347
Appendix T Barangay Level SEIR Predictions	350
Appendix U Usability Testing with San Pedro ESU Documentation.....	351

Appendix V Model Comparison.....	353
Appendix W Dataset After the Cleaning Process	354
Appendix X Object Relational Model of Web Application.....	355
Appendix Y Database Populating Code	356
Appendix Z Use Case Diagram	357
Appendix AA Example Single Page Report.....	358
Appendix BB Validation Result on All Barangay of San Pedro	359
Appendix CC Confidence Result on All Barangay of Biñan	363
Appendix DD Confidence Result on All Barangay of Cabuyao	366

List of Tables

Table 1 Sample csv columns with values after transformation	73
Table 2 BarangayTbl.....	76
Table 3 RecordTbl	77
Table 4 CaseDataTbl.....	77
Table 5 CaseCategoryTbl	77
Table 6 ModelTbl	78
Table 7 Descriptive Statistics for the Given Municipalities after data transformation.....	81
Table 8 SEIR parameters description	84
Table 9 Step by step algorithm of Simulated Annealing	90
Table 10 Gestalt Principles	103
Table 11Map Page Testcases	110
Table 12 Barangay Page Testcases	111
Table 13 Reports Page Testcases	111
Table 14 Model Testcases.....	111
Table 15 Database testing testcases	112

Table 16 ESU Usability Testing Question Statements and their equivalent ISO 9126 characteristics	114
Table 17 Model Expert Usability Testing Question Statements.....	115
Table 18 Third-party Experts Usability Testing Question Statements	117
Table 19 Cronbach's alpha and the internal consistency equivalent.....	119
Table 20 4-Point Likert Scale rating and interpretation.....	120
Table 21 Mean Score range and equivalent agreement interpretation.....	120
Table 22 Data Cleaning Processes and Results	128
Table 23 Data Dictionary of cleaned dataset	130
Table 24 Data Transformation process, iteration, and results	132
Table 25 Transformed data fields and description.....	134
Table 26 Tables used for data store	139
Table 27 Classical SEIR model iteration and hyperparameter referenced values	142
Table 28 Error Metrics of Classical SEIR model iterations	145
Table 29 Simulated Annealing optimized model hyperparameter values	157
Table 30 Classical SEIR and Simulated Annealing model error metric comparison	158

Table 31 Bayesian Inference optimized model hyperparameters	165
Table 32 Classical SEIR and Bayesian optimized model error metric values.....	166
Table 33 Training range test, number of records, and start and end dates	172
Table 34 Barangay-level training with varying time period model and error metrics.....	173
Table 35 Mean score of aggregated results from the four-time range testing on barangay-level	174
Table 36 Age Group Specific with varying time period test models and error metrics	180
Table 37 Difference between models' error metrics on age-specific test	181
Table 38 Overall training with varying time period models and error metrics	184
Table 39 Aggregated mean of error metrics for municipal-level data test	185
Table 40 Aggregated Mean Result of Model Selection Tests	187
Table 41 Optimization Process Details.....	190
Table 42 Summary of error metrics for Classical SEIR and Simulated Annealing model Confidence Test	191
Table 43 Error Reduction Rates for each barangay	194
Table 44 Average TPER and accuracy of Simulated Annealing optimized model	198

Table 45 Model Integration Conditions and Actions.....	203
Table 46 Gestalt principles and actual application on the web application.....	207
Table 47 Map Page testcases	221
Table 48 Barangay Page testcases	222
Table 49 Report Page testcases.....	222
Table 50 Model testcases.....	223
Table 51 Database test	224
Table 52 Usability testing list of respondents and position	225
Table 53 First Usability Test result with San Pedro ESU.....	227
Table 54 Second Usability Test Result with San Pedro ESU.....	229
Table 55 San Pedro Usability Test Cronbach's Alpha	232
Table 56 ISO 9126 Standard, statement codes, mean score	233
Table 57ESU Usability Test Objectives, statement code alignment, and mean score	234
Table 58 Model Expert first Usability Test Result	236
Table 59 Experts Second Usability Test Results	240
Table 60 Experts Second Usability Test Cronbach's Alpha	242

Table 61 Experts Second Usability Test ISO 9126 Characteristics, statement alignment, and mean score	243
--	-----

Table 62 Experts Second Usability Test objective, statement alignment, and mean scores	244
--	-----

List of Figures

<i>Figure 1 Conceptual Framework.....</i>	38
Figure 2 Research Framework	63
<i>Figure 3 General Prototyping Model</i>	64
<i>Figure 4 Data Preparation Phase Flow</i>	65
Figure 5 Entity Relationship Diagram	74
Figure 6 Detailed Data Collection and Transformation Process Flow	80
<i>Figure 7 Math Modeling and Parameter Experimentation Flow</i>	82
<i>Figure 8 Classical SEIR Model</i>	83
Figure 9 Classical SEIR model sample simulation.....	85
Figure 10 SEIR sample approximation of COVID-19 Cases in San Pedro City over 100 days ..	87
Figure 11 Simulated Annealing pseudocode	91
<i>Figure 12 Math Modeling and Parameter Experimentation Process</i>	98
<i>Figure 13 Model Integration and Web Application Development Flow</i>	99
Figure 14 Project File Structure.....	102
Figure 15 Testing and Analysis Flow	108

Figure 16 Test Case Classes file	110
Figure 17 Sample unit test run	112
Figure 18 Prototype Improvement diagram flow.....	122
Figure 19 Summary of methodology phases	125
Figure 20 Data Preparation Phase Flow.....	127
Figure 21 San Pedro City transformed data graph.....	136
Figure 22 Cabuyao City transformed data graph.....	137
Figure 23 Biñan City transformed data graph.....	138
Figure 24 RMSE and MAE of classical SEIR iterations 1 and 2	146
Figure 25 NMAE of classical SEIR iterations 1 and 2	146
Figure 26 SEIR model prediction for iteration 1 and 2 against actual case data	147
Figure 27 Cooling Schedule decrease over iterations.....	152
Figure 28 Score Progression across iteration.....	153
Figure 29 Beta, Sigma, Gamma Evolution across iterations	154
Figure 30 Simulated Annealing Iteration 2 Temperature decay	155
Figure 31 Simulated Annealing Iteration 2 score progression.....	156

Figure 32 Simulated Annealing Iteration 2 hyperparameter accepted values	157
Figure 33 MAE value of classical SEIR iteration 2 and Simulated Annealing optimized model	159
Figure 34 RMSE value of classical SEIR iteration 2 and Simulated Annealing optimized model	160
Figure 35 NMAE values of classical SEIR iteration 2 and Simulated Annealing optimized model	161
Figure 36 classical SEIR iteration 2 and Simulated Annealing model prediction against actual data	162
Figure 37 Bayesian Inference Prior, Likelihood, and Posterior values for Narra, San Antonio, and San Lorenzo Ruiz.....	164
Figure 38 MAE values of classical SEIR iteration 2 and Bayesian Inference model.....	167
Figure 39 RMSE values of classical SEIR iteration 2 and Bayesian Inference model.....	168
Figure 40 NMAE value of classical SEIR iteration 2 and Bayesian Inference model	168
Figure 41 classical SEIR iteration 2 and Bayesian Inference model prediction against actual data	169
Figure 42 Aggregated Mean RMSE and MAE of classical SEIR, Simulated Annealing, and Bayesian Inference model for time-range testing	175

Figure 43 Aggregated Mean NMAE of classical SEIR, Simulated Annealing, and Bayesian Inference model for time-range testing.....	175
Figure 44 Actual against prediction of Simulated Annealing and Bayesian Inference model for barangay Narra.....	176
Figure 45 Actual against prediction of Simulated Annealing and Bayesian Inference model for barangay San Antonio.....	177
Figure 46 Actual against prediction of Simulated Annealing and Bayesian Inference model for barangay San Lorenzo Ruiz.....	178
Figure 47 Mean RMSE and MAE between classical SEIR, Simulated Annealing, and Bayesian Inference on age-specific test.....	182
Figure 48 Mean NMAE between classical SEIR, Simulated Annealing, and Bayesian Inference on age-specific test.....	182
Figure 49 Aggregated mean of RMSE and MAE of classical SEIR, Simulated Annealing, and Bayesian Inference model for municipal-level data test.....	185
Figure 50 Aggregated mean of NMAE of classical SEIR, Simulated Annealing, and Bayesian Inference model for municipal-level data test	186
<i>Figure 51 Aggregated Mean RMSE and MAE of Model Selection Tests</i>	<i>188</i>
Figure 52 Aggregated Mean NMAE of Model Selection Tests	188

Figure 53 Simulated Annealing RMSE values for Confidence Test	192
Figure 54 Simulated Annealing MAE values for Confidence Test	193
Figure 55 Barangay Narra SEIR models predictions.....	195
Figure 56 Barangay San Antonio SEIR models predictions.....	195
Figure 57 Barangay San Lorenzo Ruiz SEIR models predictions.....	196
Figure 58 Barangay Santo Tomas SEIR models predictions.....	197
Figure 59 Barangay Mamatid SEIR models predictions	197
Figure 60 Django project architecture	201
Figure 61 Model Instance flowchart	204
Figure 62 Login Page.....	210
Figure 63 Map Page	210
Figure 64 Dataset Page	211
Figure 65 Django, NGINX, Gunicorn deployment architecture.....	219

List of Appendices

Appendix A San Pedro ESU Work Flowchart.....	269
Appendix B SEIR Model of Different Studies	270
Appendix C San Pedro ESU Visualization and Posts.....	278
Appendix D Prototype Graphical User Interface of Web Application	280
Appendix E San Pedro ESU Achievement and Contributions	282
Appendix F Letter of Request to San Pedro Epidemiology and Surveillance Unit.....	284
Appendix G Interview with San Pedro ESU.....	286
Appendix H San Pedro Epidemiology and Surveillance Unit Organization Chart	287
Appendix I Sample DOH COVID-19 Data Drop Record	288
Appendix J Hyperparameter Estimation Techniques Used in Various Studies.....	290
Appendix K Usability Testing Questionnaire.....	291
Appendix L Transcript of Interview	296
Appendix M Transformed Dataset Data Dictionary	303
Appendix N Simulated Annealing Validation on Barangay Level.....	306
Appendix O Screenshot of SEIR-Viz Web Application.....	309

Appendix P Model Instance Generation Flow Chart	311
Appendix Q Functional Testing Activity	312
Appendix R Usability Testing Answered Forms	329
Appendix S ERD and Data Dictionary	347
Appendix T Barangay Level SEIR Predictions	350
Appendix U Usability Testing with San Pedro ESU Documentation.....	351
Appendix V Model Comparison.....	353
Appendix W Dataset After the Cleaning Process	354
Appendix X Object Relational Model of Web Application.....	355
Appendix Y Database Populating Code	356
Appendix Z Use Case Diagram	357
Appendix AA Example Single Page Report.....	358
Appendix BB Validation Result on All Barangay of San Pedro	359
Appendix CC Confidence Result on All Barangay of Biñan	363
Appendix DD Confidence Result on All Barangay of Cabuyao	366

Introduction

Background of the Study

Epidemics are widespread occurrences of infectious diseases in a particular area at a particular time, with many outbreaks throughout history devastating populations globally (Centers for Disease Control and Prevention, 2012). More than 240 epidemics have been recorded globally, including, the most major epidemic in the history, Spanish Flu or Influenza that has billion cases annually around the world (WHO, 2023). Meanwhile, one of the most lethal epidemics that occurred in the world last 2019 was COVID-19 which has over 776 million confirmed cases and more than 7 million deaths reported globally since the beginning of the pandemic. These epidemics emphasize the critical importance of analyzing disease transmission pathways to effectively mitigate their spread.

Understanding the transmission patterns of infectious diseases at the local, regional, and national levels enhances strategies for controlling their spread (Brachman, 1996). In this context, epidemic modeling, or the application of mathematical models to epidemics, plays a crucial role in comparing, planning, implementing, evaluating, and optimizing programs aimed at detecting, preventing, treating, and managing diseases. These models can be used to design and analyze surveys, suggest key data to collect, identify trends, make general predictions, and estimate uncertainty in forecasts (Vynnycky & White, 2010; Hethcote, 2000; Larremore et al., 2021).

Computational modeling, the implementation of epidemic models in computers, enables thousands of experiments and makes complex scenarios more comprehensible (Taghizadeh & Mohammad-Djafari, 2022; National Institute of Biomedical Imaging and Bioengineering, 2020).

Unlike traditional visualization, which provides static representations of historical data, computational modeling allows dynamic simulation and prediction of complex scenarios such as the spread of disease. One example of epidemic models used to understand and simulate disease transmission is the Susceptible-Exposed-Infectious-Recovered (SEIR) model (Brauer, Castillo-Chavez, & Feng, 2019). Research has shown that epidemic models, when tailored with time-dependent parameters, offer a more accurate representation of outbreaks like COVID-19 (Gleeson, Murphy, O'Brien, & O'Sullivan, 2020; He, Peng, & Su, 2020). Computational model's predictive power and simulation capabilities are essential tools for public health units, enabling them to manage and forecast epidemic trends and play a critical role in public health crisis management (Montori et al., 2018).

Public health units such as the World Health Organization (WHO), the Department of Health (DOH), the DOH Epidemiology Bureau, and the Inter-Agency Task Force for the Management of Emerging Infectious Diseases (IATF-EID) are responsible for creating policies to mitigate the spread of diseases (Haw & Uy, 2020; WHO, n.d.). To effectively fulfill their functions, local counterparts are established to monitor and report cases in specific areas. One such local counterpart is the Epidemiology Surveillance Unit (ESU), which plays a critical role in investigating all reported notifiable diseases and health events of public health concern, particularly those with the potential for outbreaks. Additionally, ESUs are responsible for providing timely and accurate epidemiologic information to various jurisdictions.

One of the ESUs is the San Pedro Epidemiology and Surveillance Unit in San Pedro, Laguna, Philippines. The head of the unit explained that ESUs play a crucial role in providing data-driven insights for policymakers. Their responsibilities include validating daily reported cases

in the area by tests to confirm symptoms and ensure the authenticity of reports. They create clusters and classify infected individuals to create analysis and visualize the spread of disease. Additionally, they prepare daily reports for the authorities. See [Appendix A](#) for their current workflow. According to the head, units like San Pedro ESU heavily rely on epidemiologic tools that supports their mitigation process.

One of the current tools of ESUs is the Epidemic-Prone Disease Case Surveillance Information System (EDCS-IS), which serves as the primary national repository for local public health data. This centralized platform is used to report all notifiable diseases under the Epidemic-Prone Disease Case Surveillance program (Aranda & Humeau, 2022). However, while the system provides tailored forms for case investigation and reporting, it lacks the capability to visualize the current spread of epidemics and does not offer any simulation of disease transmission.

To address this issue, the Assistant Head of the San Pedro ESU developed an Excel file to generate visualizations of potential outbreaks in different areas using data from the EDCS-IS. These visualizations are used to report findings, inform policy decisions for mitigating cases, measure the scope of outbreaks, and guide responsive actions.

Ultimately, epidemiologic tools significantly support the responsibilities of public health units (Aranda & Humeau, 2022). However, the current systems lack features for visualizing epidemic spread and generating insightful reports. Furthermore, the existing systems in the Philippines do not incorporate computational models for predicting disease transmission. Nevertheless, the efforts of public health units and the tools they utilize continue to make an impact in mitigating the spread of diseases.

Statement of the Problem

While the San Pedro ESU is implementing solutions to mitigate disease spread, the need for continuous improvement in implementing solutions is important. The accurate prediction and simulation of infectious disease outbreaks is critical for public health planning and response both in local and global perspective. Among the solutions being adopted around the world, the SEIR model is widely used for simulating the spread of infectious diseases. However, the reliability of the SEIR model heavily depends on the precise estimation of its hyperparameters, namely: the infection rate (β), the incubation rate (σ), and the recovery rate (γ). These parameters play a pivotal role in determining the model's output and, ultimately, its ability to simulate real-world scenarios. To enhance the SEIR model's predictive power, computational approaches can be utilized. Through optimization techniques such as computational methods, the model's hyperparameters can be fine-tuned to more accurately reflect the disease dynamics observed in real-world data. By incorporating these computational methods, the SEIR model becomes a more robust tool for forecasting disease spread.

Despite being a vastly implemented area of research, SEIR modeling for infectious disease has yet to be fully explored. Additionally, there are limitations to existing models and their implementation of simulation and prediction of disease spread which this research aims to address. Notably, there is an evident (1) **Limited epidemiological tool and system fitted for local Epidemiology and Surveillance Unit (ESU) that can help in effective communicating and predicting disease spread as an approach to adhering to the law.** Furthermore, (2) **Existing SEIR models are made generally unoptimized for local and regional level of simulation leading to discrepancies and potential underestimation.** Lastly, there is a (3) **Limited local**

epidemic visualized simulation of disease spread as local ESU relies on data visualization which may provide limited information on disease spread analyzation.

Given the need for improved tools in monitoring, there is an evident **limited epidemiological tool and system fitted for Local Epidemiology and Surveillance Unit (ESU) that can help in effective communicating and predicting disease spread as an approach to adhering to the law.** Epidemiological models such as the SEIR math model are commonly used to simulate disease spread on a general level. The application of this computational model on a local level is limited as there is a lack of tailoring the model to a local level which may provide additional insight to provide effective communication of disease spread. Currently, San Pedro ESU implements data analysis on the collected data by performing clustering to determine cases and creation of graphs and charts using spreadsheet to visualize the spread. However, as mandated by *Republic Act No. 11332 (2018)*, also known as the *Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern Act*, Section 8 discusses the need of established local ESU to ensure the provision of timely, accurate, and reliable epidemiological information. This local DOH counterpart, established under Section 6 of R.A 11332. states that *The DOH and its local counterparts shall ensure that all surveillance and response officers have adequate capacity for mandatory reporting of notifiable diseases, risk assessment, epidemiology, disease surveillance, and response to epidemics and health events of public health concern.* There are areas of improvement for the established ESU as mandated by R.A 11332, which requires an effective system to forecast and communicate outbreaks to the public. Integrating a math model, such as the SEIR model, into their structure could help fill these gaps, offering predictive insights that go beyond relying solely on charts and statistics. This area of improvement may help the local ESU in strategizing, policy making, and decision making based on disease spread prediction as the

absence of an improved surveillance solution may hinder timely detection and response to notifiable diseases and health events of public concern.

Existing SEIR models are made generally unoptimized for local and regional level of simulation leading to discrepancies and potential underestimation. There are existing generalized SEIR models (see Appendix B for the list of SEIR models) that have been applied to simulating COVID-19 spread (López & Rodó, 2021; Peng, Yang, Zhang, Zhuge, & Hong, 2020). One of the studies that used a generalized SEIR model was López & Rodó's (2021) which explored the modification to the SEIR model while using an aggregated dataset on large regions of Spain and Italy. Their model was modified to consider movement of population to quarantining, but the lack of spatial granularity on the dataset has resulted in affecting the accuracy of simulations. In addition, other researchers applied modification on the models by either adding additional compartments or multipliers on parameters. These modifications can be applicable for the generic simulation of epidemic spread. However, studies have suggested that the application and exploration of optimization techniques on hyperparameters provide improvement on the model's accuracy and reliability (Hou & Bidkhor, 2024; Santos, 2022). It has been stated that the SEIR's practical application often faces challenges due to the variability and uncertainty in these parameters (López & Rodó, 2021; Macías, Humberto, Arroyo, & González, 2022; Yang, et al., 2020). While existing methods for parameter estimation show promise, they still exhibit limitations such as potential overestimation or underestimation when applied to real-world data which can often lead to discrepancies (López & Rodó, 2021; Mwalili, Kimathi, Ojiambo, Gathungu, & Mbogo, 2020). This highlights the need for computational models that integrate optimization techniques to refine parameter estimates and reduce discrepancies, ensuring more accurate and reliable predictions in real-world scenarios.

Limited local epidemic visualized simulation of disease spread as local ESU relies on data visualization which may provide limited information on disease spread analyzation.

Other solutions to inform the public of a pandemic have been done in the past through the DOH covid tracker which shows data visualization on active cases, recoveries, and deaths through graphs and charts (Department of Health, n.d.). Presently, San Pedro ESU provides information to the public by communicating through social media (See example of current visualization posted by San Pedro ESU in the Appendix C). Additionally, San Pedro ESU visualize disease spread by manually creating a line graph that compares current cases of a disease in line with the alert and outbreak level of that disease. Other visualization tool has been developed in the past, one of this is Mapúa Malayan Colleges Laguna's COVID-19 Case Tracker which contains case information and visualization of COVID-19 for the province of Laguna, which includes San Pedro City. However, visualization can still lack representation of data that may lead to incorrect perception and decisions (Nguyen, Jung, & Gupta, 2021). Mere data visualization will not be able to provide the spread analysis which is critical to understand for future trends especially for epidemics. Predicting the trajectory of COVID-19 has been an interest of various research, and developing a localized model can provide a more accurate projection by incorporating real-world scenarios (Santos, 2022).

Rationale of the Study

In the wake of pandemics such as COVID-19, the need for accurate epidemic modeling has become critical for public health decision-making, policy development, and resource allocation. SEIR is one of the most widely used computational mathematical models for simulating the spread of infectious diseases (Carcione, Santos, Bagaini, & Ba, 2020). However, its accuracy

and applicability are heavily dependent on the parameters governing disease transmission, incubation, and recovery dynamics, which vary across different populations and environments (López & Rodó, 2021).

Optimizing the SEIR model through estimation techniques allows researchers to fine-tune these variables (Santos G. L., 2022), enabling more precise simulations tailored to specific outbreak scenarios. Also, dynamic visualization of these simulations makes it easier for stakeholders to comprehend complex epidemiological patterns, enabling them to take informed action (López & Rodó, 2021). The groups that benefit from this study are:

Local Governments and Public Health Authorities: Optimizing the hyperparameters of the SEIR model for local contexts provides local decision-makers, DRUs, and ESUs with a tool for simulating disease spread and implementing timely interventions. This tailored model enhances planning and resource allocation, enabling faster and more effective responses to outbreaks, thereby reducing the burden on local healthcare systems. Moreover, it serves as an additional tool that complements existing technologies used in disease mitigation, offering a more comprehensive approach to managing public health crises.

Healthcare Workers and Facilities: If the solution can be fully realized, a tailored SEIR model can serve as a valuable tool for healthcare workers and facilities, enabling them to anticipate surges in cases, prepare for patient capacity needs, allocate medical supplies more efficiently and enhance overall readiness during disease outbreaks.

Citizens: Citizens will benefit from improved public health responses, as the SEIR model, tailored locally, enables more precise and effective measures such as lockdowns, quarantine

policies, and vaccination campaigns. This tailored approach will result in reduced disease transmission and better protection of the community's health. As more citizens observe the benefits of these localized solutions, a ripple effect will occur, extending the positive impact to larger populations and enhancing public health outcomes on a broader scale.

Future Researchers: The results of this study provide a valuable reference for future researchers working on disease modeling, not only for COVID-19 but also for other infectious diseases. While this knowledge can be transferred to different regions and adapted to varying environmental conditions, it is recommended that models be revalidated before application to other diseases to ensure accuracy and effectiveness. This makes the study a useful tool for improving global epidemic preparedness while emphasizing the importance of validation in diverse contexts.

Policymakers and International Organizations: Institutions such as the World Health Organization (WHO) and local policymakers can use the findings from this study to shape public health policies and ensure that future outbreaks are managed more efficiently. The solution's simulations provide valuable input for refining mitigation strategies, helping policymakers make informed decisions that can improve response efforts. Additionally, the study's alignment with SDG 3 (Good Health and Well-Being) underscores its contribution to global health goals, emphasizing the critical role of epidemic preparedness in achieving better health outcomes.

Digital Health Solution Providers: Technology developers and companies working on epidemic surveillance and digital health solutions can leverage the SEIR model's dynamic visualizations to build more intuitive tools for monitoring disease spread, improving epidemic surveillance capabilities across regions.

Thus, this study leverages estimation techniques to optimize the SEIR model for realistic epidemic simulation and visualization. By doing so, it provides a robust tool for predicting the spread of diseases and offers simulations as additional input for improving response strategies. This enables policymakers and health professionals to make more informed decisions and respond more effectively to future outbreaks. The optimized SEIR model can serve as an indispensable resource for enhancing epidemic preparedness and mitigating the impact of infectious diseases locally.

In alignment with the United Nations Sustainable Development Goals (SDGs) these initiatives contribute directly to SDG 3: Good Health and Well-Being by ensuring a better epidemic preparedness, we will create a tool that will improve epidemic surveillance.

Objectives

The study's main objective is to develop and integrate an SEIR model for visualization and simulation of local-level epidemic spread that can help support regional governments in mitigating the spread of epidemics.

The specific objectives are as follows:

- 1. Develop a SEIR model tailored to a local epidemiological and surveillance unit (ESUs) that can help in communicating and predicting disease spread to help adhere to the law.**

The SEIR model will allow local ESUs to visualize and simulate disease transmission effectively by using a dataset from the DOH COVID-19 data drop that is

filtered to the local population and epidemiological trends on the model's development. This ensures the model is highly relevant to the local context, making it an essential tool for accurate disease prediction. Integrating historical local data refines the predictions, offering a method for simulating disease spread progression.

Additionally, unlike manual processes or non-automated methods, the SEIR model may provide data-driven insights, allowing for more informed decision-making. The system is designed to provide additional adherence with legal mandates on *Republic Act No. 11332 Section 6 (a)* that states that *The DOH and its local counterparts are mandated to implement the mandatory reporting of notifiable disease and health events of public health concern*. Additionally, *Section 6 (f)* states that *The DOH and its local counterparts shall ensure that all surveillance and response officers have adequate capacity for mandatory reporting of notifiable diseases, risk assessment, epidemiology, disease surveillance, and response to epidemics and health events of public health concern*. Through this tailored approach, ESUs can meet these requirements while improving their ability to predict disease trends and ensuring public health measures are timely and accurate.

2. Optimize the SEIR model through hyperparameter experimentation that will help mitigate discrepancies on local epidemic simulation.

This objective involves enhancing the standard SEIR model through hyperparameter optimization for enhancing its predictive accuracy by adjusting the key hyperparameters such as infection rate (β), recovery rate (γ), and incubation rate (σ) using optimization techniques while using real-life data on confirmed cases and recoveries. The

SEIR optimization includes the application of Bayesian Inference and Simulated Annealing.

To evaluate the optimized performance of the SEIR model and its adaptability, validation metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Normalized Mean Absolute Error (NMAE) are applied to datasets from Cabuyao and Binan City, distinct from the training dataset of San Pedro, to evaluate its Confidence Level. An optimized model should exhibit low RMSE and MAE values, signifying minimal errors indicating strong alignment with observed trends. Consistency across these metrics between the training and validation datasets will demonstrate the model's performance and adaptability to different local contexts. This approach ensures that the model's performance is not only accurate but also generalizable, supporting its Confidence Level in providing reliable predictions across diverse scenarios.

3. Design a web application to visualize local epidemic simulation of spread by applying and integrating the optimized SEIR model.

This goal focuses on the development of a Python-based web application that can visualize simulation of epidemic spread at local levels. By applying the optimized SEIR model, the web application will generate simulations of disease spread based on real-world data, allowing local ESUs to predict potential future trends. This web application will present data in an intuitive format, including charts and maps, making it easier for health officials to interpret and act upon the model's predictions. The simulation will show a heatmap of the local area and present a time-scaled slider that the user may interact with to visualize the disease spread in each barangay. The simulation can then be interpreted by

the experts of the locale as an additional input for disease spread analyzation. Furthermore, the integration of the optimized SEIR model ensures higher accuracy and adaptability in predicting local epidemic spread. The prototype design of the web-application is seen on Appendix D .

Scope and Delimitations of the Study

This study aims to optimize the SEIR model. The optimized model will be integrated into a Python-based web application that simulates the spread of COVID-19 at a local level and to be used by local ESUs.

This study has the following scope:

- *Optimizing the SEIR Model for Local Disease Spread:* Fitting the SEIR for a local scope will include adjusting the beta, sigma, and gamma hyperparameters through experimentation. Bayesian Inference and Simulated Annealing are the techniques that will be used to adjust these parameters. To train the model, the San Pedro cases will be selected from the obtained dataset. As the locale of the study is the San Pedro ESU, it was deemed appropriate to have the San Pedro cases be the training data.
- *Validation to Determine the Optimization Techniques:* During the training of the SEIR model, error metrics will be used to determine the accuracy of the model after using each of the optimization techniques stated above. The error metrics used to determine the accuracy of the model will be Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Normalized Mean Absolute Error (NMAE). The reason these

techniques were determined to be a reliable way of determining the performance of the SEIR model are seen in Chapter 2, Math Model Validation.

- *Validating the Reliability of the Optimized SEIR Model:* To determine the ability of the SEIR model to maintain a high accuracy and confidence level, the model will be used to simulate the spread of COVID-19 in the city of Biñan and Cabuyao. The techniques that will be used to determine the accuracy and confidence level of the model will be the same as when determining the optimization techniques which are RMSE, MAE, and NMAE. Further information about how the optimization and validation techniques are used for the model is seen in Chapter 3, Phase 2: Math Modeling and Parameter Experimentation.
- *Functionalities of the Web Application:* The developed SEIR model will be integrated into a web application that will be used exclusively by the ESU members. This web application will be deployed within the servers of the ESU and will be accessed through their computers. To create simulations, the users will need to input a dataset of the current cases with the correct format to predict the progression of the disease spread. The web application will have the following tabs that will allow different functionalities:
 - *Spread Tab:* In this tab, the users will be able to see the output of the optimized SEIR model. The output will be visualized using a heat map, line graph, and ranked list. The heat map shows the overall spread of the disease within the city using a map which will be divided by barangays, with each barangay being color-coded from green to red to show the severity of the disease spread. Additionally, a line graph will illustrate the predicted progression of infected people over time. Lastly,

a ranked list will be shown that highlights the barangays with the highest case numbers for the selected date.

- *Barangay Tab*: This tab will show a more detailed view of the output. The users will be able to select a specific barangay and a line graph will show the progression of each compartment of the SEIR within the barangay. Also, several graphs will show the progression of the spread of disease that will be filtered through age. Aside from the graphs, the tab will also show the total current cases, and the total predicted cases expected in the following week. Finally, this tab will show the error rate of the output, allowing users to judge the reliability of the results.
- *Dataset Tab*: This tab will give the users the ability to view the dataset that was imported into the web application. Additionally, the users will be able to edit the information in the dataset.
- *Reports Tab*: The final tab will allow the users to generate a report from the output. This tab will allow the users to download the graphs and number of cases so that it can freely be used as reports for decision making.

To access these functionalities, the users will need to use a registered account to use the web application:

- *Supervisor Account*: This account is intended to be used by the head of the ESU. It will have the following functionalities in the web application: view the map page, view the barangay page, view the dataset page and model history, create reports, and create and delete officer accounts.

- *Officer Account:* This account will have minimum access to the web application. This account will only be able to add records in the dataset page and create reports in the report page.
- *The Dataset Used:* The data that will be used to create and determine the performance of the SEIR mathematical model will be based on the DOH COVID-19 data drop that is publicly available. The dataset contains case information from 2020-2024 in the Philippines.
- *Programming Language for Development:* Python will be utilized by researchers to create the optimized SEIR model and develop the web application. The web framework that will be used for the application is Django. Also, the Python libraries used for creating and integrating the SEIR model will be Geopandas, Scikit-learn, Pandas, NumPy, matplotlib, and SciPy.

In addition to the scope outlined above. Certain aspects of the topic are intentionally limited due to either being irrelevant to the objectives of the study or are too large in scope for the researchers to cover in the paper.

This study has the following delimitations:

- *Not Using Other Mathematical Models:* Other models such as Susceptible-Infectious-Recovered (SIR) were studied but are not used. Through analyzing multiple mathematical models, the researchers have determined that the SEIR model is the best suited for the study because of its ability to consider the incubation period of diseases through its Exposed compartment. Therefore, the specific model used for this study is the SEIR model.

- *Limiting the Optimization Techniques Used:* Many optimization techniques can be used to determine the hyperparameters of the SEIR model. However, for this study, the optimization techniques used will be limited to Bayesian Inference and Simulated Annealing. Optimizing the SEIR and validating its accuracy and reliability can take a significant amount of time. Consequently, the researchers limited the optimization technique to the two techniques, as they were determined to be sufficient techniques through the research done.
- *Specific Disease Covered in the Study:* The developed and optimized SEIR model in this study can be used in other infectious diseases through adjusting its hyperparameters. However, the specific disease used for modeling in this study is COVID-19, as it is the disease present in the dataset used. Thus, even if the SEIR model can be used in other infectious diseases, the model developed in the study will be specialized for COVID-19.
- *Limited Validations:* The optimization of the SEIR model will be based on the San Pedro cases in the dataset. To determine the performance of the model, simulation testing will be performed to determine its confidence level. For this, the City of Biñan and Cabuyao from the dataset will be used for the simulation testing. These specific cities are chosen by its proximity from San Pedro City and having similar number of barangays and cases in the dataset to San Pedro city.

Conceptual Framework

The conceptual framework diagram that the study is based on is seen on the next page (see Figure 1). The framework outlines the problems and the sources of the problems, the solutions, and how they will be developed to achieve the objectives of this study.

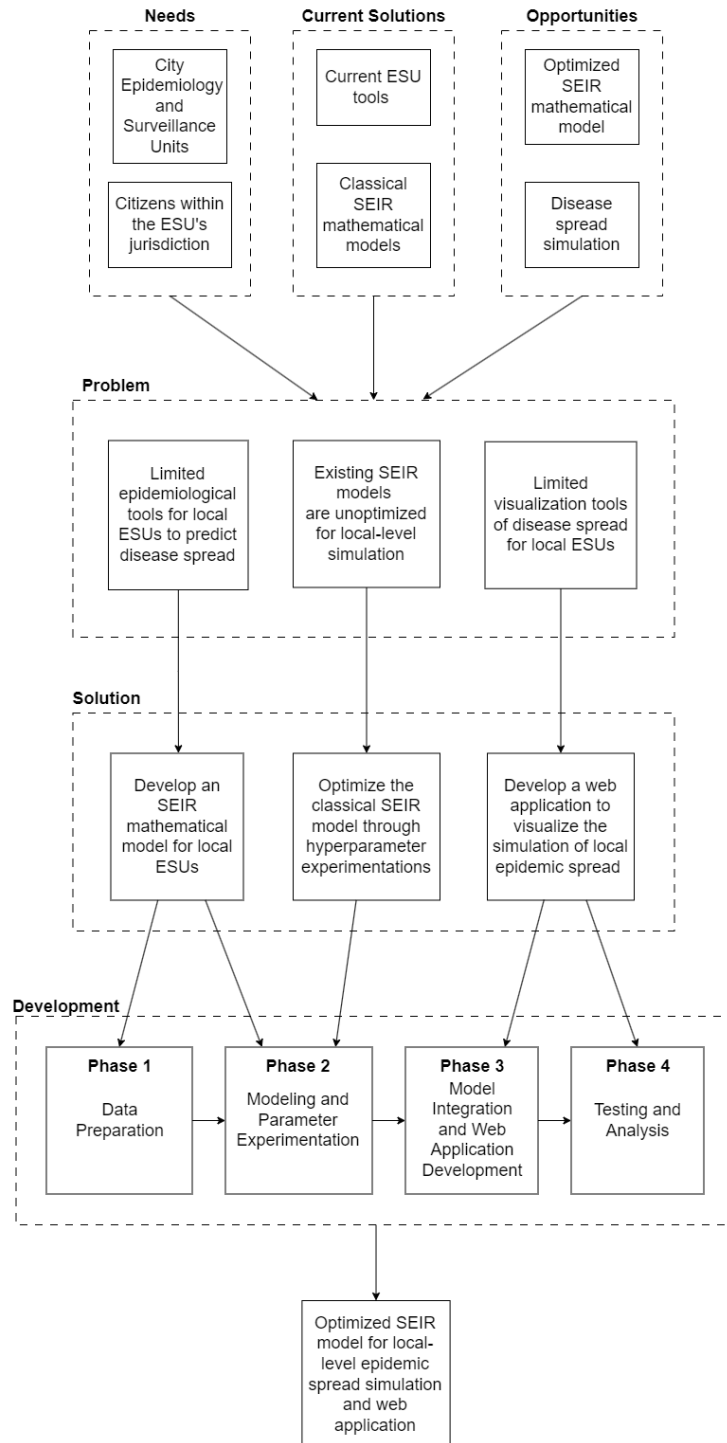


Figure 1 Conceptual Framework

The framework begins with the sources of the problems. The first source is the needs of local ESUs and the population within the ESU's jurisdiction. The second source is from the shortcomings of the current solutions which are the current ESU tools and existing classical SEIR mathematical models. The final source is the opportunities which are an optimized SEIR model and disease spread simulation.

As a result, the outlined sources generated problems which are shown in the framework. The first problem is the limited epidemiological tools for local ESUs to predict disease spread. The second problem is that existing SEIR models are not optimized for local-level simulation. The final problem is the limited visualization of disease spread for local ESUs.

Consequently, solutions have been outlined to address each of the problems stated above. Firstly, the solution for the limited epidemiological tools for local ESUs to predict disease spread is to develop an SEIR model specifically for the use of local ESUs. Next, the problem of existing SEIR models being unoptimized for local-level simulations is addressed by optimizing the classical SEIR model through estimation techniques. Finally, the problem of limited visualization of disease spread for local ESUs is addressed by developing a web application to visualize the simulation of local epidemic spread.

Finally, the development is outlined to reach the objectives of the study. The research is conducted in four phases to achieve the results of the study. The first phase is data preparation, the second phase is modeling and parameter experimentation, the third phase is model integration and web application development, and the last phase is testing and analysis.

The result of the study is an optimized SEIR model for local-level epidemic spread simulation and web application.

Review of Literature

Epidemiology

The spread of diseases is closely monitored and studied to understand their transmission patterns, with mitigation being a key focus of epidemiology. This field develops strategies for prevention, control, and treatment to reduce the impact of diseases on populations worldwide (Columbia University Mailman School of Public Health, 2021). Epidemiology plays a critical role in investigating epidemic outbreaks in any geographic area (Brachman, 1996; Columbia University Mailman School of Public Health, 2021). The spread of an epidemic is influenced by three major interconnected factors: the disease carrier, the method of transmission—whether through direct contact, common vehicles such as food or water, airborne particles, or vectors like insects—and the host who becomes infected (Brachman, 1996). Understanding these links is essential for controlling the spread and mitigating the impacts of epidemics.

One of the primary goals of epidemiology is to create an organized, problem-solving approach to managing epidemics. This process involves several key steps: (1) confirming the existence of the epidemic and verifying the diagnosis; (2) developing a case definition and collecting data on cases; (3) analyzing the data by time, place, and person; (4) formulating a hypothesis; (5) conducting further studies if necessary; (6) implementing control and prevention measures; (7) preparing and distributing a public report; and (8) evaluating the effectiveness of the control and prevention measures (Brachman, 1996).

By systematically addressing these steps, epidemiology provides a structured framework for identifying the causes of epidemics and implementing effective interventions to protect public health.

Public Health Units

Public health surveillance involves the continuous and systematic collection, analysis, and interpretation of data, which is disseminated to those responsible for preventing and managing diseases and injuries. This process is crucial for identifying public health threats and implementing timely interventions to mitigate their impact.

Several countries have developed robust public health surveillance systems to address various health challenges. For instance, China's Field Epidemiology Training Program (FETP) focuses on training public health professionals in epidemiology and outbreak investigation. This program enhances the country's ability to detect and respond to public health threats promptly. Similarly, the U.S. Agency for International Development (USAID) emphasizes using data to improve public health interventions. By leveraging data analytics, USAID aims to enhance the effectiveness of health programs and ensure better health outcomes. The primary objective of public health surveillance is to prevent the spread of epidemics and to intervene quickly to stop the transmission of diseases (Nsubuga, et al., 2006).

Resource-limited settings face significant challenges in public health surveillance, including timely detection of diseases, availability of care, contact tracing, and quarantine and isolation procedures. Limited resources can hinder the prompt identification of disease outbreaks,

delaying necessary interventions. Additionally, access to healthcare services may be restricted, affecting the ability to manage and treat diseases effectively. Efficient contact tracing is essential for controlling the spread of infectious diseases, but it can be challenging in resource-constrained environments. Implementing effective quarantine and isolation measures is crucial for containing outbreaks, yet it can be difficult in areas with limited infrastructure (Madhav, et al., 2017).

Public health units are responsible for managing and mitigating the spread of infectious diseases. These include international agencies like the World Health Organization (WHO) and local health departments in the Philippines, such as the Department of Health (DOH), the DOH-Epidemiology Bureau, and the Inter-Agency Task Force for the Management of Emerging Infectious Diseases (IATF-EID), which was established under the Philippines' Executive Order No. 168 (Haw & Uy, 2020; WHO, n.d.). Local health departments require effective monitoring and surveillance of diseases to fulfill their responsibilities.

To address this need, Disease Reporting Units (DRUs) have been established, encompassing a wide range of facilities such as Barangay Health Stations (BHS), Municipal Health Offices (MHO), City Health Offices (CHO), local hospitals, private clinics, and community and quarantine stations at seaports and airports (Aranda & Humeau, 2022; Haw & Uy, 2020).

Furthermore, *Section 8 of Republic Act No. 11332*, also known as the *Mandatory Reporting of Notifiable Diseases and Health Events of Public Health Concern Act*, established Epidemiology and Surveillance Units (ESUs) that operate at various levels, including city/municipal, provincial, and regional, ensuring a comprehensive surveillance network that can effectively monitor and respond to health threats across different areas of jurisdiction (Aranda & Humeau, 2022).

In the Philippines, COVID-19 surveillance is managed through ESUs at various administrative levels. At the regional level, ESUs oversee the collection and analysis of data from multiple provinces, ensuring a coordinated response to public health threats. Provincial ESUs gather data from cities and municipalities within their jurisdiction, providing a localized approach to disease surveillance. City and municipality-level units for comprehensive analysis and action. By establishing a multi-tiered surveillance system, Philippines aims to enhance its ability to detect, monitor, and respond to public health threats effectively (Haw, Uy, Sy, & Abrigo, 2020).

The Philippines' multi-tiered surveillance system provides a robust framework for monitoring and responding to infectious diseases, enabling data-driven public health interventions. Complementing this surveillance approach, epidemiological models serve as vital tools for interpreting and projecting the data collected at various administrative levels. By integrating surveillance data into these models, public health authorities can simulate disease dynamics, predict potential outbreaks, and evaluate the impact of interventions, such as those tailored to local or regional needs. This synergy between surveillance systems and epidemiological models strengthens the country's capacity to manage public health threats effectively, providing both real-time insights and strategic foresight.

Epidemiological Model

An epidemiological model is a mathematical model used to simulate the spread and transmission characteristics of infectious diseases within populations, such as communities, regions, or countries. These models help in understanding how diseases propagate and in comparing, planning, and evaluating strategies for disease detection, prevention, treatment, and control. By analyzing disease dynamics, epidemiological models provide insights into key data

trends, forecast disease spread, and assess the uncertainty of these predictions, aiding in better public health decision-making (Hethcote, 2000). One of the epidemiological models is the SEIR model (Larremore, et al., 2021).

Epidemiological models, as a subset of mathematical modeling, apply mathematical principles to represent and analyze the dynamics of infectious diseases within populations. These models leverage the foundational concepts of mathematical modeling—such as defining variables, parameters, and relationships—to simulate the spread of diseases and evaluate intervention strategies. By combining disease-specific data with mathematics, epidemiological models provide a structured approach to understanding and predicting the behavior of epidemics. This integration underscores the essential role of mathematical modeling in advancing public health by offering insights that guide evidence-based decision-making.

Mathematical Modeling

Mathematical modeling is the process of using mathematical equations and algorithms to represent real-world systems. It involves identifying the key variables, parameters, relationships, and constraints that govern the system and then formulating them into a mathematical model. The mathematical model is then solved using the appropriate algorithms, afterwards, the results are interpreted to make decisions or recommendations (Brauer, Castillo-Chavez, & Feng, 2019).

Mathematical modeling provides the theoretical foundation for understanding real-world systems, while computational modeling enhances its practical application by leveraging advanced computing technologies.

Computational Modeling

With mathematical modeling being used to solve real-world problems. The rapid advancements in machine learning and mathematical modeling are influencing various research fields. A crucial element of modern mathematical modeling research is its incorporation with supercomputing, which utilizes extensive parallel and distributed computing. The large volume of data and enhanced computational demands necessitate swift computational strategies (Ohue, Sasayama, & Takata, 2024). Computational modeling involves using computers to simulate and analyze complex systems through mathematics, physics, and computer science. A computational model includes numerous variables that describe the system under study. By adjusting these variables individually or in combination, scientists can observe various outcomes. This approach enables the execution of thousands of simulated experiments using a computer. The use of computational modeling is in the field of medical care and research. Tracking infectious diseases by integration of computational models are used to identify the most effective interventions and monitor and adjust these interventions to reduce the spread of the disease. Computational models aid in helping to identify interventions that can be implemented to curb the spread of disease to save lives and reduce stress on the healthcare system (National Institute of Biomedical Imaging and Bioengineering, 2020).

Computational modeling plays a crucial role in disease simulation by enabling the analysis and visualization of complex scenarios, such as the spread of infectious diseases.

Disease Simulation

Models like the SEIR models are used to simulate the spread of diseases. Proper visualization can be a powerful tool in assisting policymakers to make informed decisions on epidemics and effectively communicate its impact to the public (Blanco, Patow, & Pelechano, 2024). The importance of proper visualization of the disease spread simulation can ultimately provide meaningful insight for experts to assess a situation and determine plan of action. The SEIR mathematical model is highly technical and can be difficult for governing bodies to interpret and utilize.

Using SEIR mathematical model to simulate spread. A study by Wang, Miao, Zhang, Wei, & Li (2021) proposed a simple software that simulates spread of diseases by integrating a SEIR model. They used a data-driven model that uses actual epidemic data over a time frame to fit the model's parameters. Their software's aim is to be widely used by the public to conduct simulation and allow analysis on outbreak waves. The development of simulation software allows for visualization on development of epidemic waves which in turn can allow decisions on lockdown and lockdown lifting. Definitively, simulation software may aid in playing an important role in the global fight against infectious diseases.

Additionally, research by Md. Rahman, A. H. M. Islam, & Md. Islam (2021) explored the spatiotemporal simulation of COVID-19 spread. Their study conducted a geospatial analysis and modelling for epidemic dynamics, hotspot identification and vulnerability zoning and was analyzed using Geographical Information System. The results show that the incorporation of spatial variables proves to be useful as an analytical and decision-making tool especially because of the capability of perceiving risks geographically. The integration of space and time-based

characteristics to the model proved to be useful in aiding policy makers in prioritizing resource and formulating effective management and plan of action in order to mitigate COVID-19 problems.

In another study by Bushira and Ongala (2021) attempted to develop a spatial model to represent COVID-19 risk zones in Namibia. The addition of spatial information and GIS analysis allowed the SEIR model to simulate the spread and help identify risk zones. These risk zones aided in classifying highest to lowest risks areas within their target locale. Ultimately, the projection of risks help aid in mitigating COVID-19 spread. The study concluded with the recommendation of application of machine learning for the possibility of increasing the accuracy of the research.

Susceptible-Exposed-Infected-Recovered (SEIR) Model

Given that epidemiological model of SEIR, it compartmentalizes the population in different categories and its transition. The SEIR model is a mathematical model that is derived from the classical SIR Susceptible-Infected-Recovered (SIR) model. This model categorizes individuals into four compartments, namely: Susceptible (S): individuals susceptible to infection, Exposed (E): individuals who are exposed but not infected, Infectious (I): individuals that are infected, Recovered (R): individuals that recovered from the disease. The main difference between the SEIR and SIR is the addition of the Exposed compartment in the SEIR model. This is a compartment for the individuals that are infected but show no symptoms. As a result, SEIR is a widely used model for simulating the spread of diseases with a significant incubation period such as COVID-19 (Taghizadeh & Mohammad-Djafari, 2022).

As the SEIR model is effective in epidemiology studies, given its added compartment in comparison to other models, determining possible applications of the model for epidemiology is important.

Application of SEIR for Epidemiology

For the purpose of this study, it is important to determine the effectiveness of the SEIR model in predicting the spread of COVID-19. Trawicki (2017) explained in their paper how the SEIR has the important distinction of being more suited to modeling for diseases with a significant incubation period because of its Exposed compartment. This makes the SEIR model suitable for COVID-19 simulation, as it has been found that it has an incubation period (He, et al., 2020). However, Trawicki (2017) further explains how the SEIR falls short in completely simulating the COVID-19. Specifically, the model assumes that an individual is only contagious in the infectious period, and that the incubation period is the same for each person. These shortcomings are important to consider. Even so, SEIR provides the closest estimation to the spread of COVID-19.

The effectiveness of the SEIR model in predicting the spread of COVID-19 relies heavily on accurately defining and estimating its parameters. While the model's structure makes it suitable for diseases with significant incubation periods, such as COVID-19, its limitations, like assuming a uniform incubation period and fixed infectiousness, highlight the need for parameter modification and optimization. Adjusting parameters allows the SEIR model to better reflect real-world complexities, improving its accuracy and applicability. Through parameter estimation techniques, researchers can fine-tune the model to account for variability in disease dynamics, enhancing its predictive power and utility in epidemiological studies.

Parameter Modification, Estimation and Optimization

Parameters modification is used in fitting the model into different problems, improving the performance of model, and it makes the model more reproducible (Yang & Shami, 2020). While the parameter estimation solves an optimization problem to find the best model parameters within a permissible range (referred as solution space) by maximizing or minimizing an objective (for example, residual sum of squares) (Misra & Jin, 2021).

Parameter optimization is the problem of choosing a set of optimal hyperparameters for a learning algorithm (Yang & Shami, 2020). Listed are some of the parameter optimizations and what it does to the model. Simulated Annealing escapes local minima by allowing increases in objective function, analogous to thermal fluctuations. Additionally, Bayesian Optimization builds a probabilistic model to identify the most promising regions of the search space (Cottle & Thapa, 2017).

Simulated Annealing

To address parameter optimization problems in the SEIR modeling, optimization techniques can be applied to the model's hyperparameters. One of the optimization techniques is the Simulated Annealing. Simulated annealing is known because of the analogy between simulation of the annealing of solids and the problem solving large combinatorial optimization problems (Ohki & Teferra, 2020). Simulated annealing was created to derive and optimize unknown operational values. This optimization technique has been used in SEIR models to find the values for the hyperparameters (Mulder, 2020; Zhan, Zheng, Lai, Hao, & Li, 2020; Carcione, Santos, Bagaini, & Ba, 2020). One of the studies that applied Simulated Annealing as an

optimization technique to SEIR model is Mulder's (2020). The study explored optimization of unknown operational parameters on several datasets such as the United States of America, Canada, and South Korea. The study's results suggest that Simulated Annealing is a useful tool to derive parameter values especially for large populations of various jurisdictions that the study explored. The optimization was able to determine the values for the production rate, incubation period, and infectious period. The output model was able to acquire a correlation between the recorded data and the model with a relatively good one percent (~1%) or better relative error, aside from the South Korea dataset in the long term. The results propose that the match could show potential use of optimization when developing a model for pandemic's global and regional growth.

Another study that utilized Simulated Annealing as optimization technique is Zhan, Zheng, Lai, Hao, & Li's (2020) research of epidemic spreading dynamics of COVID-19. The integrated model that the study used was a modified model that integrated daily intercity migration. The model was used to predict COVID-19 spread in more than 300 cities in China. Having a large target of space for the model challenged it with having more than 1800 unknown parameters to be estimated from historical data. The model developed was able to gather sufficient information on the epidemic spreading including the infection rates, recovery rates, and percentage of infected population for the 367 cities in China. The study's result shows the capabilities of Simulated Annealing as a method of finding hyperparameter values in an acceptable amount of time with credible validity.

In addition, a study by Carcione, Santos, Bagaini, & Ba (2020) developed an SEIR model to compute the infected population in Lombardy, Italy. The study stated that predicting epidemic spread with high accuracy is difficult due to unknown factors such as degree of distancing,

probability of transmission, and the parameters of the epidemic; these uncertainties are mostly related to parameter beta (β). The study used Simulated Annealing algorithm to fit the model with the Lombardy, Italy's data. The paper concluded that precise determination might be challenging, especially with the lack of knowledge of an epidemic's parameters, but the calculation performed by the model can be useful for simple prediction and understanding of the dynamics of the epidemic.

Bayesian Inference

In addition to Simulated Annealing, Bayesian Inference is an optimization technique that was used to estimate hyperparameters of the SEIR model in the past (Mbuva & Marwala, 2020). Bayesian Inference is a method used to estimate transmission coefficients by combining information from both inputs and outputs, even when data is incomplete. It involves specifying a prior distribution for the input parameters, treating them as independent and using uniform distributions. This method aims to improve the estimation accuracy compared to the least square method, particularly by effectively evaluating the likelihood despite data gaps (Hotta, 2010). In addition, Poole & Raftery (2000) stated that Bayesian inference is used for inference in deterministic simulation models by combining information from both the model and the observed data. It involves melding the prior distribution for inputs and outputs then refining these distributions based on observed data. This optimization approach accounts for uncertainties both in the model input and output which is particularly useful in fields like epidemiology and environmental science where deterministic models are common.

Bayesian Inference has been utilized for epidemiological models like the SEIR model before. A study by Sánchez-Romero, Lego, Prskawetz, & Queiroz (2021) addressed the estimation

of the infection fatality rates (IFR) of COVID-19 by utilizing Bayesian Inference in their SEIR model to estimate the infected population. The technique was applied using the U.S. COVID-19 dataset for the reason of their remarkable regional diversity and size of confirmed cases and deaths. They stated that the implementation of Bayesian Inference method was also considered for the reason of the data on deaths being likely incomplete. The resulting model was able to derive the IFR values for the study to be analyzed.

Another study by Mbuva and Marwala (2020) performed Bayesian inference on parameter by utilizing the Markov Chain Monte Carlo (MCMC) methods on an SIR and SEIR epidemiological models with time varying spread rates on a South Africa dataset. The framework for the Bayesian inference performed for the model parameters allows for the posterior inference of parameters which updates the prior belief based on the data-driven likelihood. The parameter prior distribution for the parameter estimation on the model were applied to the parameters for recovery rate and spreading rate. MCMC was used to sample the posterior which approximates the solution by simulating a random sampling. The reasoning for the applied technique was that MCMC was widely used in COVID-19 parameter inferences. The result of their study performed parameter estimation based on the confirmed cases. The study's conclusion stated that the performed Bayesian inference framework for the estimation of the model allowed them to estimate the effects of government interventions on the dynamic of the pandemic. The study suggested for further improvements the application of the estimation method on regional and provincial studies as data suggests varying spread rates on more local levels of geography.

The use of optimization techniques to estimate the SEIR model's hyperparameters underscores the importance of robust statistical techniques in enhancing the reliability of the

model. By allowing for data-driven updates to parameter estimates, such methods improve the model's accuracy and enable the assessment of interventions' impacts on disease dynamics. However, as with any mathematical model, validating these parameterized models is critical to ensure their predictive accuracy and practical applicability. Incorporating rigorous validation techniques, such as statistical analyses, strengthens the credibility of models in simulating complex scenarios like COVID-19 progression and ensures their effectiveness in guiding public health decisions.

Math Model Validation

Mathematical models are powerful tools that can be applied in a wide variety of cases. However, it is important to evaluate the reliability of the model before they are implemented for practical use. Modeling COVID-19 progression is challenging; therefore, statistical analysis methods must be applied to evaluate the performance of the model (Hemmati-Sarapardeh, Larestani, & Amar, 2020; Dautel, Agyingi, & Pathmanathan, 2023). In this study, three validation techniques will be used.

Root Mean Square Error (RMSE) is a statistical metric that has been used to measure model performance for many research areas including meteorology, air quality, and climate research studies (Hodson, 2022). In RMSE, the expected output is between zero and any positive number, a higher number indicating a poorer performance of the model (Hemmati-Sarapardeh, Larestani, & Amar, 2020). However, errors are squared in RMSE, which means that large deviations have a large effect on evaluating the model. This sensitivity to outliers is a common concern for many researchers, as large errors are severely penalized. However, this also makes RMSE more effective for cases where it is important to avoid errors. A large value in RMSE also

indicates that there may be anomalies in the training dataset that should be addressed (Draxler & Chai, 2014).

Mean Absolute Error (MAE) is another statistical metric that has been widely used. It is a metric that measures the average of absolute error. Like RMSE, MAE is widely used for evaluating the performance of mathematical models. Another similarity is how the output for both can span from zero to any positive value, a higher value indicating a poorer performance (Matsuura & Willmott, 2005). However, there are critical differences between the two: for one, RMSE is more sensitive to outliers in the data and penalizes large errors. Consequently, it is used in cases where large errors can have significant implications. On the other hand, MAE is less sensitive to outliers and treats all errors equally. Therefore, MAE is described as more robust, as it doesn't over-penalize errors like RMSE (Hodson, 2022). Due to this, MAE tends to be closer to the average error, being more indicative of a typical model behavior (Draxler & Chai, 2014). While there are different advantages and disadvantages to using either method, it is a good idea to use both to determine the performance of a model (Hodson, 2022).

Epidemiologic Tools

Epidemiologic tools play a crucial role in transforming theoretical models into practical applications for public health. These tools facilitate data collection, analysis, visualization, and interpretation, enabling a deeper understanding of disease dynamics and the effectiveness of interventions. By bridging the gap between research and practice, epidemiologic tools support evidence-based decision-making and enhance the capacity to monitor, predict, and respond to health challenges. Their integration ensures that complex models can be applied effectively to address real-world public health issues.

Epidemiologic tools are methodologies, techniques, and instruments used to collect, analyze, interpret, and apply data in the field of epidemiology (Mortimer & Borenstein, 2006). These tools enable public health units to monitor and generate insights about the current state of disease spread. For instance, the World Health Organization (WHO) utilizes the Epidemic Intelligence from Open Sources (EIOS) platform to track and manage pandemic risks globally (WHO, n.d.). In the Philippines, local public health units, such as the Epidemiology and Surveillance Units (ESUs), rely on the Philippine Integrated Disease Surveillance and Response (PIDSR) system as the official platform for disease surveillance and daily case recording.

Before the implementation of PIDSR, the Philippines managed three distinct reporting systems: the Notifiable Disease Reporting System, which gathered data on various diseases; the National Epidemic Sentinel Surveillance System, a hospital-based system focused on diseases with outbreak potential; and the Vaccine-Preventable Disease Surveillance, which monitored diseases targeted for eradication through vaccination efforts. Additionally, the HIV/AIDS Registry was used to track HIV/AIDS cases through a voluntary testing program.

The PIDSR system's Epidemic-Prone Disease Case Surveillance Information System (EDCS-IS) now serves as the primary national repository for notifiable disease cases under the EDCS program (Aranda & Humeau, 2022). Additionally, local level solution for COVID-19 monitoring has been done in the past; one solution was Mapúa Malayan Colleges Laguna's COVID-19 Case Tracker that was developed by the College of Computer and Information Science alumni, students, and professors which provided visualizations of COVID-19 information on different cities in the province of Laguna (MMCL College of Computer and Information Science, 2021).

Synthesis

Mathematical modeling, particularly through epidemiological frameworks like the SEIR model, plays a critical role in understanding and mitigating infectious disease spread by simulating transmission dynamics and informing public health strategies. The SEIR model, effective for diseases with incubation periods like COVID-19, provides a structured approach to analyzing and predicting disease patterns (Taghizadeh & Mohammad-Djafari, 2022). Central to the accuracy of these models is parameter estimation and optimization, which involve methods such as Simulated Annealing and Bayesian Inference—each addressing discrepancies between predicted and observed data in unique ways. For instance, Ohki and Teferra (2020) highlight the iterative refinement in simulated annealing to manage complex parameter spaces, while Poole and Raftery (2000) find Bayesian approaches valuable in integrating uncertainty and prior knowledge.

Validation is crucial for model reliability, with metrics like Root Mean Square Error (RMSE) and Mean Absolute Error (MAE), commonly used to assess performance. RMSE, with its sensitivity to outliers, is effective in scenarios where avoiding errors is crucial but can be influenced heavily by large deviations, as Hodson (2022) and Matsuura and Willmott (2005) discuss. MAE, by contrast, provides a stable performance measure that is less affected by outliers (Chicco, Warrens, & Jurman, 2021).

Adding to the model's applicability, public health surveillance, such as that conducted by Epidemiology and Surveillance Units (ESUs) in the Philippines, further underscores the importance of systematic data collection and analysis in health crises, even amid resource constraints (Haw, Uy, Sy, & Abrigo, 2020). By integrating advanced parameter optimization, rigorous validation, spatiotemporal visualization, and epidemiological frameworks, these

approaches enhance public health resilience and enable more effective epidemic control across diverse settings.

Methodology

Research Design

Quantitative methodology involves gathering and analyzing numerical data to address scientific research inquiries. This approach also focuses on creating and implementing models derived from mathematical techniques, theories and hypotheses (Helmold, 2019).

This study used quantitative research design for the purpose of evaluating SEIR modeling which would require numerical analysis. SEIR is a math model itself that requires the use and exploration of different numerical parameters to express theoretical assumptions to real scenarios involving large sets of data. To evaluate the optimized SEIR math model, measures of agreement including accuracy and confidence levels are expected to be evaluated and assessed in the quantitative validation and simulation processes. At the final stage of this study, a Usability Testing (UT) instrument was utilized to evaluate the integrated model and the whole solution, which entails the use of numerical analysis via a Likert Scale questionnaire.

Research Locale

The locale identified for this study is the San Pedro Epidemiology and Surveillance Unit (ESU) located in San Pedro, Laguna, Philippines. It is a unit under the City Health Office of the City of San Pedro responsible for handling information on epidemiological information status, disease outbreak and other public health concern. While San Pedro ESU was selected as the locale specific for this study, the developed solution can still be applicable and used by other ESUs.

Since the study focuses on local and regional epidemic simulation, San Pedro ESU is a notable city unit that consistently responds to, and records reports of any notifiable disease such as dengue, influenza, and COVID-19. Evidently, their diligence earned them an award as an outstanding Epidemiology and Surveillance Unit for the reporting and monitoring of Adverse Event Following Immunization given by the Department of Health (City of San Pedro, 2023). In addition, San Pedro ESU's activity on conducting orientations and providing information on notifiable diseases and improving the unit's capability to provide their services and responsibilities to the public is noteworthy (see Appendix E for San Pedro ESU's recognition).

A letter of request was drafted and sent to the local surveillance unit, seeking their participation in the study. The San Pedro ESU members were needed for the study's prototype usability testing (see Appendix F for the letter of request). The researchers conducted an initial interview with the head of the locale to gather information on the San Pedro ESU's roles, efforts in surveillance, protocols, and the challenges they are facing (see Appendix G for the picture of the interview with the San Pedro ESU). In addition to the head of the ESU, the head assistant was present to give more insights from their role in the ESU (see Appendix H for San Pedro ESU's organizational chart). Their feedback in the interview was critical for determining the certain needs and struggles that the San Pedro ESU, along with other city's ESUs, are currently facing.

All things considered, the San Pedro ESU serves as an excellent reference for the researchers to gather information from and to help test the resulting optimized SEIR math model and web application from this study.

Population and Sampling Method. In this study, the sample population consists of the staff members of San Pedro ESU and third-party experts, as they are the target respondents to the

usability testing questionnaire. Currently, there are three (3) staff members within the ESU. From this population. The ESU head supervises and oversees the day-to-day operations of the ESU, while the head assistant is responsible in conducting data analysis on the data gathered, and lastly, a staff member that encodes the data.

The study uses purposive sampling to select the participating ESU staff. Purposive sampling is a non-probability sampling technique used in research to choose individuals or groups based on specific criteria pertinent to the research question or objectives. This approach is beneficial for the research as the respondents typically possess a deeper understanding of the subject and can offer valuable insights (Hassan, 2024). The researchers opted for this sampling method to gather the most information by selecting the person most involved in ESU's work. For this study, two members of the San Pedro ESU and additional third-party experts were the respondent for the usability testing. For the third-party representative, purposive sampling was also used since the epidemiology experts are knowledgeable and have a deeper understanding of the study. Ultimately, there were two (2) respondents from the ESU and two (2) additional third-party experts. A total of four (4) respondents was included in the study.

Ethical Implication of the Research

To gain some necessary information for this study, the researchers conducted an interview with the San Pedro ESU that was held in the San Pedro City Hall. Their operations can deal with sensitive information about the city, its barangays, and even individuals. Due to this, the

researchers had to exercise ethical integrity in conducting the interview and in how the information from this interview is used in the study.

As stated earlier, a letter of request was given to the ESU politely asking for their participation in the study and ensuring the confidentiality of any sensitive information shared in the interview. The letter of request is seen in Appendix F. During the interview, the members of the ESU showed some information that they stressed was not for public viewing but was still allowed for the researchers to use in the study for additional insights as long as detailed information was omitted. The interviewees, who are knowledgeable about epidemiology and the inner workings of ESUs, shared important information with the researchers which were critical in specifying key parts in the study such as the problems being faced by many of the ESUs. Also, any sensitive information was purposefully cut from the short transcript in the interview. A snippet of the transcription of the interview is seen in Appendix L. All these precautions ensured that no sensitive information was shared in this paper and that confidentiality is upheld.

Research Framework

Figure 2 shows the research framework highlighting the development which involves four phases: Data Preparation, Modeling and Parameter Experimentation, Model Integration and Web Application Development, and Testing and Analysis.

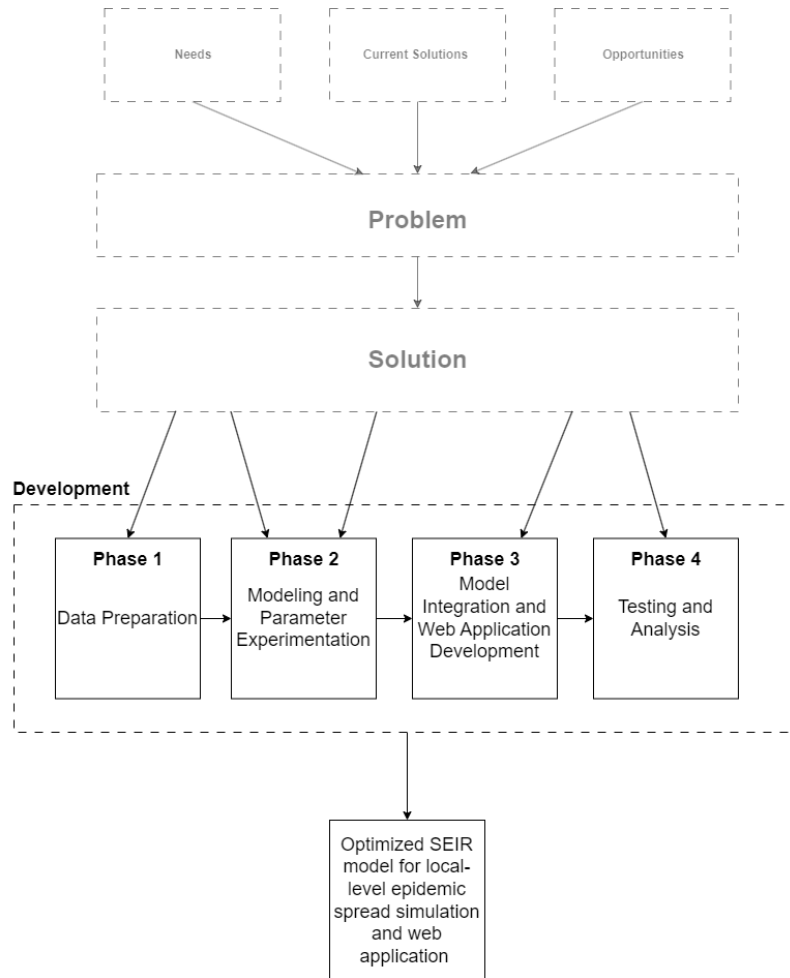


Figure 2 Research Framework

The first phase (Data Preparation) involves gathering the necessary COVID-19 dataset to be used when building the computational model. The second phase (Modeling and Parameter

Experimentation) involves the creation of the SEIR math model and the conversion of the model into an algorithm. This phase also includes using various optimization techniques for parameter experimentation. The third phase (Model Integration and Web Application Development) involves integrating the computational model into a web application. The fourth phase (Testing and Analysis) involves the functional testing of the web application and the usability testing.

The development part of the conceptual framework is anchored on the prototyping model. The prototyping model involves a continual process of developing, evaluating, and refining a prototype until a satisfactory system is developed. This approach is ideal for situations where the detailed project requirements are not fully known and involves an iterative process of trial and error (Lewis, 2023). The general prototyping model is seen in *Figure 3*. The data gathering procedure, tools and analysis are integrated on each phase of the prototyping model.

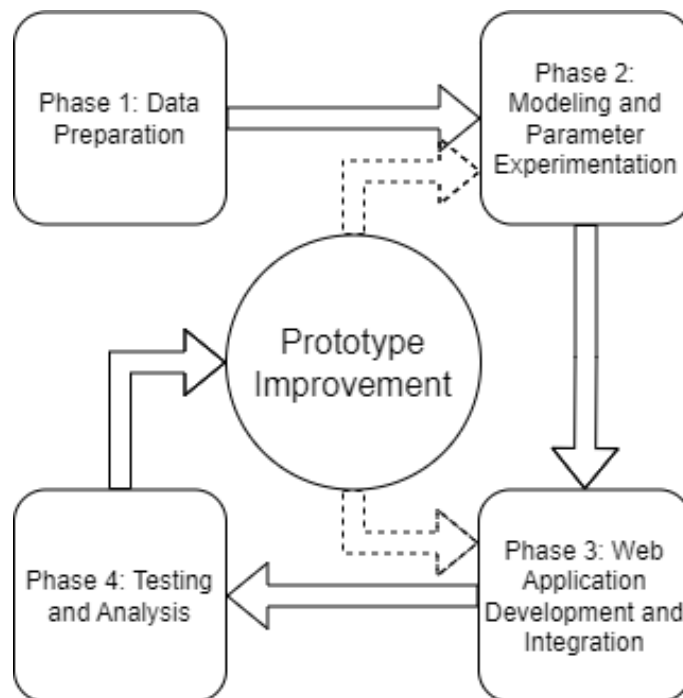


Figure 3 General Prototyping Model

Phase 1: Data Preparation

The data preparation phase includes gathering, cleaning, transformation, and storing of data (Nisbet, Miner, & Yale, 2018). The final output of this phase is to extrapolate relevant data for SEIR modeling and store it in a database (see Figure 4 for Data Preparation flow).

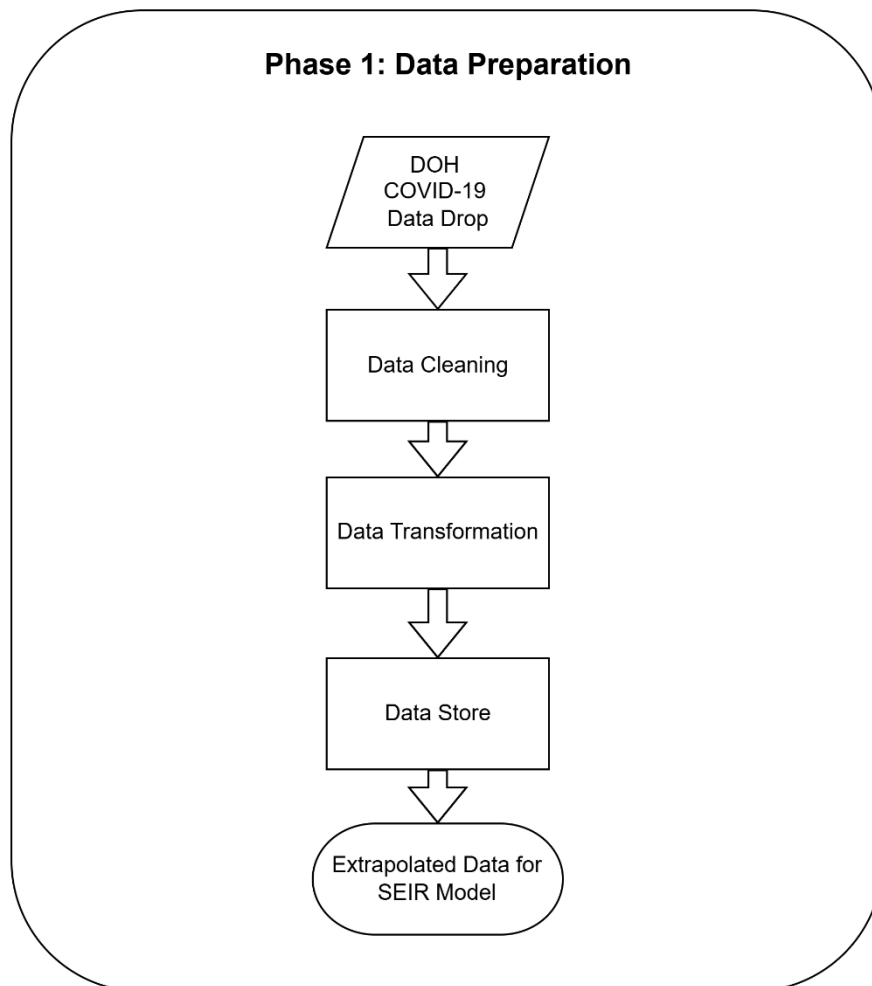


Figure 4 Data Preparation Phase Flow

Data Gathering Tools

In Phase 1, the acquisition of the dataset that was used for modeling undergo data preparation. To do this, different tools and instruments were utilized to complete the requisites needed to move to the next phase.

Dataset

The researchers utilized the publicly available COVID-19 Data Drop from the Department of Health for data collection. The dataset consists of COVID-19 case information as of January 3, 2024 (Department of Health, n.d.). It contains records of confirmed COVID-19 cases in the Philippines from January 2021 to January 2024, covering various regions, municipalities, and barangays including the target locale, which is San Pedro City, along with City of Biñan and Cabuyao for the confiden testing. The dataset also includes personal information such as age, sex, and pregnancy status. Notably, it provides the dates when individuals were exposed, infected, and recovered, which are essential for SEIR modeling. The initial dataset is derived from five batches of case information. The dataset contains 23 attributes namely: *CaseCode*, *Age*, *AgeGroup*, *Sex*, *DateSpecimen*, *DateResultRelease*, *DateRepConf*, *DateDied*, *DateRecover*, *RemovalType*, *DateRepRem*, *Admitted*, *RegionRes*, *ProvRes*, *CityMunRes*, *CityMuniPSGC*, *BarangayRes*, *BarangayPSGC*, *HealthStatus*, *Quarantined*, *DateOnset*, *Pregnanttab*, *ValidationStatus* (see Appendix I for description of fields and sample of records).

Cleaning and Transformation Tools

The data cleaning and transformation was done using Python and its library. The library utilized is Pandas. Pandas was used for data cleaning, exploration and manipulation (W3Schools, n.d.).

Data Gathering Procedure

Once the tools and instruments are loaded, several procedures were performed including Dataset Cleaning and Transformation, until Data Store which includes the process of putting the cleansed and transformed dataset into a selected database server.

Dataset Cleaning and Transformation

The initial dataset from the DOH was cleaned and transformed to a workable format for the SEIR modeling. Below is a detailed breakdown of the data cleaning and transformation process:

Data Cleaning Steps:

1. Reading and Parsing Data Fields
 - Load all five datasets, ensuring that any necessary date columns (such as *DateSpecimen*, *DateRepConf*, and *DateRepRem*) are correctly parsed and converted into the proper date format. This step ensures that these columns are properly typed as datetime for accurate handling in further analysis.

- The dataset 04 Case Information_batch_0 contains 194,574 rows, 04 Case Information_batch_<1-3> each contains 1,000,000 rows, and 04 Case Information_batch_4 contains 136,488 rows.

2. Merge Datasets

- Combine the five datasets into a single comprehensive dataset by merging them vertically, stacking rows from each dataset while aligning columns with the same names. This process results in a unified dataset that preserves all relevant data from each of the original datasets.

3. Remove Duplicate Entries

- Identify and remove duplicate rows based on the *CaseCode* column, keeping only the first occurrence to ensure each record in the dataset is unique.
- A total of 1,408 duplicates were removed, resulting in 3,329,654 remaining rows.

4. Standardize Date Format

- Validate the *DateSpecimen*, *DateRepConf*, and *DateRepRem* columns by checking if the values are acceptable. Invalid values are converted to NaN.

- A total of 701,932 rows in the *DateSpecimen*, 1,774 rows in the *DateRepRem*, and 1 row in the *DateRepConf* were converted to NaN.

5. Remove Invalid Values from *DateRepConf*

- Rows with invalid values in the *DateRepConf* column are removed.
- Only one row was removed.

6. Fill Missing Date Fields

- Fill any missing or empty date fields using domain knowledge or available data. For missing *DateSpecimen* values, researchers calculated the average difference (7 days) between valid *DateRepConf* and *DateSpecimen* and subtracted this from *DateRepConf* to estimate missing *DateSpecimen* values. Missing *DateRepRem* values were filled by adding the average difference between valid *DateRepRem* and *DateRepConf* to the *DateRepConf* (12 days). The incubation and recovery periods used to fill the blank columns are within the bounds established by several studies. According to the Centers for Disease Control and Prevention (2024), the incubation period ranges from 2 to 14 days. Additionally, a study by Lauer et al. (2020) indicates that the recovery period can range from 8.2 to 15.6 days after infection.

- A total of 704,567 missing rows were filled in the *DateSpecimen* column, and 1,774 missing rows were filled in the *DateRepRem* column.

After cleaning the dataset, the researcher selected the City of San Pedro for model training and validation, as it is the chosen locale. Additionally, municipalities with similar proximity and case characteristics, specifically, the City of Biñan and Cabuyao were used for simulation testing. Relevant features were selected from the dataset to match the existing dataset with the ESU. Based on the column descriptions (see Appendix I), The important fields selected was the: *DateSpecimen*, *DateRepConf*, *DateRepRem*, *CityMunRes*, *BarangayRes*, *AgeGroup*, and *Gender*.

Dataset Transformation Steps:

1. Select Relevant Columns of Merged Dataset:

- Select the necessary columns for analysis:
 - *CityMunRes* (City/Municipality of residence)
 - *BarangayRes* (Barangay of residence)
 - *DateSpecimen* (Probable/Exposed individuals)
 - *DateRepConf* (Confirmed/Infectious individuals)
 - *DateRepRem* (Recovered individuals)
 - *AgeGroup* (Age Bin of individuals)

- *Gender (Gender of individuals)*

2. Filter Merged Dataset for Training of SEIR Model, Validation Testing, Simulation Testing:

- Narrow down the Merged Dataset to include only records where *CityMunRes* is "San Pedro City" (for training and validation of the SEIR Model), "City of Biñan", and "Cabuyao City" (for Simulation Testing and Confidence Testing).
- Filtered data including only the 9 months in each local. The first 8 months were used in training the model in different durations, while the last month served as the test set where optimized parameters are applied to validate the model's accuracy.
- A total of 18,010 rows for City of San Pedro, 15,021 rows for Cabuyao City, and 16,692 rows for City of Biñan were selected.

3. Removal of Inconsistent Dates

- Since the SEIR model tracks the progression of individuals between compartments over time, each subsequent state should have a later date than the previous one. Therefore, the researchers removed rows where the *DateSpecimen* is later than the *DateRepConf*, or where the *DateRepConf* is later than the *DateRepRem*.
- There were 15 rows removed from City of San Pedro making it 17,994 rows in total. 13 rows removed from Cabuyao City making

it 15,008 rows in total. 15 rows are moved from City of Biñan making it 16,667 in total.

4. Aggregate and Count Occurrences by Date for Each Municipality:

- Aggregate and summarize event data for each municipality. For each municipality, the number of individuals in the exposed (*DateSpecimen*), infectious (*DateRepConf*), and removed (*DateRepRem*) compartment were counted. An individual is considered in the removed compartment if the date is equal to or later than *DateRepRem*. An individual is included in the infectious compartment if the date is equal to or later than *DateRepConf*. An individual is classified in the exposed state if the date is equal to or later than *DateSpecimen*. Also, the number of individuals in different age groups and gender were counted.
- The final output of this step is a dataset that will be fitted to the SEIR model.

The final output for data cleaning and transformation is the csv as shown in Table 1 that was stored in the database.

Table 1 Sample csv columns with values after transformation

City	Barangay	Date	Count_Exposed	Count_Infectious	Count_Recovered
San Pedro City	Landayan	2023-12-12	15	8	23
City of Biñan	Dela Paz	2023-12-12	10	5	20
Cabuyao City	Pulo	2023-12-12	5	12	18

Refer to Appendix M

Transformed Dataset Data Dictionary for the complete columns of the csv and its descriptions.

Data Store

For the data store process, csv file was stored in a relational database with the following Entity Relational Diagram as show in Figure 5. This involved normalization and creating a robust structure for the data to be ready for later SEIR modeling.

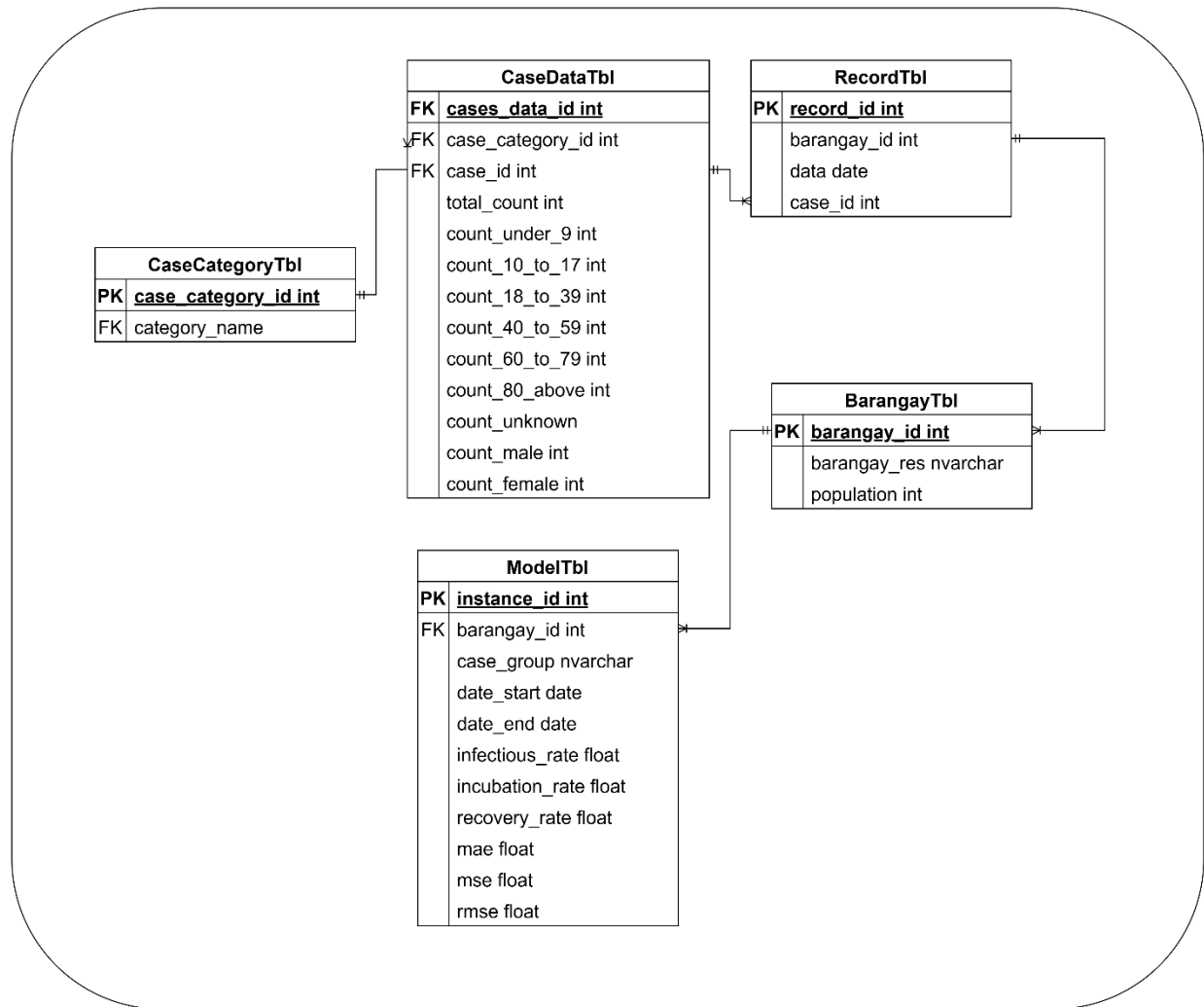


Figure 5 Entity Relationship Diagram

Database Selection

The database that was utilized is PostgreSQL. PostgreSQL was used for SEIR model data storage due to its reliability, advanced querying capabilities, and support for complex data types like JSON and time-series. Its adherence to ACID (Atomicity, Consistency, Isolation, Durability) compliance ensures data integrity, crucial for managing sensitive public health information. With powerful analytics features and seamless integration with Python, PostgreSQL enables efficient data analysis, allowing it to easily retrieve, manipulate, and visualize model outputs.

Data Import and Processing

1. Data Preparation:

- The cleaned and transformed data was prepared for data insertion, adjusting data to the acceptable format of the ERD.
- Python (using pandas) is used to read and manipulate the csv file for direct insertion into PostgreSQL.

2. Database Schema Creation:

- An ERD outlines the relationships and structure of the database, ensuring data normalization. This is illustrated in Figure 5.
- The schema consists of multiple tables to store data in a structured and non-redundant way, ensuring compliance with

Third Normal Form (3NF). This level of normalization is chosen to eliminate redundancy and ensure efficient data retrieval.

Normalization Process

The data is normalized to Third Normal Form (3NF):

1. 1NF: Ensures that each table has a primary key and that each column contains atomic, indivisible values.
2. 2NF: Ensures that each non-key column is fully dependent on the primary key.
3. 3NF: Ensures that no transitive dependencies exist, meaning each non-key column is only dependent on the primary key.

See Table 2, 3, 4, 5, and 6 for the final example table that is used for the solution.

Table 2 BarangayTbl

Barangay_id	Barangay_res
1	Landayan
2	Dela Paz
3	Pulo

Table 3 RecordTbl

Record_id	Barangay_id	Date	Case_id
1	1	2023-12-12	1
2	2	2023-12-12	2
3	3	2023-12-12	3

Table 4 CaseDataTbl

Case_data_id	Case_category_id	Case_id	Total_count	Count_under_9	Count_10_to_17
1	1	1	10	1	2
2	2	2	12	2	2
3	3	3	11	2	2

Count_40_to_59	Count_60_to_79	Count_80_above	Count_unknown	Count_male	Count_female
2	4	1	0	6	4
3	4	1	0	6	6
2	4	1	0	5	6

Table 5 CaseCategoryTbl

Case_category_id	Category_name
1	Exposed
2	Infected
3	Recovered
4	Prediction

Table 6 ModelTbl

Instance_id	Barangay_id	Case_category_id	Case_group	Date_start	Date_end
1	1	1	10_to_17	2023-12-12	2024-03-12
2	1	2	9_below	2023-12-12	2024-03-12
3	1	3	18_to_39	2023-12-12	2024-03-12

Infectious_rate	Incubation_rate	Recovery_rate	MAE	NMAE	RMSE
0.23	0.42	0.21	200.30	1.10	210.48
0.43	0.18	0.8	280.30	1.30	320.48
0.79	0.32	0.7	170.27	0.87	200.12

Data Insertion Workflow

1. Create Tables:

- SQL commands were used to create tables based on the ERD.
- Primary and foreign keys is defined to establish relationships among tables.
- Constraints were included in creating the table using SQL commands.

2. Data Import:

- Using Python, the csv file was read and converted into SQL commands to insert data into the respective tables.
- COPY command or INSERT INTO statements was used for efficient bulk insertion.

3. **Verification:**

- After insertion, SQL queries verified that data has been normalized and correctly inserted.

Once the data is normalized and stored in the database, it is ready for Phase

2. The organized data structure supports efficient querying and retrieval of required records for SEIR modeling.

To summarize the phase 1 processes, refer to Figure 6 for the process diagram.

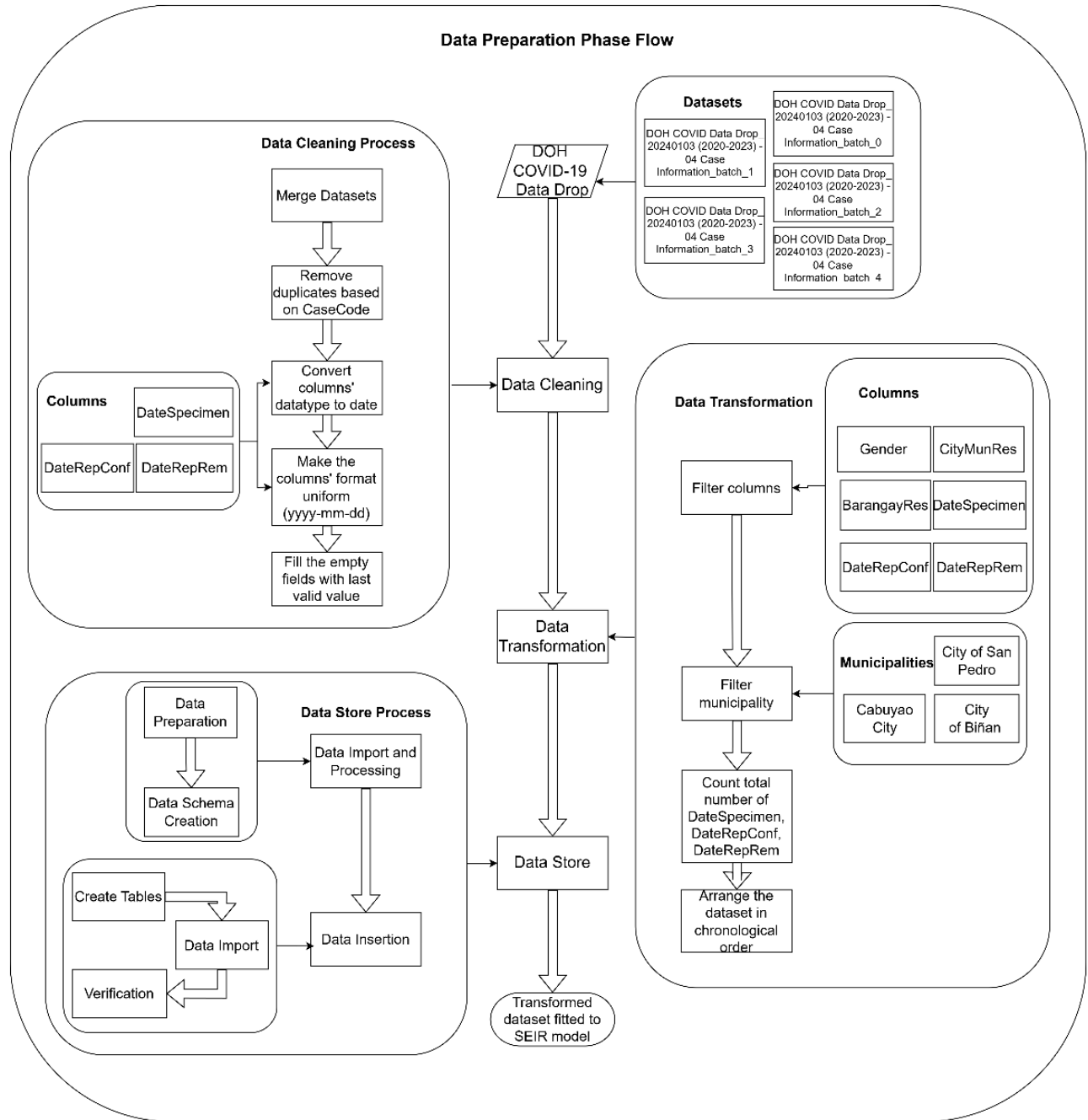


Figure 6 Detailed Data Collection and Transformation Process Flow

Data Analysis

The dataset is transformed for fitting SEIR model. San Pedro City serves as the basis for developing the SEIR model, which undergo the hyperparameter optimization and validation. In addition, City of Cabuyao and Biñan was used for simulation testing and confidence testing to evaluate the robustness and generalization of the model.

Table 7 shows the descriptive statistics for the given municipalities. It includes values on contribution on the overall case count and provide insights into the patterns of case reporting across time and geography.

Table 7 Descriptive Statistics for the Given Municipalities after data transformation

Municipality	Days	Barangays	Total Cases	Date Started of Cases	Latest Date of Cases	Mean Cases	Std. Dev of Cases
Cabuyao City	1,211	17	13,768	2020-03-08	2023-07-01	10.46	23.96
City of Biñan	1,433	20	13,841	2020-03-09	2024-02-09	6.22	18.61
City of San Pedro	1,402	27	16,359	2020-03-09	2024-01-09	6.94	23.03
Overall	1,434	64	43,968	2020-07-14	2024-02-08	7.5	21.96

Now the data is cleaned and transformed, it can move to the next phase of the study.

The data from the database was used for creating and testing the SEIR model.

Phase 2: Modeling and Parameter Experimentation

The second phase as seen in Figure 7 includes the creation of the final SEIR mathematical model to be utilized in the next phase. This phase started with the development of the model, then the hyperparameter estimation using various estimation techniques, the model validation using different validation techniques to ensure the model's accuracy in comparison to real-life data, and lastly, the simulation testing to measure the confidence of the model.

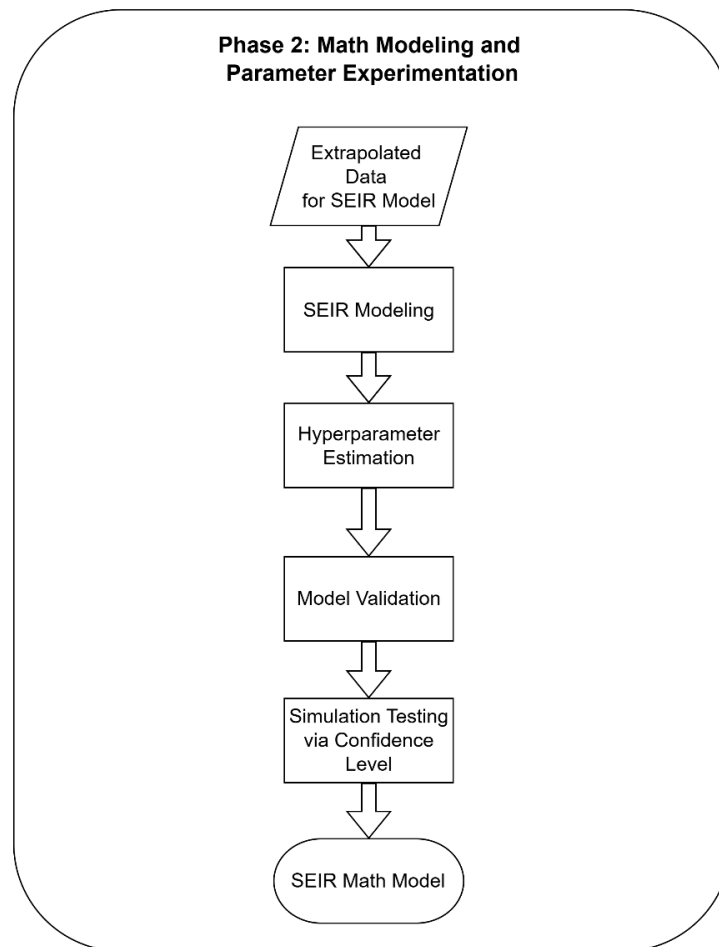


Figure 7 Math Modeling and Parameter Experimentation Flow

Data Gathering Tools

This research utilized the SEIR model for the math modeling and parameter experimentation phase. This is then subjected to the data gathering procedure for the hyperparameter estimation.

The SEIR Model

Using the classical SEIR model as the base model for predicting and simulation of the spread. The movement of individuals through each compartment in the model is seen in Figure 8. The line shows the transition from each compartment in the Susceptible – Exposed – Infectious – Recovered of the population.

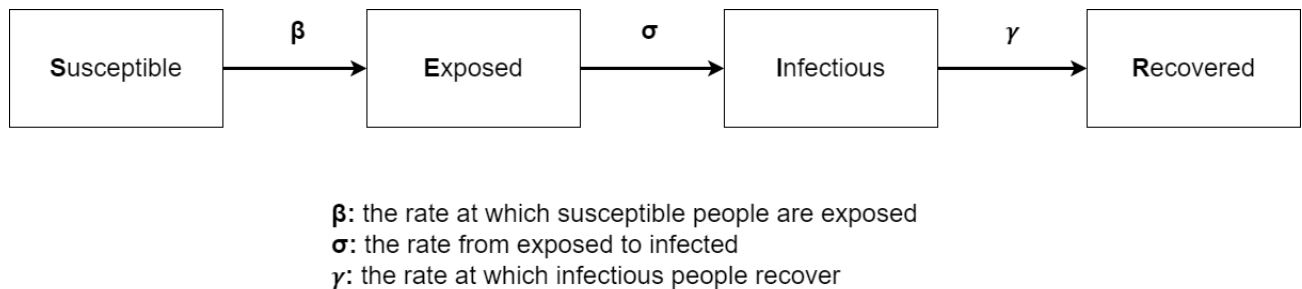


Figure 8 Classical SEIR Model

The infectious rate, β , represents the rate of spread in which the probability of transmitting the disease between a susceptible and infectious individual. The

incubation rate, σ , is the rate of an exposed individual becoming infectious. Lastly, the recovery rate, $\Upsilon = 1/D$, is determined by the average duration, D , of infection (IDM, n.d.).

In a closed population, the SEIR model in formula becomes:

$$\begin{aligned}\frac{dS}{dt} &= -\frac{\beta SI}{N} \\ \frac{dE}{dt} &= \frac{\beta SI}{N} - \sigma E \\ \frac{dI}{dt} &= \sigma E - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

Where $N = S + E + I + R$ is the total population. Table 8 gives a description of the parameters.

Table 8 SEIR parameters description

Parameter	Description
S	Susceptible individuals
E	Exposed individuals
I	Infected individuals
R	Recovered individuals
β	Infectious rate
σ	Incubation rate
Υ	Recovery rate
N	Total population

As shown in Figure 9 the classical SEIR model predicts the number of exposed, infectious, and recovered individuals over time based on the infectious

rate, incubation rate, and recovery rate hyperparameters. The figure used default values to simulate the model's predictions, with initial values for S, E, I, and R, along with default β , σ , and γ . The lines indicate the number of people in each population (susceptible, exposed, infected, recovered) over time.

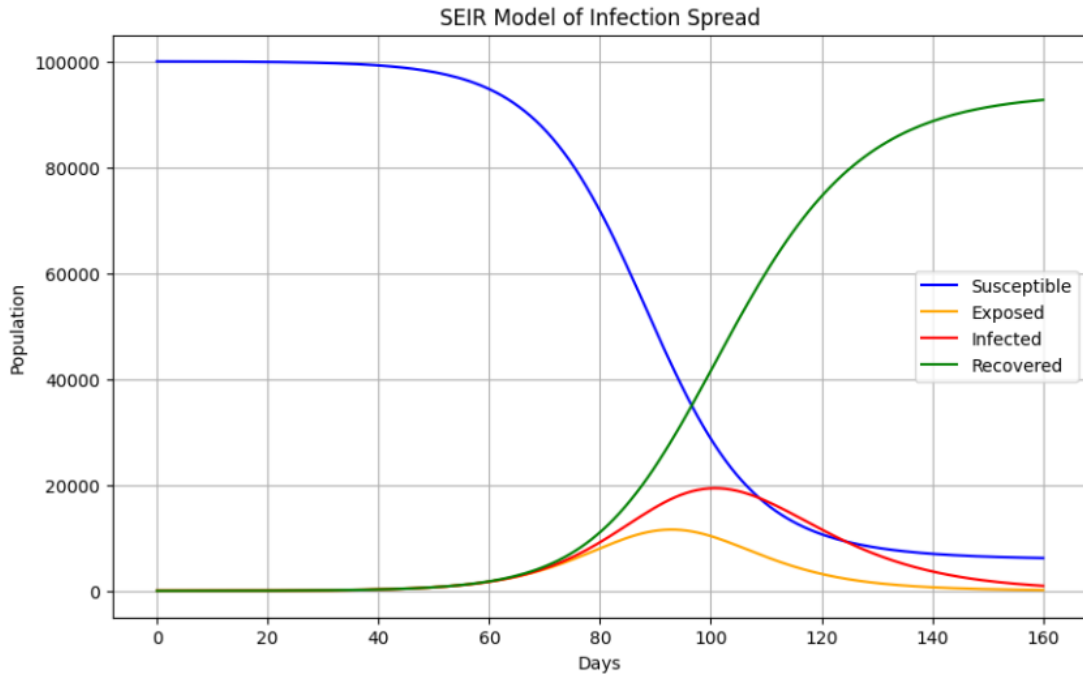


Figure 9 Classical SEIR model sample simulation

Susceptible individuals become exposed when they encounter an infected person. The rate at which this happens is governed by β , which represents the probability of transmission per contact between a susceptible and an infected person. Mathematically, the number of people moving from the Susceptible (S) compartment to the Exposed (E) compartment over time is proportional to both the

number of susceptible individuals and the number of infected individuals. This can be expressed as:

$$dS/dt = -\beta \cdot S \cdot I$$

The higher the β , the faster the susceptible population decreases, and the exposed population increases. After a certain period, the exposed individuals become infectious. The rate at which this happens is determined by σ . Mathematically, this transition is described as:

$$dE/dt = \beta \cdot S \cdot I - \sigma \cdot E$$

A higher σ means a shorter latent period (the disease incubates quickly), causing exposed individuals to become infected more rapidly. Infected individuals can spread the disease. Over time, they either recover or pass away (in the SEIR model, all infections lead to recovery for simplicity). The rate at which infected individuals recover is determined by γ , which is the reciprocal of the average duration of infection ($1/\gamma$). Mathematically, the rate of recovery is:

$$dI/dt = \sigma \cdot E - \gamma \cdot I$$

A higher γ means faster recovery, the infected population decreases more quickly as individuals move to the recovered compartment.

Figure 10 illustrates how the SEIR model can approximate the actual number of COVID-19 cases in San Pedro City over 100 days. While the model provides an estimate of the overall trend, real-world complexities such as changing

public health policies, contact tracing, and community behavior can cause deviations from the predicted outcomes (Maged, Ahmed, Haridy, Baker, & Xie, 2023). Nevertheless, the SEIR model serves as a valuable tool for forecasting and understanding the potential spread of infectious diseases within a population (Girardi & Gaetan, 2023).

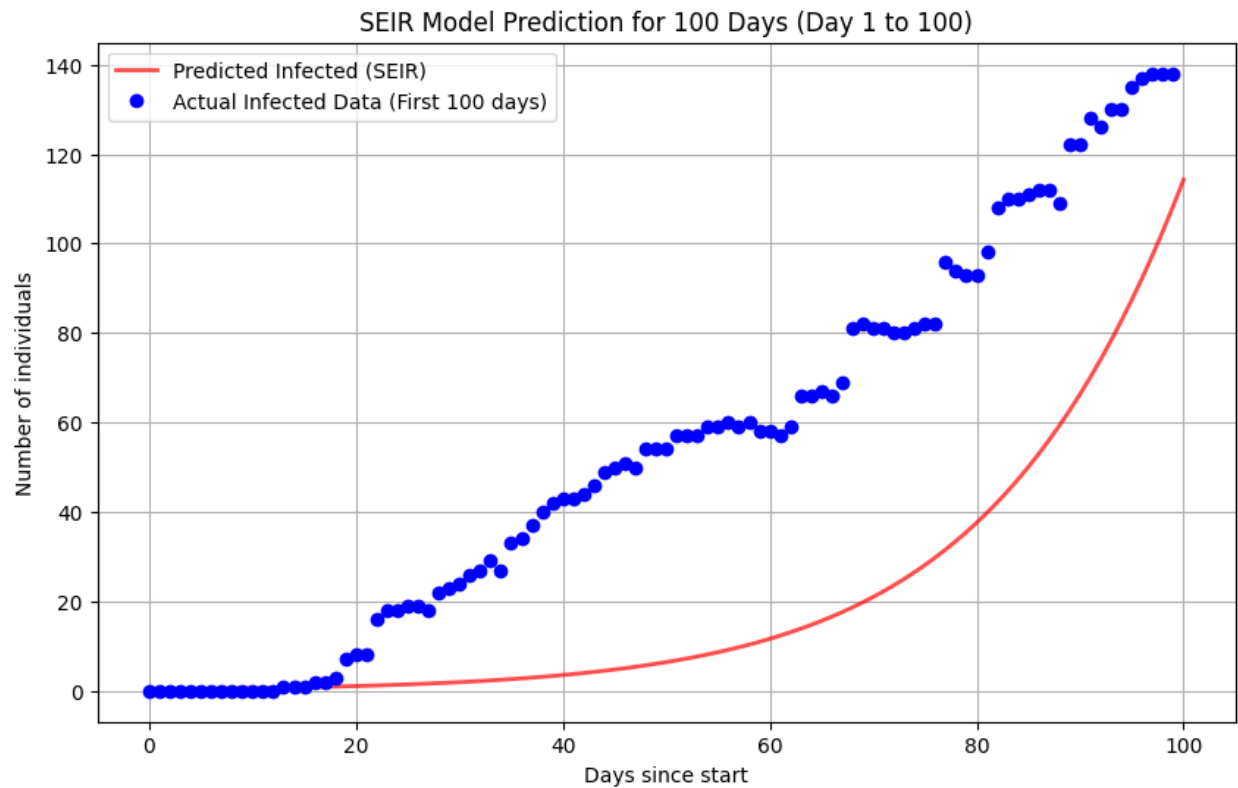


Figure 10 SEIR sample approximation of COVID-19 Cases in San Pedro City over 100 days

While the classical SEIR model can predict these values, its accuracy requires improvement. To enhance the model's accuracy, hyperparameters should not rely on default values. Instead, more accurate hyperparameter estimation methods should be applied to improve the precision of the classical SEIR model

(Pinter, Felde, Mosavi, Ghamisi, & Richard, 2020). The focus of Phase 2 is the hyperparameter estimation by experimenting on various optimization techniques to improve the model's accuracy.

Data Gathering Procedure

The data gathering procedure for Phase 2 involves the estimation of hyperparameters using optimization techniques applied to the classical SEIR model. This step is crucial for conducting accurate and reliable data analysis in this phase.

Hyperparameter Estimation. Optimization techniques was used to estimate the hyperparameters of the SEIR model (Lopes, Bello Moreira, & Schmidt, 1998; West, 1992). The uncertainty inherent in the hyperparameters, specifically the infectious rate, incubation rate, and recovery rate, was addressed through optimization techniques. By leveraging optimization methods, these key parameters was fine-tuned to reflect the most accurate representation of the observed data. This enhanced the precision of the SEIR model and ensured robust results for subsequent analysis. Optimization techniques that was used are Simulated Annealing (SA) and Bayesian Inference. See [Appendix J](#) for the studies that use these optimization techniques.

Simulated annealing (SA) is an optimization algorithm inspired by the physical process of annealing, where a material is heated and then slowly cooled to reduce defects (Ohki & Teferra, 2020). In SA, a solution's "energy" is minimized by exploring possible solutions, allowing occasional increases in energy (uphill

moves) to escape local optima. Over time, the probability of accepting worse solutions decreases as the "temperature" lowers. In an SEIR model, SA can optimize parameters like infection rates by minimizing error between predicted and observed data.

Simulated annealing (SA) starts with an initial set of SEIR model parameters (such as infection rate β , incubation rate σ , and recovery rate γ). The algorithm sets an initial value of temperature T_0 and perturbs these parameters to generate new solutions. The error, calculated as the difference between model predictions and observed data, is used to evaluate the solution's quality. Worse solutions can be accepted based on a probability determined by

$$P = e^{-\Delta E/kT}$$

which is compared to a random value R , where ΔE is the difference in the objective function (error) between current and new solutions, T is the current solution that controls the acceptance of worse solutions, k is the Boltzmann constant, and R is a randomly generated value between 0 and 1. If $R \leq P$ the worse solution is accepted (Kirkpatrick, Gelatt Jr., & Vecchi, 1983; Hosseinabadi, Alavipour, Band, & Balas, 2016). The accepted values' error is gradually lowered, and the process repeats until the system converges to an optimal solution, providing the best-fit parameters for the SEIR model. Table 9, on the next page, shows the step-by-step algorithm of simulated annealing while Figure 11 shows the pseudocode.

Table 9 Step by step algorithm of Simulated Annealing

Step	Description	Current Parameters	ΔE	Temperature (T)	Acceptance Probability	Accept or Reject
1	Initialize starting parameters, compute error	Initial values	N/A	Starting T_0	N/A	Initial values are set
2	Perturb parameters to create a new solution	Current values	$\Delta E = \text{New Error} - \text{Current Error}$	T_0	$P = e^{-\Delta E/T}$,	Accept if $R \leq P$
3	Evaluate new solution: calculate New Error, ΔE , and P	Updated current values	ΔE	Decrease T	If $\Delta E \leq 0$, $P=1$; else, calculate and compare	Accept if $R \leq P$ or reject
4	Acceptance Decision: accept/reject based on R and P	Keep accepted parameters	N/A	Continue to decrease T	N/A	Update current values if accepted
5	Repeat: Perturb parameters, calculate errors, ΔE , and P for each iteration until stopping criteria	Latest accepted values	ΔE	Continue cooling T	Adjust acceptance based on temperature	Stop if criteria met (convergence T threshold, max iterations)

Initialize parameters;

S = generate initial solution ();

T = T₀;

While (T < T_{final})

{

Until (N ≤ I – Iter)

 {

 Generate solution S' in the neighborhood of S

If f(S') < (f(S))

 S ← S'

else

$\Delta = f(S') - f(S)$

 r = random();

 if (r < exp(- $\Delta/k \cdot T$))

 S ← S'

 }

 T = a × T;

}

Figure 11 Simulated Annealing pseudocode

Bayesian Inference is a statistical method for combining expert knowledge with data in mathematical models. It works by treating both model inputs (parameters) and outputs (predictions) as uncertain quantities and applies Bayesian inference to update beliefs about these inputs based on observed data (Hotta, 2010). In an SEIR model, Bayesian inference can be used to incorporate prior knowledge about disease infection rates, incubation periods, and recovery rates, then update these parameters as new data on infection rates becomes available, yielding more accurate predictions of epidemic dynamics.

Bayesian Inference utilizes Bayes' Theorem to integrate prior knowledge about the model parameters with observed data to derive updated, or posterior distributions for those parameters. The core formula for this process is expressed as:

$$P(\theta|D) \propto P(D|\theta)P(\theta)$$

In this equation, $P(\theta|D)$ represents the posterior distribution of the parameters θ (which includes β, σ, γ), given the observed data D . The $P(D|\theta)$ denotes the likelihood, reflecting how well the model predicts the observed data given the parameters. Lastly $P(\theta)$, signifies the prior distribution, encapsulating the initial beliefs about the parameters before observing the data.

The acceptance probability α of the posterior distribution will be established through sampling method using Markov Chain Monte Carlo (MCMC). MCMC provides a mechanism for approximating the posterior distribution by generating

samples that represent the parameter space, allowing for a comprehensive exploration of the uncertainties associated with the model parameters (Columbia Mailman School of Public Health, n.d.).

The MCMC method, particularly the Metropolis-Hastings algorithm, was utilized to sample from the posterior distribution of parameters in the Bayesian Inference process. The first step involved initializing the parameters using their prior distributions. Next, for many iterations, the new parameter values will be computed based on the SEIR model and the prior distribution of the parameters. For each set of computed values, the log-likelihood of the observed data given these predictions will be evaluated. The Metropolis-Hastings criterion will then be applied to determine whether to accept or reject the new parameter values, calculated using the formula:

$$\alpha = \min \left(1, \frac{P(D|\theta_n)P(\theta_n)}{P(D|\theta_c)P(\theta_c)} \right)$$

Once the new parameters are accepted, they will become the current values for the next iteration; if not, the current values will be retained. This iterative process will continue, producing a chain of samples that approximate the posterior distribution of the parameters. Finally, the collected samples will be analyzed to extract insights about the parameter estimates and their uncertainty, enhancing the model's predictive capability and reliability.

Data Analysis

After applying estimation techniques, it is essential to conduct a thorough analysis through validation and simulation to ensure the robustness and accuracy of the estimated values. Estimation techniques often rely on assumptions, approximations, or simplified models that may not fully capture real-world complexities (Selic & Gerard, 2014). Validation is performed to assess the accuracy of the parameter estimations against actual test set of the same dataset, verifying their alignment. Simulation then provides a controlled, variable-rich environment that mirrors real-world conditions, allowing for assessment of the confidence in these techniques across diverse scenarios. This combined approach of validation and simulation is crucial for confirming that the estimations are not only theoretically sound but also practically applicable (The National Academies Press, 1998).

Model Validation. Statistical validation ensures accuracy across various measurements and establishes the credibility of the mathematical model (Smith, 1998). The optimization techniques applied to the SEIR model was validated to confirm their accuracy. Various statistical validation metrics will be computed, including *Root Mean Square Error (RMSE)*, *Mean Absolute Error (MAE)*, and *Normalized Mean Absolute Error (NMAE)* to assess the model's performance.

The RMSE measures the square root of the average squared differences between the predicted and observed values, penalizing larger errors more severely (Hemmati-Sarapardeh, Larestani, & Amar, 2020). RMSE can be calculated by

$$\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2},$$

where y_i is the actual (observed) value, $\hat{y}_i$ is the predicted value from the model, n is the number of observations.

Based on the study "A multi-stage SEIR model to predict the potential of a new COVID-19 wave in KSA after lifting all travel restrictions" (Khedher, Lioua, & Alsaif, 2021) , the RMSE of the model using optimized hyperparameter was 419 with 95% confidence. This threshold will be used to set an acceptable accuracy rate for the model, ensuring predictions remain within a reasonable error range.

MAE measures the average absolute difference between predicted and actual values, without considering the direction of the error (Chugh, 2020). MAE is computed as follows

$$\frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

To ensure model reliability, the MAE were assessed, with predictions considered acceptable if they fall within a 324 MAE threshold, ensuring that, on average, the prediction error remains minimal and aligned with observed values.

The results of these validation metrics was recorded and compared across SEIR models that use different optimization techniques to identify the ideal model for simulating disease spread. After calculating statistical measures for each SEIR model with optimization techniques, both simulated annealing and Bayesian inference optimization

techniques was evaluated using RMSE, MAE and NMAE metrics. These comparisons determined which model provides the most accurate fit and was selected for simulation testing phase.

Simulation Testing via Confidence Level. The model with the highest accuracy will undergo simulation testing to determine its confidence level, which refers to the degree of certainty that the model's predictions are reliable and consistent when applied to new data (Gass & Joel, 1981). This phase involves applying the model, with optimized parameters, to the datasets of the City of Biñan and Cabuyao. During simulation testing, various validation metrics such as RMSE, MAE, and NMAE will be computed to assess the model's predictive accuracy which was also used in model validation. For the model to qualify for use in Phase 3, it must maintain the acceptable rate used in validation. This ensures that the model's confidence level is high enough to make reliable predictions in subsequent phases.

In summary, the ideal SEIR model was selected after rigorous validation and hyperparameter optimization to yield the most accurate fit to the DOH COVID-19 dataset. Hyperparameters, including infection rate, recovery rate, and incubation rate, was fine-tuned to maximize predictive performance. This process involved iterating through possible values to minimize RMSE and MAE and achieve a acceptable NMAE value.

An acceptable RMSE demonstrates that the deviation between predicted and actual cases is within acceptable limits, while an acceptable MAE confirms that the average error remains within established bounds. Upon achieving an acceptable NMAE, the model progressed to further testing. To ensure reliability, the model was applied to data from the

City of Biñan and Cabuyao in a simulation testing phase. During this phase, the model is expected to maintain low RMSE and MAE values and affirming its high predictive accuracy across different datasets.

Once optimized, this SEIR model is anticipated to meet the established accuracy threshold for Phase 3, confirming its confidence level and suitability for future predictions. The model's consistent performance across varied datasets will reinforce its robustness, allowing it to serve as a reliable tool in simulating and predicting disease spread dynamics for more informed decision-making and response strategies (See Figure 12 for the summary of the Phase 2 process).

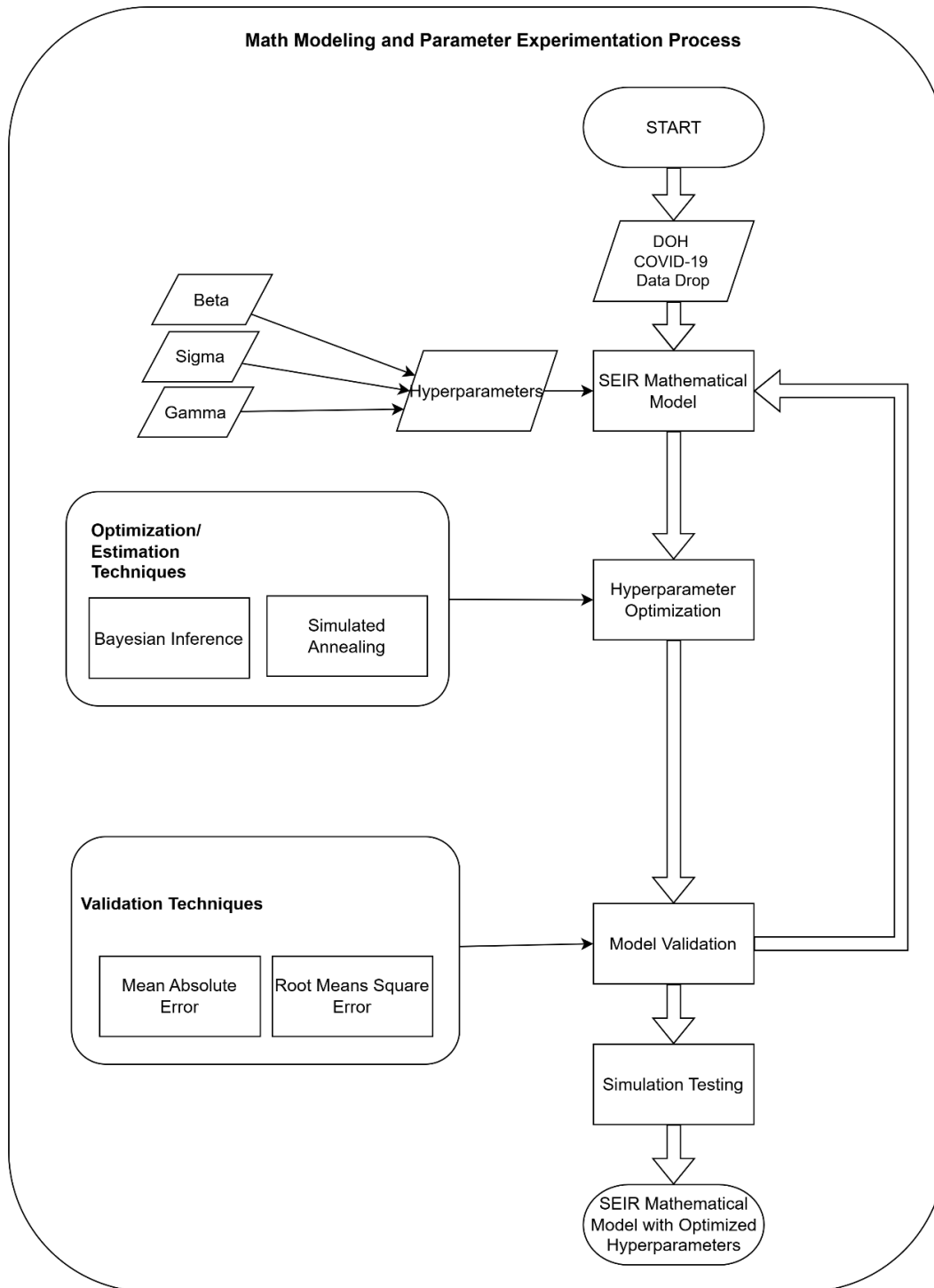


Figure 12 Math Modeling and Parameter Experimentation Process

Phase 3: Model Integration and Web Application Development

Once the data preparation and math modeling and parameter experimentation phases are finished, the study continues with the model integration and web application development phase. This phase involved integrating the developed math model into a working prototype web application. Development of this application follows a modified prototype methodology where the development was looped until the application was deemed ready for deployment. Figure 13 illustrates the steps of developing the prototype.

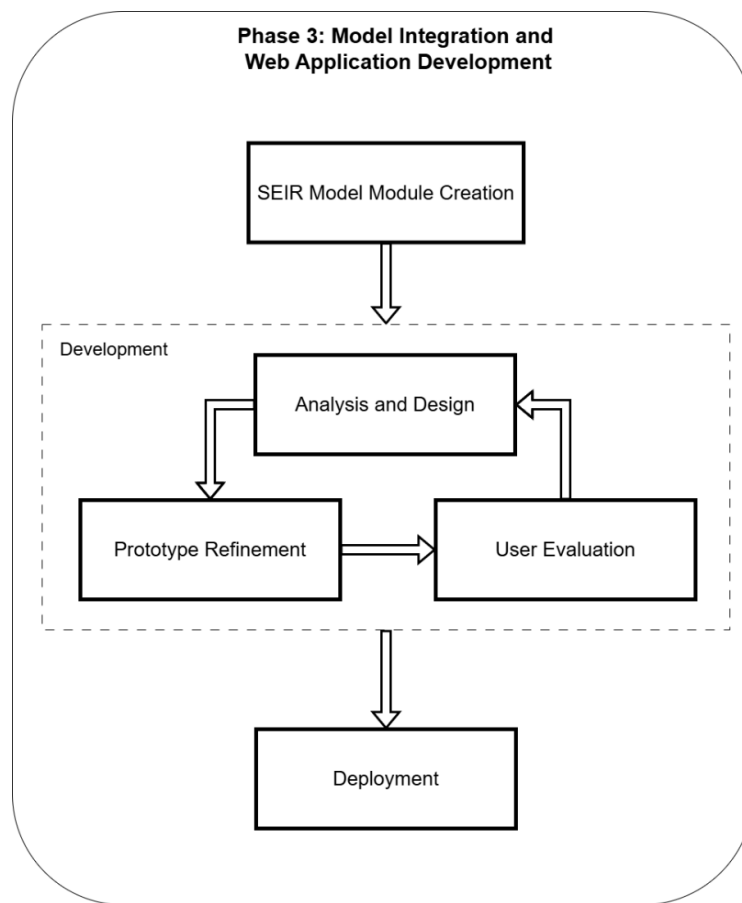


Figure 13 Model Integration and Web Application Development Flow

SEIR Model Module Creation

The first step in this phase was to create an SEIR model module based on the model developed in the previous phase. This module contains the optimized SEIR model that simulates and predicts the progression of infectious diseases. Additionally, this module calculates the daily counts for each compartment: susceptible, exposed, infectious, and recovered for every barangay based on the entered case counts by the user. As such, this model expects a properly formatted dataset with the correct fields and data types. As a result, the output is the predicted progression of the disease which is visualized in the web application as charts and through the heat map. With this in place, the development of the web application then proceeded.

Development

The **development tools** used were primarily Python-based. The application utilized Django as the web framework, while GeoPandas was integrated to handle geospatial features. Django is a free, open-source web framework that encourages rapid development and offers tools for easy integration of features into the web application (Django Software Foundation, n.d.). Meanwhile, GeoPandas is an extension of the Pandas library that adds support for geospatial data processing, enabling the processing of location-based information for the web application (GeoPandas, n.d.).

During development, with this web application framework, the Django backend was responsible for processing client-side requests, responding to user interactions, and managing the application's data flow. The server-side operations utilized Django views, which handle both GET and POST requests, acting as intermediaries between the frontend and the underlying data processing logic. When a user interacts with the web page,

JavaScript was used to trigger specific events that send asynchronous requests to Django. These requests were then received and processed by the Python-based Django views, enabling the server to dynamically retrieve or update data and send responses back to the client. Moreover, PostgreSQL was used as the database for the application. It is regarded as a powerful open-source object-relational database that is often recommended with Django applications.

For the frontend design, Tailwind CSS was employed to structure and style the user interface, ensuring a responsive and modern visual experience. Tailwind's utility-first approach to CSS enables concise, reusable styling, which enhances maintainability and simplifies customization across different components of the application.

In terms of data visualization, the application integrated Plotly to render dynamic charts and graphs directly within the user interface. JavaScript controlled the communication between the client-side and Django views, fetching the data necessary for these visualizations. Upon receiving data from the backend, JavaScript updates the Plotly elements, allowing the graphs to reflect the requested information.

Additionally, spread map data visualization was implemented using GeoPandas and Plotly Choropleth map to display map-based data based on a shapefile format. GeoPandas processed geospatial data on the server side, while Plotly was utilized to create an interactive map, which is then rendered within the web page as an HTML element. By leveraging Django's server-side capabilities and the interactive JavaScript frontend, this application effectively combines map-based visualizations with real-time data updates, enhancing both user interactivity and data interpretability. Figure 14 shows the project

folder structure. The main project folder contains the project folder: SEIR_Web which contains the project settings, while the maps folder is the app folder that integrates the map and disease spread visualization for the view. Also, the accounts folder handles the account-based functionalities such as login and editing user data. Moreover, the theme folder contains the app for the Django-tailwind library that is utilized for the frontend. Lastly, the static folder contains the static files such as css, svgs, and html templates that are accessed by the other apps.

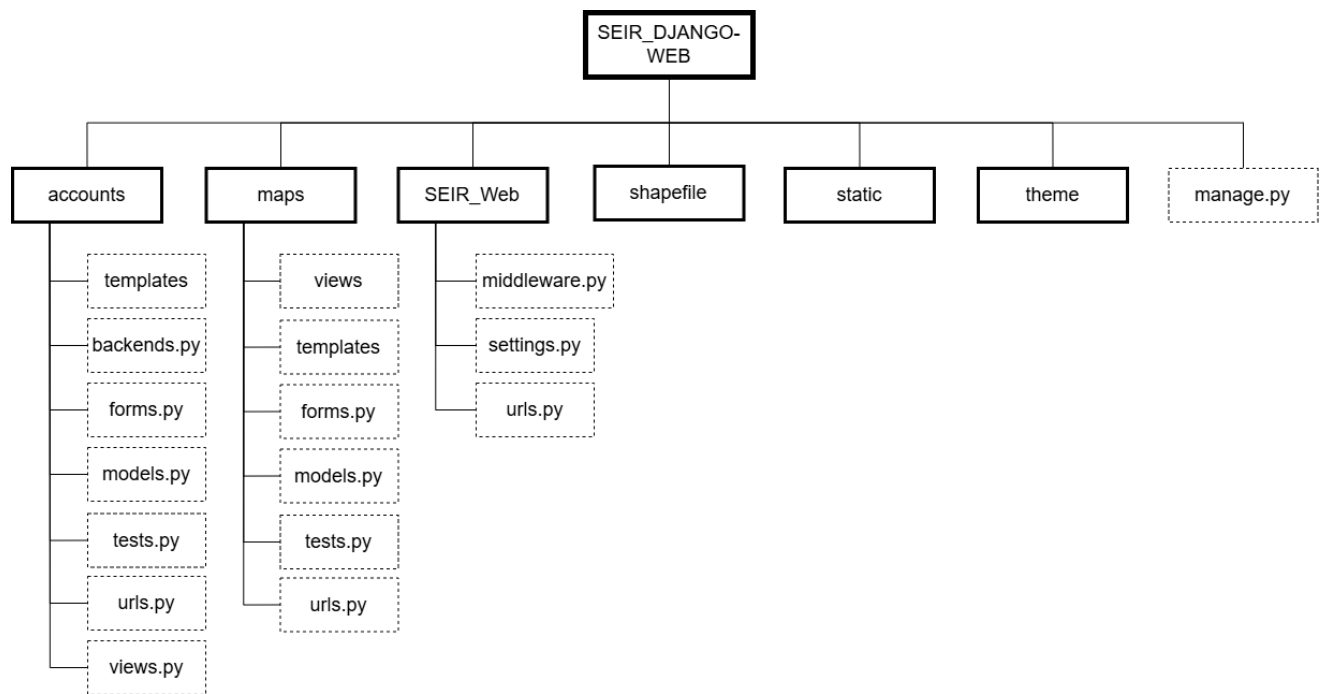


Figure 14 Project File Structure

Analysis and Design

The first step of the development cycle is the Analysis and Design step where the researchers analyzed the requirements for the web application, which

includes determining the functionalities that are needed for the users to utilize the web application. The overall design of user interface followed the Gestalt principles. These principles explain how the human mind perceives and organizes visual information, with the key idea being how the whole is more important than the individual parts. The guiding principles for Gestalt can be used for all forms of art and design (Garg, 2019). For this paper, the researchers utilized the Gestalt principles as a guide on creating a comprehensive design for the web application. Table 10 shows the Gestalt Principles and the description of how it was implemented.

Table 10 Gestalt Principles

UI Principle	Description
Proximity	Elements placed together are perceived as related
Similarity	Similar elements (color, size, shape) are grouped
Continuity	Items aligned on a line or curve are seen as connected
Figure-Ground	Differentiates the figure from the background
Pragnanz	Simplifies complex visuals into intuitive patterns
Symmetry	Balanced design elements create order and appeal
Connectedness	Visually linked elements are perceived as related
Common Region	Elements in the same bounded area are seen as related
Focal Point	Unique design elements draw user attention
Common Fate	Elements moving together appear related

In addition to the guiding principles for the user interface, the researchers adhered to the Model-View-Controller (MVC) pattern of development. The MVC

pattern separates the application into three components: Model, View and Controller. This modularity eases development by ensuring separation between the application layer and business layer. In the MVC pattern, the model handles the data and the business logic, the view displays the data to the user, and the controller is the connection between the two other compartments. This separation makes it easier to work on complex applications and for adding new features without affecting the entire application (GeeksforGeeks, 2025). In Django, the MVC pattern is followed, but with some slight differences. Django follows a Model-Template-View (MTV) pattern. The model still handles the data logic like in MVC. The template consists of the HTML output that the user sees. The view acts as the controller that connects the other compartments and defines what will be displayed to the user. The main difference between the two patterns is that the Django view acts as both the MVC view and controller, handling what the user sees while also containing the underlying logic in the pages. While there are some slight differences in MTV and MVC, both patterns result in the separation between presentation layer and business layer (Yong, 2021).

The listed principles above guided the researchers in developing the web application in a structured way and ensured that early development went smoothly. If this was a succeeding development cycle, the researchers also reviewed and addressed any issues or shortcomings found in previous versions to improve the application's overall performance and user experience. Appendix D shows the prototype design.

Prototype Refinement

In the second step of the cycle, the researchers focused on refining the prototype web application. The requirements outlined from the previous step are incorporated into the prototype, ensuring that the application aligns with the needs of the users. If this was a succeeding development cycle, the researchers addressed any issues identified in earlier versions, implementing necessary fixes. Additionally, adjustments were made to parts of the prototype that required changes. If new requirements were requested and are within the scope of the study, these were integrated into the prototype as new features.

User Evaluation

The last step of the cycle had the researchers conduct a user evaluation by testing the prototype web application. The primary focus was on identifying and listing any existing or potential problems within the application. Also, the researchers thoroughly tested the prototype to assess its overall usability and functionality, ensuring that it meets the intended requirements. The researchers determined that Django's built-in module for unit testing served as a sufficient tool for testing the application's functionalities. After extensive testing and issue identification, the process cycled back to the analysis and design phase, incorporating any necessary changes or improvements.

Deployment

Once the web application was deemed to be fit for ESUs to use, the researchers finalized the web application. Users were able to access and interact with the web application through the web browser on their computers, enabling them access to the simulation and data visualization tools.

To ensure optimal functionality and performance of the web application, specific hardware requirements were met by the computer where the application was deployed. The web application was built using Django with PostgreSQL as the database management system. Given the nature of the SEIR model and the application's data visualization requirements, the researchers outlined the server and client requirements to ensure that the application's usability is not negatively affected by system specifications.

Server Requirements

The server is responsible for running the Django-based application, for this purpose the following minimum specifications are recommended (GeeksForGeeks, 2024):

1. Processor: Multi-core, 2.5 GHz or above
2. Memory (RAM): At least 8 GB (16 GB recommended for data-intensive ESUs)
3. Storage: 100GB SSD or more

Client Requirements

The client computers access the server-hosted application to interact with the data and visualizations. Minimum requirements are as follows:

1. Processor: Dual-core processors, such as Intel I3
2. Memory (RAM): At least 4GB of RAM
3. Web Browser: Any modern web browser (e.g., Google Chrome, Mozilla Firefox, Microsoft Edge).

Phase 4: Testing and Analysis

This phase includes functional testing and usability testing, with the output being the feedback from San Pedro ESU members and third-party experts. The functional testing consisted of unit testing to validate the functionality of individual modules of the system. The phase flow is seen in Figure 15.

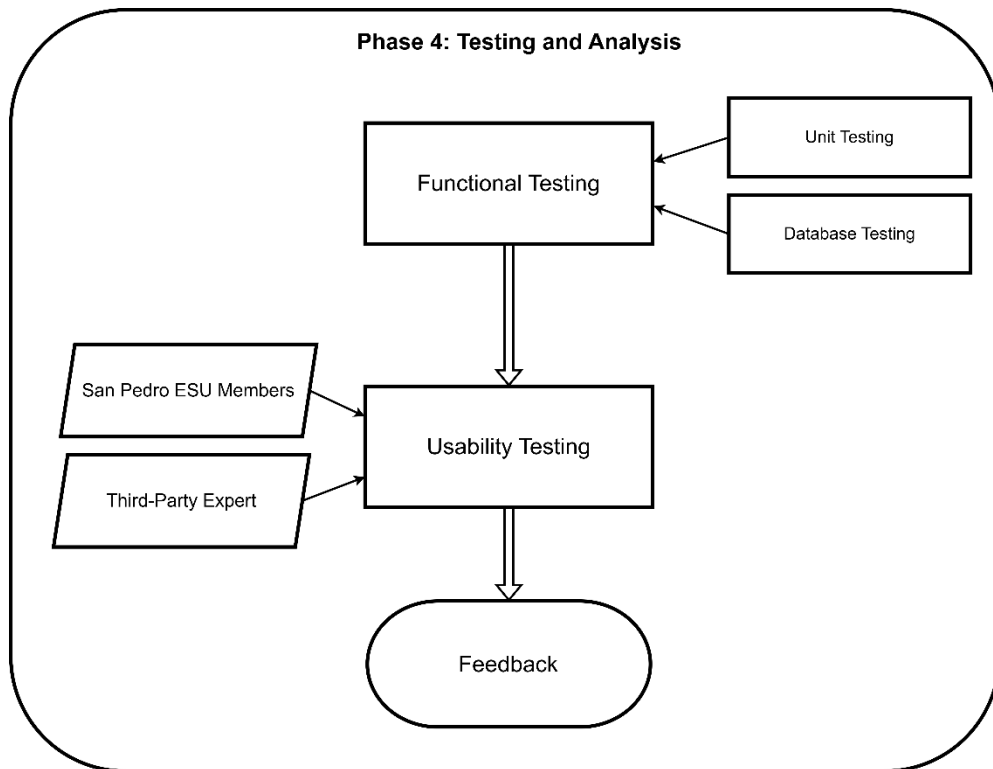


Figure 15 Testing and Analysis Flow

Data Gathering Tools

Functional Testing involves evaluating software to ensure it meets specified requirements (Jorgensen, 2013). The researchers conducted functional testing to verify that the application functions correctly. Unit testing was performed for this process.

Database Testing. Database testing involves validating the database function such as the connection with the web application and accuracy of the record on create and retrieve functions (Kaur & Singh, 2014). Database testing was performed using *PyUnit* and the *unittest* library.

Unit testing involves individually testing software components to identify any issues (Anwar & Kar, 2019). This method allows for early detection of bugs and errors. The researchers will utilize python's built in library for testing, *PyUnit* which uses python's *unittest* library. This module defines the tests by using a class-based approach. These tests were executed every time code changes occur.

To perform the unit testing, the researchers first created a test class for each page that will be tested. These classes were placed within the project folder of the source code (see Figure 16)

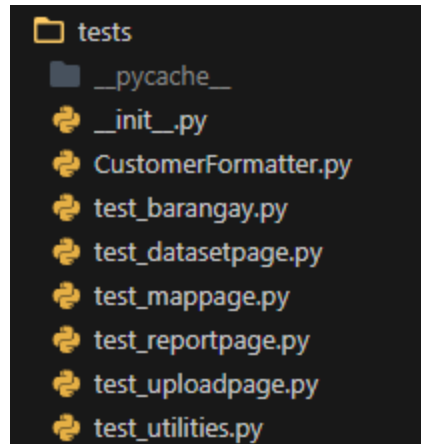


Figure 16 Test Case Classes file

After the creation of the test classes, the test cases for each module were created, the test cases are created as a method in the class. The test methods test specific function in the page. Test cases can be seen in Table 11-14.

Table 11Map Page Testcases

Test Case
Test if map current and percentage data is fetched accurately
Test if map total data is fetched accurately
Test if top barangay cases count is accurate
Test if map page load shapefile loads the choropleth map
Test if create choropleth with frame data is fetched properly

Table 12 Barangay Page Testcases

Test Case
Test if all barangay Exposed count based on date are correct
Test if all barangay Infected count based on date are correct
Test if all barangay Recovered count based on date are correct
Test if all age groups in barangay's exposed, infected, and recovered are accurate
Test if barangay page SEIR graph renders properly
Test if barangay page age distribution chart is accurate

Table 13 Reports Page Testcases

Test Case
Test if Positive Surveillance Testing reports are generated
Test if Prediction Reports are generated
Test if download csv report is functional

Table 14 Model Testcases

Test Case
Test if prediction will proceed with more than 90 data
Test if prediction will not proceed with 90 or less data
Test if prediction will proceed if there are 30 days projected cases
Test if model instance will populate upon prediction

The methods implemented for the test cases followed the Arrange-Act-Assert (AAA) pattern. In the AAA pattern, the researchers first initialized the necessary inputs and modules and acted by implementing the inputs to the methods of the modules. Lastly, the researchers asserted by evaluating the output if it is the same as the expected output.

After the creation of test methods, each test class was run by the researchers using Visual Studio Code. The researchers must pass all test cases to accomplish the unit test (An example of unit test is shown in Figure 17)

```
Found 8 test(s).
Creating test database for alias 'default'...
System check identified no issues (0 silenced).
2025-03-20 15:46:02,908 - maps.tests.test_mappage - INFO - test_create_choropleth succeeded (test_mappage.py:275)
.2025-03-20 15:46:02,920 - maps.tests.test_mappage - INFO - test_getActualPredictedGraph succeeded (test_mappage.py:409)
.2025-03-20 15:46:02,930 - maps.tests.test_mappage - INFO - test_getTopBarangay succeeded (test_mappage.py:320)
.2025-03-20 15:46:02,940 - maps.tests.test_mappage - INFO - test_get_barangay_data succeeded (test_mappage.py:472)
.2025-03-20 15:46:02,950 - maps.tests.test_mappage - INFO - test_get_current_choropleth_data succeeded (test_mappage.py:241)
.2025-03-20 15:46:02,967 - maps.tests.test_mappage - INFO - test_get_total_choropleth_data_async succeeded (test_mappage.py:157)
.2025-03-20 15:46:02,984 - maps.tests.test_mappage - INFO - test_get_total_choropleth_data_async_percentage succeeded (test_mappage.py:200)
.2025-03-20 15:46:05,276 - maps.tests.test_mappage - INFO - test_load_shapefile succeeded (test_mappage.py:114)
.
-----
Ran 8 tests in 3.086s
OK
```

Figure 17 Sample unit test run

The database testing was performed similarly to the unit testing. The test cases for the database testing involve the connections and the selecting, inserting, and updating of data in the database. A dataset test class was created containing methods that test each function that relates to the database and the system. Table 15 shows the test cases that was performed.

Table 15 Database testing testcases

Test Case
Test if case data on prediction will populate prediction
Test San Pedro records upon upload data
Test if the cases on San Pedro are correct on different dates
Test if exposed and infected records are correct
Test if recovered cases are accumulating

Usability testing involves users performing tasks on a product to identify issues and improvements (Fox, 2015) . It was done to evaluate the usability of the web application by the San Pedro ESU members.

After the San Pedro ESU members and the third-party expert tested the application, they were asked to answer a questionnaire to give their feedback. The usability testing performed consist of three (3) types of questionnaires: one for the San Pedro ESU, one for the third-party model expert, and one for the public health expert and model expert. The Usability testing was also iterated based on the feedback.

For the San Pedro ESU, the usability testing questionnaire was formulated with ISO 9126 as the standard. It consists of a four-point Likert scale about the web application. This includes statements about the accuracy of the model, UI design, ease of use, and usefulness of the solution. As seen in Table 16, each statement is aligned with a software quality characteristic from ISO/IEC 9126.

Table 16 ESU Usability Testing Question Statements and their equivalent ISO 9126 characteristics

Statement Code	Statement	ISO/IEC 9126 Quality Characteristics
S1	The SEIR math model produced consistent and reliable results across multiple simulations	Reliability
S2	The web app was stable and did not crash or experience significant issues while running	Reliability
S3	The web app consistently produced results without discrepancies across different runs.	Reliability
S4	The web app performed well under different data sizes and stress conditions.	Reliability
S5	The web application was easy to learn and use	Usability
S6	The user interface design was intuitive and easy to navigate	Usability
S7	The color scheme and contrast of the UI made it easy to distinguish different elements.	Usability
S8	The tooltips, labels, and descriptions provided sufficient context for understanding the functionalities.	Usability
S9	The visualizations and graphs representing the SEIR model results were clear and easy to understand	Usability
S10	Adding records and generating prediction based on the SEIR math model was efficient and required minimal effort	Efficiency
S11	The layout and process flow minimized unnecessary steps in using the SEIR model	Efficiency
S12	The web app responded quickly to user interactions without noticeable delays.	Efficiency
S13	The time taken to generate results was reasonable and acceptable	Efficiency
S14	The steps involved in running the SEIR model and visualizing were logically arranged and efficient.	Efficiency
S15	If errors occurred, the error messages were helpful in resolving the issues.	Reliability
S16	The application provided clear logs or feedback for debugging errors.	Reliability
S17	The web app functioned well on different browsers and devices without compatibility issues.	Portability
S18	The simulation and prediction results of the SEIR model provided enough information for analyzation	Functionality
S19	The web app was able to successfully visualize the SEIR prediction without issues	Functionality
S20	The generated report was sufficient and accurate to requirements	Functionality
S21	The web app correctly handled different input values and scenarios without unexpected behavior.	Functionality
S22	The web app provided sufficient options for customizing the visualization generated.	Functionality
S23	Overall, I found the web application to be a useful tool for epidemic spread visualization and decision-making.	Functionality

For the third-party model expert, the usability testing questionnaire was also formulated with ISO 9126 as the standard. This includes statements about the accuracy of the model and the technicality of the model's application. Table 17 shows the usability testing statements provided to the model expert.

Table 17 Model Expert Usability Testing Question Statements

Statement Code	Statement	ISO/IEC 9126 Quality Characteristics
S1	The methodology for optimizing the SEIR math model using Bayesian Inference and Simulated Annealing is a clearly defined and well-justified choice	Usability
S2	The use of Bayesian Inference for parameter estimation improves the accuracy and reliability of the SEIR math model's prediction compared to the actual observed data.	Functionality
S3	The use of Simulated Annealing for parameter estimation improves the accuracy and reliability of the SEIR math model's prediction compared to the actual observed data.	Functionality
S4	The Bayesian Inference process effectively balances accuracy and computational efficiency	Efficiency
S5	The Simulated Annealing process effectively balances accuracy and computational efficiency	Efficiency
S6	The metrics used to validate the optimized model using RMSE are appropriate and clearly presented	Functionality
S7	The metrics used to validate the optimized model using MAE are appropriate and clearly presented	Functionality
S8	The application of Simulated Annealing demonstrates a clear improvement in predictive performance compared to the classical model, as evidenced by the validation metrics	Functionality
S9	The application of Bayesian Inference demonstrates a clear improvement in predictive performance compared to the classical model, as evidenced by the validation metrics	Functionality
S10	Based on the results, Simulated Annealing is better than Bayesian in optimizing parameters.	Functionality
S11	The application of Bayesian Inference in the classic model maintains predictive stability when tested with different datasets or varying initial conditions	Reliability
S12	The application of Simulated Annealing in the classic model maintains predictive stability when tested with different datasets or varying initial conditions	Reliability
S13	When applied to different age groups, Simulated Annealing maintains prediction stability	Reliability
S14	When applied to different age groups, Bayesian Inference maintains prediction stability	Reliability
S15	When applied to different training durations, Bayesian Inference maintains prediction stability	Reliability

S16	When applied to different training durations, Simulated Annealing maintains prediction stability	Reliability
S17	The model validation process sufficiently accounts for overfitting and ensures generalizability.	Usability
S18	The optimized models' predictions align with historical or real-world epidemic data, supporting its practical applicability.	Functionality
S19	The optimized SEIR model is robust when applied to datasets with varying characteristics and different epidemiologic assumptions.	Usability
S20	The parameters optimized by Bayesian Inference are within acceptable bounds	Functionality
S21	The parameters optimized by Simulated Annealing are within acceptable bounds	Functionality
S22	The application of Bayesian Inference to classical model can be used to predict the future cases	Functionality
S23	The application of Simulated Annealing to classical model can be used to predict the future cases	Functionality

Lastly, for the third-party health worker and model expert, the usability testing questionnaire was also formulated with ISO 9126 as the standard. This includes statements about the usability of the model and applicability for general use. Table 18 below shows the usability testing statements provided to the third-party experts. All of the Usability Testing Questionnaire can be seen on Appendix K

Table 18 Third-party Experts Usability Testing Question Statements

Statement Code	Statement	ISO/IEC 9126 Quality Characteristics
S1	The developed optimized Model provides acceptable prediction of the disease spread	Functionality
S2	The web application correctly presents useful data in assessing the spread of the disease	Usability
S3	The developed optimized model can easily fit well in different epidemic phenomenon	Reliability
S4	The optimized SEIR model used provides acceptable prediction results against actual data	Functionality
S5	The web application can accurately provide useful visualization on how likely the disease can spread	Usability
S6	The parameters and the model used in predicting disease spread are applicable and correctly align to requisites in epidemiology	Functionality
S7	The optimization techniques applied can clearly present closer approximation of prediction against actual data	Efficiency
S8	The overall web application can generally be helpful in augmenting the way epidemiologists and related public health personnel identify and observe prediction spread overtime	Usability
S9	The developed optimized model can handle various scales of data	Efficiency
S10	The optimized SEIR model incorporated the optimization techniques effectively	Functionality
S11	The model maintained consistent predictive performance when applied to different data	Reliability
S12	The web application provided sufficient visualization incorporating the developed model	Usability
S13	The model can be applied to new emerging infectious disease	Reliability
S14	The web application effectively integrates predictive modeling to assess disease spread.	Functionality
S15	The model can be effectively utilized for pandemic preparedness.	Functionality

Data Gathering Procedure

The functional testing was performed by the researchers. The unit testing consists of tests in each iteration of the development. Unit testing ensured the functionality of each component of the solution. Similar to the unit testing, the database testing was performed after each iteration to validate the consistency of the values in the database.

The usability testing was performed with the San Pedro ESU members and the third-party experts to a total of four (4) testers. It was performed after the prototype development phase and after ensuring all functional testing testcase has been passed.

Data Analysis

Cronbach's Alpha for Reliability

To ensure the usability questionnaire reliability measures user perceptions, the researchers calculated Cronbach's alpha. This measure is used to assess reliability by comparing variance among items of an instrument to the overall variance. The instrument is noted as reliable if there is a great deal of covariance among the items relative to the variance (Collins, 2007). The formula for the Cronbach's alpha is defined as:

$$\alpha = \frac{N \times \bar{c}}{\bar{v} + (N - 1)\bar{c}}$$

Where:

- N = number of items

- $\bar{c}$ = average inter-item covariance
- $\bar{v}$ = average variance

Cronbach's alpha is interpreted as follows (see Table 19):

Table 19 Cronbach's alpha and the internal consistency equivalent

Cronbach's alpha	Internal Consistency
$\alpha \geq 0.9$	Excellent
$0.9 > \alpha \geq 0.8$	Good
$0.8 > \alpha \geq 0.7$	Acceptable
$0.7 > \alpha \geq 0.6$	Questionable
$0.6 > \alpha \geq 0.5$	Poor
$0.5 > \alpha$	Unacceptable

An alpha value of 0.7 or higher is generally considered acceptable for internal consistency in the usability testing (StatisticsHowTo, n.d.). A high Cronbach's alpha validated the reliability of the questionnaire items, indicating that the questions are consistent in measuring the targeted aspects of usability.

Usability Testing Analysis

The usability of the web application was assessed using a 4-point Likert scale, allowing the participants to rate their experience across key usability factors. The Likert scale provides a quantifiable measure of user satisfaction, where each point represents a level of agreement with a given statement (see Table 20 for the 4-point Likert Scale Interpretation).

Table 20 4-Point Likert Scale rating and interpretation

Rating	Interpretation
4	Strongly Agree
3	Agree
2	Disagree
1	Strongly Disagree

The interpretation of the Likert Scale results required calculating the mean score for each usability factor and mapping these scores to a corresponding agreement level (Pimintel, 2010). Table 21 provides the criteria for interpreting mean scores:

Table 21 Mean Score range and equivalent agreement interpretation

Mean Score Interval	Difference	Description
3.28 – 4.00	0.72	Strongly Agree
2.52 – 3.27	0.75	Agree
1.76 – 2.51	0.75	Disagree
1.00 – 1.75	0.75	Strongly Disagree

After computing the mean scores for each factor, the researchers categorized the overall usability experience based on the corresponding interpretation ranges. This analysis provides insight into the application's usability from the users' perspective (Joshi, Kale, Chandel, & Pal, 2015).

Prototype Improvement

The feedback from the Usability Testing was used to determine if the prototype needs improvement. If some aspects of the prototype are to receive unsatisfactory feedback, the improvements were made by the researcher on those parts of the prototype.

After the prototype improvement is done, the researchers conducted another usability test to gather new feedback and decided if another prototype improvement is necessary.

Methodology Summary

Figure 18 on the next page shows the summary of the phases in the methodology of the study. These phases are data preparation, math modeling and parameter experimentation, web application development and integration, and testing and analysis. A prototype improvement was also done after phase 4, which went back to phases two or three.

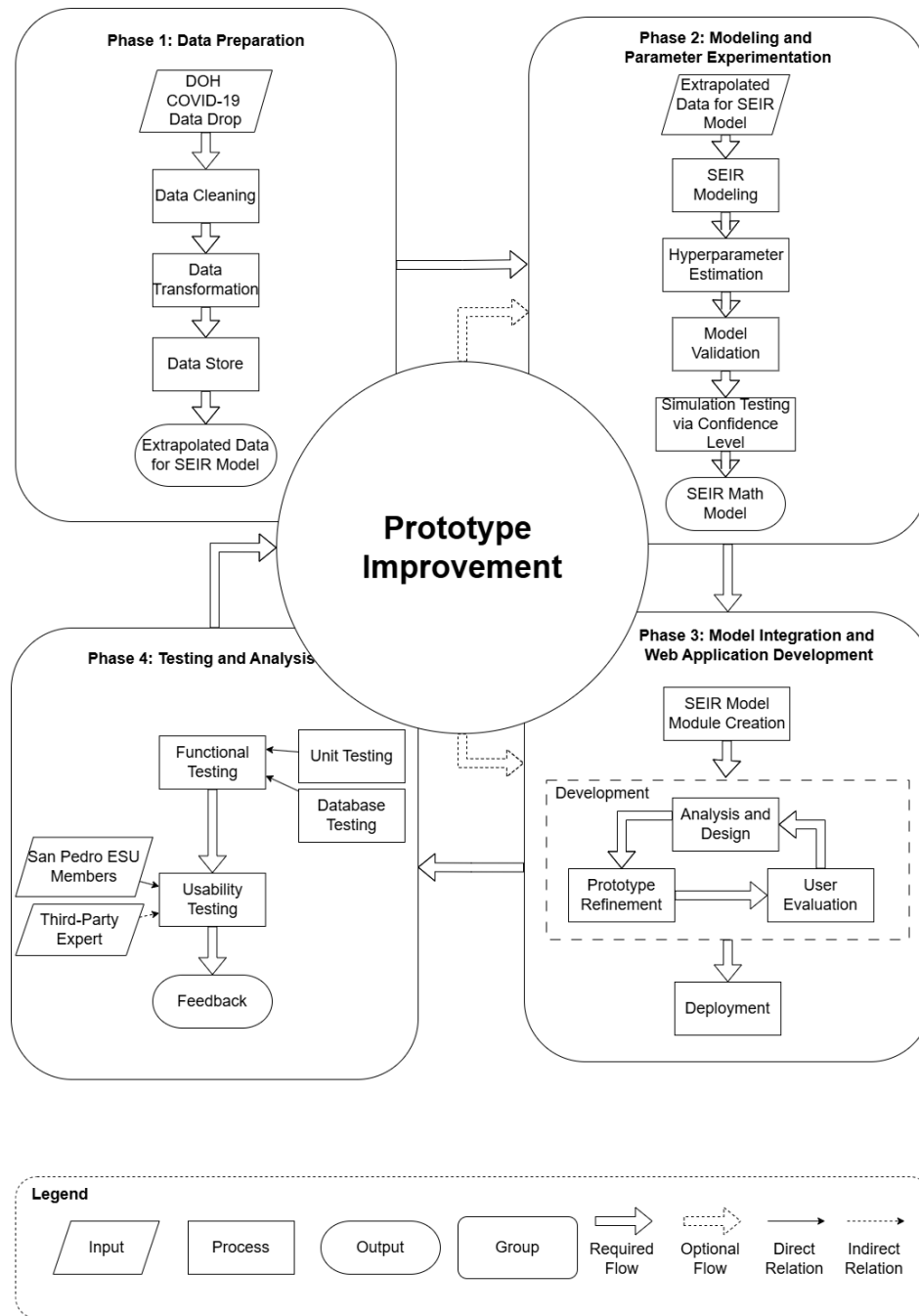


Figure 18 Prototype Improvement diagram flow

In the data preparation phase, the researchers collected the dataset from the DOH COVID-19 data drop, performed data cleaning methods using *python*, performed data transformation for the SEIR mathematical model, and performed data store to store the dataset in a database for the web application. The output of this phase was a transformed dataset stored in a database to be utilized by the SEIR model.

In the math modeling and parameter experimentation phase, the researchers created a SEIR mathematical model based on the classical SEIR model. The model was then optimized using Simulated Annealing and Bayesian Inference Simulated Annealing and Bayesian Inference optimization techniques while utilizing the transformed dataset to fit the model. The model was then validated using validation techniques such as the RMSE, MAE, and NMAE and a model was selected between the two optimized models with the acceptable accuracy. Simulation testing was performed on two (2) other different cities, the City of Biñan and Cabuyao.

In the model integration and web application development phase, the web application was developed based on the planned requirements for the study. The SEIR mathematical model was then integrated to the web application to be able to perform the spread visualization. The researchers evaluated the developed application and improved the prototype as needed.

In the testing and analysis phase, the researchers performed functional testing including unit testing to evaluate the developed application and database testing to validate functions of the database with the application. A usability testing was then performed along with the San Pedro ESU members and third-party experts to evaluate the solution for its overall functionality and usability. Afterwards, the feedback was collected from the respondents of the usability testing.

During the prototype improvement phase, the researchers improved the prototype based on unsatisfactory feedback received during the usability testing. These improvements may involve various phases of the development, making this step comprehensive and all-encompassing.

Results and Discussion

The results of the study were discussed based on the methodology used by the researchers. The phases included in the methodologies are Data Preparation, Modeling and Parameter Experimentation, Model and Integration and Web Application Development, and Testing and Analysis (See Figure 19)

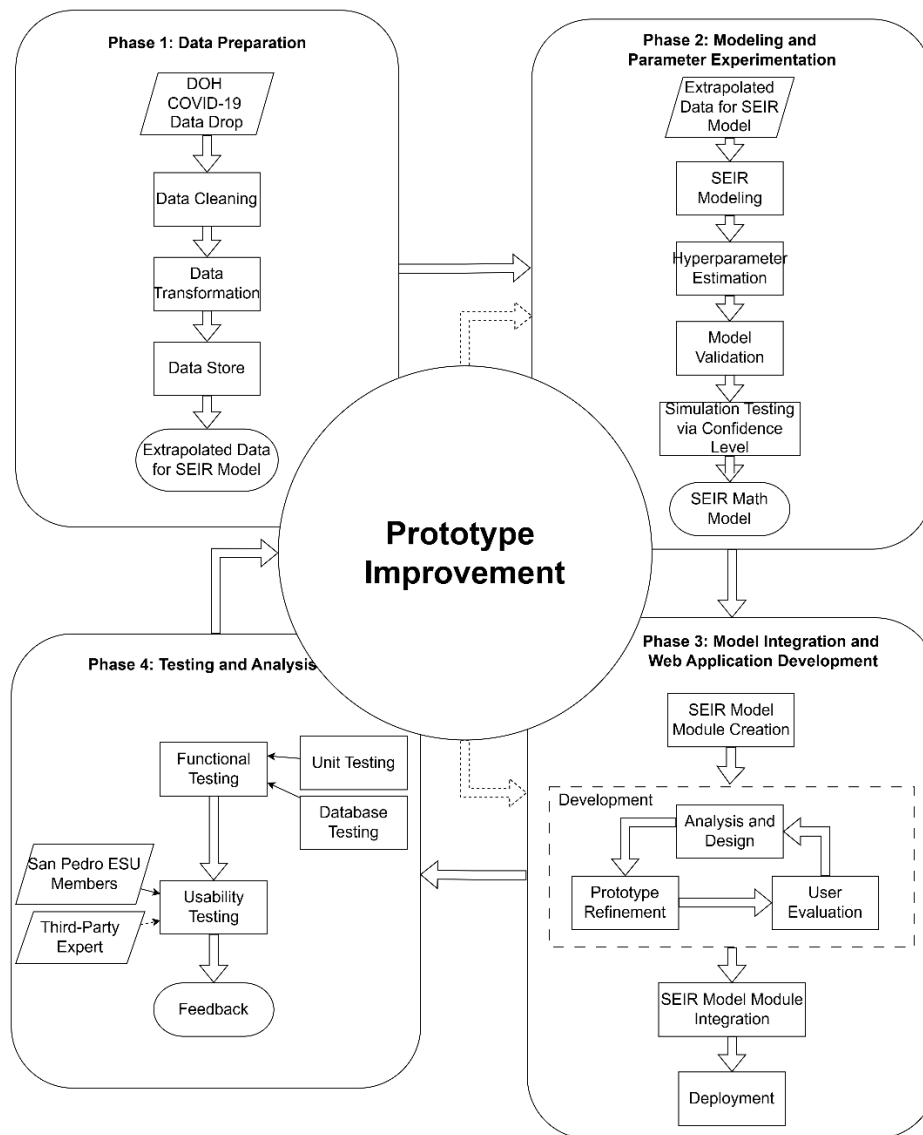


Figure 19 Summary of methodology phases

Phase 1: Data Preparation

This phase includes data cleaning, data transformation, and data storage as shown in Figure 20, which were performed using Python libraries such as Pandas and NumPy. The process ensures that the data from the DOH is properly structured and refined for further analysis. The final output of this phase consists of three datasets from San Pedro, Cabuyao, and Binan City, which are crucial for the subsequent modeling and optimization of hyperparameters. Additionally, this phase populates the database required for the web application development in the next phase. The steps done in this phase was shown in Figure 20 and was explained in the Phase 1 of the Methodology.

Specifically, data from San Pedro, focusing on barangays San Antonio, Narra, and San Lorenzo Ruiz, was used for validating the SEIR model and optimizing its hyperparameters. Meanwhile, data from Binan (Santo Tomas) and Cabuyao (Mamatid) will serve as the confidence set, ensuring robustness and reliability in the modeling process. Furthermore, the barangay data from San Pedro will be integrated into the web application to display real-time data and predictive insights for the ESU.

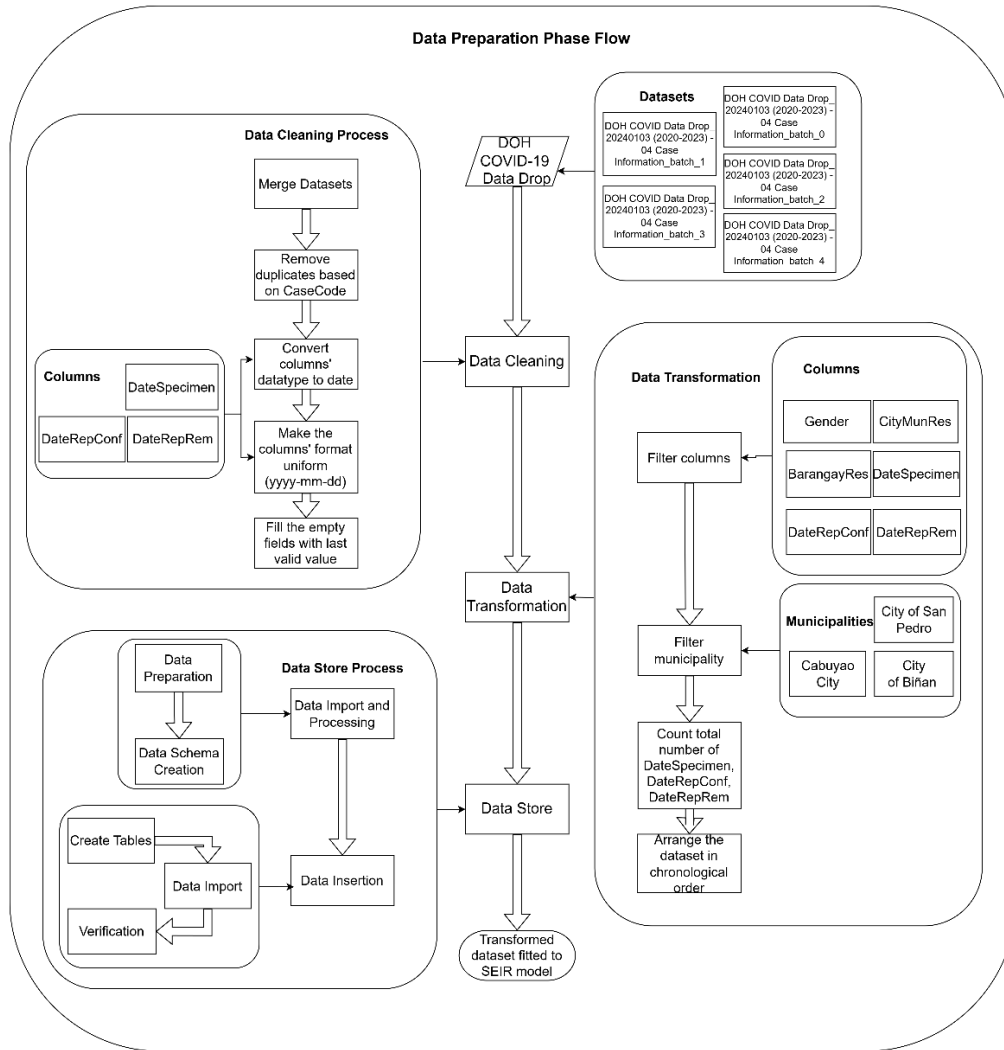


Figure 20 Data Preparation Phase Flow

Data Cleaning

Using the dataset from DOH mentioned in the data preparation of the development phase, the cleaned dataset was created for data transformation. The cleaning phase includes data import, data merging, duplicates removal, data type conversion, and filling of empty values. These steps were taken to ensure to have a complete dataset to be transformed for the modeling phase. During the data cleaning process, the researchers performed it using a

Jupyter notebook. The list of processes accomplished in the data cleaning phase are shown in Table 22.

Table 22 Data Cleaning Processes and Results

Process	Description of Process	Result
Data Import	The 5 DOH Covid Data Drops nationwide was imported to Jupyter notebook using pandas.	The data drop 04 Case Information_batch_0 contains 194,574 rows, 04 Case Information_batch_<1-3> each contains 1,000,000 rows, and 04 Case Information_batch_4 contains 136,488 rows.
Data Merging	The data imported was merged in a single data frame.	The merged dataset contains 3,331,062 rows.
Duplicates Removal	The cases with same casecode from the data frame were removed.	A total of 1,408 duplicates were removed, resulting in 3,329,654 remaining rows.
Data Type Conversion and Date Formatting	The dates from the data frame were converted from string to date and removed the time included in the date and retaining only the date.	A total of 701,932 rows in the <i>DateSpecimen</i> , 1,774 rows in the <i>DateRepRem</i> , and 1 row in the <i>DateRepConf</i> were converted to NaN.
Remove Invalid Values from <i>DateRepConf</i>	Invalid values of <i>DateRepConf</i> were removed for the next step	1 row was removed in the records. resulting in 3,329,653 remaining rows.
Fill Empty Columns with Valid Value	The dates from column <i>DateSpecimen</i> (date exposed) and <i>DateRepRem</i> (date removed) was filled with values. <i>DateSpecimen</i> Filled Values: Subtracting the mean of the differences of all valid <i>DateSpecimen</i> to the <i>DateRepConf</i> (date infected) to the <i>DateRepConf</i> with missing <i>DateSpecimen</i> . <i>DateRepRem</i> Filled Values: Adding up the mean of the differences of all valid <i>DateRepRem</i> to the	A total of 704,567 missing rows were filled in the <i>DateSpecimen</i> column, and 1,774 missing rows were filled in the <i>DateRepRem</i> column.

DateRepConf to the *DateRepConf*
with missing *DateRepConf*.

$$\text{DateSpecimen} = \text{DateRepConf} - \text{average valid (DateRepConf - DateSpecimen)}$$
$$\text{DateRepRem} = \text{DateRepConf} + \text{average valid (DateRepRem - DateRepConf)}$$

The computed average incubation rate used in filling the empty DateSpecimen values is 7 days, and it is also the stated rate by the assistant head. As for the average recovery rate, the computed value is 12 days and it is within the recovery rate of studies (Institute of Clinical Epidemiology, n.d.; Benisek, 2024).

From the 5 data drops of COVID cases from the DOH, the data has been merged and cleaned to contain complete values of the needed features to be transformed into usable dataset.

The columns needed are filled with values and ready for transformation. Table 23 shows the data dictionary of the cleaned set of data. Appendix W shows the dataset after the data cleaning process.

Table 23 Data Dictionary of cleaned dataset

Field	Description
CaseCode	Random code assigned for labelling cases; does not equate to the unique case number assigned by DOH
Age	Age
AgeGroup	Five-year age group
Sex	Sex
DateSpecimen	Date when specimen was collected
DateResultRelease	Date of release of result
DateRepConf	Date publicly announced as confirmed case
DateDied	Date died
DateRecover	Date recovered
RemovalType	Type of removal (recovery or death)
Admitted	Binary variable indicating patient has been admitted to hospital
RegionRes	Region of residence
ProvRes	Province of residence
CityMunRes	City of residence
CityMuniPSGC	Philippine Standard Geographic Code of Municipality or City of residence
BarangayRes	Barangay of residence
BarangayPSGC	Philippine Standard Geographic Code of Barangay of residence
HealthStatus	Known current health status of patient (asymptomatic, mild, severe, critical, died, recovered)
Quarantined	Ever been home quarantined, not necessarily currently in home quarantine
DateOnset	Date of Onset of Symptoms
Pregnanttab	Yes/No if patient is pregnant at any point during COVID-19 condition

From five raw datasets obtained from the DOH, the researchers applied data cleaning steps to consolidate them into a single dataset. This cleaned dataset was essential for the data transformation process, which would be utilized in subsequent phases.

Data Transformation

After completing the data cleaning process, the researchers proceeded with data transformation to prepare the dataset for analysis. The goal of this phase was to structure the data in a way that accurately represented the number of exposed, infected, and recovered individuals across different age groups and gender in different cities.

As mentioned in Chapter 3 Phase 1, San Pedro City was designated as the validation set, ensuring that the model's predictions could be evaluated against real-world data. Meanwhile, Biñan City and Cabuyao City served as the confidence sets, providing additional data to assess the model's reliability and generalizability.

Following the prototype methodology, the researchers adopted an iterative approach involving continuous development, evaluation, and refinement of the dataset. This process ensured that the transformed data met the study's analytical requirements while incorporating feedback for improvement. As a result, data transformation was performed in multiple iterations to enhance accuracy and relevance.

The first iteration of data transformation was conducted after the data cleaning phase and before usability testing. In this phase, the researchers structured the dataset to capture the number of exposed, infected, and recovered individuals across different age groups in San Pedro City, Biñan City, and Cabuyao City. This initial transformation ensured that key variables were properly formatted and organized for preliminary analysis.

After usability testing, the researchers conducted a second iteration of data transformation. Based on feedback from the ESU, refinements were made to improve the

dataset's analytical value. One significant modification was the addition of a gender column, as it was deemed crucial for understanding demographic differences in infection trends. This enhancement allowed for a more comprehensive analysis aligned with the ESU's recommendations.

Table 24 shows the processes, description of process in iteration 1 and 2, and the results of the process.

Table 24 Data Transformation process, iteration, and results

Process	Iteration 1	Iteration 2	Results
Feature Selection	The data frame was filtered by columns CityMunRes, BarangayRes, DateSpecimen, DateRepConf, DateRepRem, and Age	The data frame was filtered by columns CityMunRes, BarangayRes, DateSpecimen, DateRepConf, DateRepRem, Age and Gender	Refer to Table 25
Selecting Municipality	The data frame was then filtered by CityMunRes. San Pedro City, Cabuyao City, and Biñan City.		A total of 18,010 rows for City of San Pedro, 15,021 rows for Cabuyao City, and 16,692 rows for City of Biñan were selected.
Empty Barangay Removal	The rows with no barangay are removed.		There were 15 rows removed from City of San Pedro making it 17,994 rows in total. 13 rows removed from Cabuyao City making it 15,008 rows in total. 15 rows are moved from City of Biñan making it 16,667 in total.
Dataset Creation	Populate the dataset in Appendix W by looping through the data frame.		There were 3 csv files generated from 3 different municipalities.

The transformed dataset resulted in a time series file that tracks case counts over time for San Pedro City, Cabuyao City, and Biñan City. The inclusion of age and sex-specific breakdowns allows for more granular analysis, making the dataset more valuable for trend assessment and forecasting models. The exposed and infected case counts are the active cases on that barangay of specific date while the recovered count is the accumulated number of cases up to that date. The actual values associated with the fields were shown in Table 25.

Table 25 Transformed data fields and description

Field	Description
city_Mun_Res	City Residence
barangay_Res	Barangay Residence
Date	Date case occurred
Count_Exposed	Total number of exposed individuals
Count_Infectious	Total number of infectious individuals
Count_Recovered	Total number of recovered individuals
Count_Exposed_age_9_below	Number of exposed individuals ages 9 and below
Count_Exposed_age_10_17	Number of exposed individuals ages 10 to 17
Count_Exposed_age_18_39	Number of exposed individuals ages 18 to 39
Count_Exposed_age_40_59	Number of exposed individuals ages 40 to 59
Count_Exposed_age_60_79	Number of exposed individuals ages 60 to 79
Count_Exposed_age_80_above	Number of exposed individuals ages 80 and above
Count_Exposed_Unknown	Number of exposed individuals of unknown ages
Count_Exposed_Male	Number of male exposed individuals
Count_Exposed_Female	Number of female exposed individuals
Count_Infectious_age_9_below	Number of infected individuals ages 9 and below
Count_Infectious_age_10_17	Number of infected individuals ages 10 to 17
Count_Infectious_age_18_39	Number of infected individuals ages 18 to 39
Count_Infectious_age_40_59	Number of infected individuals ages 40 to 59
Count_Infectious_age_60_79	Number of infected individuals ages 60 to 79
Count_Infectious_age_80_above	Number of infected individuals ages 80 and above
Count_Infectious_Unknown	Number of infected individuals of unknown ages
Count_Infectious_Male	Number of male infected individuals
Count_Infectious_Female	Number of female infected individuals
Count_Recovered_age_9_below	Number of recovered individuals ages 9 and below
Count_Recovered_age_10_17	Number of recovered individuals ages 10 to 17
Count_Recovered_age_18_39	Number of recovered individuals ages 18 to 39
Count_Recovered_age_40_59	Number of recovered individuals ages 40 to 59
Count_Recovered_age_60_79	Number of recovered individuals ages 60 to 79
Count_Recovered_age_80_above	Number of recovered individuals ages 80 and above
Count_Recovered_Unknown	Number of recovered individuals of unknown ages
Count_Recovered_Male	Number of recovered male individuals
Count_Recovered_Female	Number of recovered female individuals

Additionally, Figure 21, Figure 22, and Figure 23 shows the number of active cases in exposed and infected case type while in recovered cases shown the accumulated number of cases on a regular basis. The graphs shown the counts on different age groups and different sex. The number of cases in each age group and gender shows a pattern having a peak number of cases in January 2022 in all cities.

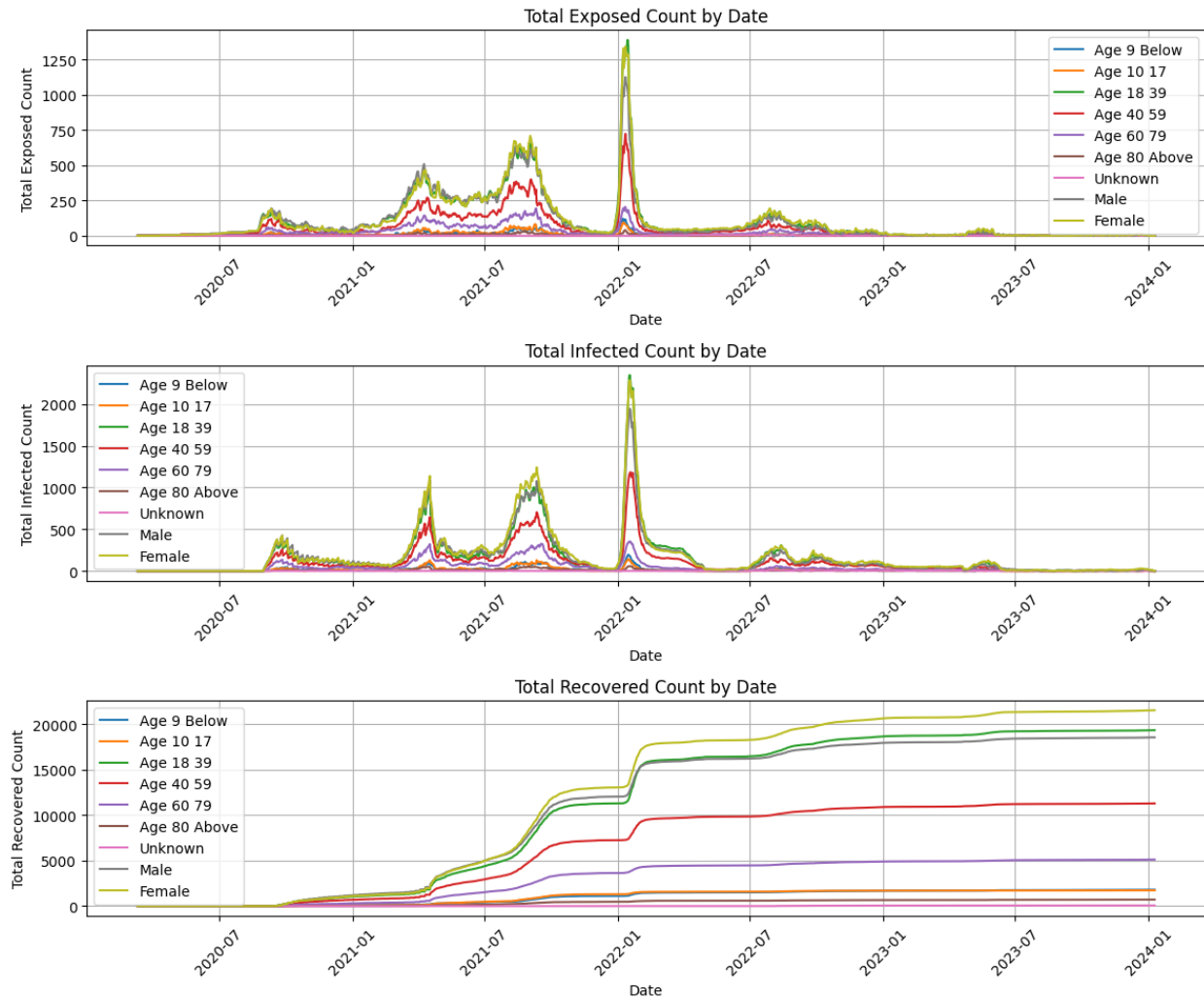


Figure 21 San Pedro City transformed data graph

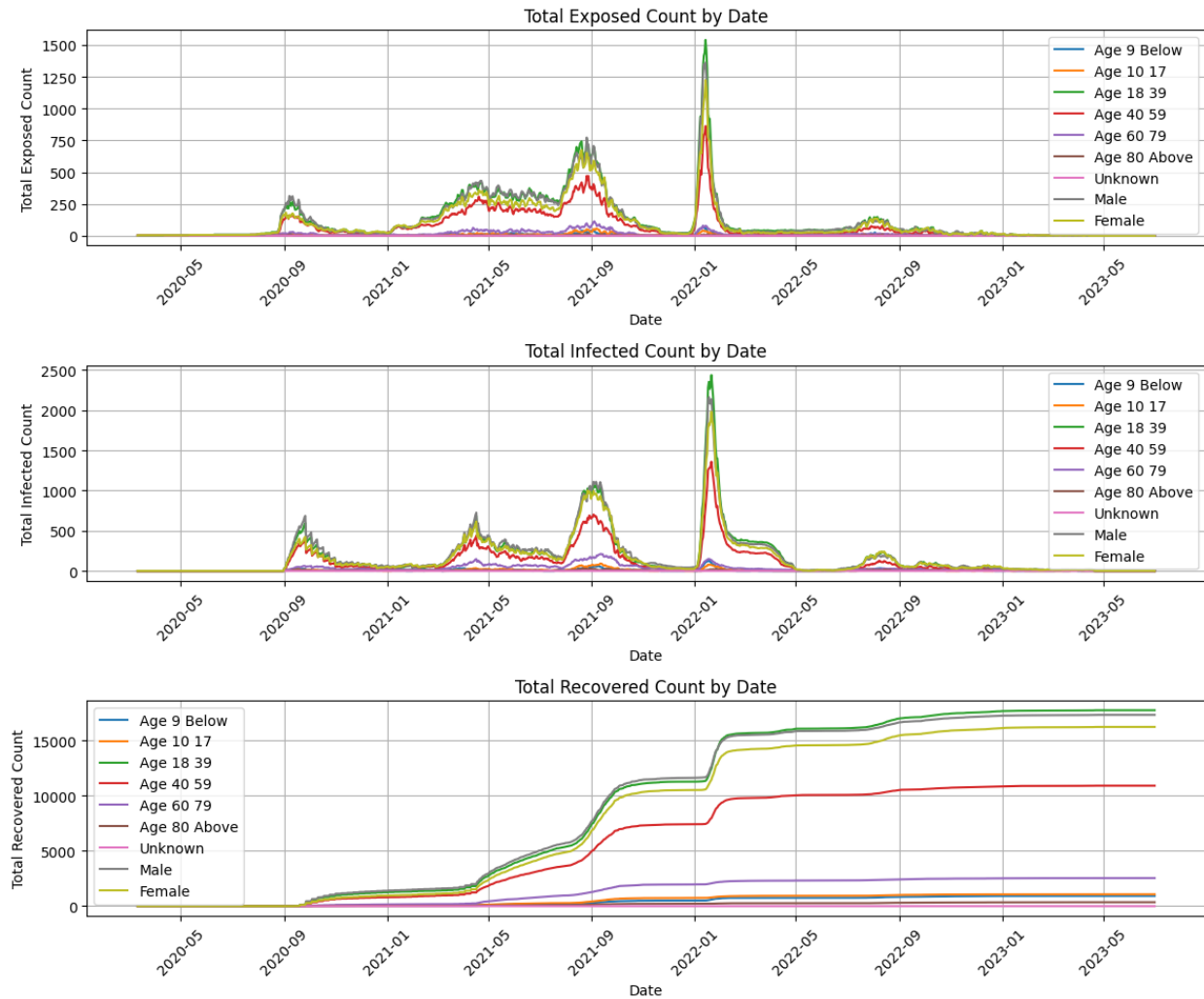


Figure 22 Cabuyao City transformed data graph

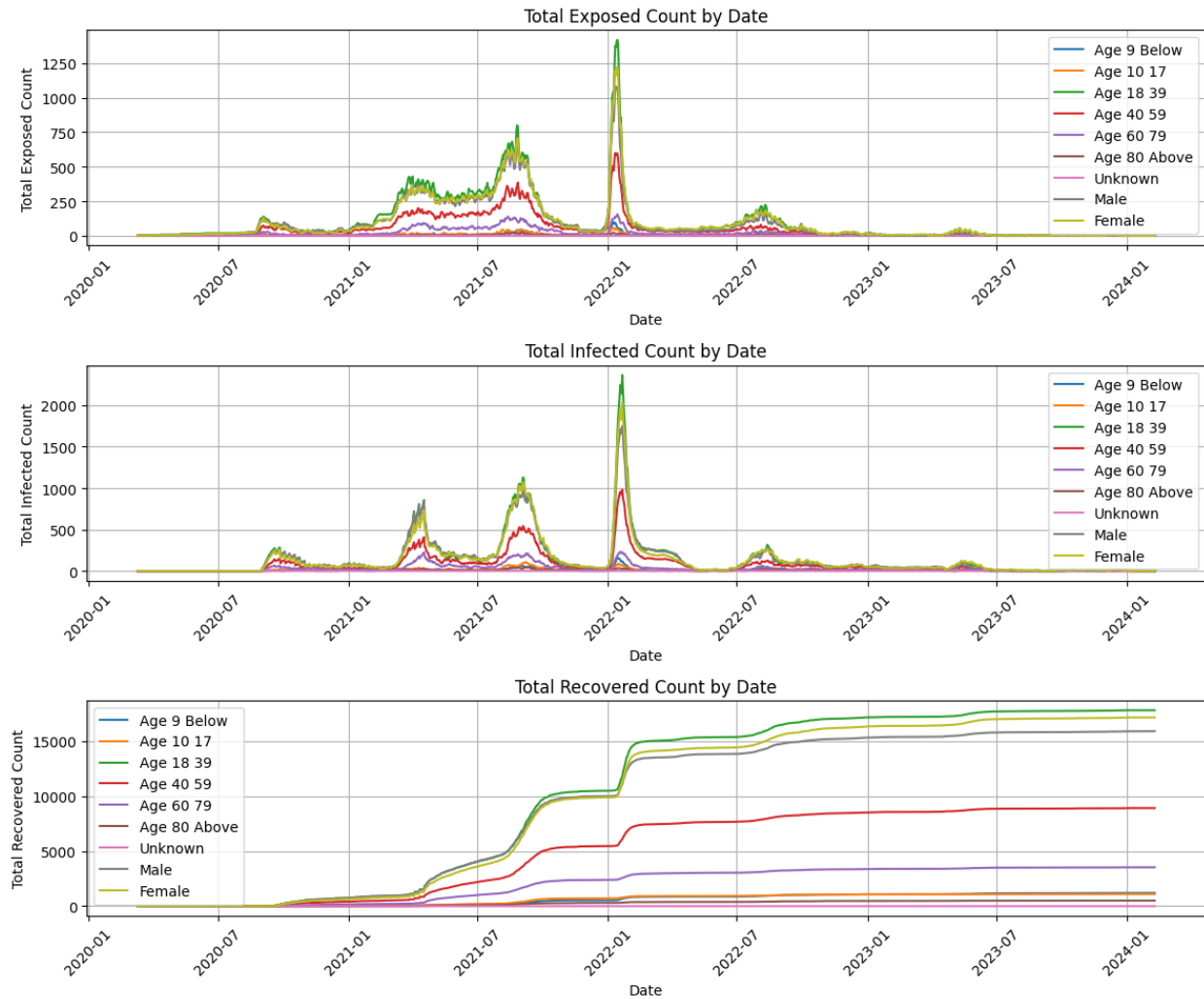


Figure 23 Biñan City transformed data graph

This phase resulted in the creation of files necessary for web development and validating the SEIR model.

Data Store

After the data has undergone cleaning and transformation, the processed dataset is stored in a PostgreSQL database using Django ORM (Object-Relational Mapping) shown

in the Appendix X. This ensures efficient data management, retrieval, and integration with the web application.

In storing the transformed data to database, the researchers utilized Django models to represent different data entities such as Barangay, Record, and CaseData. The transformed dataset is then inserted into these models shown in Appendix Y, populating rows of tables with values, ensuring data consistency and referential integrity.

These are the tables used by the web application in storing the number of cases that will be shown in the user interface of the web. The columns and data dictionary associated with these tables are displayed in Table 26. See Appendix S for the ERD and Data Dictionary.

Table 26 Tables used for data store

Table	Description	Records
Barangay	Stores all the barangay and their population	27
CaseCategory	Stores all types of cases (Exposed, Infected, Recovered) and Prediction	4
Record	Stores the case date on barangay and its information in CaseData	17,994
CaseData	Stores the count of age groups and gender for all case types and prediction	53,982
Model Instance	Stores the information about the model used in predicting actual cases	

With the successful completion of the Data Preparation phase, the necessary datasets have been structured and stored, ensuring data consistency and accessibility for further analysis. These datasets serve as the foundation for the next phase, where mathematical modeling and optimization techniques are applied.

Phase 2: Modeling and Parameter Experimentation

As detailed in Chapter 3, phase 2 focuses on developing and training the computational model to estimate its hyperparameters while applying optimization techniques to minimize prediction error. The transitions between compartments and the underlying formulas of the SEIR model was also explained in Chapter 3.

Phase 2 includes the computational model development and optimization, where the classical SEIR model, using the optimized values of the parameters based on research, is plotted against the values from the transformed data. The optimized values from existing research are compared to the hyperparameters obtained through Simulated Annealing and Bayesian Inference. To select the best model, four tests with different date ranges were conducted. The results of these optimization techniques are validated against the classical SEIR model to determine which approach yields better accuracy. The best-performing model is then integrated into the web development module.

To validate the models, barangays in San Pedro City, San Antonio, San Lorenzo Ruiz, and Narra to be specific was used for training. These barangays were chosen based on the number of populations, San Antonio having the greatest population in all barangay in San Pedro, San Lorenzo Ruiz the median of the list, and Narra has the lowest population recorded. After training, the model was used to predict the next month of the last data and error metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) were computed to compare the SEIR model's predictions against observed data. This ensures a robust performance evaluation.

After selecting the best model, additional validation was performed using datasets from Mamatid, Cabuyao and Santo Tomas, Biñan City to assess reliability and generalizability. These barangays have the greatest number of individuals in the City. The final model developed in this phase is intended for web application integration.

SEIR Model Development and Optimization

The classical SEIR model was trained to estimate the hyperparameters: β (infection rate), σ (incubation rate from exposed to infectious), and γ (recovery rate). Two optimization techniques were applied to improve model performance: Simulated Annealing (SA) and Bayesian Inference (BI).

To validate the result of the models, the researchers fit the data from Narra, San Antonio, and San Lorenzo Ruiz in San Pedro where the starting date was 05/13/21 until 01/13/22. The optimal hyperparameters resulted in the training were used to predict the data ranging from 01/14/22 to 02/13/22. The results of error metrics of each barangay were averaged for comparing the results of the models.

Classical SEIR Model

The selection and application of hyperparameters in this study were guided by the need to accurately model disease transmission within a population that closely resembles the target locale. The SEIR model's parameters: infection rate (β), incubation rate (γ), and recovery rate (σ), were adjusted across iterations to align with real-world epidemiological data from different regions, ensuring that the model captured the dynamics of disease spread relevant to the study area.

Existing studies on the use of the SEIR model for COVID-19 simulations were referenced as baseline data for classical SEIR Model. The researchers selected hyperparameters that correlated with the current dataset.

For Iteration 1, the values were derived from optimized hyperparameters obtained using a genetic algorithm, based on the study that used same dataset from the Department of Health (DOH). The genetic algorithm was applied at both the country level and the city level in Quezon City, Philippines. The initial values were based on a population of over two million; however, for this study, the researchers adapted the values for a smaller population.

For Iteration 2, the hyperparameters were obtained from a modified SEIR model used in Vo', Italy, with a population of 3,275 (Lavazzo, 2020). This study focused on the suppression of SARS-CoV-2 and the identification of infection rates across different age groups.

Table 27 presents the hyperparameters used for the classical SEIR model iterations.

Table 27 Classical SEIR model iteration and hyperparameter referenced values

Iteration	Reference	Area of Study	Population (2020)	Infection Rate (β)	Incubation Rate (γ)	Recovery Rate (σ)
Iteration 1	Santos G.L., 2022	Quezon City, Philippines	2,900,000	3.12	0.28	0.33
Iteration 2	Lavazzo, 2020	Vo', Italy	3,250	0.198	0.3	0.1389

In the first iteration, the hyperparameters were derived from optimized result of the model from observed epidemiological data in Quezon City, Philippines

(Santos G. L., 2022), being densely populated urban areas in the country, provided a robust dataset for initial parameter estimation. The hyperparameters are the optimized values obtained from genetic algorithm applied in city-level. The reasoning behind applying these values was that Quezon City's infection dynamics could serve as a reasonable approximation for the broader population within the same country. Since the target locale is also situated in the Philippines, it was assumed that factors such as healthcare infrastructure, population mobility, and public health policies would be comparable, making these parameters relevant at a larger scale.

Additionally, the data used in this research was obtained from DOH Data Drop, ensuring that the parameter estimation was based on official and up-to-date epidemiological records. This further reinforced the reliability and applicability of the chosen hyperparameters for modeling disease transmission in the study area.

Recognizing that large urban centers like Quezon City may not fully represent the infection dynamics of smaller communities, the second iteration incorporated hyperparameters from Vo', Italy (Enrico Lavezzo, 2020). Vo', a municipality with a population of approximately 3,250, shares a demographic scale more like barangays in San Pedro, Laguna. By adapting the hyperparameters from a smaller, well-documented outbreak, the model accounted for factors such as localized transmission rates, population density, and social interactions that are more reflective of the target study area.

The decision to use these specific datasets was driven by the need to balance generalizability and specificity in modeling disease spread. The first iteration provided an initial benchmark based on national-scale data, ensuring the model aligned with known epidemiological trends in the Philippines. The second iteration refined these estimates by incorporating data from a similar sized population, allowing for a more precise representation of infection dynamics at the barangay level. As seen on Table 28 the values of error metrics between iteration shows significant difference. For this study, this percentage difference of error will be called **Error Reduction**, which refers to the percentage difference in error metrics between the baseline (classical SEIR) model and the optimized model. This metric highlights the improvement in predictive accuracy achieved through optimization.

$$\text{Error Reduction} = \frac{\text{Baseline} - \text{Current}}{\text{Baseline}} \times 100$$

Table 28 compares the error metrics of the SEIR model iterations. Iteration 1 serves as the baseline, while Iteration 2 shows the comparison from the baseline. MAE, RMSE, NMAE all exhibit dramatic reductions. The MAE threshold of 324 and RMSE threshold of 419, as discussed in Chapter 3 Phase 2, serves as a benchmark, highlighting how Iteration 2 achieves a much lower error than the threshold.

Table 28 Error Metrics of Classical SEIR model iterations

Iteration	MAE	MAE Threshold	RMSE	RMSE Threshold	NMAE	Difference in Error Metrics (Error Reduction)
Iteration 1	2003.59	324	2752.51	419	43.05	Baseline
Iteration 2	143.17	324	174.34	419	1.50	Significant improvement post tuning with ~92.85% difference on MAE, ~93.67% difference on RMSE, and ~96.51% difference on NMAE.

These improvements (shown in Figure 24 and Figure 25 emphasize better performance of Iteration 2 values on barangay level using SEIR, demonstrating a significant reduction in forecasting errors.

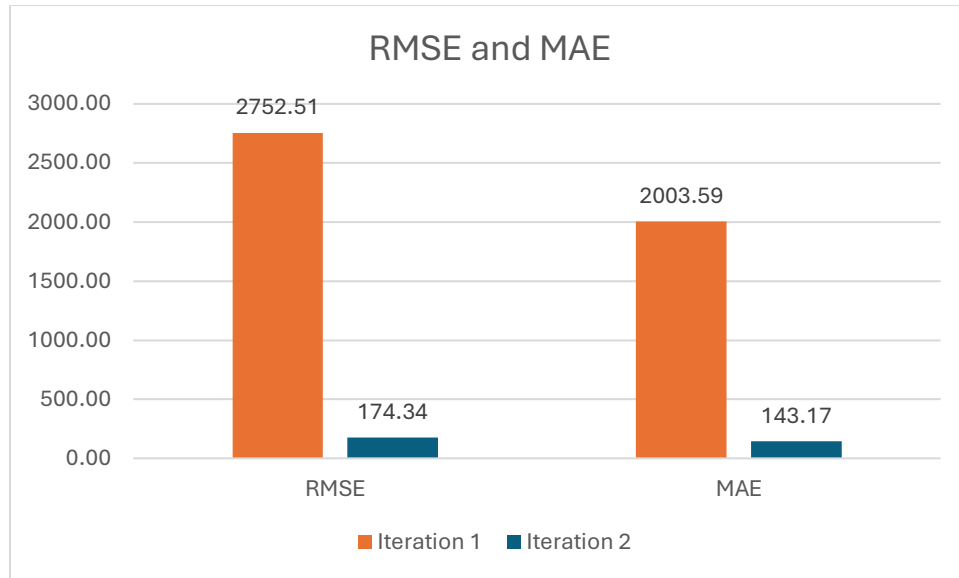


Figure 24 RMSE and MAE of classical SEIR iterations 1 and 2

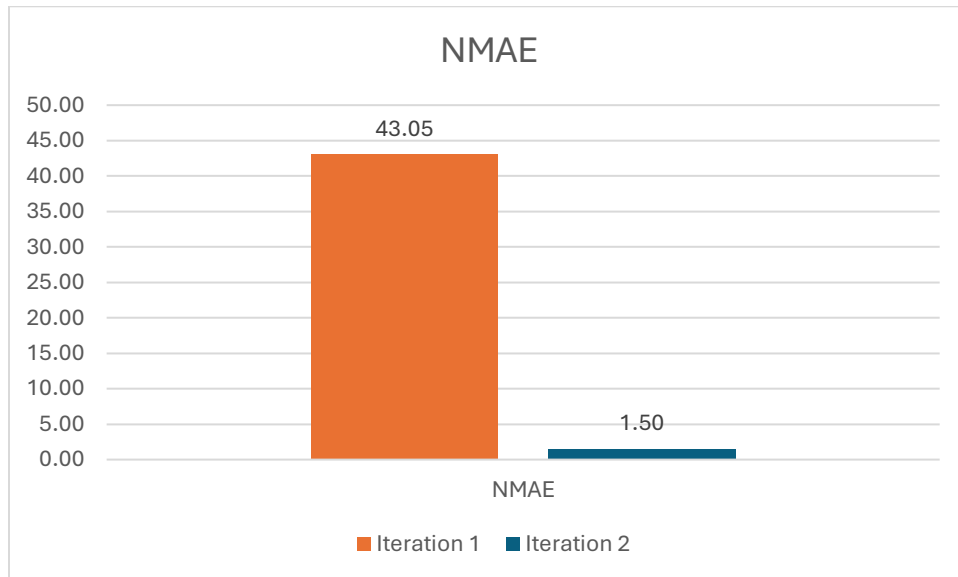


Figure 25 NMAE of classical SEIR iterations 1 and 2

The SEIR model predictions were plotted against actual data shows in Figure 26 to further see the difference in the iterations. The actual cases, represented by blue markers, remain relatively low and stable throughout the time period, while the SEIR model predictions exhibit peaks at different times depending on the

iteration used. The figure also shows the cases predicted by Iteration 1 and 2 against actual values of Narra from May 13, 2021, to January 13, 2022. This shows the significant differences between the iterations in predicting values (also see Appendix T).

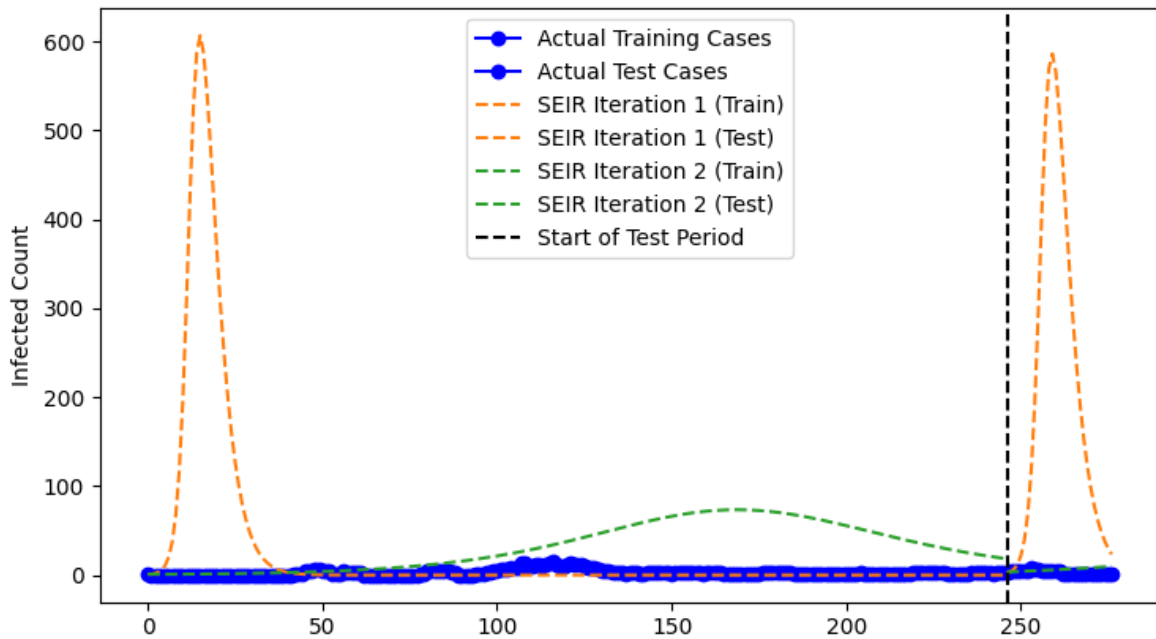


Figure 26 SEIR model prediction for iteration 1 and 2 against actual case data

Iteration 1 (orange dashed lines) demonstrates a much more aggressive infection trajectory with a higher peak, suggesting that it overestimates the infection count compared to the actual data. This is likely due to its higher infection rate ($\beta = 3.12$), which causes the model to predict rapid disease spread and exaggerated outbreak peaks. However, the actual cases remain relatively low, meaning that this iteration does not accurately reflect real-world transmission dynamics.

Iteration 2 (green dashed lines) presents a much more subdued infection trajectory, with significantly lower peaks that align more closely with the actual case data. The lower infection rate ($\beta = 0.198$) leads to a slower spread, which is more consistent with observed trends. Additionally, the slightly higher incubation rate ($\gamma = 0.3$) and lower recovery rate ($\sigma = 0.1389$) contribute to a more extended infection duration, making the predictions more stable over time.

As the graph shows, Iteration 2 is the better model to use since its predictions are more in line with the actual case data, capturing a more realistic spread of infections. Iteration 1, with its high infection rate, leads to an overestimation of cases, making it unsuitable for accurately modeling the outbreak dynamics in Barangay Narra. By this result, Iteration 2 was used as the baseline of comparison to other models.

Overall, Iteration 2 demonstrates significantly better performance than Iteration 1, as evidenced by the error metrics. The MAE and RMSE for Iteration 2 are drastically lower than those of Iteration 1, with reductions of approximately 92.85% and 93.67%, respectively. Additionally, the NMAE is reduced from 43.05 to 1.50, indicating a substantial improvement in prediction accuracy.

Although Iteration 2 demonstrated that the model remained within acceptable thresholds, there was still room for further refinement, as it retained a noticeable degree of error and deviation from actual case data. Fine-tuning the hyperparameters using optimization techniques could enhance the model's accuracy and reliability in predicting infection trends. To assess the effectiveness

of these optimization techniques, Iteration 2 was used as the baseline for comparison. The error metrics of the optimized models were compared to those of the classical SEIR model to determine whether improvement had occurred. This initial optimization step serves as a critical benchmark for the subsequent evaluation of more advanced optimization techniques, which will be explored in the following sections.

SEIR Optimization Techniques

To further enhance the accuracy of the SEIR model, optimization techniques were employed to fine-tune its parameters. These techniques aim to reduce prediction errors by systematically refining key epidemiological parameters such as the infection rate (β), incubation rate (σ), and recovery rate (γ). By optimizing these parameters, the model can more effectively capture the underlying dynamics of disease transmission and improve predictive performance.

Two prominent optimization techniques were explored: Bayesian Inference and Simulated Annealing. These optimization techniques were assessed by comparing their predictions to those of the classical SEIR model. The goal was to determine whether the optimized models truly improved prediction accuracy. The error metrics of the optimized SEIR models were analyzed against the classical SEIR predictions, measuring improvements in Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Normalized MAE (NMAE). A significant reduction in these error metrics would indicate successful optimization, demonstrating that the refined models provide more reliable

estimates of infection trends. The following sections delve into the application of these techniques

SEIR Optimization via Simulated Annealing Optimization

The first optimization technique used for the classical SEIR is Simulated Annealing (SA), the dual annealing algorithm from the SciPy library was employed. This approach was chosen due to its ability to efficiently explore a broad search space while avoiding local minima, leading to a more refined parameter estimation.

The optimization process aimed to estimate three key parameters of the SEIR model: infection rate (β), incubation rate (σ), and recovery rate (γ). The objective function used for optimization was formulated based on the mean squared error (MSE) between the observed number of infectious individuals and the corresponding predictions generated by the SEIR model. To prevent extreme parameter values and improve generalization, a regularization term was incorporated into the objective function. This term penalizes overly large parameter values, thereby ensuring stability in the learned parameters. The results demonstrated a substantial decrease in error metrics compared to previous iterations.

The optimization process begins by defining an objective function, which evaluates how well a given set of parameters fits the observed data. This function computes the mean squared error (MSE) between predicted and actual cases, with

an added regularization term to prevent extreme values. Then, SA is initialized with initialized chosen parameter values within predefined bounds. The algorithm maintains a temperature (T) that gradually decreases over iterations, controlling the likelihood of accepting suboptimal solutions to escape local minima.

At each iteration, a new candidate solution is generated by slightly modifying the current parameters. If the new solution improves the objective function (i.e., reduces the error), it is accepted. Otherwise, it may still be accepted with a probability proportional to the temperature, allowing occasional exploration of worse solutions. This probability is computed using the Metropolis criterion:

$$\exp\left(\frac{-\Delta}{T}\right)$$

where delta (Δ) is the change in the objective function value. As the temperature decreases, the acceptance probability of worse solutions diminishes, leading to convergence toward an optimal set of parameters.

The algorithm terminates when the temperature falls below a predefined threshold, or the maximum number of iterations is reached. The final output consists of the best parameters found, the corresponding error, and a history of all attempted and accepted parameter updates. Through this iterative refinement, SA effectively optimizes SEIR model parameters, balancing exploration and exploitation to achieve an accurate fit to observed data.

The Figure 27 below shows the values in each iteration, including both accepted and rejected solutions. This provides insights into how the algorithm explores the search space and converges toward an optimal solution. This plot represents the cooling schedule, which controls how the temperature (T) decreases over iterations. The graph shows how the cooling schedule decreases the temperature over iterations, following an exponential decay pattern. The high initial temperature allows for broader exploration, meaning even worse solutions have a chance of being accepted to escape local minima. As the temperature lowers, the algorithm becomes more selective, reducing the likelihood of accepting worse solutions. Eventually, the temperature reaches a point where the search process stabilizes, leading to convergence toward an optimal solution.

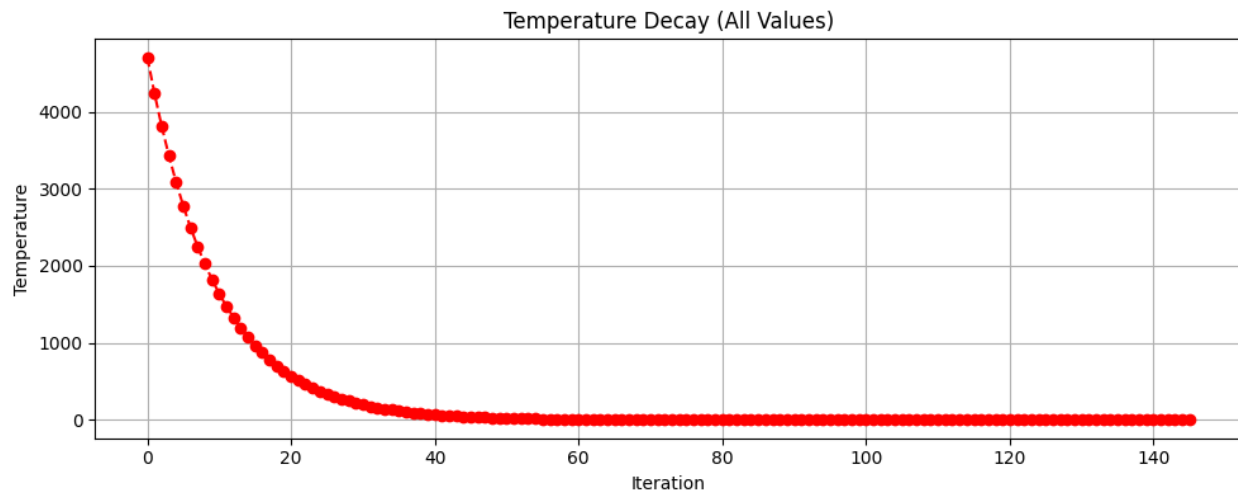


Figure 27 Cooling Schedule decrease over iterations

The Figure 28 below illustrates how the objective function value changes over iterations. The score is based on the Mean Squared Error (MSE) between the observed and predicted infectious cases with an added regularization term. Initially,

the score is high, indicating that the early parameter choices are suboptimal. A rapid decline in the score suggests that the algorithm quickly finds better solutions. Fluctuations in the middle phase show the exploration process, where some worse solutions are evaluated and occasionally accepted. As the temperature decreases, the score stabilizes, signifying that the optimization process has identified an optimal or near-optimal solution. Since all values—both accepted and rejected—are plotted, occasional spikes appear where worse solutions were considered but not selected.

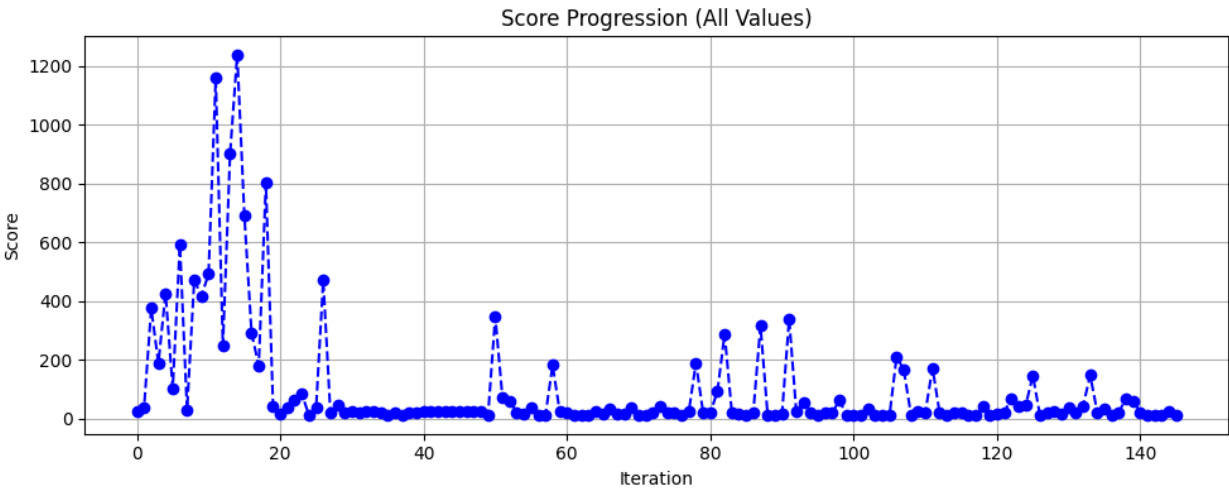


Figure 28 Score Progression across iteration

The Figure 29 below tracks how the three SEIR model parameters change throughout the optimization process. The infection rate (β) starts with high variability, indicating broad exploration in the early iterations. As the temperature decreases, β stabilizes, suggesting convergence toward an optimal value. The incubation rate (σ) and recovery rate (γ) show less fluctuation, meaning they require less adjustment or were already near their optimal values from the start. Minor

fluctuations in the later stages reflect the fine-tuning process. Since all values are plotted, this visualization provides a complete view of how each parameter evolves, demonstrating the balance between exploration and exploitation before reaching stability.

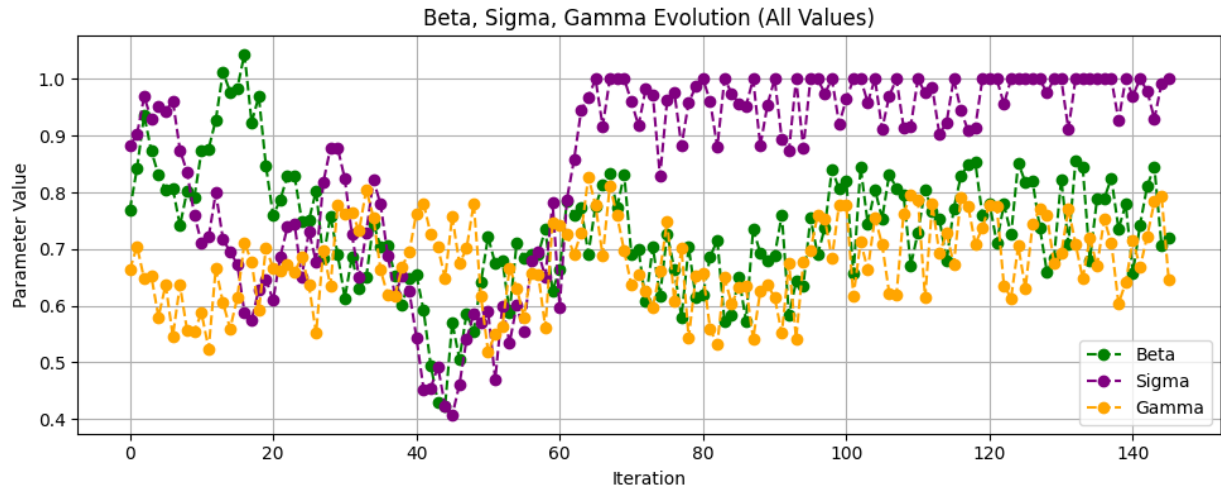


Figure 29 Beta, Sigma, Gamma Evolution across iterations

The beta, sigma, and gamma line illustrates the progression of the SA optimization process for the SEIR model parameters. Each graph reflects a crucial aspect of the algorithm's behavior of accepted values.

The plots of accepted values provide a filtered view of the optimization process, showing only the solutions that were retained at each step. The Figure 30 now displays smoother progression since only accepted values are recorded. The trend remains an exponential decay, but unlike the previous plot showing all values, there are fewer points, highlighting the cooling process and decision-making behind solution acceptance.

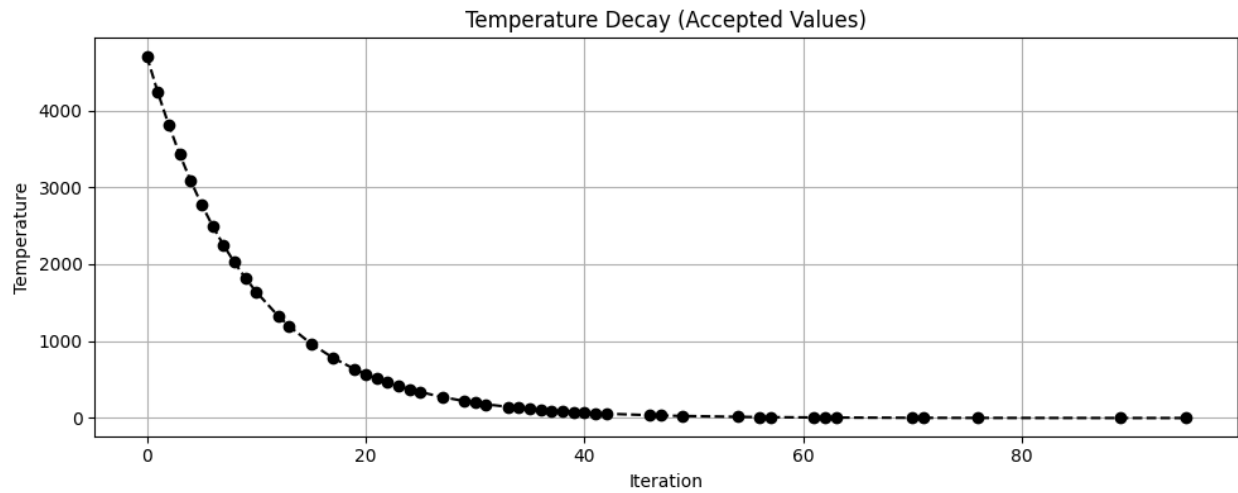


Figure 30 Simulated Annealing Iteration 2 Temperature decay

The Figure 31 now reveals a clearer trend of improvement, as only accepted solutions are included. The score consistently decreases with fewer fluctuations compared to the previous plot, indicating that the optimization process gradually refines the solution while minimizing unnecessary jumps. Since rejected solutions are omitted, the plot highlights the steady convergence toward an optimal score without distractions from temporary deviations.

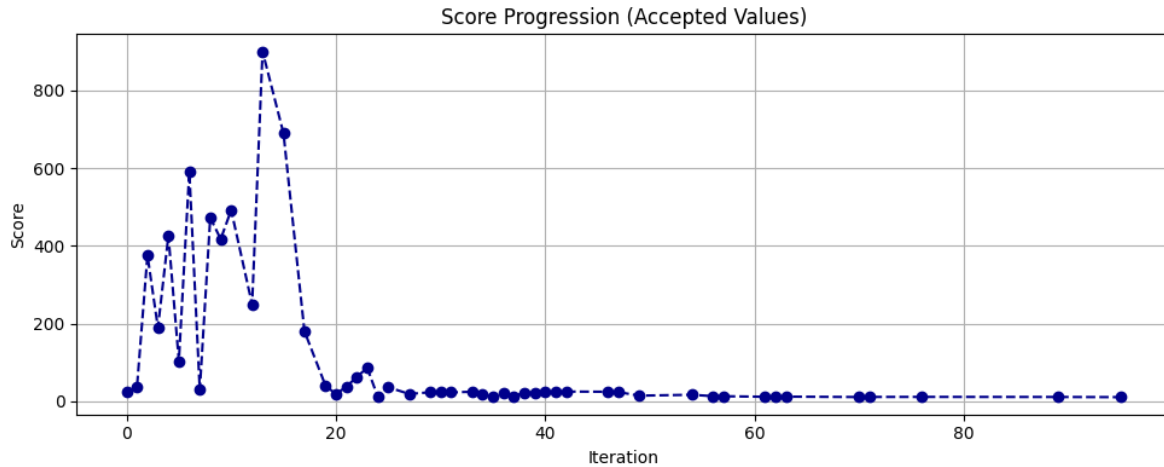


Figure 31 Simulated Annealing Iteration 2 score progression

The Figure 32 now presents a more structured trajectory for parameter tuning. Unlike the previous version where all values were plotted, this plot captures only the parameters corresponding to accepted solutions. Beta (infection rate) exhibits a steady decrease before stabilizing, while Sigma (incubation rate) and Gamma (recovery rate) show minor adjustments before settling. This visualization emphasizes how SA selectively refines parameters over iterations, demonstrating the controlled and gradual convergence of the algorithm.

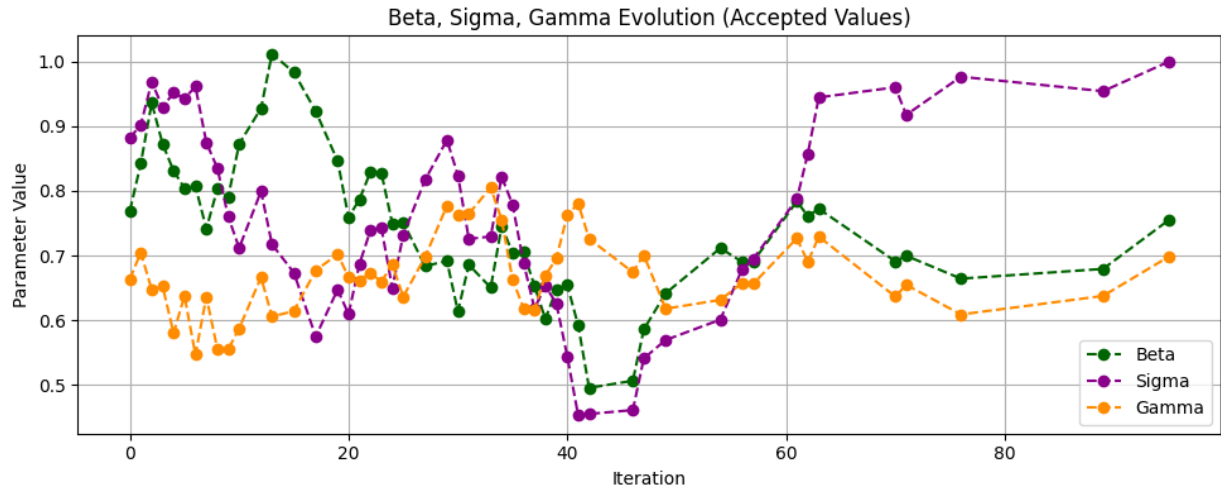


Figure 32 Simulated Annealing Iteration 2 hyperparameter accepted values

The optimized hyperparameters using Simulated Annealing resulted in the values in Table 29.

Table 29 Simulated Annealing optimized model hyperparameter values

Barangay	Infectious Rate	Incubation Rate	Recovery Rate
San Antonio	1.24	0.20	1.00
San Lorenzo Ruiz	1.18	0.23	1.00
Narra	1.19	0.27	0.99

Once the Simulated Annealing algorithm optimizes the hyperparameters, these refined values are applied to the classical SEIR model to predict future cases. The predictions generated using the optimized values for β (infection rate), σ (incubation rate), and γ (recovery rate) are then compared against the baseline model's error metrics to evaluate whether the optimization leads to improved performance.

Simulated Annealing Validation

Table 30 provides a comparative analysis of the error metrics between the Classical SEIR model and the Simulated Annealing optimized version when applied to the San Pedro dataset on a municipal level. The results show substantial improvements across all metrics. The results show a significant reduction in error across all three metrics after applying the optimization.

Table 30 Classical SEIR and Simulated Annealing model error metric comparison

Model Values	MAE	RMSE	NMAE	Error Reduction
Classical SEIR Values	143.17	174.34	1.50	Baseline
Simulated Annealing Values	47.33	58.75	0.69	~66.94% in MAE, ~66.30% difference in RMSE, and ~54.00% difference in NMAE

As shown in Table 30 the optimized SEIR model significantly outperforms the classical SEIR model in terms of predictive accuracy. The error metrics for the Simulated Annealing model are reduced by approximately **66.94% in MAE, 66.30% in RMSE, and 54.00% in NMAE**, demonstrating the substantial improvement gained from the optimization. This reduction in error can be interpreted as an improvement in the model's ability to capture the actual spread of the infectious disease, highlighting the effectiveness of optimization methods in enhancing predictive performance.

The MAE decreased from 143.17 in the Classical SEIR model to 47.33 with Simulated Annealing. This significant reduction indicates that the optimized model's predictions are much closer to the actual-observed values, thereby enhancing its overall reliability (see Figure 33)

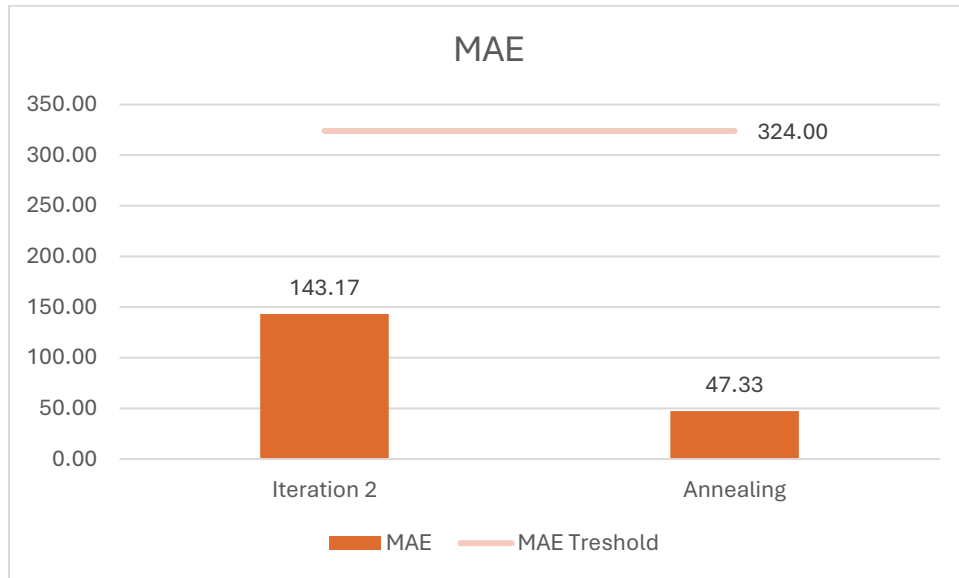


Figure 33 MAE value of classical SEIR iteration 2 and Simulated Annealing optimized model

Similarly, the RMSE was reduced from 174.34 to 58.75. Given that RMSE penalizes larger errors more heavily than MAE, this reduction suggests that not only are the average prediction errors lower, but the occurrence of large errors has also been minimized, resulting in a more robust forecasting model (see Figure 34)

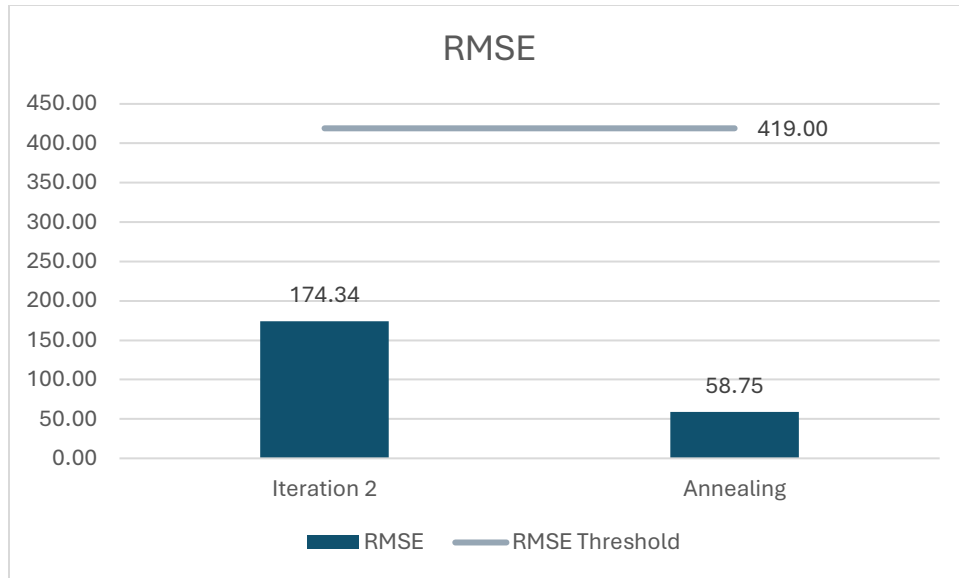


Figure 34 RMSE value of classical SEIR iteration 2 and Simulated Annealing optimized model

The NMAE, which was also improved, dropped from 1.50 to 0.69. This scale-independent metric confirms that the optimized model performs significantly better across the entire range of data values, offering a more consistent prediction quality irrespective of the data scale (see Figure 35).

The validation results of each barangay was shown in Appendix BB where the 3 and other barangays was used to validate the result of the model. All the barangay shows significant improvement when Simulated Annealing was applied in parameter estimation compared to the values set by Iteration 2.

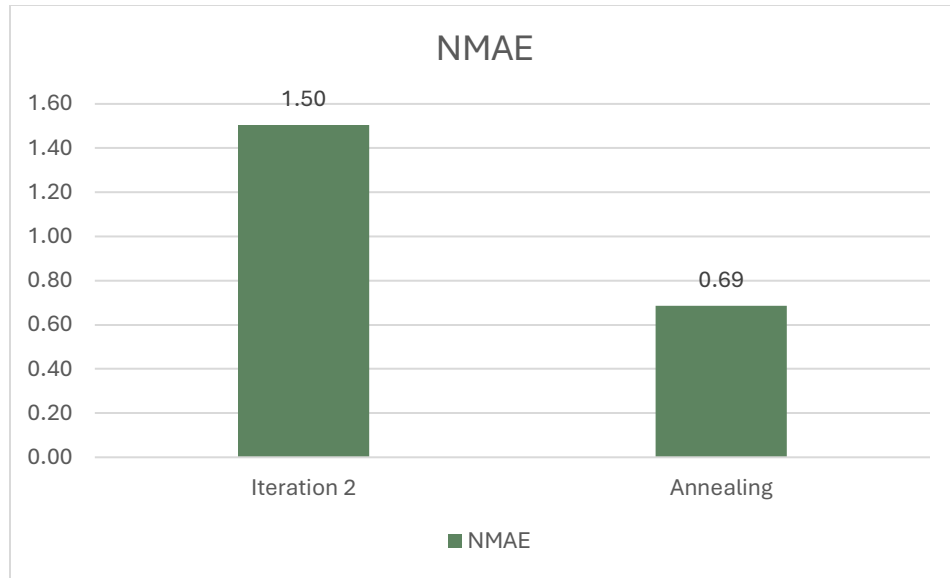


Figure 35 NMAE values of classical SEIR iteration 2 and Simulated Annealing optimized model

The error metrics show the great difference between the error rates of the baseline model and simulated annealing optimized model. To further show the difference in prediction, Figure 36 shows the predicted value of used by Iteration 2 and Simulated Annealing against actual values of Narra. In the figure, SEIR Iteration 2 (green dashed line) overestimates the infection peak significantly, predicting a sharp rise in cases that is not reflected in the actual data. This suggests that the hyperparameters used in Iteration 2 fail to capture the real-world transmission dynamics accurately, leading to inflated infection counts. While Simulated Annealing (orange dashed line) follows a much closer trajectory to the actual data (blue dots), effectively capturing both the timing and magnitude of the infection peaks. By fine-tuning the hyperparameters through SA, the model

achieves smoother and more realistic progression of cases, leading to a notable reduction in prediction error.

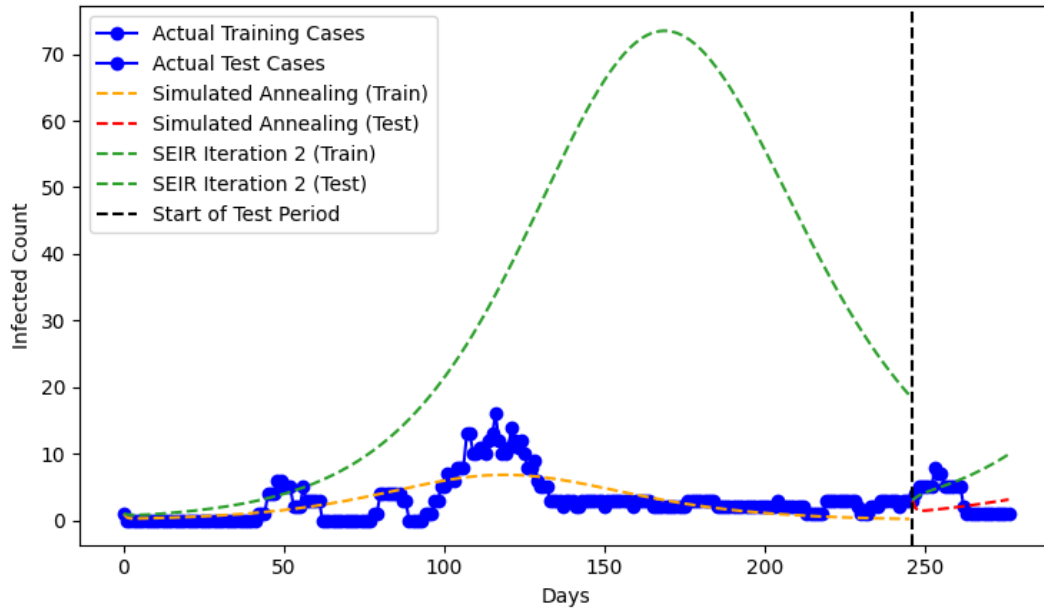


Figure 36 classical SEIR iteration 2 and Simulated Annealing model prediction against actual data

This contrast highlights the advantage of hyperparameter optimization using Simulated Annealing, which refines the model’s ability to replicate observed trends. The lower error rates and improved alignment with actual values confirm that the optimized SEIR model offers a more accurate result.

SEIR Optimization via Bayesian Inference Model

Another optimization applied to SEIR is the Bayesian inference through a Stan model. The optimization process employs Markov Chain Monte Carlo (MCMC) sampling, with Hamiltonian Monte Carlo (HMC) as the primary

algorithm for exploring the posterior distribution of the model parameters. This approach allows for robust parameter estimation and uncertainty quantification.

The first step in Bayesian inference is defining the prior distributions for the hyperparameters. These priors encode our initial beliefs about the possible values of the hyperparameters (β , σ , γ) before considering any observed data.

The model integrates the SEIR ODEs over time using `integrate_ode_rk45`, which numerically solves the differential equations with specified tolerances. The model then computes the daily incidence, which is a transformed version of the SEIR states adjusted by the reporting factor. This transformation converts the continuous model output into an observable metric that can be compared with the actual reported cases.

This Bayesian inference model leverages the strength of HMC in Stan to efficiently sample from the posterior distribution of SEIR model parameters. By incorporating uncertainty quantification and employing a robust likelihood framework (via the negative binomial distribution), the model provides a powerful tool for understanding and forecasting the spread of infectious diseases.

Figure 37 illustrates the process of Bayesian inference in refining parameter estimates. The prior distribution, shown in blue, represents the initial belief about the parameter's possible values before incorporating any data. It covers a broad range, indicating uncertainty. The likelihood distribution, in green, reflects the information derived from observed data, narrowing the range of plausible values.

Finally, the posterior distribution, in red, represents the updated belief, integrating both the prior knowledge and the likelihood of producing a refined estimate.

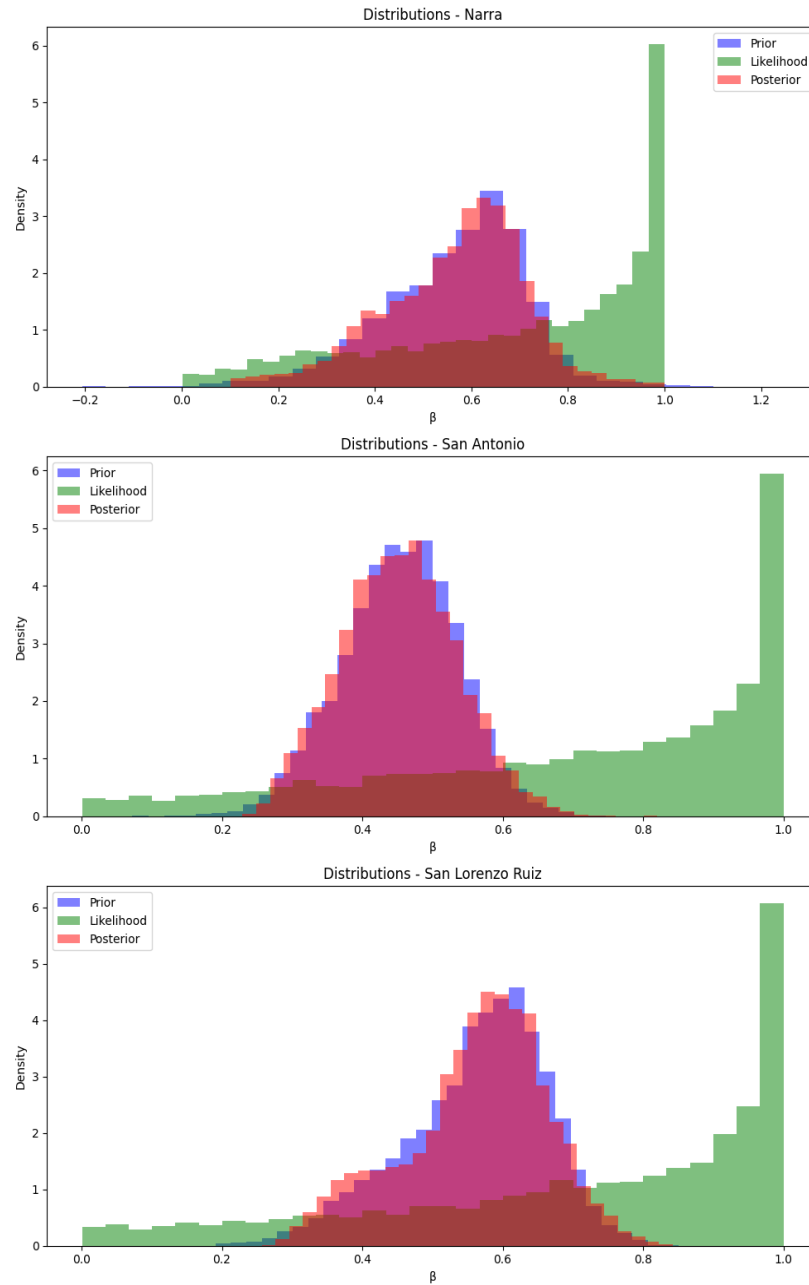


Figure 37 Bayesian Inference Prior, Likelihood, and Posterior values for Narra, San Antonio, and San Lorenzo Ruiz

This visualization highlights how Bayesian inference, particularly when implemented with Hamiltonian Monte Carlo (HMC), balances prior information with empirical data to yield well-calibrated parameter estimates. The posterior distribution aligns more closely with the likelihood while still retaining some influence from the prior, preventing overfitting. Such an approach is especially crucial in epidemiological modeling, where accurate parameter estimation directly impacts decision-making and public health policies.

The result of the optimization using Bayesian Inference was shown in Table 31

Table 31 Bayesian Inference optimized model hyperparameters

Barangay	Infectious Rate	Incubation Rate	Recovery Rate
San Antonio	0.39	0.31	0.34
San Lorenzo Ruiz	0.44	0.33	0.35
Narra	0.46	0.28	0.37

These values were applied to the classical SEIR model to validate whether improvement of error rate happened compared to the values obtained in SEIR Iteration 2.

Bayesian Inference Validation

Table 32 presents the error metrics for both iterations on the aggregated results from San Antonio, San Lorenzo Ruiz, and Narra, highlighting the improvements in model accuracy.

Table 32 Classical SEIR and Bayesian optimized model error metric values

Model Values	MAE	RMSE	NMAE	Error Reduction
Classical SEIR Values	143.17	174.34	1.50	Baseline
Bayesian Inference Optimized Values	70.52	81.09	0.93	~50.74% difference in MAE, ~53.49% difference in RMSE, and ~38.00% difference in NMAE

The Bayesian optimization through Stan, which utilizes MCMC with Hamiltonian Monte Carlo, yields an improvement of approximately 39–54% across all metrics, demonstrating the model's enhanced accuracy and robustness in capturing the epidemic dynamics within San Pedro City.

Figure 38, 39, and 40 shows the error reduction achieved by Bayesian inference is substantial, with a **50.74% decrease in MAE**, a **53.49% drop in RMSE**, and a **38.00% improvement in NMAE**. These improvements demonstrate that Bayesian inference not only enhances overall prediction accuracy but also reduces large deviations to the actual values. The lower RMSE indicates that

Bayesian inference effectively minimizes significant errors, ensuring that extreme deviations from actual case counts are far less frequent than in the classical SEIR model. Additionally, the decrease in NMAE suggests that the improvements are consistent across different scales, meaning that the Bayesian-optimized model maintains its accuracy whether predicting high or low infection counts.

The results of other barangays were also shown in Appendix BB, the results highlighted that Bayesian Inference greatly enhanced the result compared to classical SEIR.

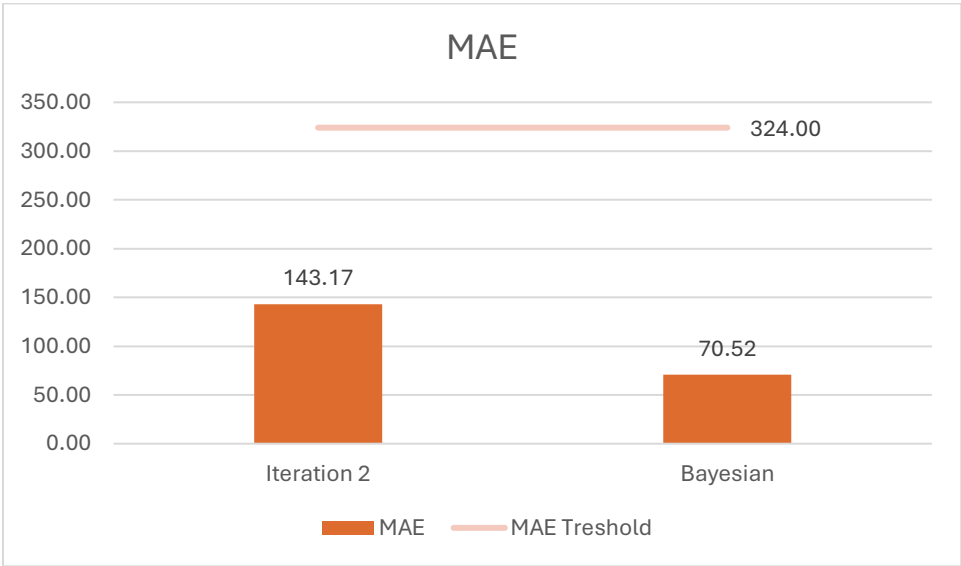


Figure 38 MAE values of classical SEIR iteration 2 and Bayesian Inference model

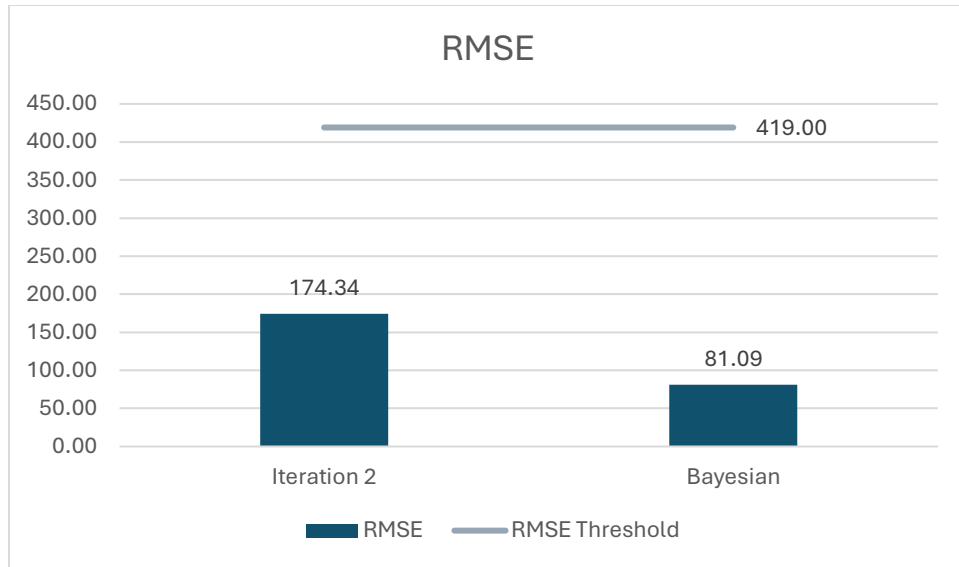


Figure 39 RMSE values of classical SEIR iteration 2 and Bayesian Inference model

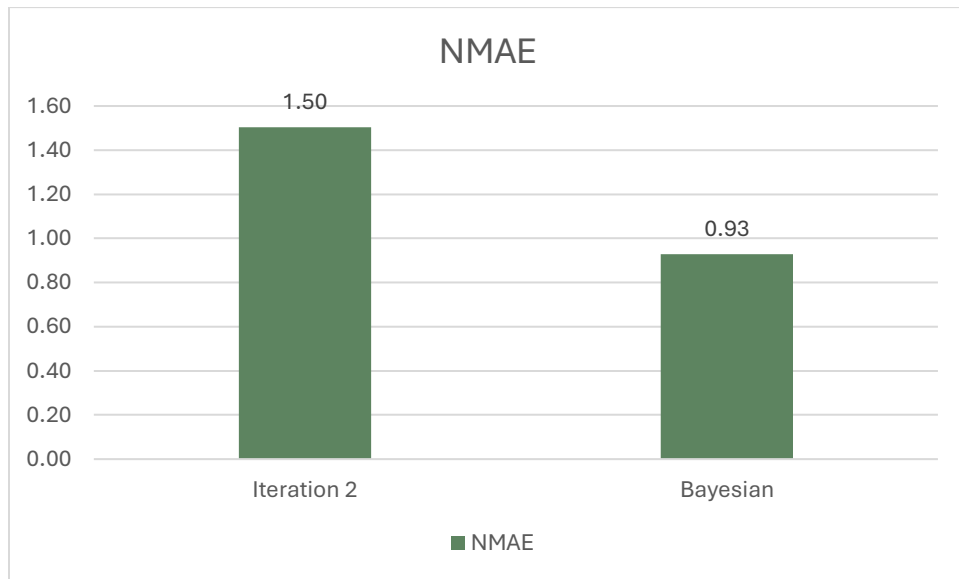


Figure 40 NMAE value of classical SEIR iteration 2 and Bayesian Inference model

To further show the difference between the prediction made by the parameters from Iteration 2 and the optimized parameters of Bayesian Inference,

Figure 41 shows the prediction of values using the parameters. This shows the number of infected individuals over time; it was compared to the actual date.

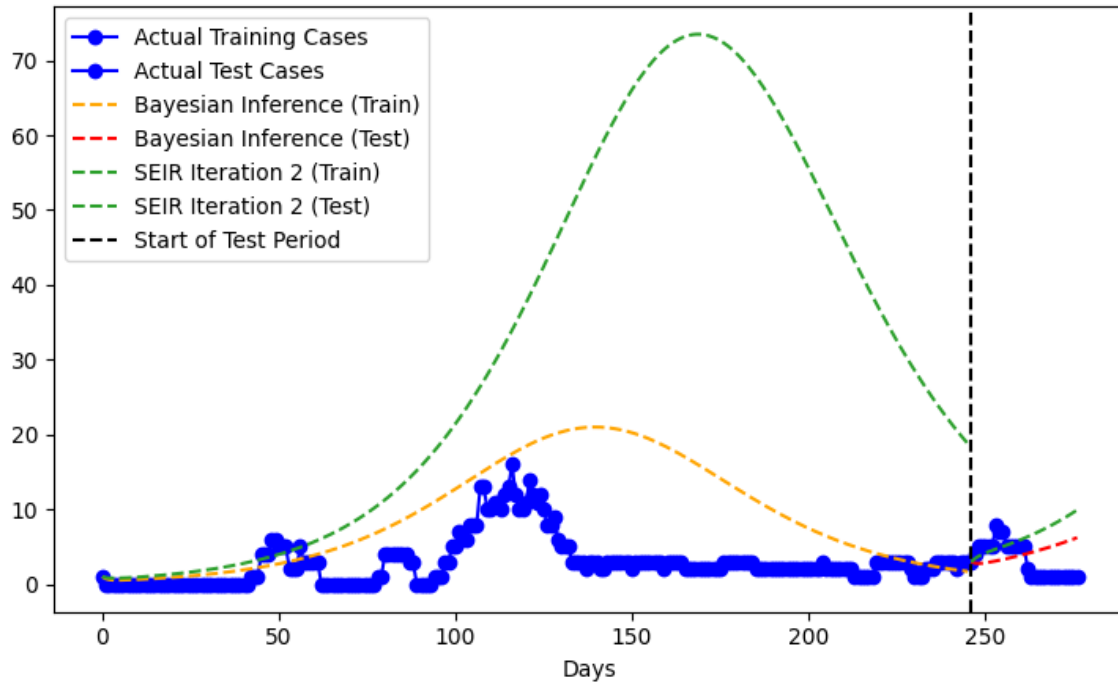


Figure 41 classical SEIR iteration 2 and Bayesian Inference model prediction against actual data

The Bayesian Inference model provides a more data-driven approach by incorporating uncertainty and optimizing parameters based on observed data compared to the prediction using the hyperparameters of Iteration 2. The Bayesian model's predictions (represented by the orange dashed line for training and red dashed line for testing) follow the actual recorded cases more closely, demonstrating its ability to adapt to real-world fluctuations. Unlike the deterministic SEIR model, which relies on fixed parameter estimates, Bayesian Inference dynamically updates its predictions, making it more robust when

generalizing unseen data. In the test period, its estimates remain closer to the actual cases, showing a more accurate and reliable forecast compared to the SEIR model.

Both the Simulated Annealing and the Bayesian Inference model consistently outperformed the baseline hyperparameters across different test cases. In every testing scenario, these models produced lower errors and better predictive accuracy compared to the classical SEIR model with its initial hyperparameter settings. Appendix V provides a detailed breakdown of the model performances in three barangays: Narra, San Lorenzo Ruiz, and San Antonio, before aggregating the results.

Overall, both models demonstrated superior predictive performance, with lower errors and better alignment with actual infection trends compared to the baseline SEIR model. These results confirm that refining hyperparameters through Bayesian Inference and Simulated Annealing leads to more reliable epidemiological forecasts, making them valuable alternatives to traditional deterministic models. Refer to Appendix BB to show the summary of validation results of each parameter estimation technique compared to the classical SEIR.

Now that both models were proven better than the baseline model, the researchers conducted more tests to identify which optimization technique will be integrated in the web application.

Model Selection

The section above discussed the implementation of the classical SEIR, Simulated Annealing optimized, and Bayesian Inference optimized models. For the model selection, the researchers conducted various tests by using varying date ranges for the training of the model. The tests performed were the (1) integration of different age groups, (2) different granularity of the dataset by using Barangay level datasets, and (3) test on the overall San Pedro municipality. For the different tests, the same dataset of San Pedro was used and for the sake of showcasing the training results. Finally, the model selected was based on the lowest RMSE, MAE, and NMAE values after aggregating via mean the results of all the tests performed.

Training Range Variations

For the training range variations testing, the goal was to evaluate how well the SEIR model can predict the infectious spread with varying length of training time range. Table 33 shows the start and end date used for each test. The training sets below were used to train model and validated using the date from 01/14/2022 until 02/13/2022 since these date range consist of the peak cases during the pandemic time.

Table 33 Training range test, number of records, and start and end dates

Tests	Number of Records	Start Date	End Date
Test 1	245	05/13/2021	01/13/2022
Test 2	214	06/13/2021	01/13/2022
Test 3	153	08/13/2021	01/13/2022
Test 4	92	10/13/2021	01/13/2022

Each optimization technique underwent iterative refinement, with error metrics compared between iterations to achieve optimal results.

Barangay Level Tests

The researchers performed the testing on the varying training time period on a barangay-level cases. This was performed to test the capability and consistency of the model when given data is smaller than the municipality level. The chosen barangays from the San Pedro municipality were San Lorenzo Ruiz, San Antonio, and Narra as the sample barangay. Table 34 shows the summary of the model validation for the barangay level tests.

Table 34 Barangay-level training with varying time period model and error metrics

Models	RMSE	MAE	NMAE
Test 1: 8 Months Training San Lorenzo Ruiz, San Antonio, Narra			
Classical SEIR	174.34	143.17	1.50
Simulated Annealing	58.75	47.33	0.69
Bayesian Inference	81.09	70.52	0.93
Test 2: 7 Months Training San Lorenzo Ruiz, San Antonio, Narra			
Classical SEIR	174.34	143.17	1.50
Simulated Annealing	56.48	50.46	0.68
Bayesian Inference	86.31	74.33	1.70
Test 3: 5 Months Training San Lorenzo Ruiz, San Antonio, Narra			
Classical SEIR	174.34	143.17	1.50
Simulated Annealing	44.67	41.33	0.62
Bayesian Inference	65.51	58.56	0.78
Test 4: 3 Months Training San Lorenzo Ruiz, San Antonio, Narra			
Classical SEIR	174.34	143.17	1.50
Simulated Annealing	67.74	49.81	0.61
Bayesian Inference	78.25	68.35	0.75

The Simulated Annealing and Bayesian Inferenced optimized model shows great difference in comparison to the classical SEIR model. Table 35 shows the mean score of the aggregated results from the four tests that were performed for the Barangay-level test.

Table 35 Mean score of aggregated results from the four-time range testing on barangay-level

Models	Mean RMSE	Mean MAE	Mean NMAE	Error Reduction
Classical SEIR	174.34	143.17	1.50	Baseline
Simulated Annealing	56.91	47.23	0.65	~67.36% difference in RMSE, ~67.01% difference in MAE, and ~56.98% difference in NMAE
Bayesian Inference	77.79	67.94	1.04	~55.38 % difference in RMSE, ~52.54 % difference in MAE, and ~30.78% difference in NMAE

The Simulated Annealing model reduced achieved a ~67.36% error reduction in the RMSE value, ~67.01% reduction in the MAE, and a ~56.98% reduction on the NMAE value. While the Bayesian Inference optimized model achieved a ~55.38 % error reduction on RMSE, ~52.54 % reduction in MAE, and ~30.78 % reduction in NMAE. These substantial improvements in comparison to the baseline classical SEIR model indicates improvement on the optimized models. Figure 42 and 43 show the difference between the RMSE, MAE, and NMAE values for the models.

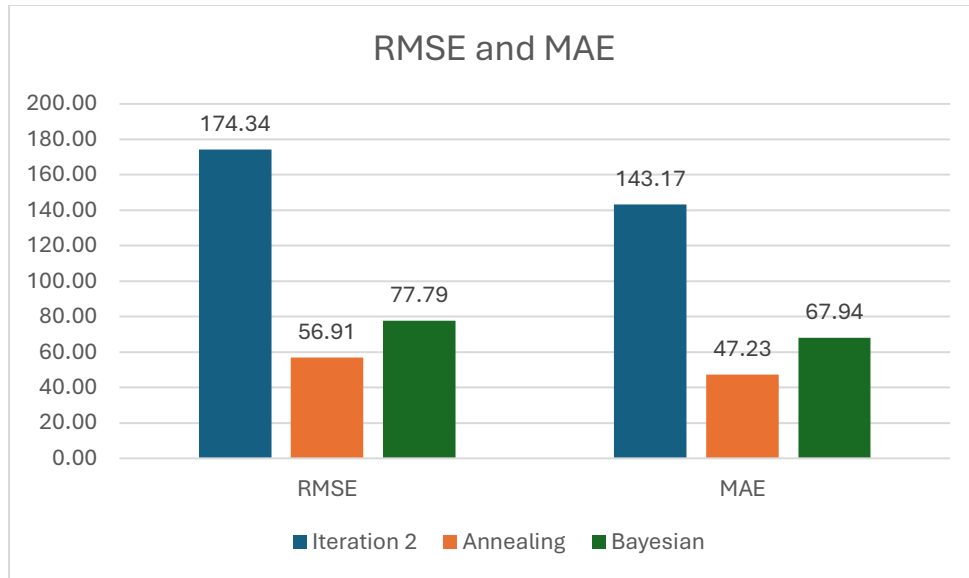


Figure 42 Aggregated Mean RMSE and MAE of classical SEIR, Simulated Annealing, and Bayesian Inference model for time-range testing

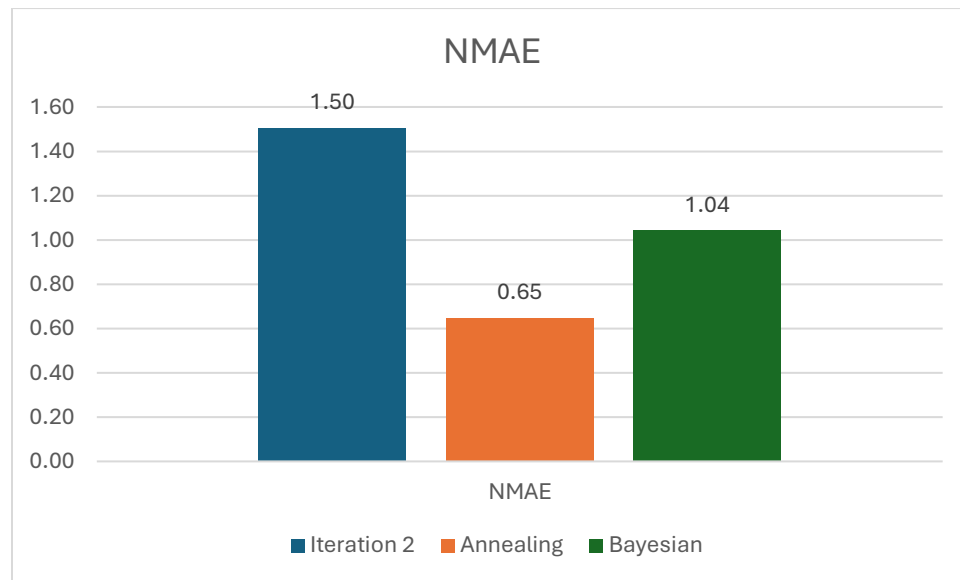


Figure 43 Aggregated Mean NMAE of classical SEIR, Simulated Annealing, and Bayesian Inference model for time-range testing

The SEIR model's predictions for Barangays Narra, San Antonio, and San Lorenzo Ruiz vary based on the inference method used. Figure 44 on the next page shows the actual

values against the prediction of the Simulated Annealing and Bayesian Inference optimized model for barangay Narra.

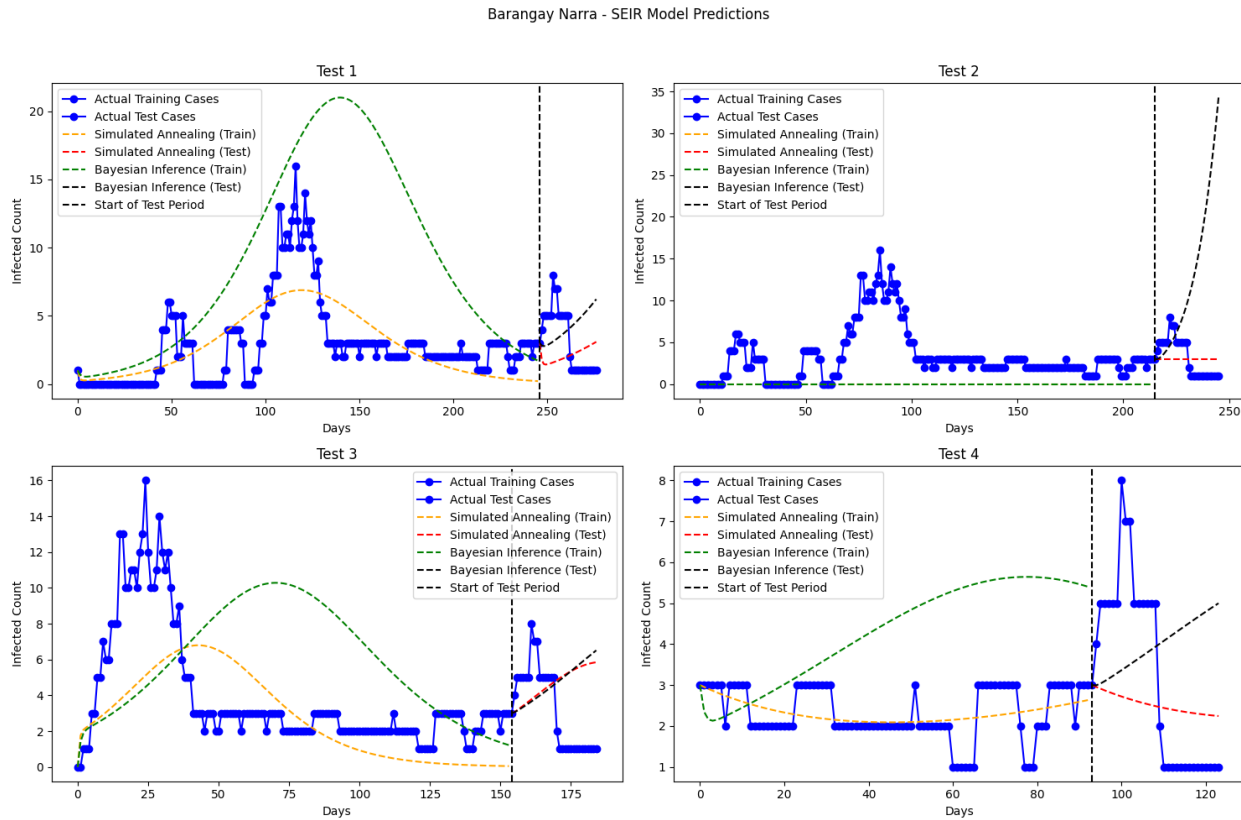


Figure 44 Actual against prediction of Simulated Annealing and Bayesian Inference model for barangay Narra

Simulated Annealing provides smoother and more stable forecasts but tends to underestimate peak infection cases. In contrast, Bayesian Inference is more reactive, capturing sudden increases but often overestimating case counts. In Barangay San Antonio, seasonality in cases poses a challenge, with Bayesian Inference being more aggressive in its forecasts, sometimes predicting sharper rises than observed (as seen in Figure 45).

Barangay San Antonio - SEIR Model Predictions

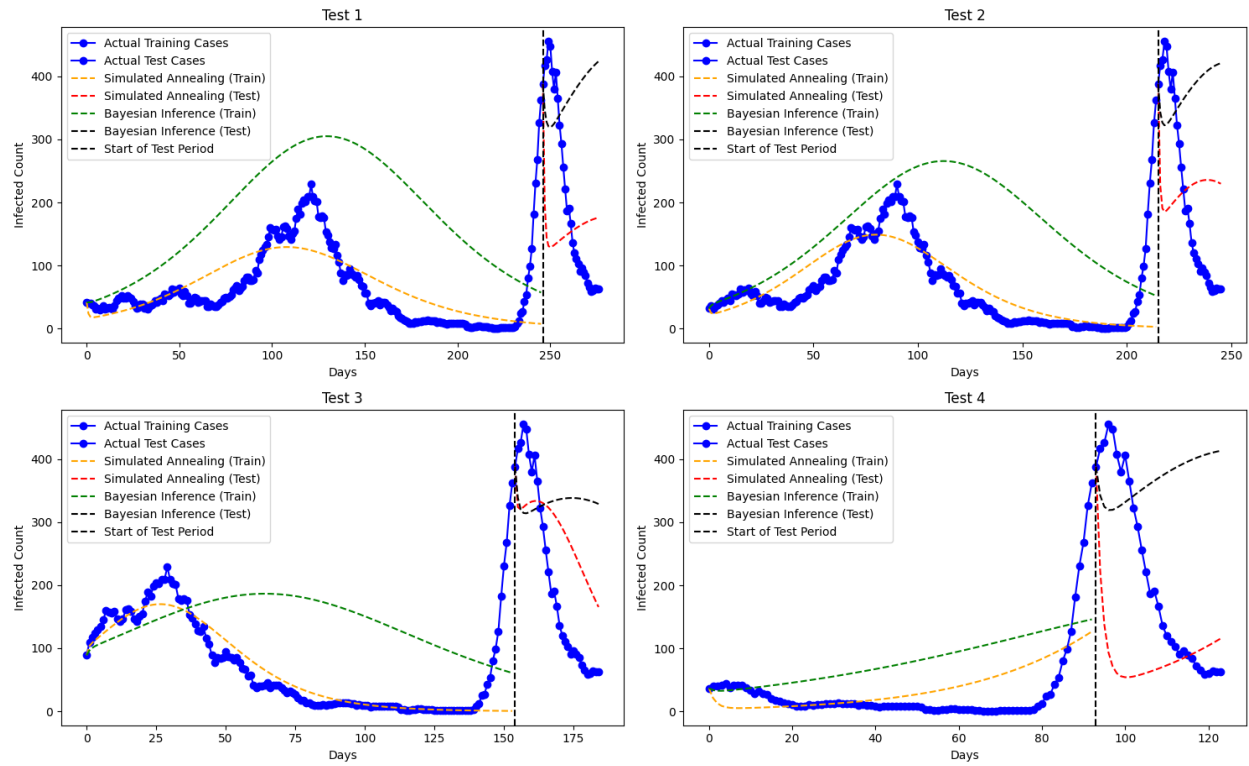


Figure 45 Actual against prediction of Simulated Annealing and Bayesian Inference model for barangay San Antonio

In Barangay San Lorenzo Ruiz, both methods align well with overall trends but struggle with capturing sudden outbreaks accurately (as seen in Figure 46). Overall, Simulated Annealing is more conservative, while Bayesian Inference is more dynamic but prone to variability.

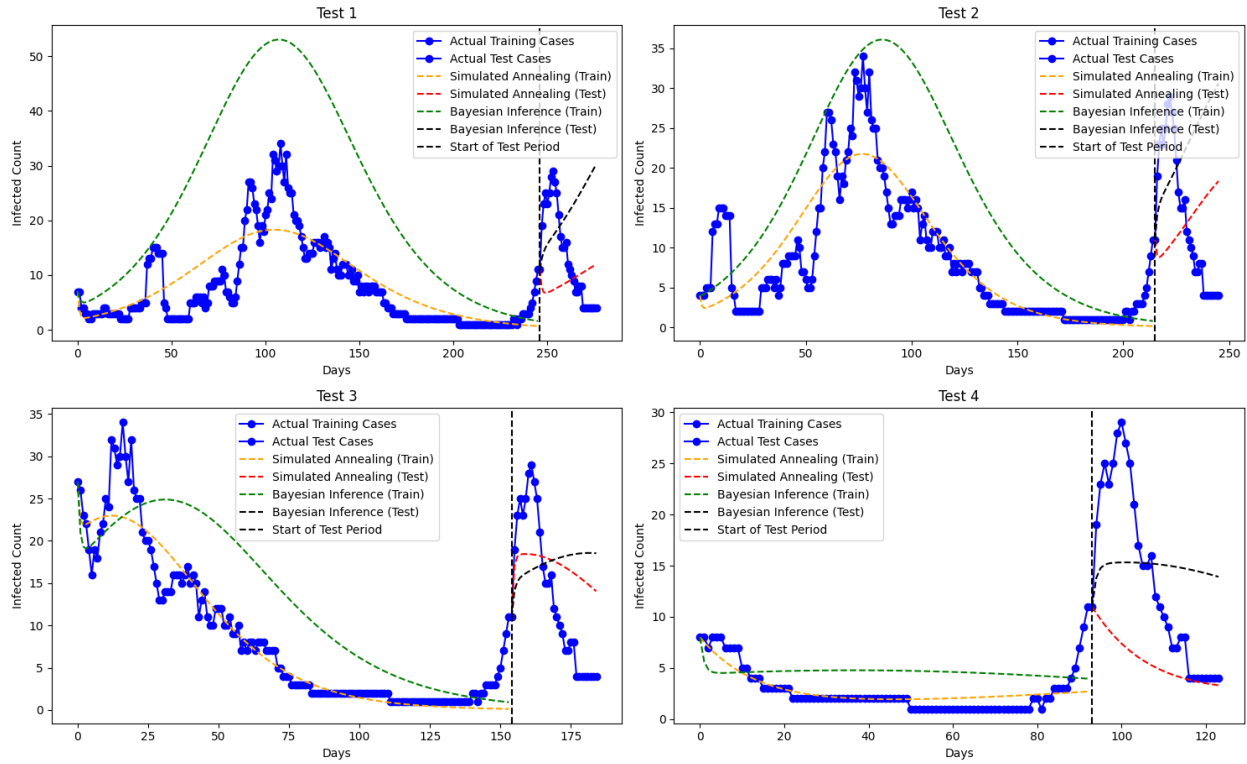


Figure 46 Actual against prediction of Simulated Annealing and Bayesian Inference model for barangay San Lorenzo Ruiz

Ultimately, the barangay-level tests confirms that both the Simulated Annealing and the Bayesian Inference optimization methods substantially enhances the predictive accuracy of the SEIR model on a barangay-level data. It is noted that the Simulated Annealing optimized model yielded greater error reduction in all of the error metrics in comparison to the Bayesian Inference optimized model.

Age Group Tests

The researchers incorporated age group variations into the model to better capture the heterogeneity in transmission dynamics across different segments of the population. The age groups were segmented into six categories: nine and below, 10 to 17, 18 to 39, 40 to 59, 60 to 79, and 80 and above. This stratification enables the models to account for differences in social behavior, susceptibility, and contact patterns among age groups, which are critical factors in the spread of an epidemic.

Table 36 on the next page presents the performance of the three SEIR model evaluated on the San Pedro dataset across six age groups, using an 8-month training period with a 1-month prediction window. The table reports the error metrics for each age group: 9 and below, 10–17, 18–39, 40–59, 60–79, and 80 and above. Overall, the optimized models, Simulated Annealing and Bayesian Inference, consistently yield lower error values than the baseline Classical SEIR model.

Table 36 Age Group Specific with varying time period test models and error metrics

Models	RMSE	MAE	NMAE
Test 1: 8 Months Training San Lorenzo Ruiz, San Antonio, Narra			
Classical SEIR	32.89	26.86	1.74
Simulated Annealing	11.70	10.59	0.90
Bayesian Inference	15.33	13.21	1.94
Test 2: 7 Months Training San Lorenzo Ruiz, San Antonio, Narra			
Classical SEIR	32.89	26.86	1.74
Simulated Annealing	12.80	11.28	0.90
Bayesian Inference	16.05	13.73	2.11
Test 3: 5 Months Training San Lorenzo Ruiz, San Antonio, Narra			
Classical SEIR	32.89	26.86	1.74
Simulated Annealing	14.39	12.76	0.88
Bayesian Inference	12.45	10.78	1.64
Test 4: 3 Months Training San Lorenzo Ruiz, San Antonio, Narra			
Classical SEIR	32.89	26.86	1.74
Simulated Annealing	11.95	8.88	0.67
Bayesian Inference	17.26	14.57	1.74

The Simulated Annealing and Bayesian Inference optimized model appears to acquire lower error metrics on age specific data. Table 37 demonstrates the difference between each model.

Table 37 Difference between models' error metrics on age-specific test

Models	Mean RMSE	Mean MAE	Mean NMAE	Error Reduction
Classical SEIR	32.89	26.86	1.74	Baseline
Simulated Annealing	12.71	10.88	0.84	~61.35% difference in RMSE, ~59.51% difference in MAE, and ~51.91% difference in NMAE against the Baseline
Bayesian Inference	15.27	13.07	1.86	~53.55% difference in RMSE, ~51.33% difference in MAE, and ~-6.66% difference in NMAE against the Baseline

Both the Simulated Annealing and Bayesian Inference optimized model achieved great error reduction in comparison to the baseline SEIR model. The difference suggests the importance of considering age as a part of the dataset on optimizing the SEIR model. Figure 47 and 48 show the difference between the Simulated Annealing and Bayesian Inference model's error metrics.

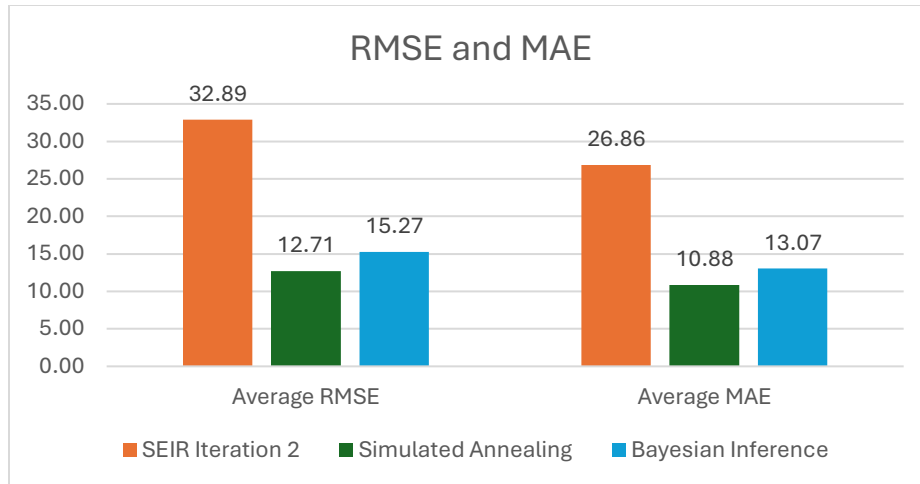


Figure 47 Mean RMSE and MAE between classical SEIR, Simulated Annealing, and Bayesian Inference on age-specific test

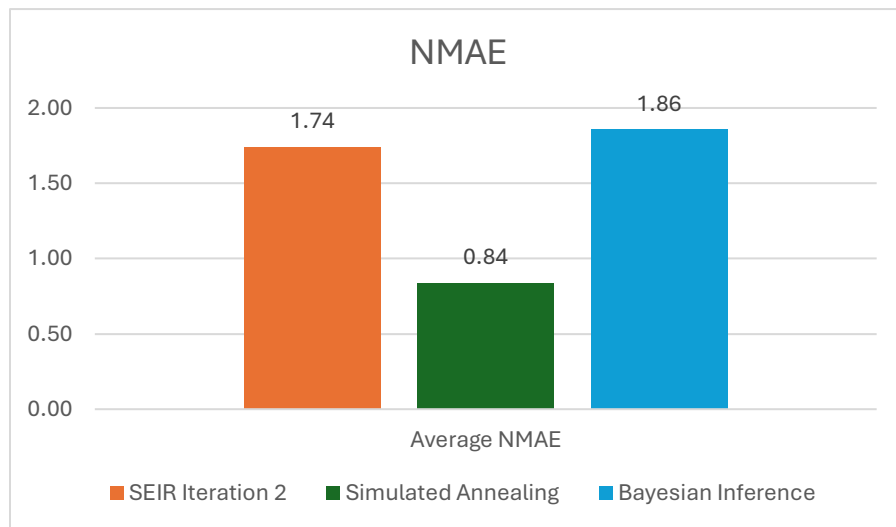


Figure 48 Mean NMAE between classical SEIR, Simulated Annealing, and Bayesian Inference on age-specific test

The results clearly demonstrate that both the Simulated Annealing and Bayesian Inference models outperform the baseline SEIR Iteration 2 model in terms of error reduction when applied to age groups. Among the three models, Simulated Annealing

achieves the best performance, with a 61.35% reduction in RMSE, 59.51% reduction in MAE, and 51.91% reduction in NMAE compared to SEIR. This significant improvement indicates that Simulated Annealing is the most effective approach to optimization. Meanwhile, Bayesian Inference also shows notable improvements, reducing RMSE by 53.55% and MAE by 51.33%. However, it exhibits a 6.66% increase in NMAE, suggesting that while it enhances accuracy in some respects, there may be trade-offs in model generalization.

The SEIR Iteration 2 model serves as the baseline, with the highest error rates, emphasizing the need for optimization techniques. The comparison highlights the effectiveness of incorporating age-specific data into the model, as both Simulated Annealing and Bayesian Inference achieve significant error reductions. The findings suggest that advanced optimization techniques can substantially improve SEIR models' predictive accuracy, making them more reliable for epidemiological forecasting.

Overall San Pedro Test

The models was tested using the total cases for the San Pedro municipality. The researchers tested to validate the usability of the model, aside from the barangay-level data to a municipal-level of data.

Table 38 on the next page presents the mean error metrics evaluated on the overall San Pedro dataset. The table compares performance across four tests with different training durations (8, 7, 5, and 3 months). The Classical SEIR model serves as the baseline with

constant error values, while both optimized models show marked improvements. Notably, the Simulated Annealing model consistently yields the lowest errors.

Table 38 Overall training with varying time period models and error metrics

Models	RMSE	MAE	NMAE
Test 1: 8 Months Training San Pedro Overall			
Classical SEIR	2339.43	1899.49	1.75
Simulated Annealing	779.70	633.83	0.58
Bayesian Inference	1039.89	899.29	0.83
Test 2: 7 Months Training San Pedro Overall			
Classical SEIR	2339.43	1899.49	1.75
Simulated Annealing	759.33	686.61	0.63
Bayesian Inference	1055.57	913.59	0.84
Test 3: 5 Months Training San Pedro Overall			
Classical SEIR	2339.43	1899.49	1.75
Simulated Annealing	963.92	830.29	0.77
Bayesian Inference	795.78	713.40	0.66
Test 4: 3 Months Training San Pedro Overall			
Classical SEIR	2339.43	1899.49	1.75
Simulated Annealing	960.20	723.54	0.67
Bayesian Inference	907.64	799.97	0.74

Even on a municipal-level data, the optimized models maintain the consistency on the improvement of error metrics against the baseline model. Specifically, the Simulated Annealing model shows promising improvement in the error reduction as seen in Table 39.

Table 39 Aggregated mean of error metrics for municipal-level data test

Models	Mean RMSE	Mean MAE	Mean NMAE	Error Reduction
Classical SEIR	2339.43	1899.49	1.75	Baseline
Simulated Annealing	865.79	718.57	0.66	~62.99% difference in RMSE, ~62.17% difference in MAE, and ~62.17% difference in NMAE against the Baseline
Bayesian Inference	949.72	831.56	0.77	~59.40% difference in RMSE, ~56.22% difference in MAE, and ~56.22% difference in NMAE against the Baseline

Figure 49 and 50 shows the difference between the Simulated Annealing and Bayesian Inference optimized model's error metrics to the classical SEIR.

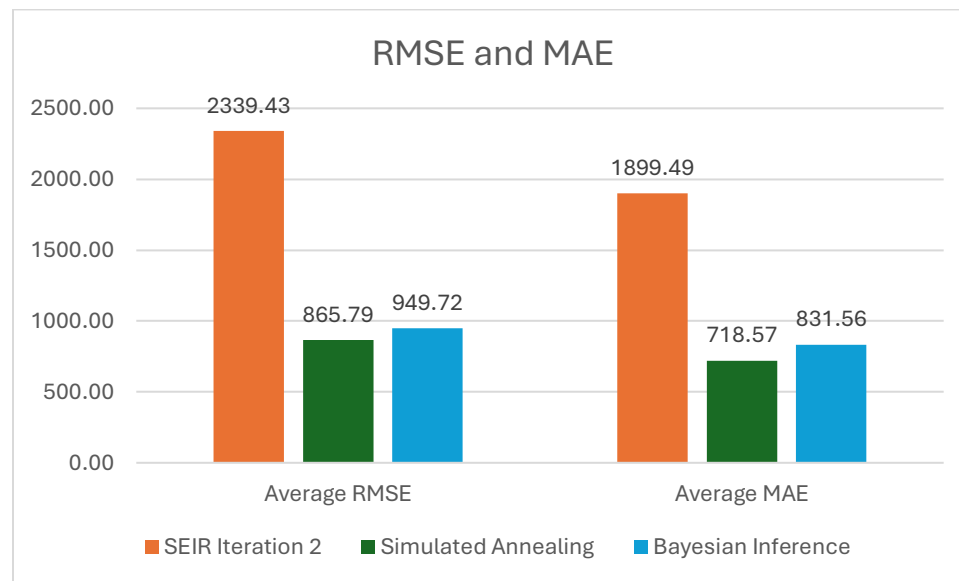


Figure 49 Aggregated mean of RMSE and MAE of classical SEIR, Simulated Annealing, and Bayesian Inference model for municipal-level data test

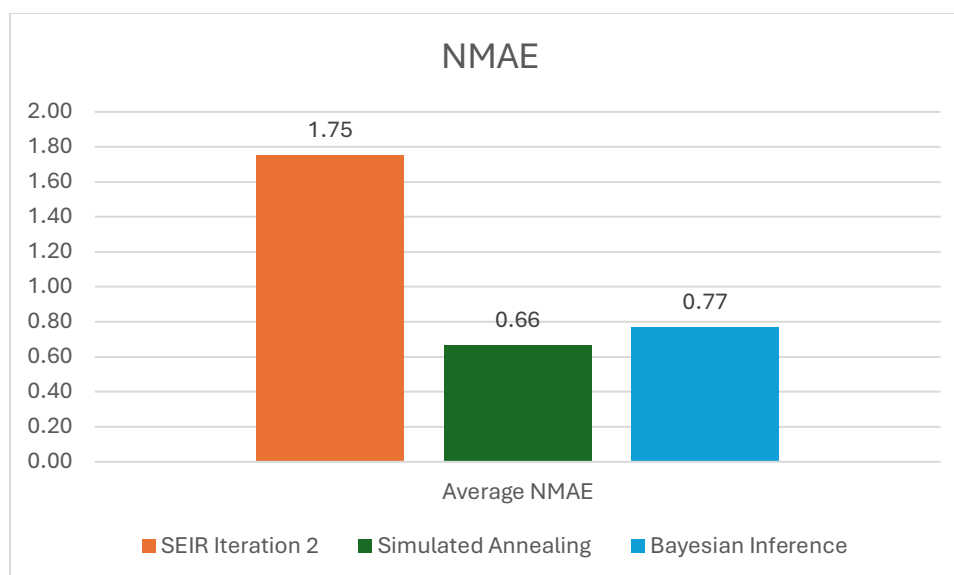


Figure 50 Aggregated mean of NMAE of classical SEIR, Simulated Annealing, and Bayesian Inference model for municipal-level data test

As seen from the results of the Overall San Pedro Test, the models' performance are consistent even when applied to a municipal-level data.

Aggregated Mean Tests Error

Table 40 summarizes the aggregated mean error metrics—Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and Normalized Mean Absolute Error (NMAE)—calculated across all tests: Barangay Overall, Barangay Age Group, and San Pedro Overall.

Table 40 Aggregated Mean Result of Model Selection Tests

Model	Mean RMSE	Mean MAE	Mean NMAE	Error Reduction
Classical SEIR	838.49	642.62	32.39	Baseline
Simulated Annealing	308.58	243.36	22.95	~63.20% difference in RMSE, ~62.13% difference in MAE, and ~29.13% difference in NMAE against the Baseline
Bayesian Inference	344.31	281.89	19.33	~58.94% difference in RMSE, ~56.13% difference in MAE, and ~40.30% difference in NMAE against the Baseline

Compared to the Classical SEIR model, both optimized models demonstrate considerable improvements. Figure 51 and 52 shows the differences between the Simulated Annealing and Bayesian Inference model.

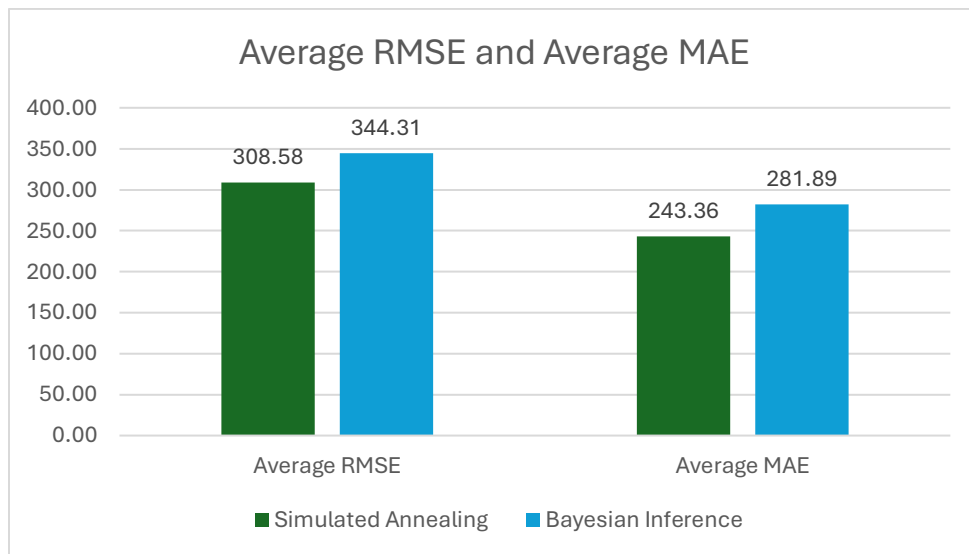


Figure 51 Aggregated Mean RMSE and MAE of Model Selection Tests

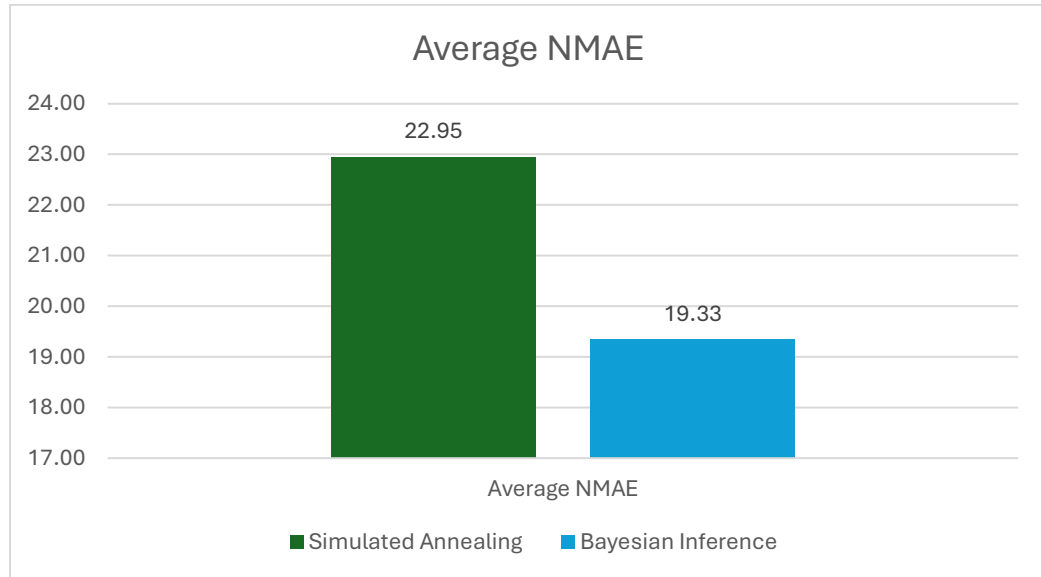


Figure 52 Aggregated Mean NMAE of Model Selection Tests

Notably, the Simulated Annealing model exhibits the lowest aggregated errors of Mean MAE = 243.36, Mean RMSE = 305.58, and NMAE of 22.95 value.

Final Model

The model selection process involved a comprehensive evaluation of three SEIR model variants: Classical SEIR, Simulated Annealing optimized, and Bayesian Inference optimized across multiple tests. These tests included: Overall Training with Varying Time Periods, Age Group-Specific Analysis, Barangay-Level Testing, and lastly aggregating the mean metrics.

Based on comprehensive testing across various temporal and spatial dimensions, the Simulated Annealing optimized model consistently provided superior performance by significantly reducing forecasting errors. Therefore, the Simulated Annealing model is selected as the final model for epidemic forecasting in this study, underscoring the efficacy of advanced optimization techniques in enhancing the predictive accuracy of epidemiological models.

When predicting future cases using Simulated Annealing, the model starts with an initial set of conditions, determined from observed case data. These include the initial number of exposed, infectious, and recovered individuals, with the remaining population assumed to be susceptible. The model then numerically integrates the SEIR differential equations over a given time horizon to estimate how these compartments evolve.

Simulated Annealing optimizes beta, sigma, and gamma by iteratively adjusting these parameters to minimize the difference between predicted and actual cases. The objective function, which the algorithm seeks to minimize, measures the mean squared error between predicted and actual infections. A regularization term is included to prevent extreme values that could lead to unrealistic predictions. Figure <num> shows the results on each process.

To prevent unrealistic projections, the model enforces non-negative constraints on the predicted values and maintains biologically plausible parameter ranges. By leveraging Simulated Annealing's capability to navigate complex parameter spaces, the model achieves enhanced predictive accuracy, reducing forecasting errors compared to traditional SEIR models.

Table 41 shows more details on the optimization process and prediction of the Simulated Annealing.

Table 41 Optimization Process Details

Category	Details
Optimization Technique	Simulated Annealing
Hyperparameter Bounds	β : 0.0001 - 2.0 σ : 0.0001 - 1.0 γ : 0.0001 - 1.0
Objective Function	Reduce Mean Squared Error (MSE)
Error Minimized	Prediction Error (MSE + Regularization)
Initial Condition	S,E,I,R initial values
Search Strategy	Global Search with Temperature-based probability
Values Predictor	SEIR Mathematical Model
Library	Dual Annealing (Scipy)
Dual Annealing Parameters	Initial Temperature: 5230.0 Max Iterations: 1000 Max Objective Function Calls: 10,000,000
Acceptance Criteria	Initial Acceptance: -5.0 Acceptance Boundary: (-1e4, -5]

Model Confidence Testing

For the model confidence testing, the researchers used the dataset from the municipality of Biñan and Cabuyao and its barangays. Specifically, Mamatid was selected as the sample barangay for Cabuyao, while Santo Tomas was chosen for Biñan. To validate and compare the results acquired from both the model validation from the San Pedro dataset and the Confidence testing dataset. Table 42 shows the summary of error metrics generated by the Simulated Annealing optimized model with the classical SEIR model's error metrics as baseline.

Table 42 Summary of error metrics for Classical SEIR and Simulated Annealing model Confidence Test

Model	Barangay	RMSE	MAE	NMAE
Classical SEIR	San Lorenzo Ruiz (from San Pedro)	24.58	19.70	1.44
	San Antonio (from San Pedro)	518.70	426.18	2.00
	Narra (from San Pedro)	5.00	3.89	1.21
	Mamatid (from Cabuyao)	263.35	204.72	1.41
	Santo Tomas (from Biñan)	403.11	322.79	1.73
Simulated Annealing	San Lorenzo (from San Pedro)	11.46	9.07	0.66
	San Antonio (from San Pedro)	159.53	129.27	0.61
	Narra (from San Pedro)	2.52	2.90	0.78
	Mamatid (from Cabuyao)	103.06	83.48	0.58
	Santo Tomas (from Biñan)	132.05	108.31	0.58

Figures 53 and 54 show depicts the error RMSE and MAE values with the reference threshold of 419 for the RMSE and 324 for the MAE.

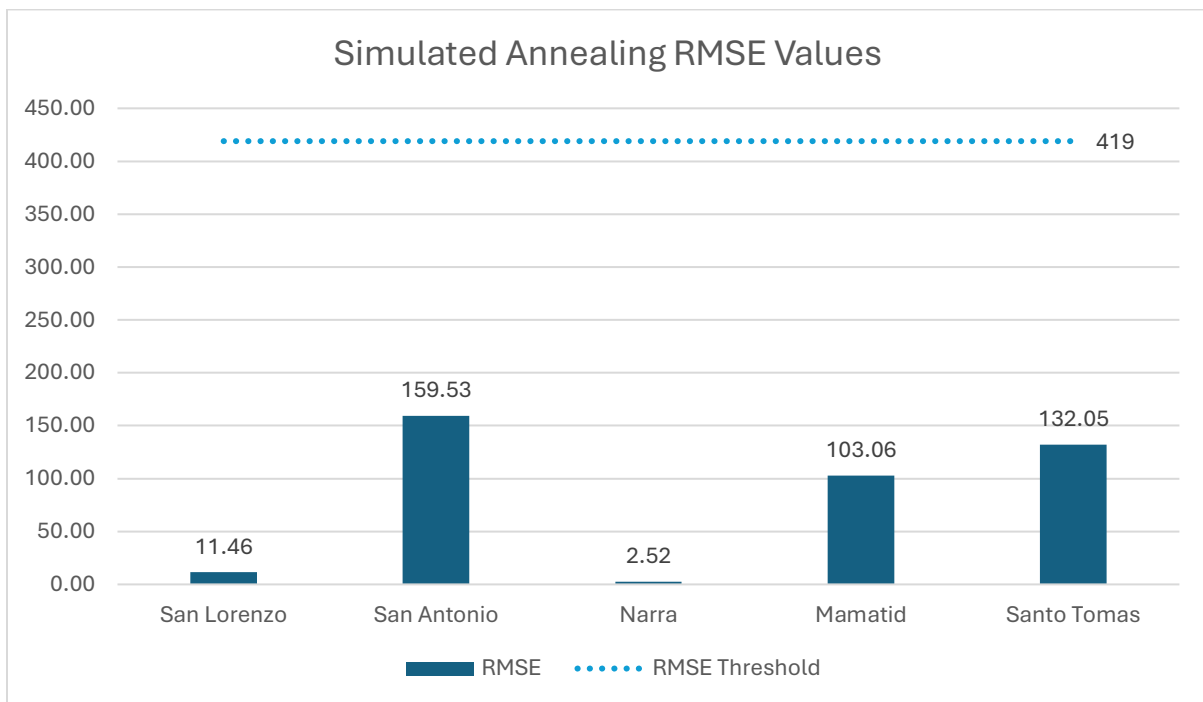


Figure 53 Simulated Annealing RMSE values for Confidence Test

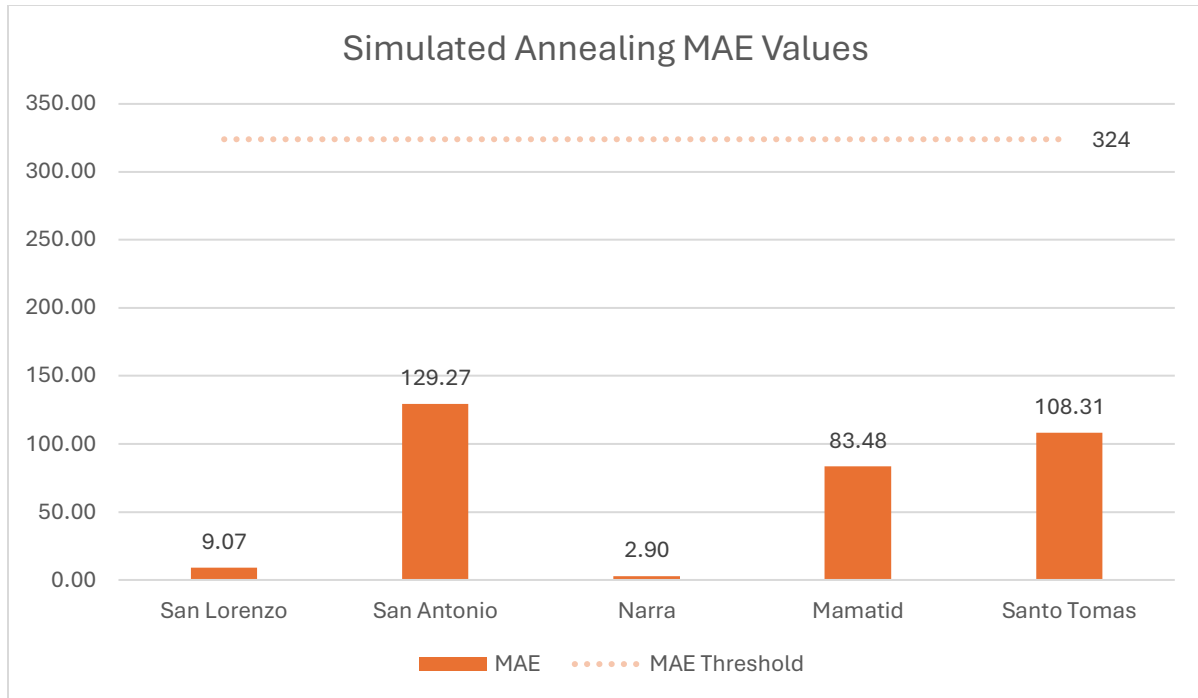


Figure 54 Simulated Annealing MAE values for Confidence Test

As observed, the Simulated Annealing optimized model's RMSE and MAE values are below the maximum threshold even on the confidence testing dataset's barangay Mamatid and Santo Tomas.

Additionally, the model hold a consistent error reduction value across the tested barangays. Table 43 shows the error reduction rates on each error metric for the different barangays.

Table 43 Error Reduction Rates for each barangay

Barangay	RMSE Error Reduction	MAE Error Reduction	NMAE Error Reduction
San Lorenzo Ruiz	53.38%	53.99%	53.99%
San Antonio	69.24%	69.67%	69.67%
Narra	42.03%	35.13%	35.13%
Mamatid	60.87%	59.22%	59.22%
Santo Tomas	67.24%	66.45%	66.45%

The confidence testing results demonstrate the reliability of the Simulated Annealing optimized SEIR model in reducing error metrics across various barangays. The model consistently outperforms the classical SEIR model, with significant reductions in RMSE, MAE, and NMAE values, ensuring that predictions remain within the predefined accuracy thresholds. These findings validate the model's robustness and its applicability for barangay-level infectious disease prediction. With consistent performance across multiple datasets, the optimized model proves to be a viable tool for improving predictive accuracy in infectious disease modeling and public health decision-making.

The results of other barangay was shown in Appendix BB, Appendix CC, and Appendix DD that will further shows the significant improvement done by the Simulated Annealing in predicting future infectious disease in barangay level compared to the classical SEIR.

The Figures 55, 56, and 57 below shows the actual values against the prediction of the classical SEIR and Simulated Annealing model for barangays Narra, San Antonio, and San Lorenzo Ruiz from the Municipality of San Pedro.

Barangay Narra - SEIR Model Predictions

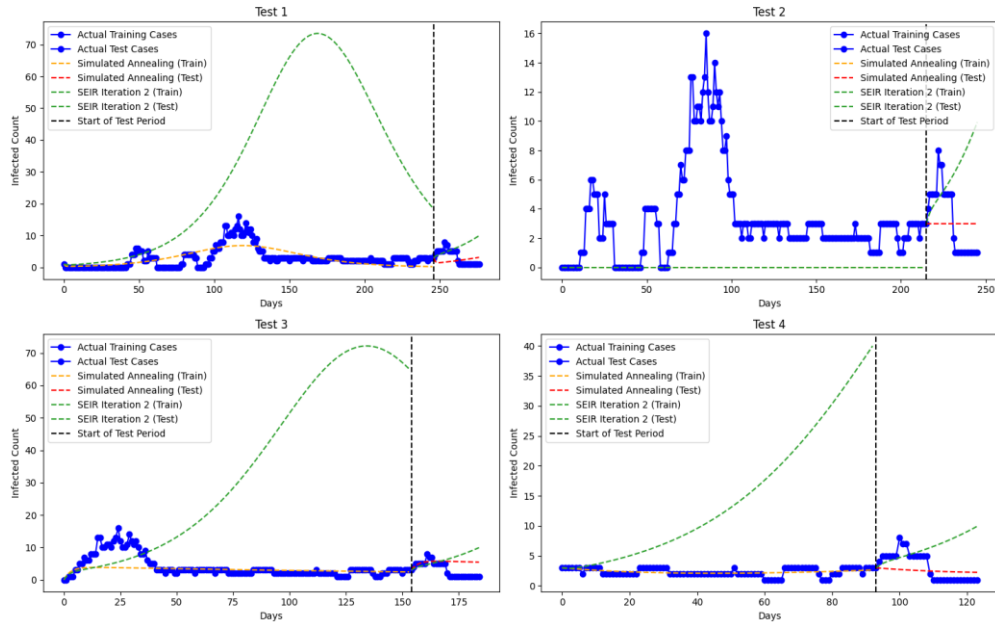


Figure 55 Barangay Narra SEIR models predictions

Barangay San Antonio - SEIR Model Predictions

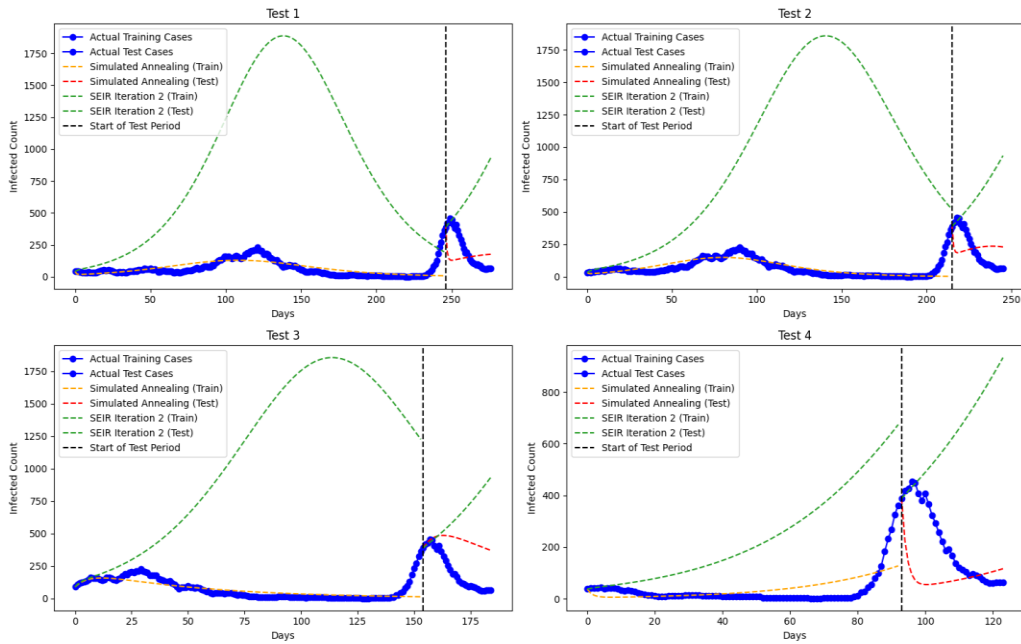


Figure 56 Barangay San Antonio SEIR models predictions

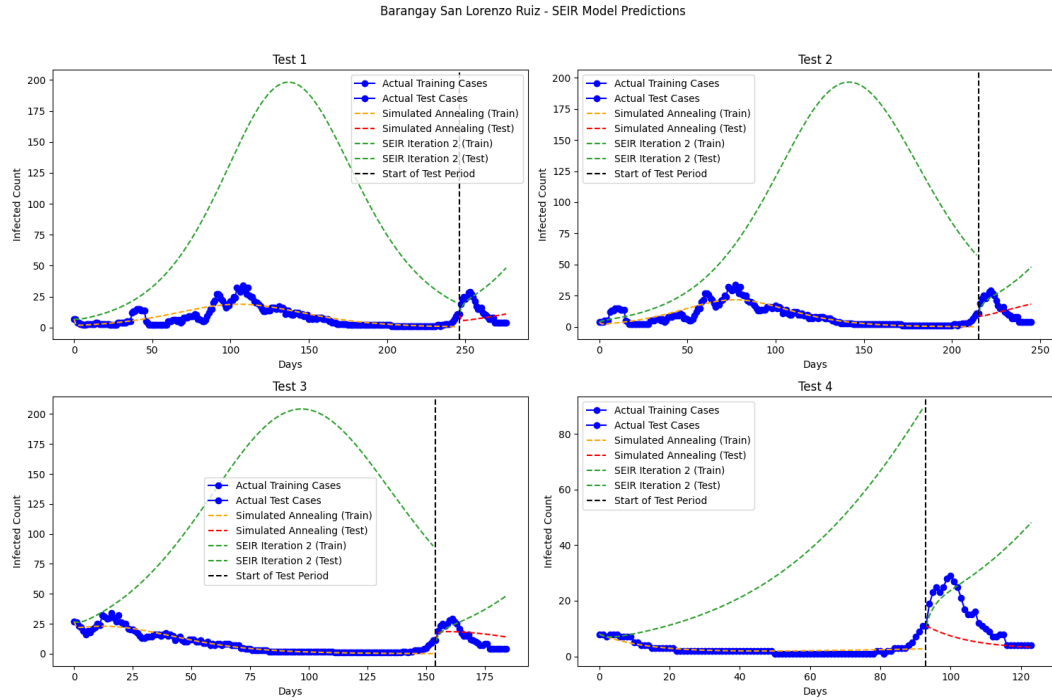


Figure 57 Barangay San Lorenzo Ruiz SEIR models predictions

The Figure 58 and 59 below shows the actual values against the prediction of the classical SEIR and Simulated Annealing model for barangays Santo Tomas and Mamatid.

Barangay Santo Tomas (Calabuso) - SEIR Model Predictions

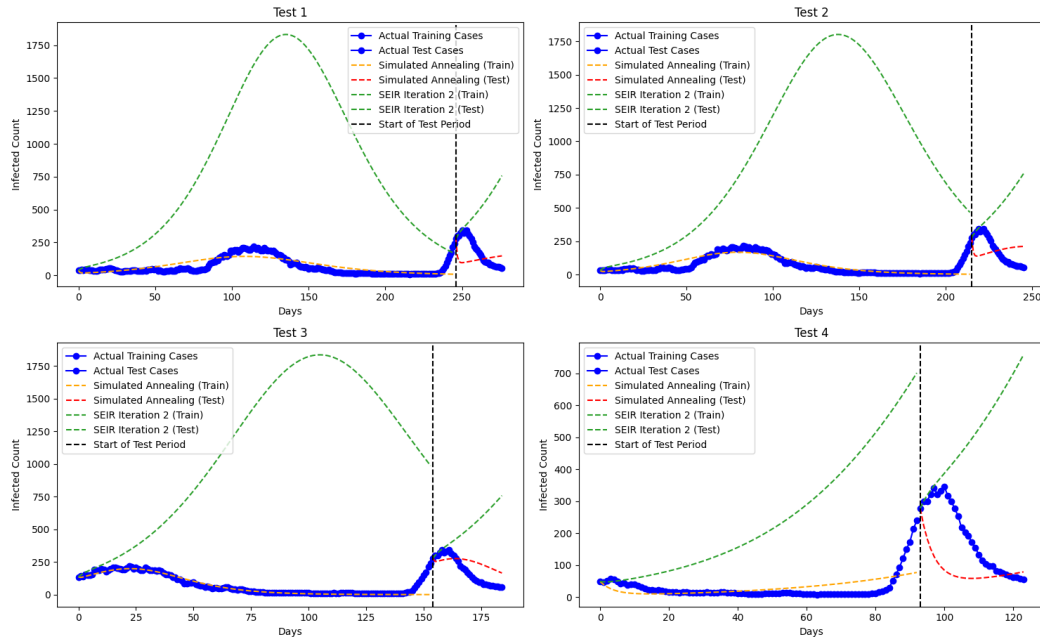


Figure 58 Barangay Santo Tomas SEIR models predictions

Barangay Mamatid - SEIR Model Predictions

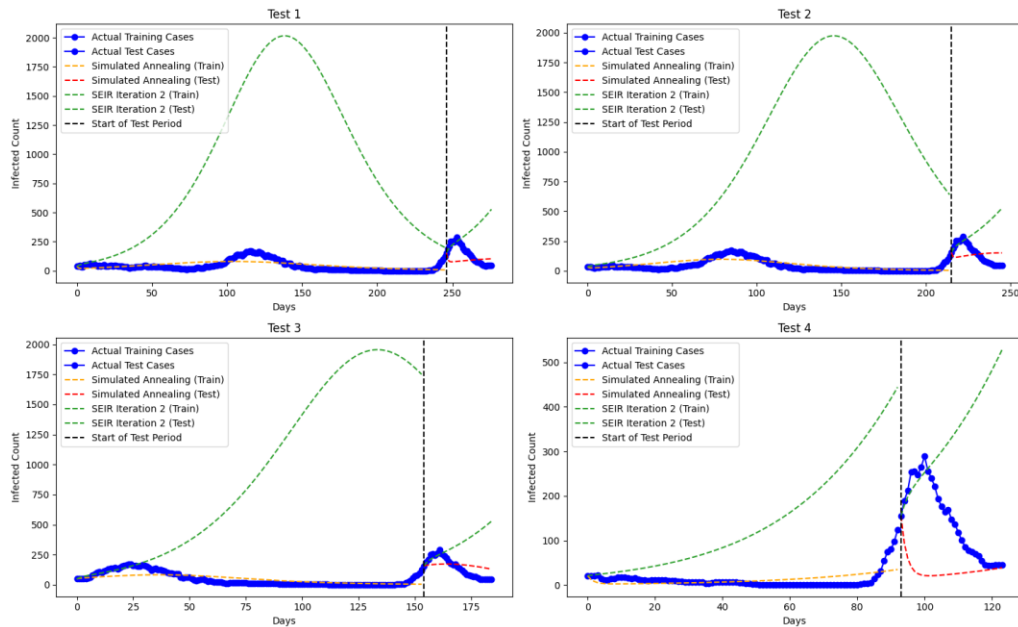


Figure 59 Barangay Mamatid SEIR models predictions

After conducting multiple validation tests and confidence evaluations, Simulated Annealing demonstrated superior performance compared to both the classical SEIR model and Bayesian Inference.

The TPER represents the percentage of incorrect predictions made by the model over time

$$TPER = \frac{Predicted - Actual}{Actual} \times 100$$

while Accuracy indicates the percentage of correct predictions, calculated as (100 - TPER).

In addition, Table 44 presented the average Time Plot Error Rate (TPER) and Accuracy of a model optimized using the Simulated Annealing algorithm when applied to test sets from different barangays. By evaluating the model's performance across multiple barangays, the table provides insights into its reliability and effectiveness in different locations.

Table 44 Average TPER and accuracy of Simulated Annealing optimized model

Barangay	Average TPER	Accuracy
San Antonio	29.77	70.23
San Lorenzo Ruiz	8.24	91.76
Narra	40.26	59.74
Santo Tomas	5.85	94.15
Mamatid	1.28	98.72
Average	17.08	82.92

Among the five barangays analyzed, Mamatid recorded the best performance, achieving a TPER of just 1.28% and an accuracy of 98.72%. Similarly, Santo Tomas and San Lorenzo Ruiz also demonstrated accuracy of 94.15% and 91.76%, respectively, indicating that the model performed consistently well in these areas with minimal error rates.

On the other hand, the model faced challenges in certain barangays, particularly Narra, which had the highest TPER of 40.26%, resulting in an accuracy of only 59.74%. This indicates a significant proportion of incorrect predictions, suggesting potential issues such as data inconsistencies, unique patterns that the model struggled to learn, or insufficient optimization for this specific area. San Antonio also exhibited a relatively high TPER of 29.77%, leading to an accuracy of 70.23%, showing that while the model performed moderately well, there is room for improvement.

Overall, the average TPER across all barangays was 17.08%, leading to an average accuracy of 82.92%. This indicates that while the model generally performs well, its effectiveness varies by location. The disparities in accuracy could be attributed to differences in data distribution, the complexity of patterns in each barangay, or the need for further tuning of the hyperparameters.

With these results, Simulated Annealing was chosen for implementation in the model integration phase of web development, where it will be used to predict future cases in the barangays of San Pedro.

Phase 3: Model Integration and Web Application Development

As mentioned in the previous chapter, phase 3 includes integrating the model from the previous phase into a prototype web application. This phase followed a modified prototype methodology. As such, there are multiple iterations in developing the prototype, with each iteration fixing mistakes and implementing additional requirements gathered from the previous iteration. To create this prototype, Django was used as a web framework for the application, with Tailwind and Plotly being utilized for creating a user-friendly interface. Because of this, the programming language was primarily Python in the back end, HTML as the front end, and JavaScript to facilitate most of the transferring of data in-between.

As stated in Chapter 3, the development of the web application adhered to the Model-Template-View (MTV) pattern for Django. The standard architecture for Django projects is shown in Figure 60. Firstly, the figure shows that the clients use their browser to access the web app. Django has URL files that store the path to each web page that is accessible to the user. These URLs redirect the users to the different views of the application. Figure 60 then shows that the views fetch the corresponding template file that serves as the frontend that the user sees. Next, the templates consist of HTML files to show the retrieved data to the user. Meanwhile, the static files provide the styling and assets that are used in the user interface such as the shape file for the heat map and the various graphs. Next, the figure illustrates how the view accesses the different database models to perform data logic. Finally, model files act as the connection that enables query operations to the PostgreSQL Database. As a result of following this structure, the researchers were able to conduct smooth development of the application.

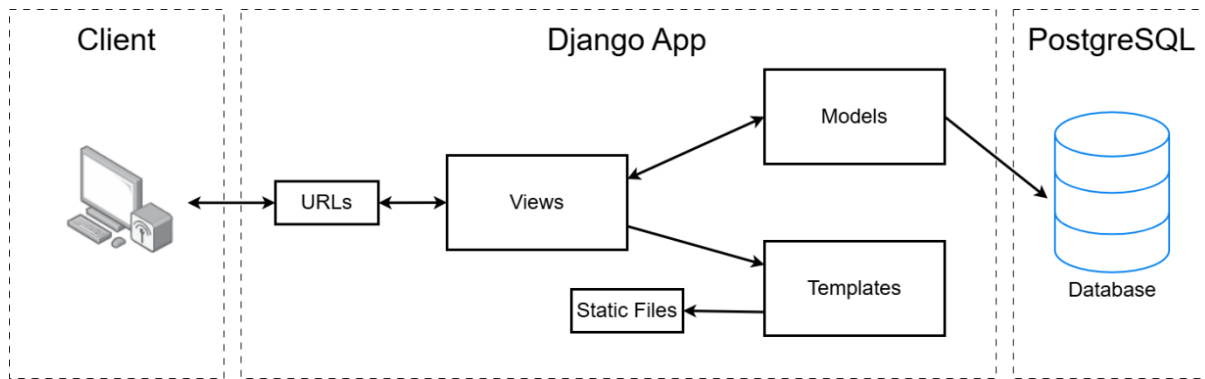


Figure 60 Django project architecture

SEIR Model Integration

The first step in this phase was to create a module for the optimized SEIR model using Simulated Annealing to be called for generating predictions. The class *SimulatedAnnealing* was designed for SEIR model parameter optimization using the Simulated Annealing algorithm. It performs four key functions: initializing the model, optimizing hyperparameters, making predictions, and evaluating error rates.

The class is initialized with the initial conditions of the SEIR model and defines parameter bounds ensuring that optimization stays within reasonable biological constraints. To estimate the optimal values for hyperparameters, the model requires real active case data. Using Dual Annealing, the algorithm minimizes the error between predicted and actual infections, guided by an objective function that incorporates mean squared error (MSE) and a regularization term to prevent extreme values. The resulting optimized parameters provide a refined SEIR model for better accuracy.

Once optimized, the estimated parameters were used in the SEIR function to simulate the progression of the disease. The model can predict trained cases using optimized parameters and forecast future infections for a specified number of days based on the learned parameters. Lastly, it will compute the error rates MSE, MAE, and RMSE to validate the model. And with that, the model instance is created.

As a result, predicted disease progression in acceptable format which will be visualized in the web application as charts and through the heat map. With this in place, the development of the web application can then proceed.

Model instances are created whenever the user inputs new data. After saving the new data into the database, the application checks whether there are at least 90 records within a five(5)-month period before generating predictions. If predictions already exist, the class verifies whether the most recent prediction is more than 30 days ahead of the current date. If so, new predictions are created; otherwise, the system forecasts the next 30 days using the *SimulatedAnnealing* class. Additionally, if actual cases extend beyond existing predictions due to many new records being inserted, the system runs the *SimulatedAnnealing* class again to generate predictions for the existing records. Finally, for the last recorded case, the optimized parameters from the most recent model instance are used to predict the next 30 days, ensuring continuity and accuracy in forecasting. Refer to Table 45 for the conditions before creating a new model for instance and predicting.

Table 45 Model Integration Conditions and Actions

Condition	Action
Less than 90 records in 5 months	No model instance created
More than 90 records in 5 months	Check if predictions exist
No prior predictions	Create model instances and generate predictions
Predictions exist but outdated	Create new model instances and refresh predictions
Predictions exist and still relevant	No new model instance needed

Figure 61 shows the model instance flow chart. The full process of creating a new model using five(5) months of data is shown in Appendix P

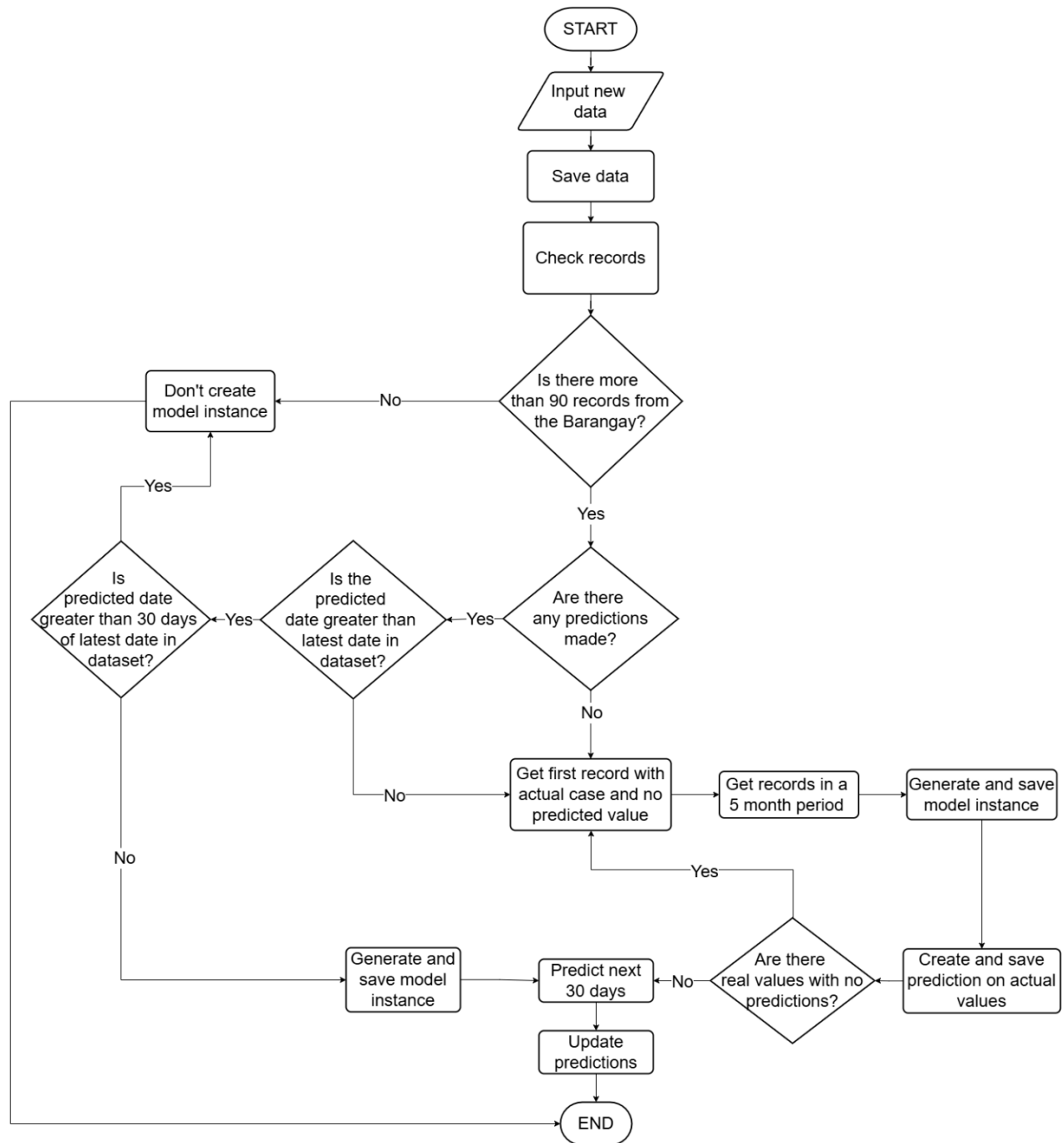


Figure 61 Model Instance flowchart

Prototype Development

The next step in this phase is the development of the SEIR-Viz web application around the SEIR model. The prototype web application that was designed to run on a web browser was created. It was developed using the *Django* framework and used the model developed from the previous phase to simulate and predict infectious diseases. In addition, other tools were utilized, which were outlined in chapter 3, to ensure that the researchers achieve the third objective of creating a web application that provides intuitive way to view data for the health officials.

PostgreSQL as the database management system provided an easy way to insert, store, and retrieve data. Also, Django provided many built-in tools and features that facilitated the smooth development that the researchers required. Its *MVC* structure made the process of fetching data from *PostgreSQL* to showing data to the user easy to manage. Additionally, *Tailwind* was used to create a concise user interface that also provided features for responsiveness in screens with different aspect ratio. Lastly, *Plotly* provided many features which ensured that data can be presented in a consistent and easy-to-read way.

By effectively utilizing the tools listed above, the resulting web application achieved favorable responses from the ESU as outlined in Phase 4: Testing and Analysis. As discussed in Chapter 3, the study uses the prototyping model as the development methodology. As such, the development of the prototype consisted of looping through several steps until the prototype is finalized. The steps in the development cycle are analysis and design, prototype refinement, and the user evaluation.

Analysis and Design

The first step of the prototype development cycle is the analysis and design of the web application. For the initial requirements analysis, the researchers went to the locale to conduct an initial interview with the ESU members and collected information on what the unit needs for the application to assist them. Succeeding interviews resulted in some changes in design and requirements which will be detailed further in the prototype refinement step. However, the scope and limitations of the study were upheld, and the resulting web application was received positively, with the results being seen in the Usability Testing of Phase 4 in Chapter 4.

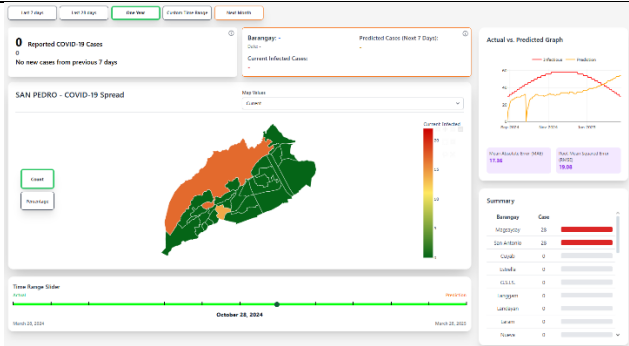
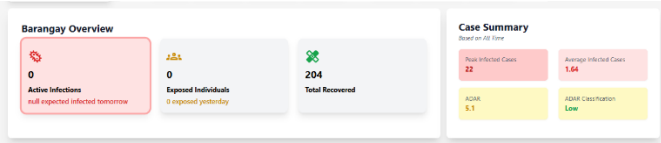
After the initial interview, the researchers started with the development iteration. For the design part of the analysis and design, the researchers conducted a design session to identify the specific data visualizations that the locale can utilize in the web application. During the prototyping refinement phase in the first iteration, the researchers developed the initial prototype consisting of the map page, barangay page, dataset page, and account management page. After the prototype development phase, the researchers evaluated the overall solution and identified bugs and necessary refinements.

After the development process of phase 3, the SEIR module integration was done by the researchers by integrating the developed model's simulation and prediction, as well as populating the database of the web application. The integrated

model was then made as a module within the web application to provide data for the visualizations in the web app.

The third objective of the study stated that the web application should present data in an intuitive format and make it easier for the health officials to interpret the data visualization provided to them. Therefore, Gestalt principles of design were applied to the development of the SEIR web application to ensure a user-friendly and visually appealing interface. These principles emphasize how users perceive and organize visual information, enabling intuitive navigation and efficient interaction with the application. Table 46 outlines the Gestalt principles used, their descriptions, and how they were practically implemented in the SEIR web application.

Table 46 Gestalt principles and actual application on the web application

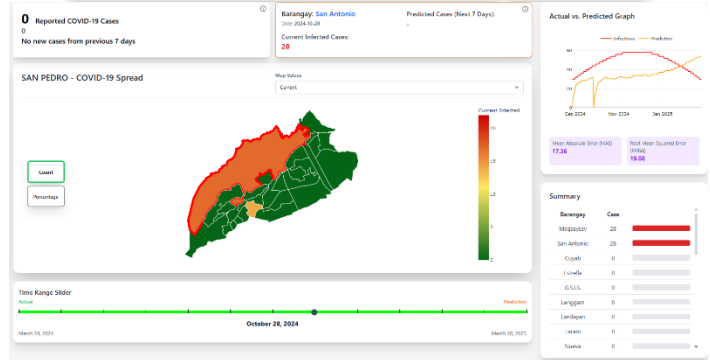
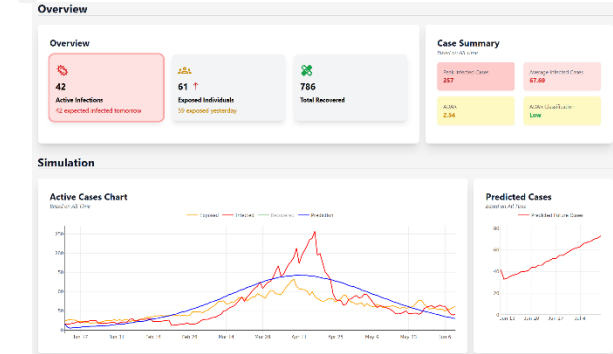
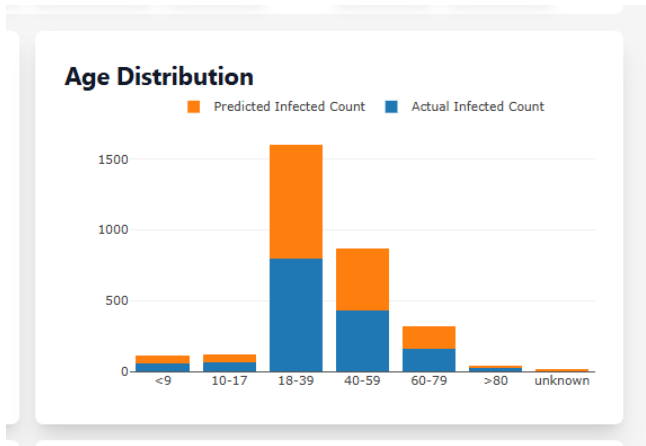
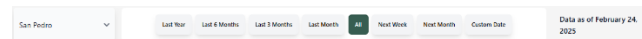
UI Principle	Description	Actual Application
Proximity	Elements placed together are perceived as related	
Similarity	Similar elements (color, size, shape) are grouped	

Continuity
Items aligned on a line or curve are seen as connected

Figure-Ground
Differentiates the figure from the background

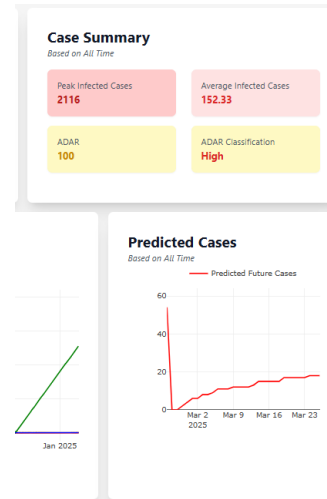
Symmetry
Balanced design elements create order and appeal

Connectedness
Visually linked elements are perceived as related



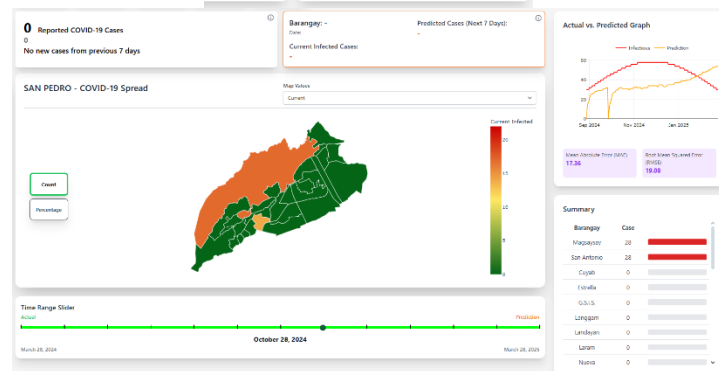
Common Region

Elements in the same bounded area are seen as related



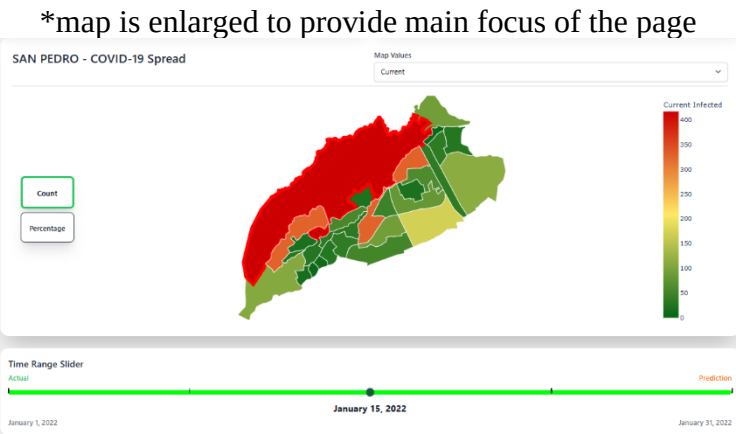
Focal Point

Unique design elements draw user attention



Common Fate

Elements moving together appear related



*dropdown and radio controls move and adjust data on related elements

Figure 62 shows the landing page of the user where they have to input email and password to go through the rest of pages.

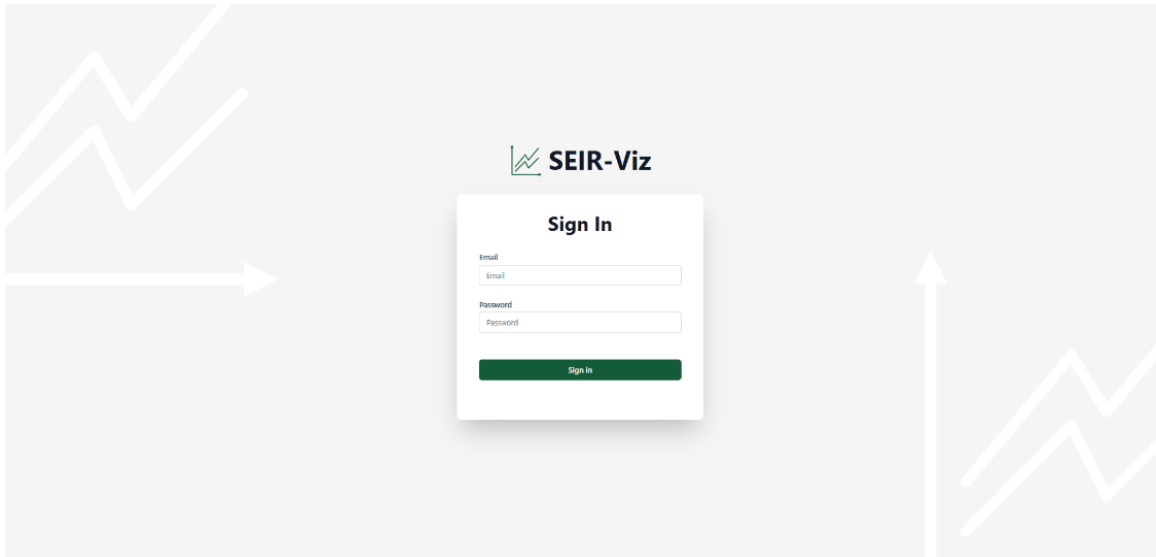


Figure 62 Login Page

Figure 63 shows the spread map page that shows the visualization of infectious disease spread on different barangay.

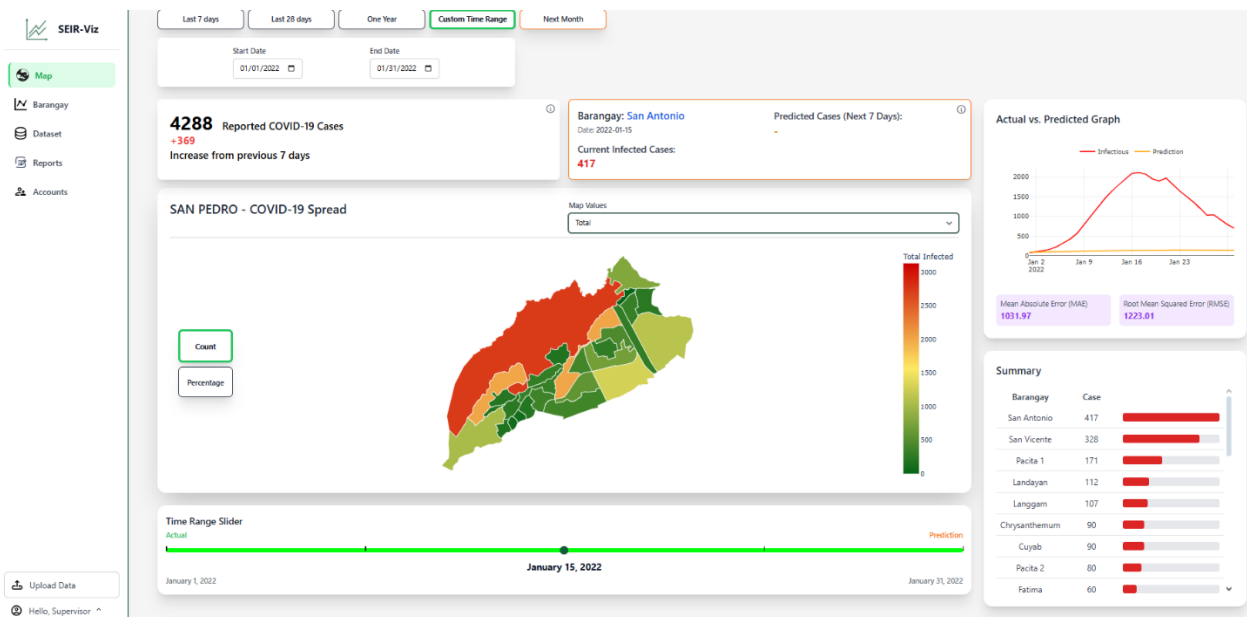


Figure 63 Map Page

Figure 64 shows the dataset page that shows the current data stored within the system’s database.

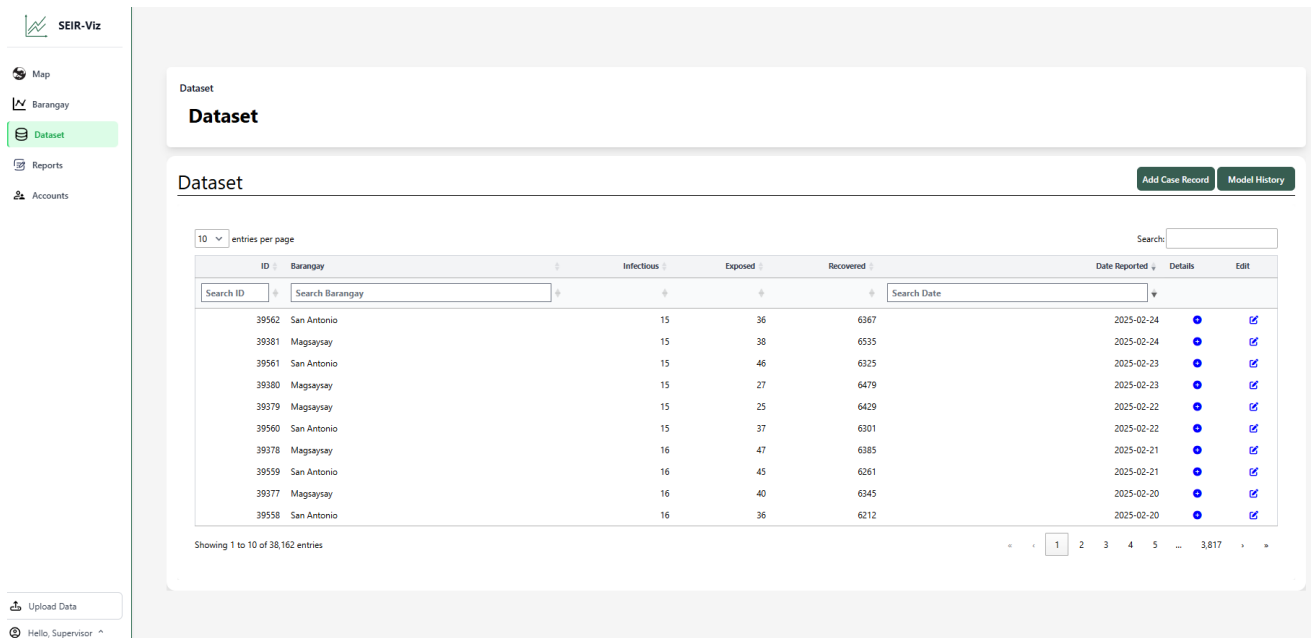


Figure 64 Dataset Page

Prototype Refinement

The prototype refinement step in the development cycle consisted of incorporating the requirements from the interviews conducted with the ESU. An initial interview was conducted to outline the basic requirements the ESU members expect, and the design principles outlined in the previous step were used to create the initial web application. The contents of each subsequent interview consisted of the ESU members scrutinizing the web application until it is deemed workable for the users.

First Iteration

Once the initial prototype application was completed with the initial requirements, an informal presentation was conducted to show the progress of the application and to gather additional requirements from the ESU members. The main features the ESU members outlined were additional information in the data shown. Namely, the ESU requested tracking of the age group for the records which resulted in the seven columns for ages as seen in Phase 1. The ESU members commented that the distribution of ages is important in analyzing the individuals that are the most vulnerable.

Additionally, the ESU members commented on some improvements for the user interface such as adding more graphs for better readability. The ESU members commented that visualizations are important to them so that they can immediately see critical information such as highlighting notable

barangays. Also, the ESU members requested easier manipulation of the time slider by adding time slices, rather than sliding day-by-day. The comments of the ESU members emphasized the importance of data visualization for showing critical information and conveying disease spread.

This meeting with ESU also described the different types of accounts that will be used. The supervisor account can access all the pages within the application and can create and delete officer accounts. Meanwhile, the officer account has the least amount of access in the application, only having access to the dataset page for encoding new records and the reports page for creating reports. The Use Case diagram in Appendix Z visualizes how each of the expected users interact with the application.

The last additional information gathered from this meeting was details for the design and contents of the report page. The ESU members provided an example single-page report that the researchers could use as reference for the reports page in the application. The example report given to the researchers can be seen in Appendix AA. Moreover, the ESU gave the researchers the option to freely add or change any part of the report to their discretion.

Second Iteration

The second meeting with the ESU members resulted in less overall changes in the web application, as most of their requirements were met in the first prototype refinement iteration. After changing the heat map's control by adding time slices, the ESU members were satisfied with the functionality of the page. Additionally, the members appreciated that a line graph showing the prediction of cases and table for the top barangay by cases showed additional insight into current and predicted cases.

The researchers gathered more information from additional comments about the current situation in tracking diseases. During the years when COVID-19 was more widespread, contact tracing was more reliable which resulted in more accurate tracking of exposed and infected individuals. However, tracking the exposed individuals became more difficult after 2024, as people are less inclined to report disease cases. This challenge could cause difficulty in providing the properly formatted input required for accurate case predictions.

During the presentation, the ESU members voiced some missing requirements that the researchers missed in the first iteration. One notable addition was adding case counts for genders in the dataset. The ESU members commented on how it is important to track both the age and gender distribution for analyzing the most vulnerable group of people in the population. The last notable addition was to add the Average Daily Attack

Rate (ADAR) to track the most at-risk barangays. For this, the researchers followed the formula for calculating the ADAR from the Centers for Disease Control and Prevention (CDC) (Centers for Disease Control and Prevention, 2012). Additionally, the researchers followed the revised alert level metrics for categorizing the ADAR levels as stated by Inter-Agency Task Force (IATF) and referenced by the given one-pager report in Appendix AA.

Third Iteration

The third meeting with the ESU also marked the last testing session for the web application. In this meeting, most of the changes requested by the ESU were for easier readability of key information. One such change was on the color coding for the map page. While the researchers used yellow to show the least severe and purple as the most severe, the ESU members requested a more traditional green-to-red color coding to show the severity of the number cases. Also, additional color coding in the report page was requested to more easily differentiate between the predicted and actual cases.

The ESU members voiced additional comments about how key information should be understood immediately at a glance. Other than the color coding, the ESU stated how their audience for reporting did not need to be shown technical details such as the error rate of the prediction. The

ESU members stated that the graphs and tables provided are sufficient in showing the essential information.

In every interview with the ESU, the researchers conducted a Usability Testing to identify the exact areas in the application that need improvements. This feedback proved critical in the improvement of the web application. After the last testing, the feedback from the ESU marked an end to the development cycle, as the results from the UT showed a high score which indicated that the application was ready for deployment. The results from these Usability Testing are outlined in phase 4 of chapter 4.

User Evaluation

As stated in defining the final step of the development cycle, the prototype web application is tested to identify problems in critical functions. The tests were conducted using Django's TestCase class in the built-in django.test module. This module provided many tools to simulate test environments by creating mock database instances and sending mock API requests. In setting up the unit tests, the researchers followed the basic structure of testing using Django. Testing using this module usually consists of a setup function. In testing the web application, the setup function is where the database instances for required fields were created: Barangay, Case Category, Record ID, and Case Data. Afterward, each function in the web application was tested one-by-one with the appropriate parameter inputs while

keeping in mind the expected response from the function. The overall results of these tests are seen in the Functional Testing in phase 4 of chapter 4.

The map page displays data with dates from previous and future dates which the heat map and various charts visualize. As such, the setup function was created with the appropriate fake data from previous dates up to future dates to test if the functions in the map page returned the data. Additionally, the page was tested to confirm that the shape file is loaded properly. Also, the researchers tested for the color coding for each Barangay in the map, as it is a critical component in visualizing disease spread. Furthermore, each element in the page that consists of data was tested to confirm if they return the expected response. Lastly, the time slider was tested to ensure that it would return the expected data for the selected date.

The Barangay page also requires data that is from previous dates and future dates. Because of this, the setup for this page had records that are for future dates. To test the functionality of this page, tests were done to ensure that the page returns the correct exposed, infected, and recovered case counts for every barangay based on the date selected. Other than the case counts, the age groups are also tested to be accurate, as they are critical for the data analyst in ESU. Finally, the graphs and charts on the page were tested to ensure that they were rendered properly and the data within them were accurate.

Like the previous pages, the reports page also needs data from past and future dates. It was critical to ensure that the response from the functions in the

page returned accurate values for the page. Since the only parameters needed for this page are the start and end date, each function was tested to properly handle valid dates, invalid dates, and missing dates. Moreover, the researchers ensured that the generated reports would be properly downloaded into a CSV file.

The model page requires a minimum of 90 days' worth of records. As such, the set up for testing for the model page consisted of creating a sufficient number of fake records. At first, the page was tested to ensure that when there is sufficient data, the page will properly create predictions. On the other hand, when there is insufficient data, the page was tested to not create predictions, as insufficient data will result in inaccurate predictions. Finally, the page was tested to see if it would properly create a model instance, so that previous models can be revisited.

In testing the dataset page, the researchers ensured that the page could properly handle invalid inputs. At first, the page was tested to ensure it properly added new records. Then, invalid inputs were tested which resulted in a 302 response. This is to ensure that user mistakes did not result in an error or an improper recording of data. In addition to testing for user inputs, this page was also used to test the functionalities of the database such as retrieving, adding, and changing data.

In addition to using the built-in Django test module, the researchers also validated the results of the Django queries by comparing the returned data to the direct PostgreSQL query. This added validation ensured that the Django ORM present within the application's code perfectly aligns with the PostgreSQL query.

Deployment

The developed web application, built using the Django framework, was deployed on a Linux-based System (Ubuntu) to ensure stability, scalability, and performance. Deployment is a critical phase in web application development as it makes the system accessible to end-users. The deployment process involved setting up a Gunicorn (Green Unicorn) application server and NGINX (Engine-X) as a reverse proxy to handle incoming HTTP requests, ensuring efficient request handling and load balancing. Figure 65 shows how the web application was deployed.

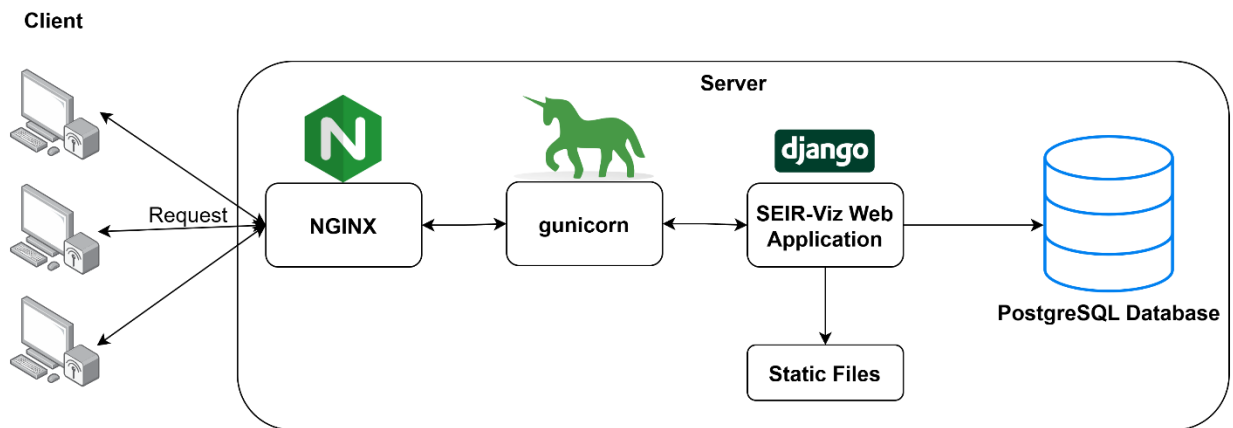


Figure 65 Django, NGINX, Gunicorn deployment architecture

Deployment Environment. The web application was deployed on an Ubuntu 22.04 LTS, a widely used and stable operating system for deploying web applications. Ubuntu was chosen due to its strong community support, security features, and compatibility with the Django framework. The deployment process ensured that the application could handle concurrent requests, minimize downtime, and remain accessible through a stable domain.

Application Server. The *Gunicorn* (Green Unicorn) application server was used to serve the Django web application. *Gunicorn* is a Python WSGI HTTP Server designed to serve Python applications in production environments. It was chosen for its lightweight nature, ability to handle multiple requests simultaneously, and seamless integration with Django. *Gunicorn* efficiently translates client requests into Python function calls, allowing the web application to remain responsive under concurrent requests (gunicorn, n.d.).

Proxy. To enhance the application's performance and security, *NGINX* was configured as a reverse proxy server. The primary role of *NGINX* in this setup was to: (1) Forward HTTP requests from the client to the *Gunicorn* server. (2) Handle static files such as images, CSS, and JavaScript files, reducing the load on *Gunicorn*. (3) Improve security by hiding the *Gunicorn* server from direct exposure to the internet.

Deployment Process. The deployment process involved the following key steps: (1) Setting up the virtual environment for the Django application. (2) Installing *Gunicorn* and configuring it to run as a service on the Ubuntu server. (3) Configuring *NGINX* to act as a reverse proxy, ensuring that requests were routed to *Gunicorn*. (4) Setting up a firewall (UFW) to allow HTTP and HTTPS traffic while restricting access to unnecessary ports.

Phase 4: Testing and Analysis

As mentioned in Chapter 3, this phase includes the functional testing and usability testing with the expert and San Pedro ESU.

Functional Testing

The functionality of the key functions of the web applications were tested to verify if they are functioning as intended under various test cases.

Table 47, Table 48, Table 49, and Table 50 shows the results of the unit testing conducted on different functions of the web applications. These include tests on the different pages, namely: Map, Barangay, Reports page, and the model integration and functions within the pages. See Appendix Q for the test case activity

Table 47 Map Page testcases

Test Case No.	Test Case	Description	Remarks (Pass/Fail)
1	Test if map current and percentage data is fetched accurately	Test is map shows current and percentage infected counts	Pass
2	Test if map total data is fetched accurately	Test if map total infected count data is correct	Pass
3	Test if top barangay cases count is accurate	Test if top barangay cases count is presented and matches map data.	Pass
4	Test if map page load shapefile loads the choropleth map	Test is heatmap displaces different colors on the map page	Pass
5	Test if create choropleth with frame data is fetched properly	Test if heatmap colors changes on time slider change	Pass

Table 48 Barangay Page testcases

Test Case No.	Test Case	Description	Remarks (Pass/Fail)
6	Test if all barangay Exposed count based on date are correct	Test if exposed counts for 3, 6, and all date counts are outputted	Pass
7	Test if all barangay Infected count based on date are correct	Test if infected counts for 3, 6, and all date are outputted	Pass
8	Test if all barangay Recovered count based on date are correct	Test if recovered counts for 3, 6, and all date are outputted	Pass
9	Test if all age groups in barangay's exposed, infected, and recovered are accurate	Test if age group data for infected, exposed, and recovered for 3, 6, and all date filter is outputted	Pass
10	Test if barangay page SEIR graph renders properly	Test if barangay page exposed, infected, and recovered graph renders	Pass
11	Test if barangay page age distribution chart is accurate	Test if barangay page age distribution chart renders	Pass

Table 49 Report Page testcases

Test Case No.	Test Case	Description	Remarks (Pass/Fail)
12	Test if Positive Surveillance Testing reports are generated	Test if Section A of one-pager report is generated	Pass
13	Test if Prediction Reports are generated	Test if Section B of one-pager report is generated	Pass
14	Test if download csv report is functional	Test if CSV is downloaded on click	Pass

Table 50 Model testcases

Test Case No.	Test Case	Description	Remarks (Pass/Fail)
15	Test if prediction will proceed with more than 90 data	Test if prediction is populated on barangay with sufficient data	Pass
16	Test if prediction will not proceed with 90 or less data	Test If no prediction is populated given insufficient data	Pass
17	Test if prediction will proceed if there are 30 days projected cases	Test if prediction will be retained for projected cases	Pass
18	Test if model instance will populate upon prediction	Test if model instance is added to database	Pass

To evaluate the predictive capabilities of the SEIR model, the researchers conducted tests using mock data and uploaded the dataset to the web application. This process ensures that the model can accurately generate predictions and handle various input conditions. The test cases outlined provide insights into the performance and reliability of the prediction requirements.

For the purpose of testing, the researchers utilized a mock dataset containing sample case data record from the San Pedro dataset. This dataset was then uploaded to the web application, where the model processed the input and generated the predictions. After separating the dataset and running the prediction model, the system successfully executed the tests. The model identified when sufficient data was available for prediction and prevented execution when data constraints were not met.

Table 51 on the next page shows the result of the database tests. This test was performed to validate the function of the database and its communication with the web application. (see Appendix Q for the sample execution of test cases)

Table 51 Database test

Test Case No.	Test Case	Description	Remarks (Pass/Fail)
19	Test if case data on prediction will populate prediction	Test if case data prediction is generated and populated on the database	Pass
20	Test San Pedro records upon upload data	Test is San Pedro records are added or updated on the database	Pass
21	Test if the cases on San Pedro are correct on different dates	Test if San Pedro record dates are accurate and populated on the database	Pass
22	Test if exposed and infected records are correct	Test if exposed and infected case data is populated on the database	Pass
23	Test if recovered cases are accumulating	Test if recovered case data records are accumulating on the database	Pass

Usability Testing

For the usability testing, two (2) members of the ESU were the respondents of the usability testing questionnaire. Initially, three (3) ESU members were supposed to answer the usability test but due to scheduling issues, one member was not able to participate in the usability test. A different questionnaire was provided to the two (2) experts with the questions focusing on the usability of the model. Table 52 shows the list of respondents and their position.

Table 52 Usability testing list of respondents and position

Respondent	Position
San Pedro ESU Supervisor	Epidemiology expert
San Pedro ESU Assistant Supervisor	IT, Data analyst
Expert 1	Scientist, Model Expert
Expert 2	Public Health Officer

The researchers conducted an in-person usability test with the ESU at the San Pedro City Hall. This was a guided usability test, wherein the researchers directly interacted with the ESU officers to explain the web application and the SEIR model's functionality. The researchers demonstrated the key features of the system, collected feedback on its usability, and documented the interaction. See Appendix U for the usability testing picture documentation with the ESU.

Due to constraints, usability testing with the expert was conducted remotely. The researchers coordinated multiple meetings to facilitate the process and ensure a comprehensive evaluation. The following materials were provided to the expert:

A *Google Colab Notebook* containing the SEIR model, its validation results, and sample performance data, a detailed write-up explaining the results of the model's optimization, and a video demonstration showcasing the functionality and workflow of the web application.

San Pedro ESU Usability Testing

For the first ESU usability test, the ESU respondent answered via digital questionnaire using google forms. During the first iteration of the prototype, only one (1) ESU member was available to answer the questionnaire. The ESU member has the role of supervisor which means they were able to test all the functionality of the web application (see Appendix R for the answered questionnaire).

The score of each statement for the first usability test of the ESU is shown in Table 53 on the next page.

Table 53 First Usability Test result with San Pedro ESU

Statement Code	Statement	Score	Interpretation
S1	The SEIR math model produced consistent and reliable results across multiple simulations	3.00	Agree
S2	The web app was stable and did not crash or experience significant issues while running	3.00	Agree
S3	The web app consistently produced results without discrepancies across different runs.	3.00	Agree
S4	The web app performed well under different data sizes and stress conditions.	3.00	Agree
S5	The web application was easy to learn and use	4.00	Strongly Agree
S6	The user interface design was intuitive and easy to navigate	4.00	Strongly Agree
S7	The color scheme and contrast of the UI made it easy to distinguish different elements.	2.00	Disagree
S8	The tooltips, labels, and descriptions provided sufficient context for understanding the functionalities.	3.00	Agree
S9	The visualizations and graphs representing the SEIR model results were clear and easy to understand	3.00	Agree
S10	Adding records and generating prediction based on the SEIR math model was efficient and required minimal effort	2.00	Disagree
S11	The layout and process flow minimized unnecessary steps in using the SEIR model	2.00	Disagree
S12	The web app responded quickly to user interactions without noticeable delays.	3.00	Agree
S13	The time taken to generate results was reasonable and acceptable	4.00	Strongly Agree
S14	The steps involved in running the SEIR model and visualizing were logically arranged and efficient.	3.00	Agree
S15	If errors occurred, the error messages were helpful in resolving the issues.	3.00	Agree
S16	The application provided clear logs or feedback for debugging errors.	3.00	Agree
S17	The web app functioned well on different browsers and devices without compatibility issues.	2.00	Disagree
S18	The simulation and prediction results of the SEIR model provided enough information for analyzation	3.00	Agree
S19	The web app was able to successfully visualize the SEIR prediction without issues	3.00	Agree
S20	The generated report was sufficient and accurate to requirements	3.00	Agree
S21	The web app correctly handled different input values and scenarios without unexpected behavior.	3.00	Agree
S22	The web app provided sufficient options for customizing the visualization generated.	3.00	Agree
S23	Overall, I found the web application to be a useful tool for epidemic spread visualization and decision-making.	3.00	Agree
Average		2.96	Agree

The table shows that the respondent disagree that the web application: (1) color scheme and contract of the UI made it easy to distinguish between elements, (2) adding of records was efficient and required minimal effort, (3) the layout and process flow minimized unnecessary steps in using the SEIR model, and (4) the web application functioned well on different browser and devices. Meanwhile, the respondents agree and strongly agree on other aspects of usability, functionality, efficiency, and reliability of the web application. The overall average of the score was 2.96 giving it an Agree interpretation for the first iteration of the usability testing with the ESU.

Other verbal feedback was given during the testing and was only noted by the researchers. The main concern was the capability of the system to accept data that is catered to their excel format. While the system's capability of adding record and accepting CSV was originally formatted to accept data that was similar to the data collection during the COVID-19 case tracking, the ESU's concern was being able to use the system for the current data collection format that they were employing. The ESU also noted the availability of gender for the visualization and data.

For the second usability test with the ESU, after the changes were made, the researchers conducted the second usability testing for the ESU. Table 54 on the next page shows the result of the second usability testing.

Table 54 Second Usability Test Result with San Pedro ESU

Statement Code	Statement	Mean Score	Interpretation
S1	The SEIR math model produced consistent and reliable results across multiple simulations	4.00	Strongly Agree
S2	The web app was stable and did not crash or experience significant issues while running	3.50	Strongly Agree
S3	The web app consistently produced results without discrepancies across different runs.	3.00	Agree
S4	The web app performed well under different data sizes and stress conditions.	3.50	Strongly Agree
S5	The web application was easy to learn and use	4.00	Strongly Agree
S6	The user interface design was intuitive and easy to navigate	4.00	Strongly Agree
S7	The color scheme and contrast of the UI made it easy to distinguish different elements.	3.00	Agree
S8	The tooltips, labels, and descriptions provided sufficient context for understanding the functionalities.	4.00	Strongly Agree
S9	The visualizations and graphs representing the SEIR model results were clear and easy to understand	3.50	Strongly Agree
S10	Adding records and generating prediction based on the SEIR math model was efficient and required minimal effort	3.50	Strongly Agree
S11	The layout and process flow minimized unnecessary steps in using the SEIR model	3.50	Strongly Agree
S12	The web app responded quickly to user interactions without noticeable delays.	3.50	Strongly Agree
S13	The time taken to generate results was reasonable and acceptable	4.00	Strongly Agree
S14	The steps involved in running the SEIR model and visualizing were logically arranged and efficient.	3.50	Strongly Agree
S15	If errors occurred, the error messages were helpful in resolving the issues.	3.50	Strongly Agree
S16	The application provided clear logs or feedback for debugging errors.	3.50	Strongly Agree
S17	The web app functioned well on different browsers and devices without compatibility issues.	3.50	Strongly Agree
S18	The simulation and prediction results of the SEIR model provided enough information for analyzation	4.00	Strongly Agree
S19	The web app was able to successfully visualize the SEIR prediction without issues	4.00	Strongly Agree
S20	The generated report was sufficient and accurate to requirements	4.00	Strongly Agree
S21	The web app correctly handled different input values and scenarios without unexpected behavior.	4.00	Strongly Agree
S22	The web app provided sufficient options for customizing the visualization generated.	3.50	Strongly Agree
S23	Overall, I found the web application to be a useful tool for epidemic spread visualization and decision-making.	4.00	Strongly Agree
Average		3.67	Strongly Agree

The usability testing conducted with the ESU members of resulted in an average rating of 3.67 (Strongly Agree), indicating strong positive feedback on the web application's performance and usability. The SEIR model was acknowledged for producing consistent and reliable results (S1), and the web application was found to be stable, responsive, and easy to use (S2, S5, S6). The visualizations and data representations were generally well-received, with minor areas for improvement in UI contrast and element distinction (S7). Additionally, the system was praised for its efficiency in running simulations and generating predictions with minimal effort (S10, S13). While there were no major usability concerns, continued refinement in UI clarity and error messaging could further enhance the user experience. Overall, the testing validated the system's effectiveness in aiding public health monitoring and decision-making.

During testing, verbal feedback was provided and recorded by the researchers. One key suggestion was to adjust the text and visualization colors to align with the charts and graphs for better consistency and readability. While the overall design was well-received, this minor refinement was recommended to improve the user experience and was applied by the researchers to the suggested areas. Additionally, the ESU emphasized the need for the system to support their specific data format for case reporting. Although originally designed for infectious case tracking format, they highlighted the importance of making the system adaptable for other infectious diseases. This feedback was considered for future improvements to enhance flexibility.

The survey questionnaires' consistency and reliability was also assessed using Cronbach's Alpha. Table 55 presents the data collected from the ESU respondents along with the Cronbach's Alpha calculation. With a Cronbach's Alpha of 0.81, an internal consistency of Good, the results demonstrate that the questionnaires are reliable and consistent for the study.

Table 55 San Pedro Usability Test Cronbach's Alpha

	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12
Respondent 1	4	4	3	4	4	4	3	4	3	4	4	4
Respondent 2	4	3	3	3	4	4	3	4	4	3	3	3
Variances	0	0.25	0	0.25	0	0	0	0	0.25	0.25	0.25	0.25

	S13	S14	S15	S16	S17	S18	S19	S20	S21	S22	S23	Total
Respondent 1	4	3	4	4	4	4	4	4	4	4	4	88
Respondent 2	4	4	3	3	3	4	4	4	4	3	4	81
Variances	0	0.25	0.25	0.25	0.25	0	0	0	0	0.25	0	
No. of Items	23											
Sum of Variances	2.75											
Variance of Total Scores	12.25											
Cronbach's Alpha	0.81											

In this iteration, the model's scores were evaluated using ISO 9126 quality characteristics. The summarized scores in Table 56 show that the application complies with ISO 9126 standards, with respondents strongly agreeing on portability, efficiency, functionality, usability, and reliability.

Table 56 ISO 9126 Standard, statement codes, mean score

ISO/IEC 9126 Quality Characteristics	Statement Codes	Mean Score	Interpretation
Portability	S17	3.50	Strongly Agree
Functionality	S18, S19, S20, S21, S22, S23	3.92	Strongly Agree
Efficiency	S10, S11, S12, S13, S14	3.60	Strongly Agree
Usability	S5, S6, S7, S8, S9	3.70	Strongly Agree
Reliability	S1, S2, S3, S4, S15, S16	3.50	Strongly Agree

The average scores of the statements in relation to the objectives were computed, as detailed in Table 57 on the next page. The table reveals that the average scores for each objective's statements surpass the minimum acceptable score of 3. Consequently, all objectives have been achieved.

Table 57 ESU Usability Test Objectives, statement code alignment, and mean score

Objectives	Statement Codes	Mean Score	Interpretation
Develop a SEIR model tailored to a local epidemiological and surveillance unit (ESUs) that can help in communicating and predicting disease spread to help adhere to the law.	S1, S18	4.00	Strongly Agree
Optimize the SEIR model through hyperparameter experimentation that will help mitigate discrepancies on local epidemic simulation.	S3, S4, S10, S11, S12, S13, S14, S21	3.56	Strongly Agree
Design a web application to visualize local epidemic simulation of spread by applying and integrating the optimized SEIR model.	S2, S5, S6, S7, S8, S9, S15, S16, S17, S19, S20, S22, S23	3.69	Strongly Agree

Expert Usability Testing

For the usability testing of the expert, two (2) types of questionnaires specific to the SEIR model was prepared and provided. Expert 1 is a scientist and a model expert, while Expert 2 is a public health officer.

The first usability testing questionnaire was focused on the technicalities of math modeling. Expert 1 responded to the questionnaire and answered via digital form using google forms. The question statements focused on evaluating the effectiveness, reliability, and applicability of the optimization techniques used in the SEIR model. Specifically, the statements aimed to assess the clarity and justification of the chosen optimization methods, namely Bayesian Inference and Simulated Annealing, in improving the accuracy of the SEIR model predictions. It

also explored whether these methods could effectively balance predictive accuracy and computational efficiency and if the metrics used for model validation, such as RMSE and MAE, were appropriate. The overall goal of the technical usability testing was to gather expert insights regarding the optimization process applied to the SEIR model, particularly the use of Bayesian Inference and Simulated Annealing, and to validate whether these techniques significantly improved the model's predictive performance while maintaining stability and practical applicability. Appendix R shows the answered questionnaire.

The score of each statement for the usability test of the expert is shown in Table 58 on the next page.

Table 58 Model Expert first Usability Test Result

Statement Code	Statement	Mean Score	Interpretation
S1	The methodology for optimizing the SEIR math model using Bayesian Inference and Simulated Annealing is a clearly defined and well-justified choice	4.00	Strongly Agree
S2	The use of Bayesian Inference for parameter estimation improves the accuracy and reliability of the SEIR math model's prediction compared to the actual observed data.	4.00	Strongly Agree
S3	The use of Simulated Annealing for parameter estimation improves the accuracy and reliability of the SEIR math model's prediction compared to the actual observed data.	4.00	Strongly Agree
S4	The Bayesian Inference process effectively balances accuracy and computational efficiency	3.00	Agree
S5	The Simulated Annealing process effectively balances accuracy and computational efficiency	3.00	Agree
S6	The metrics used to validate the optimized model using RMSE are appropriate and clearly presented	3.00	Agree
S7	The metrics used to validate the optimized model using MAE are appropriate and clearly presented	4.00	Strongly Agree
S8	The application of Simulated Annealing demonstrates a clear improvement in predictive performance compared to the classical model, as evidenced by the validation metrics	3.00	Agree
S9	The application of Bayesian Inference demonstrates a clear improvement in predictive performance compared to the classical model, as evidenced by the validation metrics	4.00	Strongly Agree
S10	Based on the results, Simulated Annealing is better than Bayesian in optimizing parameters.	4.00	Strongly Agree
S11	The application of Bayesian Inference in the classic model maintains predictive stability when tested with different datasets or varying initial conditions	3.00	Agree
S12	The application of Simulated Annealing in the classic model maintains predictive stability when tested with different datasets or varying initial conditions	3.00	Agree
S13	When applied to different age groups, Simulated Annealing maintains prediction stability	4.00	Strongly Agree
S14	When applied to different age groups, Bayesian Inference maintains prediction stability	4.00	Strongly Agree
S15	When applied to different training durations, Bayesian Inference maintains prediction stability	4.00	Strongly Agree
S16	When applied to different training durations, Simulated Annealing maintains prediction stability	4.00	Strongly Agree

Statement Code	Statement	Mean Score	Interpretation
S17	The model validation process sufficiently accounts for overfitting and ensures generalizability.	2.00	Disagree
S18	The optimized models' predictions align with historical or real-world epidemic data, supporting its practical applicability.	4.00	Strongly Agree
S19	The optimized SEIR model is robust when applied to datasets with varying characteristics and different epidemiologic assumptions.	3.00	Agree
S20	The parameters optimized by Bayesian Inference are within acceptable bounds	4.00	Strongly Agree
S21	The parameters optimized by Simulated Annealing are within acceptable bounds	4.00	Strongly Agree
S22	The application of Bayesian Inference to classical model can be used to predict the future cases	4.00	Strongly Agree
S23	The application of Simulated Annealing to classical model can be used to predict the future cases	4.00	Strongly Agree
Average		3.61	Strongly Agree

The first usability testing for the expert was made to solidify the technicalities in the method of developing and optimizing the SEIR math model. The first usability test had a 3.61 average score with the interpretation being Strongly Agree with the math model's optimization methodology. The respondent acknowledged the clarity and justification of the optimization methods, the appropriateness of the validation metrics, and the robustness of the model in handling different datasets and epidemiologic assumptions. Notably, Simulated Annealing was rated favorably in terms of maintaining prediction stability across different age groups and training durations. However, concerns regarding overfitting and generalizability (S17) suggests that further refinements may be necessary to enhance the model's adaptability to real-world scenarios.

Verbal feedback was also given by the expert when discussing the model. Concerns noted by the researchers were: (1) validity of the time range of the model and (2) model's accuracy and history because of the model's dynamic computation when given new data. The concerns were remedied by the researchers by (1) providing a way to view the current model's error metrics on the web application and (2) providing a view the model's history containing information about the data range it used to estimate the hyperparameters and the error metrics of the model.

For the second usability testing of the expert, a different questionnaire was given that focuses on the SEIR modeling and its usability and validity. Two (2) experts, both Expert 1 and Expert 2, responded to the questionnaire and answered via digital form using google forms. See Appendix R for the answered forms.

The statements for the second usability testing with the experts focused on the applicability of the model and validity of its predictions in relation to actual data. The first part of the questionnaire aimed to assess the performance and accuracy of the optimized SEIR model in predicting the spread of the disease, as well as its ability to handle different epidemic scenarios. This included statements such as the model's capability to provide acceptable predictions, its applicability to different epidemic phenomena, and its effectiveness in predicting the spread of the disease based on actual data. The overall goal of the second expert usability test was to determine if the developed optimized SEIR model, along with the web application, could provide reliable and accurate predictions of disease spread while

offering clear and informative visualizations. Additionally, it aimed to evaluate if the model and application could be adaptable to various scales of data and potentially applicable to emerging infectious diseases.

The score of each statement for the usability test of the expert is shown in Table 59 on the next page.

Table 59 Experts Second Usability Test Results

Statement Code	Statement	Mean Score	Interpretation
S1	The developed optimized Model provides acceptable prediction of the disease spread	3.50	Strongly Agree
S2	The web application correctly presents useful data in assessing the spread of the disease	4.00	Strongly Agree
S3	The developed optimized model can easily fit well in different epidemic phenomenon	3.00	Agree
S4	The optimized SEIR model used provides acceptable prediction results against actual data	3.50	Strongly Agree
S5	The web application can accurately provide useful visualization on how likely the disease can spread	4.00	Strongly Agree
S6	The parameters and the model used in predicting disease spread are applicable and correctly align to requisites in epidemiology	3.00	Agree
S7	The optimization techniques applied can clearly present closer approximation of prediction against actual data	3.50	Strongly Agree
S8	The overall web application can generally be helpful in augmenting the way epidemiologists and related public health personnel identify and observe prediction spread overtime	3.50	Strongly Agree
S9	The developed optimized model can handle various scales of data	3.00	Agree
S10	The optimized SEIR model incorporated the optimization techniques effectively	3.50	Strongly Agree
S11	The model maintained consistent predictive performance when applied to different data	3.00	Agree
S12	The web application provided sufficient visualization incorporating the developed model	4.00	Strongly Agree
S13	The model can be applied to new emerging infectious disease	3.50	Strongly Agree
S14	The web application effectively integrates predictive modeling to assess disease spread.	3.50	Strongly Agree
S15	The model can be effectively utilized for pandemic preparedness.	3.50	Strongly Agree
Average		3.47	Strongly Agree

The average mean score of the overall usability test is 3.47 (Strongly Agree), showing that the experts strongly agree with the statements regarding the model and applicability and validity. The results indicate that the model provides an acceptable prediction of disease spread and aligns well with epidemiological principles. The web application was particularly well-rated for its ability to present useful data and visualizations (S2, S5, S12), suggesting that it effectively aids in assessing epidemic trends. Additionally, the optimization techniques were recognized for improving predictive accuracy, making the model applicable for pandemic preparedness (S15). However, areas for improvement were noted in ensuring that the model consistently maintains performance across different datasets and scales (S3, S9, S11). Despite these minor limitations, the results highlight the model's potential for real-world epidemiological applications.

Using Cronbach's Alpha, the survey questionnaires' consistency and reliability were calculated. Refer to Chapter 3 Phase 4 for the interpretation of the Cronbach's Alpha values. The data collected from the experts' respondents and the calculation for the Cronbach's alpha is shown in Table 60. Based on the value of the Cronbach's alpha of 0.83, internal consistency of Good, the questionnaires are reliable and consistent for the study.

Table 60 Experts Second Usability Test Cronbach's Alpha

	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	Total
Respondent 1	4	4	3	4	4	3	4	4	3	3	3	4	4	4	4	55
Respondent 2	3	4	3	3	4	3	3	3	3	4	3	4	3	3	3	49
Variances	0.25	0	0	0.25	0	0	0.25	0.25	0	0.25	0	0	0.25	0.25	0.25	
No. of Items	15															
Sum of	2															
Vairances																
Variance of	9															
Total Scores																
Cronbach's	0.83															
Alpha																

Table 61 shows the summary of the ISO 9126 guidelines that apply to each statement. The model and integration with the web application based on the experts' respondents shows that they agree on the functionality, efficiency, and reliability, while strongly agreeing with the usability of the solution.

Table 61 Experts Second Usability Test ISO 9126 Characteristics, statement alignment, and mean score

ISO/IEC 9126 Quality Characteristics	Statement Codes	Mean Score	Interpretation
Functionality	S1, S4, S6, S10, S14, S15	3.42	Strongly Agree
Efficiency	S7, S9	3.25	Agree
Usability	S2, S5, S8, S12	3.88	Strongly Agree
Reliability	S3, S11, S13	3.17	Agree

The average scores of the statements with alignment to each objective was also calculated as depicted in Table 62 on the next page. The table indicates the average scores of each objective statement and the interpretation showing agreement with the first objective while strongly agreeing with the second and third objective.

Table 62 Experts Second Usability Test objective, statement alignment, and mean scores

Objectives	Statement Codes	Mean Score	Interpretation
Develop a SEIR model tailored to a local epidemiological and surveillance unit (ESUs) that can help in communicating and predicting disease spread to help adhere to the law.	S6, S15	3.25	Agree
Optimize the SEIR model through hyperparameter experimentation that will help mitigate discrepancies on local epidemic simulation.	S1, S3, S4, S7, S9, S10, S11, S13	3.31	Strongly Agree
Design a web application to visualize local epidemic simulation of spread by applying and integrating the optimized SEIR model.	S2, S5, S8, S12, S14	3.80	Strongly Agree

In summary, the Phase 4 Testing and Analysis involved the performance of functionality testing by performing unit and database testing. Usability testing was performed with experts and the San Pedro ESU members which both yielded agreeable results that was overall deemed by the researchers acceptable.

Conclusion

This study aimed to optimize the predictive performance of the Classical SEIR (Susceptible-Exposed-Infectious-Recovered) Model for forecasting the spread of infectious diseases in selected barangays of San Pedro City. Specifically, the researchers aimed to (1) Develop a SEIR model tailored to a local epidemiological and surveillance unit (ESUs) that can help in communicating and predicting disease spread to help adhere to the law, (2) Optimize the SEIR model through hyperparameter experimentation that will help mitigate discrepancies on local epidemic simulation, and (3) Design a web application to visualize local epidemic simulation of spread by applying and integrating the optimized SEIR model.

The first phase of the study is data preparation, the focus was on preparing the dataset to support the SEIR model and web application development. Data from the Department of Health (DOH) was gathered and structured to ensure accuracy and reliability for further analysis. The dataset was divided into two categories: a **validation set** using data from **San Pedro City** and a **confidence set** from **Biñan City and Cabuyao City** to enhance the robustness of the model. The dataset was then stored to ensure efficient management and retrieval. This structured approach allowed for seamless integration with the web application, which visualize real-time case trends and provide predictive insights. The successful completion of Phase 1 established a solid foundation for the next phase, where the SEIR model will be validated, and its parameters optimized for improved prediction.

The second phase of the study was the modeling and parameter experimentation, The study focused on minimizing prediction errors by using Bayesian Inference and Simulated Annealing adjusting model parameters to fit the actual observed data. Three primary error metrics were

considered in evaluating the models: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Normalized Mean Absolute Error (NMAE). Results revealed a substantial reduction in prediction errors when the SEIR model underwent hyperparameter optimization. Specifically, Simulated Annealing-optimized SEIR model exhibited approximately a **66.30% reduction in RMSE, 66.94% reduction in MAE, and 54.35% reduction in NMAE** compared to the classical SEIR model.

Among the two optimization techniques, Simulated Annealing consistently demonstrated the lowest error metrics across different testing scenarios, including barangay-level, age-group-specific, and municipality-wide datasets. When plotted against actual observed data, the predictions generated by the Simulated Annealing model closely followed real case trends, minimizing overestimations and underestimations. The optimized model provided smoother, more stable forecasts while maintaining the ability to capture infection peaks more accurately than both the classical SEIR model and the Bayesian Inference-optimized model.

Additionally, confidence testing using datasets from different barangays further validated the reliability of Simulated Annealing in maintaining low prediction errors. Across various locations, the model consistently aligned with real-world infection dynamics, demonstrating its robustness and adaptability. Based on a usability testing conducted among the experts, the model received a **mean rating of 3.47**, indicating strong agreement that it has the potential to be used in real-life cases and for monitoring other epidemics. However, its application has not yet been implemented due to a lack of sufficient real-time data.

With its superior performance in reducing prediction errors and closely matching actual values, Simulated Annealing was selected as the final model for integration into the web-based

prediction system. Future work can focus on gathering more comprehensive datasets to enable real-world deployment and further refinement of the model for broader epidemiological applications.

The third phase of the study is the model integration and web application development phase. In this phase, the researchers developed a web application that serves as a supplementary tool for the San Pedro ESU to visualize current and predicted cases around San Pedro. The methodology for development followed a prototyping methodology that resulted in three iterations.

The first step of this phase is integrating the optimized SEIR model. In this step, the finalized SEIR model from the previous phase was integrated and is configured to be called to predict 30 days ahead of time after 3 months' worth of data has been entered into the database. After this, the next part of the phase is creating the web application. In creating the web application, the researchers utilized Django as the core framework for development. Additionally, PostgreSQL was used as the database management system. Other tools that were used were *Tailwind CSS* and *Plotly* to ensure that the design of the application and the data visualization were user friendly and visually appealing. Using these tools, the researchers followed three steps in developing the web application.

First, the Analysis and Design step was started with an initial interview with the San Pedro ESU members to collect information about their needs and expectations in the application. Succeeding interviews lead to changes that are outlined in the prototype refinement step while ensuring that the scope and limitations of the study were upheld. Also, the design of the application followed Gestalt principles to ensure a user-friendly and visually appealing user interface. Second,

the Prototype Refinement step consisted of implementing changes that were asked of the ESU in every prior interview. In the first interview, the comments from the ESU members emphasized the importance of intuitive data visualizations for conveying disease spread. Also, the ESU asked for the addition of tracking the ages in the database records. In the next interview, the main changes were adding the gender of each record and showing the Average Daily Attack Rate (ADAR). In the last interview, most of the ESU's requested changes were on how color was used to show key information and omitting the technical details that their audience generally does not need.

In the final step of the development cycle, the researchers utilized the built-in Django module for testing. Each key function in the application was rigorously tested to ensure that they were robust in handling errors. The results of the Usability Testing and Functional Testing are shown in further detail in Phase 4.

The final part of this phase is the deployment of the application. After development, the web application was deployed on Ubuntu to ensure stability, scalability, and performance. Also, the deployment process involved setting up a Gunicorn application server and NGINX to handle incoming HTTP requests, ensuring efficient request handling and load balancing. The success of this phase ensured that the researchers achieved the study's third objective of integrating the optimized SEIR model into a web application to visualize local epidemic simulation spread.

The fourth phase of the study is the testing and analysis. In this phase, the researchers first performed functionality tests including unit and database tests on the web application to ensure that the application was working as intended. After performing the tests, the researchers performed usability testing with experts and the San Pedro ESU. As discussed in Chapter 3, the tests must attain a high Cronbach's Alpha score to verify the questionnaire's consistency and validity and

achieve a high and agreeable interpretation with the average of the mean scores of the tests. The researchers utilized three (3) types of questionnaire with the first test, for the expert, containing statements to validate the technical aspects of the model and its integration, the second test for the experts with statements on the usability and validity of the model, and the last test for the San Pedro ESU members to measure the usability of the web application with the integrated model as a tool that can aid them in their activities. Notable feedback gathered from the expert was the time range validity of the model which the researchers addressed by providing a view of the historical models that was generated by the system to provide insight into the hyperparameters estimated and error metrics of each model. While feedback from the San Pedro ESU mainly focused on the overall readability of the visualizations as well as the capability to accept data input catered to their case-by-case basis.

Overall, the results from the usability testing both with the expert and the ESU members demonstrated a strong positive response to the developed system. The developed model was found to provide acceptable predictions of disease spread, aligning well with epidemiological principles and supporting the study's objective of developing a tailored model for local epidemiological surveillance. The optimization techniques implemented in the study were recognized for improving predictive accuracy, addressing the discrepancies in epidemic simulations, and enhancing the model's applicability for infectious disease preparedness. Additionally, the web application was highly rated for its ability to effectively visualize epidemic trends, fulfilling the objective of designing an accessible and informative application. While minor areas for improvement were noted such as refining UI and ensuring consistent model performance, the system was validated as an effective tool for public health monitoring and decision-making. These

findings highlight the model's potential for real-world epidemiological applications and suggest that further refinements could improve its adaptability to broader disease surveillance efforts.

Having achieved all the specific research goals, the researchers successfully fulfilled their primary aim of developing and optimizing an SEIR math model and integrating it into a web application as a tool for prediction and visualization of infectious disease spread.

Recommendations

The researchers aspire that the study will serve as foundational data for future work in the fields of SEIR math modeling optimization, SEIR model's application, and infectious disease monitoring. Consequently, the researchers have offered recommendations aimed at guiding future researchers toward new directions or enhancing the study's methods.

In this study, the researchers optimized the classical SEIR math model to be applied to barangay-level data. However, the optimization techniques used were limited to Bayesian Inference and Simulated Annealing. Given the improvements observed in the results of the optimization, it is evident that the model's predictive capabilities were enhanced. That being said, the researchers recommend the application of other optimization techniques to the SEIR model. Future studies could explore the use of machine learning-based approaches, such as genetic algorithms, reinforcement learning, or deep learning methods, to further refine predictions. Additionally, hybrid optimization techniques may yield more accurate and adaptive forecasting models. Expanding the dataset to include more barangays, longer timeframes, or additional epidemiological factors could also improve the model's applicability and reliability in real-world scenarios.

Furthermore, the researchers recommend implementing a more robust system for dynamic model creation and metric validation. One of the concerns raised during expert evaluation was the accuracy and historical tracking of the model due to its dynamic computation when processing new data. While the study addressed this by providing a way to view error metrics and model history, future research could enhance this feature by automating model validation against past performance and incorporating real-time monitoring of error trends. This could be achieved

through adaptive thresholding for error metrics, version control for model parameters, and visual analytics to compare past and present model behavior. By improving the model's transparency and accountability, researchers and policymakers can make more informed decisions based on the model's evolving predictions.

Additionally, the researchers recommend testing and optimizing the SEIR model on datasets from other epidemics. Since the study was conducted using COVID-19 data, its applicability to other infectious diseases remains unverified. While the mathematical results indicate that the optimization techniques improved the SEIR model's predictive performance, further validation on different epidemic datasets would help confirm its generalizability. This could provide deeper insights into the model's usability across various disease dynamics and improve its adaptability for future public health applications.

References

- Ala'raj, M., Majdalawieh, M., & Nizamuddin, N. (2021). Modeling and forecasting of COVID-19 using a hybrid dynamic model based on SEIRD with ARIMA corrections. In *Infectious Disease Modelling*.
- Anwar, N., & Kar, S. (2019). Review Paper on Various Software Testing Techniques & Strategies. *Global Journal of Computer Science and Technology: CSoftware & Data Engineering Volume 19*.
- Aranda, C., & Humeau, E. (2022). Disease Surveillance and Monitoring in the Philippines. In *Disease Surveillance and Monitoring in the Philippines: Building resilience through mobile and digital technologies*.
- AWS. (n.d.). What is PostgreSQL.
- Baba, I., Baba, B., & Esmaili, P. (2020). A Mathematical Model to Study the Effectiveness of Some of the Strategies Adopted in Curtailing the Spread of COVID-19. In P.-H. Tsui (Ed.), *Computational and Mathematical Methods in Medicine*. doi:10.1155/2020/5248569
- Benisek, A. (2024). Coronavirus Recovery. *WebMD*.
- Betancourt-Cravioto, M., Conyer, R. T., Cravioto, M. B., Martinez, R. S., Murguia, L. M., & Rincon, H. G. (2013). Strengthening vaccination policies in Latin America: An evidence-based approach. In G. A. Poland (Ed.), *Vaccine*. doi:10.1016/j.vaccine.2012.12.062

- Blanco, R., Patow, G., & Pelechano, N. (2024). Simulating real-life scenarios to better understand the spread of diseases under different contexts. *Sci Rep*, 2694. doi:<https://doi.org/10.1038/s41598-024-52903-w>
- Brachman, P. S. (1996). Chapter 9 Epidemiology. In S. Baron (Ed.), *Medical Microbiology*. 4th edition.
- Brauer, F., Castillo-Chavez, C., & Feng, Z. (2019). Mathematical Models in Epidemiology. *Texts in Applied Mathematics*. doi:10.1007/978-1-4939-9828-9
- Bushira, K. M., & Ongala, J. O. (2021). Modeling Transmission Dynamics and Risk Assessment for COVID-19 in Namibia Using Geospatial Technologies. *Transactions of the Indian National Academy of Engineering*, 377-394. doi:<https://doi.org/10.1007/s41403-021-00209-y>
- Carcione, J. M., Santos, J. E., Bagaini, C., & Ba, J. (2020). A Simulation of a COVID-19 Epidemic Based on a Deterministic SEIR Model. *Frontiers in Public Health*. doi:10.3389/fpubh.2020.00230
- CDC. (n.d.). Symptoms of COVID-19.
- Centers for Disease Control and Prevention. (2012, May 18). *Principles of Epidemiology*. Retrieved from CDC: https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson3/section2.html

- Centers for Disease Control and Prevention. (2012, May 18). *Section 11: Epidemic Disease Occurrence*. Retrieved from CDC: https://archive.cdc.gov/www_cdc_gov/csels/dsepd/ss1978/lesson1/section11.html
- Chen, Z., Feng, L., Lay jr., H. A., Furati, K., & Khaliq, A. (2022, August). SEIR model with unreported infected population and dynamic parameters for the spread of COVID-19. *Mathematics and Computers in Simulation*, 198, 31-46. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0378475422000787>
- Chicco, D., Warrens, M. J., & Jurman, G. (2021, July 5). The coefficient of determination R-squared is more informative than SMAPE, MAE, MAPE, MSE and RMSE in regression analysis evaluation. *PeerJ Computer Science*, 7, e623. doi:<https://doi.org/10.7717/peerj-cs.623>
- Chugh, A. (2020, December 8). *MAE, MSE, RMSE, Coefficient of Determination, Adjusted R Squared — Which Metric is Better?* Retrieved from Medium: <https://medium.com/analytics-vidhya/mae-mse-rmse-coefficient-of-determination-adjusted-r-squared-which-metric-is-better-cd0326a5697e>
- City of San Pedro. (2023). *Congratulations to the San Pedro City Epidemiology and Surveillance Unit*. Retrieved from cityofsanpedrolaguna: <https://cityofsanpedrolaguna.gov.ph/congratulations-to-the-san-pedro-city-epidemiology-and-surveillance-unit/>
- Collins, L. (2007). Research Design and Methods. *Encyclopedia of Gerontology (Second Edition)*, 433-442. doi:<https://doi.org/10.1016/B0-12-370870-2/00162-1>

Columbia Mailman School of Public Health. (n.d.). Markov Chain Monte Carlo.

Columbia University Mailman School of Public Health . (2020, October 21). *What is Epidemiology*. Retrieved from Columbia Public Health: <https://www.publichealth.columbia.edu/news/what-epidemiology>

Columbia University Mailman School of Public Health . (2021, February 19). *Epidemic, Endemic, Pandemic: What are the Differences?* Retrieved from Columbia Public Health: <https://www.publichealth.columbia.edu/news/epidemic-endemic-pandemic-what-are-differences>

Conyer, R. T., Martinez, R. S., Cravioto, M. B., Murguia, L. M., & Rincon, H. G. (2013). Strengthening vaccination policies in Latin America: An evidence-based approach. In *Vaccine*. doi:10.1016/j.vaccine.2012.12.062

Cottle, R. W., & Thapa, M. N. (2017). Linear and Nonlinear Optimization. *International Series in Operations Research & Management Science*. doi:10.1007/978-1-4939-7055-1

Dautel, K. A., Agyingi, E., & Pathmanathan, P. (2023). Validation framework for epidemiological models with application to COVID-19 models. *PLoS Comput Biol*. doi:10.1371/journal.pcbi.1010968

Department of Health. (n.d.). *COVID-19 Case Tracker*. Retrieved from DOH: <https://doh.gov.ph/diseases/covid-19/covid-19-case-tracker/>

Django Software Foundation. (n.d.). *Django: The web framework for perfectionists with deadlines*. Retrieved from Django Software Foundation: <https://www.djangoproject.com/>

- Draxler, R., & Chai, T. (2014, June 30). Root mean square error (RMSE) or mean absolute error (MAE)? – Arguments against avoiding RMSE in the literature. *Geoscientific Model Development*, 7(3), 1247–1250. doi:<https://doi.org/10.5194/gmd-7-1247-2014>
- Enrico Lavezzo, E. F.-D. (2020). *Suppression of a SARS-CoV-2 outbreak in the Italian municipality of Vo'.* Vo'.
- Farrell, S., & Lewandowsky, S. (2018). Basic Parameter Estimation Techniques. In S. Farrell, & S. Lewandowsky, *Computational Modeling of Cognition and Behavior* (pp. 47-71). Cambridge University Press.
- Fox, J. E. (2015). The Science of Usability Testing. *Proceedings of the 2015 Federal Committee on Statistical Methodology (FCSM) Research Conference*.
- Garg, C. (2019, August 2). *Principles of Design: Visual Perception and Gestaltism*. Retrieved from Medium: <https://charchitgarg27.medium.com/principles-of-design-visual-perception-and-gestaltism-2b0c4206834a>
- Gass, S. I., & Joel, L. S. (1981). Concepts of Model Confidence. In *Computers & Operations Research*. doi:10.1016/0305-0548(81)90019-8
- GeeksforGeeks. (2024, January 10). *Data Driven Testing in Software Testing*. Retrieved from GeeksforGeeks: <https://www.geeksforgeeks.org/data-driven-testing/>
- GeeksForGeeks. (2024, August 19). *What are the Minimum Hardware Requirements for Python Programming?* Retrieved from GeeksForGeeks: <https://www.geeksforgeeks.org/what-are-the-minimum-hardware-requirements-for-python-programming/>

GeoPandas. (n.d.). *GeoPandas*. Retrieved from GeoPandas: <https://geopandas.org/en/stable/>

Girardi, P., & Gaetan, C. (2023). An SEIR Model with Time-Varying Coefficients for Analyzing the SARS-CoV-2 Epidemic. *Risk Anal*, 144-155. doi:10.1111/risa.13858

Gleeson, J. P., Murphy, T. B., O'Brien, J. D., & O'Sullivan, D. J. (2020). A population-level SEIR model for COVID-19 scenarios.

GSM Association. (2022). Disease surveillance and monitoring in the Philippines. In *Disease Surveillance and Monitoring in the Philippines: Building resilience through mobile and digital technologies*.

gunicorn. (n.d.). *gunicorn*. Retrieved from gunicorn: <https://gunicorn.org/>

Hassan, M. (2024, March 25). *Purposive Sampling – Methods, Types and Examples*. Retrieved from researchmethod: <https://researchmethod.net/purposive-sampling/>

Hatami, F., Chen, S., Paul, R., & Thill, J.-C. F. (2022, 11). Simulating and Forecasting the COVID-19 Spread in a U.S. Metropolitan Region with a Spatial SEIR Model. *International Journal of Environmental Research and Public Health (IJERPH)*. doi:10.3390/ijerph192315771

Haw, N. J., Uy, J., Sy, K. T., & Abrigo, M. R. (2020). Epidemiological profile and transmission dynamics of COVID-19 in the Philippines. *Epidemiology and infection*, 148. doi:<https://doi.org/10.1017/S0950268820002137>

He, S., Peng, Y., & Su, K. (2020). SEIR modeling of the COVID-19 and its dynamics. *Nonlinear Dynamics*. doi:<https://doi.org/10.1007/s11071-020-05743-y>

- He, X., Lau, E. H., Wu, P., Deng, X., Wang, J., Hao, X., . . . Leung, G. M. (2020). Temporal dynamics in viral shedding and transmissibility of COVID-19. *Nat Med*, 672–675. doi:<https://doi.org/10.1038/s41591-020-0869-5>
- Helmold, M. (2019). Excellence in PM. *Progress in Performance Management*, 39-49. doi:[10.1007/978-3-030-20534-8_3](https://doi.org/10.1007/978-3-030-20534-8_3)
- Hemmati-Sarapardeh, A., Larestani, A., & Amar, M. N. (2020). *Applications of Artificial Intelligence Techniques in the Petroleum Industry*.
- Hethcote, H. W. (2000). The Mathematics of Infectious Diseases. *SIAM Review*, 599-653. doi:[10.1137/S0036144500371907](https://doi.org/10.1137/S0036144500371907)
- Hodson, T. O. (2022, July 19). Root-mean-square error (RMSE) or mean absolute error (MAE): when to use them or not. *Geoscientific Model Development*, 15(14), 5481–5487. doi:<https://doi.org/10.5194/gmd-15-5481-2022>
- Hosseiniabadi, A. A., Alavipour, F., Band, S. S., & Balas, V. E. (2016). A Novel Meta-Heuristic Combinatory Method for Solving Capacitated Vehicle Location-Routing Problem with Hard Time Windows. In *Advances in Intelligent Systems and Computing*. Springer international Publishing Switzerland 2017, Transportation Systems, Advances in Intelligent Systems and Computing .
- Hotta, L. K. (2010). Bayesian Melding Estimation of a Stochastic SEIR Model. *Mathematical Population Studies: An International Journal of Mathematical Demography*, 101-111. doi:[10.1080/08898481003689528](https://doi.org/10.1080/08898481003689528)

- Hou, Y., & Bidkhori, H. (2024). Multi-feature SEIR model for epidemic analysis and vaccine prioritization. doi:<https://doi.org/10.1371/journal.pone.0298932>
- IDM. (n.d.). *SEIR and SEIRS models*. Retrieved from IDM: <https://docs.idmod.org/projects/emod-generic/en/latest/model-seir.html>
- Institute of Clinical Epidemiology, National Institutes of Health, UP Manila. (n.d.). Philippine COVID-19 Living Clinical Practice Guidelines. In N. I. Institute of Clinical Epidemiology, *Philippine COVID-19 Living Clinical Practice Guidelines*.
- Jorgensen, P. C. (2013). *Software Testing*. CRC Press Taylor & Francis Group.
- Joshi, A., Kale, S., Chandel, S., & Pal, D. K. (2015). Likert Scale: Explored and Explained. *British Journal of Applied Science & Technology* 7, 396-403. doi:10.9734/BJAST/2015/14975
- Kaur, T., & Singh, B. (2014). Testing of Databases. *International Journal of Innovative Science, Engineering & Technology*.
- Kirkpatrick, S., Gelatt Jr., C. D., & V. M. (1983). Optimization by Simulated Annealing. *Science* Volume 220, 671-680. doi:10.1126/science.220.4598.671
- Larremore, D. B., Fosdick, B. K., Bubar, K., Zhang, S., Kissler, S. M., Metcalf, J. E., . . . Grad, Y. H. (2021). Estimating SARS-CoV-2 seroprevalence and epidemiological parameters with uncertainty from serological surveys. *eLife*. doi:<https://doi.org/10.7554/eLife.64206>

- Lauer, S. A., Grantz, K. H., Bi, Q., Jones, F. K., Zheng, Q., Meredith, H. R., . . . Lessler, J. (2020). The Incubation Period of Coronavirus Disease 2019 (COVID-19) From Publicly Reported Confirmed Cases: Estimation and Application.
- Lavezzo, E., Franchin, E., & Ciavarella, C. (2020). Suppression of a SARS-CoV-2 outbreak in the Italian municipality of Vo'. *nature*, 425–429.
- Lewis, S. (2023, June). *Prototyping Model*. Retrieved from TechTarget: <https://www.techtarget.com/searchcio/definition/Prototyping-Model>
- Lopes, H. F., Bello Moreira, A. R., & Schmidt, A. M. (1998). Hyperparameter estimation in forecast models. *Computational Statistics & Data Analysis*, 387-410. doi:[https://doi.org/10.1016/S0167-9473\(98\)00078-4](https://doi.org/10.1016/S0167-9473(98)00078-4)
- López, L., & Rodó, X. (2021). A modified SEIR model to predict the COVID-19 outbreak in Spain and Italy: Simulating control scenarios and multi-scale epidemics. *Results in Physics*. doi:<https://doi.org/10.1016/j.rinp.2020.103746>
- Macías, R. Z., Humberto, G.-P., Arroyo, E. A., & González, A. P. (2022). Geographical network model for COVID-19 spread among dynamic epidemic regions. *Mathematical Biosciences and Engineering*. doi:10.3934/mbe.2022196
- Madhav, N., Oppenheim, B., Gallivan, M., Mulembakani, P., Rubin, E., & Wolfe, N. (2017). Pandemics: Risks, Impacts, and Mitigation. In *Disease Control Priorities: Improving Health and Reducing Poverty. 3rd edition*. Washington (DC): The International Bank for Reconstruction and Development.

- Maged, A., Ahmed, A., Haridy, S., Baker, A. W., & Xie, M. (2023). SEIR Model to address the impact of face masks amid COVID-19 pandemic. *Risk Anal*, 129-143. doi:10.1111/risa.13958
- Matsuura, K., & Willmott, C. J. (2005, December 19). Advantages of the mean absolute error (MAE) over the root mean square error (RMSE) in assessing average model performance. *Climate Research*, 30, 79–82. doi:doi:10.3354/cr030079
- Mbuyha, R., & Marwala, T. (2020). Bayesian Inference of COVID-19 Spreading Rates in South Africa. doi:<https://doi.org/10.1371/journal.pone.0237126>
- Md. Rahman, R., A. H. M. Islam, H., & Md. Islam, N. (2021). Geospatial modelling on the spread and dynamics of 154 day outbreak of the novel coronavirus (COVID-19) pandemic in Bangladesh towards vulnerability zoning and management approaches. *Modeling Earth Systems and Environment*, 2059-2087. doi:<https://doi.org/10.1007/s40808-020-00962-z>
- Misra, S., & Jin, Y. (2021). Multifrequency electromagnetic data interpretation using particle swarm optimization and ant colony optimization methods. *Multifrequency Electromagnetic Data Interpretation for Subsurface Characterization*, 331-349. doi:<https://doi.org/10.1016/B978-0-12-821439-8.00009-4>
- MMCL College of Computer and Information Science. (2021). *COVID-19 Case Tracker for the Province of Laguna*. Retrieved from COVID-19 Case Tracker for the Province of Laguna: <https://mcl-covid19tracker-laguna.netlify.app/>

- Montori, V. M., Hargraves, I., McNellis, R. J., Ganiats, T. G., Genevro, J., Miller, T., . . . Bierman, A. S. (2018). The Care and Learn Model: a Practice and Research Model for Improving Healthcare Quality and Outcomes. *J Gen Intern Med.*, 154–158. doi:10.1007/s11606-018-4737-7
- Mortimer, J. A., & Borenstein, A. R. (n.d.). Tools of the Epidemiologist. In *Alzheimer Disease & Associated Disorders*, 20 (Supplement 2). doi:10.1097/00002093-200607001-00004
- Mulder, L. (2020). Use of simulated annealing to determine the operational parameters of the SEIR model for the coronavirus for various jurisdictions. doi:10.2471/BLT.20.260513
- Murray_smith, D. J. (1998). Methods for the external validation of continuous system simulation models: a review. *Mathematical and Computer Modelling of Dynamical Systems*, 5–31. doi:https://doi.org/10.1080/13873959808837066
- Mwalili, S., Kimathi, M., Ojiambo, V., Gathungu, D., & Mbogo, R. (2020). SEIR model for COVID-19 dynamics incorporating the environment and social distancing. *BMC Res Notes*, 352. doi:10.1186/s13104-020-05192-1
- National Institute of Biomedical Imaging and Bioengineering. (2020, May). *Computational Modeling*. Retrieved from National Institute of Biomedical Imaging and Bioengineering: <https://www.nibib.nih.gov/science-education/science-topics/computational-modeling>
- National Institute of Biomedical Imaging and Bioengineering. (2020). Computational Modeling.

- Nguyen, V. T., Jung, K., & Gupta, V. (2021). Examining data visualization pitfalls in scientific publications. *Visual Computing for Industry, Biomedicine, and Art*. doi:<https://doi.org/10.1186/s42492-021-00092-y>
- Nisbet, R., Miner, G., & Yale, K. (2018). Handbook of Statistical Analysis and Data Mining Applications. In R. M. Nisbet, *The Data Mining and Predictive* (pp. 39-54).
- Nsubuga, P., White, M. E., Thacker, S. B., Anderson, M. A., Blount, S. B., Broome, C. V., . . . Trostle, M. (2006). Public Health Surveillance: A Tool for Targeting and Monitoring Interventions. In *Disease Control Priorities in Developing Countries. 2nd edition*. Washington DC: The International Bank for Reconstruction and Development.
- Ohki, Y., & Teferra, D. (2020). Simulated Annealing: Theory and Applications .
- Ohue, M., Sasayama, K., & Takata, M. (2024). Mathematical modeling and problem solving: from fundamentals to applications. *The Journal of Supercomputing Volume 80*, 14116-14119. doi:<https://doi.org/10.1007/s11227-024-06007-x>
- Ozili, P. K. (2023). The Acceptable R-Square in Empirical Modelling for Social Science Research. *Social Research Methodology and Publishing Results*.
- Peng, L., Yang, W., Zhang, D., Zhuge, C., & Hong, L. (2020). Epidemic analysis of COVID-19 in China by dynamical modeling. *medRxiv* 2020.02.16.20023465. doi:<https://doi.org/10.1101/2020.02.16.20023465>
- Pimintel, J. L. (2010, December). A note on the usage of Likert Scaling for research data analysis. *USM R & D*, 18, 109-112. Retrieved from

https://www.researchgate.net/publication/331231816_A_note_on_the_usage_of_Likert_Scaling_for_research_data_analysis

Pinter, G., Felde, I., Mosavi, A., Ghamisi, P., & Richard, G. (2020). COVID-19 Pandemic Prediction for Hungary; A Hybrid Machine Learning Approach. In *Mathematics 2020*.

Poole, D., & Raftery, A. E. (2000). Inference for deterministic simulation models: The Bayesian melding approach. *Journal of the American Statistical Association*. doi:<https://doi.org/10.1080/01621459.2000.10474324>

Presidential Communications Officer. (2022, February 24). *IATF revises Alert Level Metrics*. Retrieved from Presidential Communications Officer: https://pco.gov.ph/news_releases/iatf-revises-alert-level-metrics/

Republic Act No. 11332. (2018). *An Act Providing Policies and Prescribing Procedures on Surveillance and Response to Notifiable Diseases, Epidemics, and Health Events of Public Health Concern, and Appropriating Funds*. Official Gazette.

Sánchez-Romero, M., Lego, V. d., Prskawetz, A., & Queiroz, B. L. (2021). An indirect method to monitor the fraction of people ever infected with COVID-19: An application to the United States. *PLoS One*. doi:[10.1371/journal.pone.0245845](https://doi.org/10.1371/journal.pone.0245845)

Santos, G. L. (2022). Parameter Estimation for a Modified SEIR Model of the COVID-19 Dynamics in the Philippines using Genetic Algorithm. doi:<https://doi.org/10.1101/2022.05.17.22275187>

Santos, G. L. (n.d.). *Parameter Estimation for a Modified SEIR Model of the COVID-19 Dynamics in the Philippines using Genetic Algorithm.*

Selic, B., & Gerard, S. (2014). Foundations for Model-Based Analysis. In *Modeling and Analysis of Real-Time and Embedded Systems with UML and MARTE*. doi:10.1016/B978-0-12-416619-6.00009-2

StatisticsHowTo. (n.d.). *Cronbach's Alpha: Definition, Interpretation, SPSS*. Retrieved from StatisticsHowTo: <https://www.statisticshowto.com/probability-and-statistics/statistics-definitions/cronbachs-alpha-spss/>

Taghizadeh, E., & Mohammad-Djafari, A. (2022). SEIR Modeling, Simulation, Parameter Estimation, and Their Application for COVID-19 Epidemic Prediction. *Physical Sciences Forum*, 18. doi:<https://doi.org/10.3390/psf2022005018>

The National Academies Press. (1998). Chapter 9: Using Modeling and Simulation in Test Design and Evaluation. In *Statistics, Testing, and Defense Acquisition: New Approaches and Methodological Improvements*.

Trawicki, M. B. (2017). Deterministic SEIRS epidemic model for modeling vital dynamics, vaccinations, and temporary immunity. *Mathematics*, 7. doi:<https://doi.org/10.3390/math5010007>

Vynnycky, E., & White, R. (2010). *An Introduction to Infectious Disease Modelling*. OUP Oxford.

w3schools. (n.d.). w3schools. Retrieved from https://www.w3schools.com/python/pandas/pandas_intro.asp

- W3Schools. (n.d.). W3Schools. Retrieved from https://www.w3schools.com/python/pandas/pandas_intro.asp
- Wang, H., Miao, Z., Zhang, C., Wei, X., & Li, X. (2021). K-SEIR-Sim: A simple customized software for simulating the spread of infectious diseases. *Computational and Structural Biotechnology Journal*, 1966-1975. doi:<https://doi.org/10.1016/j.csbj.2021.04.004>
- West, M. (1992). Hyperparameter estimation in Dirichlet process mixture models. Retrieved from <https://www2.stat.duke.edu/~mw/MWextrapubs/West1992alphaDP.pdf>
- WHO. (2023, October 3). *Influenza (Seasonal)*. Retrieved from who: [https://www.who.int/news-room/fact-sheets/detail/influenza-\(seasonal\)](https://www.who.int/news-room/fact-sheets/detail/influenza-(seasonal))
- WHO. (2024). *2022-24 Mpox (Monkeypox) Outbreak: Global Trends*. Retrieved from worldhealthorg: https://worldhealthorg.shinyapps.io/mpx_global/
- WHO. (n.d.). PART I. THEORY AND PRACTICE OF EPIDEMIOLOGICAL MODELS.
- WHO. (n.d.). *Surveillance in emergencies*. Retrieved from who: <https://www.who.int/westernpacific/emergencies/surveillance>
- Yang, L., & Shami, A. (2020). On hyperparameter optimization of machine learning algorithms: Theory and practice. *Neurocomputing*, 295-316. doi:<https://doi.org/10.1016/j.neucom.2020.07.061>

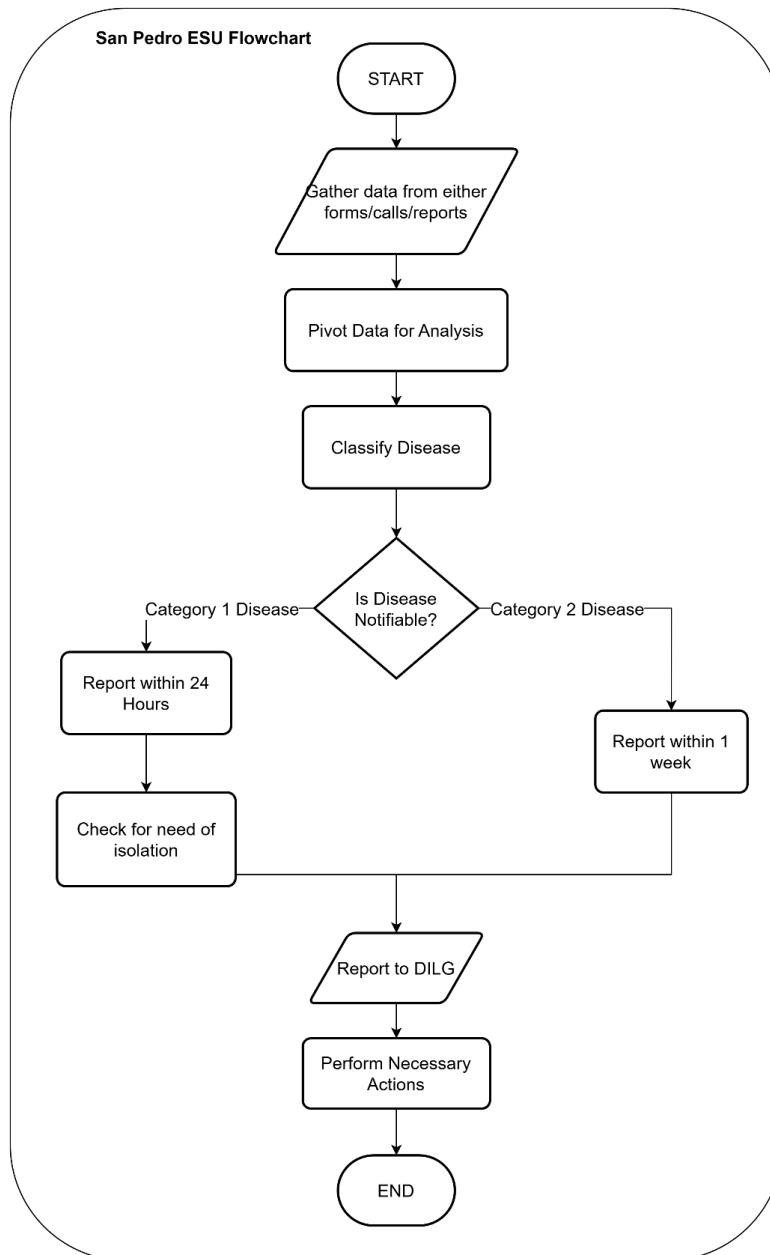
Yang, Z., Zeng, Z., Wang, K., Wong, S.-S., Liang, W., Zanin, M., . . . He, J. (2020). Modified SEIR and AI prediction of the epidemics trend of COVID-19 in China under public health interventions. *Journal Of Thoracic Disease*, 165-174. doi:doi:10.21037/jtd.2020.02.64

Zhan, C., Zheng, Y., Lai, Z., Hao, T., & Li, B. (2020). Identifying epidemic spreading dynamics of COVID-19 by pseudocoevolutionary simulated annealing optimizers. *Neural Comput Appl*. doi:10.1007/s00521-020-05285-9

Appendices

Appendix A

San Pedro ESU Work Flowchart



Appendix B

SEIR Model of Different Studies

Title	Year	Type of SEIR Model		Modified Parameters	Optimization Technique/Parameter Estimation	How was the model fitted?	Description	Limitations noted
A modified SEIR model to predict the COVID-19 outbreak in Spain and Italy: Simulating control scenarios and multi-scale epidemics	2021	Modified	SEIR Model	Quarantine Rate	trust-region-reflective with a maximum number of iterations of 500 and a step size of $6.4e-7$	The model was fitted in a non-linear approach by calculating the normalized least-squares error of the model approximation and the infected reported cases	The model evaluates the confinement rate at the early stages of the epidemic to minimize incidence and mortality. It highlights the importance of timely and drastic containment measures. However, the model's accuracy is limited by uncertainties in case reporting and the evolving nature of the pandemic.	The study suggests that drastic social distancing interventions significantly reduce the number of infections and deaths. Assumptions on SEIR Compartment can lead to underestimation to real data

SEIR model for COVID-19 dynamics incorporating the environment and social distancing	2020	Extended Model	SEIR	Transmission rate, recovery rate	N/A		Model Equations are solved numerically using fourth and fifth Runge-Kutta methods	The model is designed to assess the impact of social distancing and environmental factors on the spread of COVID-19. It provides insights into how these factors can alter the transmission dynamics and inform public health strategies.	Limitations: The model does not describe disease severity and death, and assumptions made at the onset of the pandemic may lead to overestimations
Modified SEIR and AI prediction of the epidemics trend of COVID-19 in China under public health interventions	2020	Modified Model	SEIR	Infection rate, recovery rate	N/A (model fitting using real-data)		Using epidemic data from Hubei, we modeled the skewed SEIR model to determine the probability of transmission (b) which was used to derive β and the probability of recovery or death (γ).	This paper presents a modified SEIR model that integrates artificial intelligence to predict the trends of COVID-19 epidemics in China. It analyzes the impact of	The model does not describe disease severity and death, and assumptions made at the onset of the pandemic may lead to overestimations

Epidemic analysis of COVID-19 in China by dynamical modeling	2020	Generalized SEIR model	Protection rate, infection rate, average latent time, average quarantine time, cure rate, mortality rate	Lease-square regression combined with the simulated annealing algorithm	Fitted using daily official reports of quarantined, recovered, and closed cases from the National Health Commission of China	various public health interventions, such as social distancing and lockdown measures, on the spread of the virus. The model is adapted to improve accuracy in forecasting by incorporating real-time data and adjustments based on intervention effectiveness.	Data Limitations: The study acknowledges the limitations due to the lack of official data and changes in diagnostic criteria during the early stages of the outbreak ¹ . This
						The generalized SEIR model includes seven states: susceptible, insusceptible, exposed, infectious, quarantined, recovered, and closed (death). It incorporates the effects of	

self-protection and quarantine measures to estimate key epidemic parameters and predict the epidemic's trajectory

affected the accuracy of parameter estimation and predictions. Overfitting Concerns: The authors mention the risk of overfitting due to the complexity of the model and limited data. They suggest that simpler models might sometimes be more effective.

Analysis and prediction of improved SEIR transmission dynamics model: taking the second outbreak of COVID-19 in Italy as an example	2023	Improved SEIR Model	Contact rate, detection capability, recovery rate, infection rate	Regression Algorithm and heuristic algorithms (Genetic Algorithm)	Model Parameters were calibrated using data from February to May 2020, and the cases from June to July were used to evaluate the predictive performance	The improved SEIR model incorporates asymptomatic or mildly symptomatic infections, medical track quarantine, quarantine treatment, and distributed delays to define parameters such as contact rates and testing capabilities.	Model Assumptions: The model assumes strict compliance with anti-epidemic measures, which may not reflect real-world scenarios in all regions.
SEIR Modeling of the Italian Epidemic of SARS-CoV-2 Using Computational Swarm Intelligence	2020	Generalized SEIR Mmodel	Time-dependent infection rate, protection rate, quarantine rate, recovery rate, mortality rate	Particle Swarm Optimization	Model parameters were fitted using a stochastic approach with the PSO algorithm	The SEIR model simulates the epidemic dynamics by categorizing the population into Susceptible, Exposed, Infective, and Recovered groups. It incorporates time-dependent parameters to	Model Limitations: The model assumes recovered individuals are immune, which may not be accurate. It also doesn't account for partial infection ability or differences in testing policies. Data Accuracy:

						reflect changes in infection rates due to lockdowns and other measures. The PSO algorithm helps in finding the best-fit parameters by exploring the solution space adaptively.	The official data used may not be accurate enough for detailed studies, affecting the model's reliability
SEIR Modeling, Simulation, Parameter Estimation, and Their Application for COVID-19 Epidemic Prediction	2022	Time-dependent parameter SEIR model	Time-dependent transmission rate	Nonlinear least squares (NLS) optimization and Bayesian estimation method	Model was fitted using real data from Johns Hopkins University GitHub repository, adapting the parameter estimation to the available data	The SEIR model used in this study includes four compartments: Susceptible, Exposed, Infectious, and Removed. The model was adapted to include time-dependent parameters to better fit real data and predict the spread of COVID-19.	Fixed-Parameter Model Limitations: The fixed-parameter model did not provide satisfactory results on real data. The authors recommend using a time-dependent parameter model to better capture the

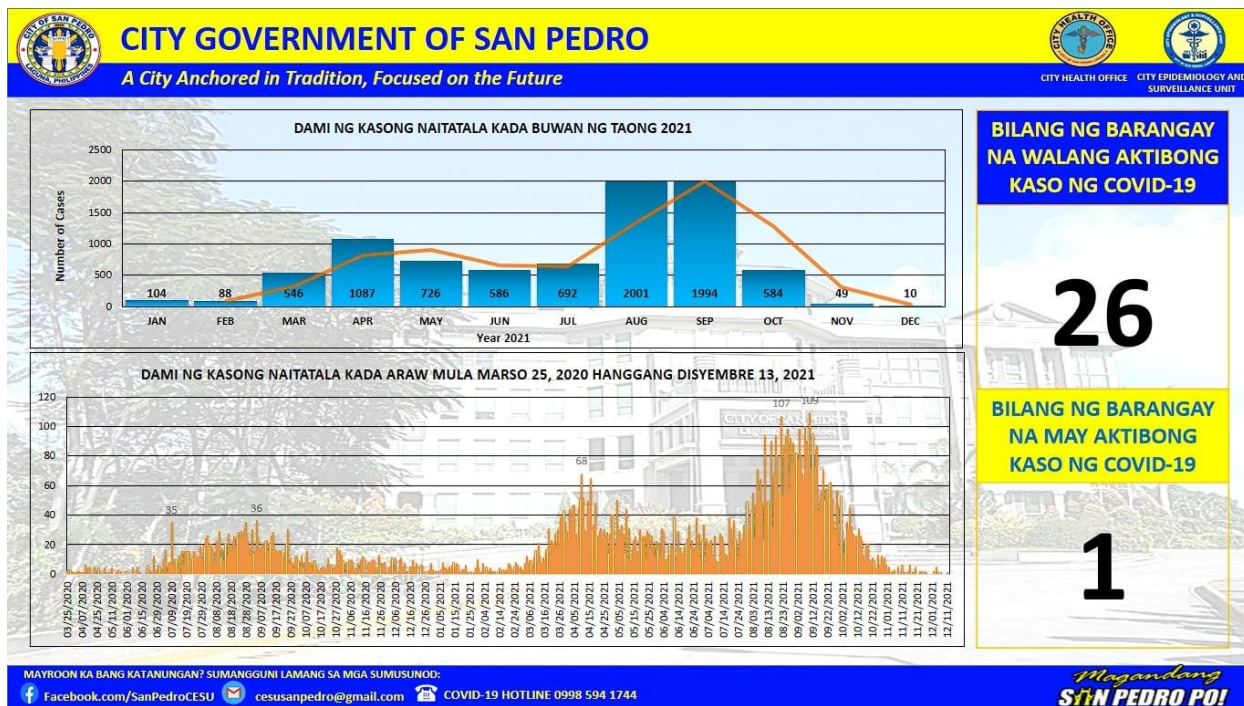
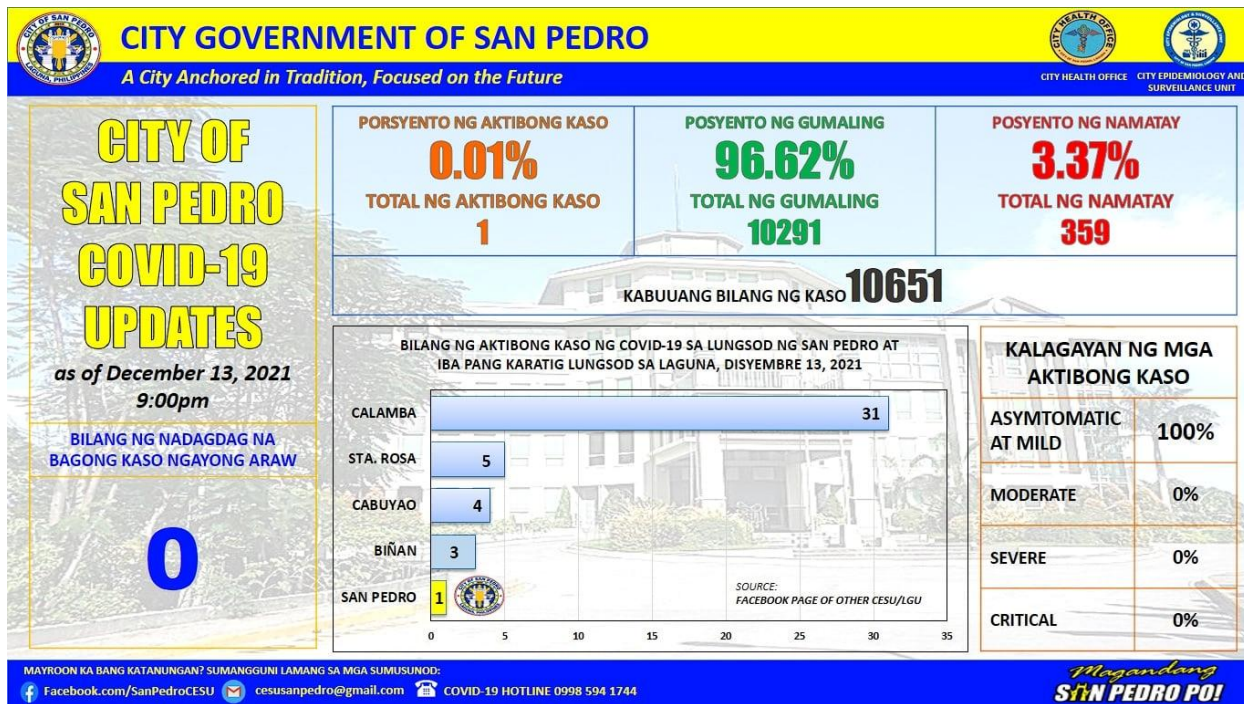
dynamics of the epidemic.

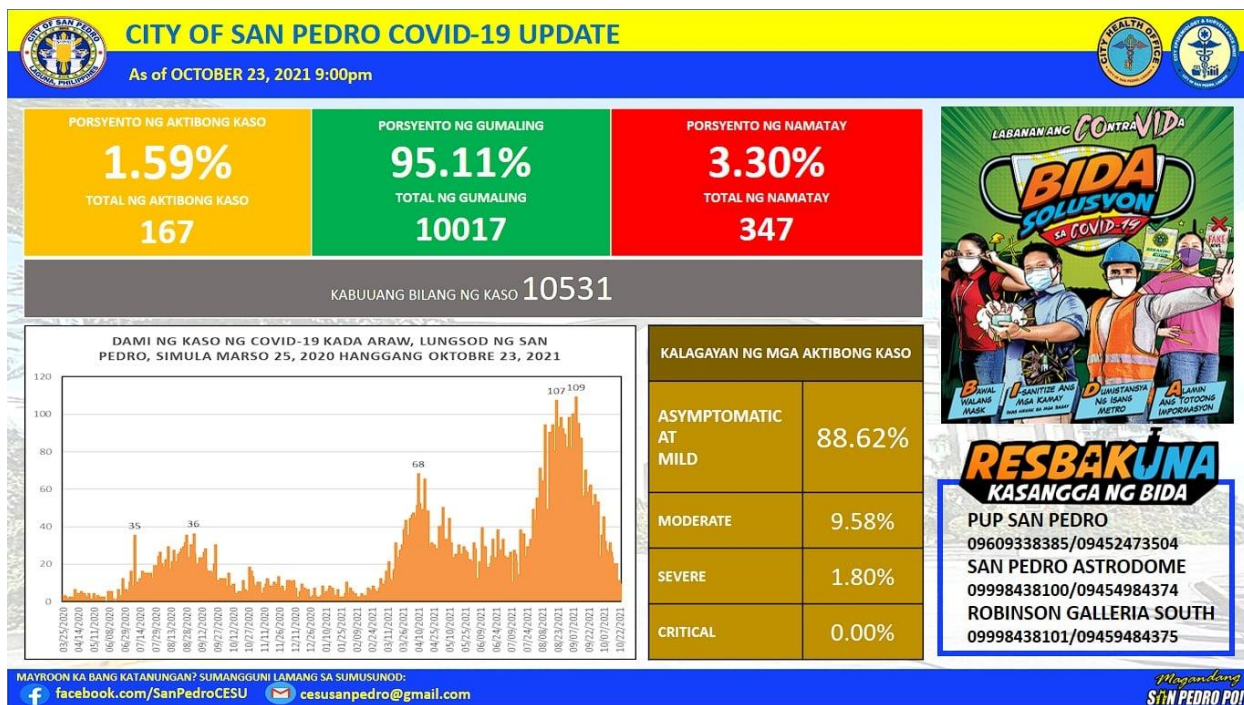
Multi-feature SEIR model for epidemic analysis and vaccine prioritization	2024	Multi-feature SEIR model	Contact rate	Formulated optimization problems for vaccine distribution strategies, including chance-constraint optimization problem (CCP) using Conditional Value-at-Risk (CVaR)	Validated using confirmed COVID-19 cases from Allegheny County (Pennsylvania, USA) and Hamilton County (Ohio, USA)	The model incorporates heterogeneity in health conditions and social activity levels to provide a more accurate prediction of disease dynamics and evaluate different vaccine prioritization strategies	The study proposes using chance-constraint optimization to account for uncertainties in disease transmission and vaccination effectiveness, ensuring more robust and reliable intervention strategies.
Parameter Estimation for a Modified SEIR Model of the COVID-19 Dynamics in the Philippines using Genetic Algorithm	2022	Age-Stratified, Quarantine-modified SEIR model with Nonlinear Incidence Rates	Age-stratification, quarantine modification, nonlinear incidence rates	Genetic Algorithm	Parameters generated by the genetic algorithm were used to fit the model to actual data	The model uses nonlinear incidence rates and genetic algorithms to estimate parameters, providing a more efficient	The peak daily infection projections are significantly lower than previous estimates, suggesting a need for further

method than validation. The the shooting study suggests method. The refining the model was model and applied to both exploring other country-level optimization (Philippines) techniques to and city-level improve (Quezon City) accuracy and data, showing reliability. different dynamics of virus spread.

Appendix C

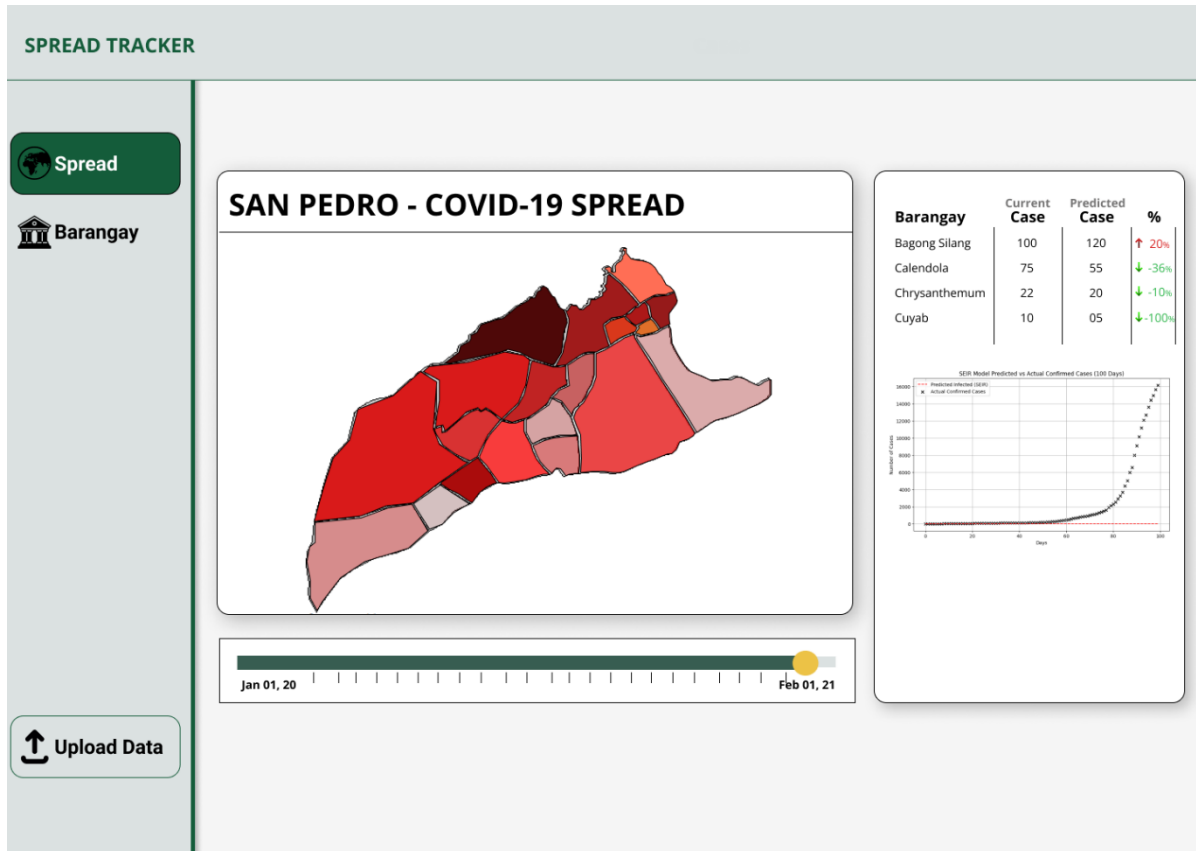
San Pedro ESU Visualization and Posts





Appendix D

Prototype Graphical User Interface of Web Application



SPREAD TRACKER

Spread

Barangay

Upload Data

Barangay

Landayan

100

Current Cases
As of January 04, 2025

4

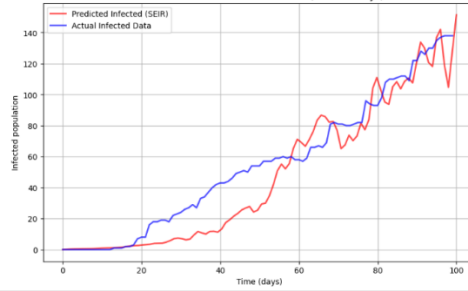
Recovered

+12

Cases
Expected Next Week

LANDAYAN

SEIR Model Prediction vs Actual Data (First 100 Days)



SPREAD TRACKER

Spread

Barangay

Upload Data

ADD FILE

Sanpedro_nov06.csv



Sample Data

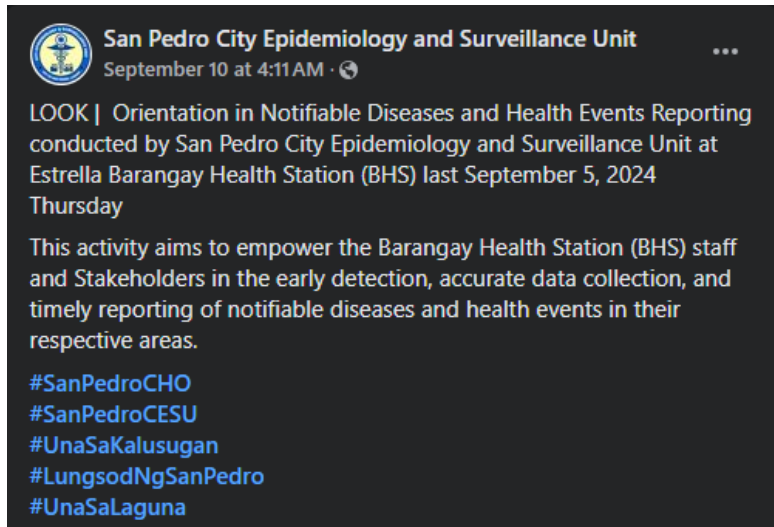
City	Barangay	Date	Exposed	Infected	Removed
San Pedro	Landayan	10/29/24	123	32	2
San Pedro	Cuyab	10/29/24	22	1	3
San Pedro	Bagong Silang	10/29/24	12	2	5
San Pedro	Calendola	10/29/24	55	23	1

UPLOAD

Appendix E

San Pedro ESU Achievement and Contributions





Appendix F

Letter of Request to San Pedro Epidemiology and Surveillance Unit



Mapúa Malayan Colleges Laguna
South Point Subdivision, Pulo-Diezmo Road,
Cabuyao City, Laguna 4025 Philippines

September 21, 2024

Rai Arboleda
San Pedro City Epidemiology and Surveillance Unit
New City Hall Bldg., Brgy. Poblacion,
City of San Pedro, Laguna

Subject: REQUEST TO CONDUCT INTERVIEW

Greetings,

We are 4th year Computer Science students currently pursuing our thesis from **Mapúa Malayan Colleges Laguna**. We would like to humbly ask if your organization would like to be part of our study. The working title of our thesis is **"Modeling and Simulation of Epidemic Spread Using an Optimized SEIR Model Through Parameter Experimentation for Dynamic Visualization."**

We seek your kind permission to allow us to conduct an interview with your San Pedro City Epidemiology and Surveillance Unit Expert. We would like to ask for your most comfortable date and time of availability to have a sit-down discussion and collect information that will help in commencing and directing our study. The said interview could help us gain a better insight and understanding of gaps in epidemic surveillance.

We look forward to a quick and positive response from you. We anticipate your kind consideration and approval of this request. For any queries, feel free to contact us using the contact information provided.

/jbsahagun03
 +63 926-886-9073
 2021ibesahagun@live.mcl.edu.ph



Mapúa Malayan Colleges Laguna
South Point Subdivision, Pulo-Diezma Road,
Cabuyao City, Laguna 4025 Philippines

All the information that you provide will be treated as confidential and will only be used for research purposes. We do not use any of the information you provide for direct marketing or other non-research activities. Also, if you have any questions, please do not hesitate to contact us by email (2021jbsahagun@live.mcl.edu.ph) or mobile (09268869073).

Sincerely,


John Bernard G. Sahagun
Researcher/Student


Jon Peter P. Pomperada
Researcher/Student


Jeremy Aidan L. Genato
Researcher/Student

Noted by:


Dennis A. Martillano
Thesis Adviser Professor
College of Computer and Information Science


Khristian G. Kikuchi
College Dean/Course Coordinator
College of Computer and Information Science

Approved by:

Signature Over Printed Name

Title

Date

 /jbsahagun03
 +63 926-886-9073
 2021jbsahagun@live.mcl.edu.ph

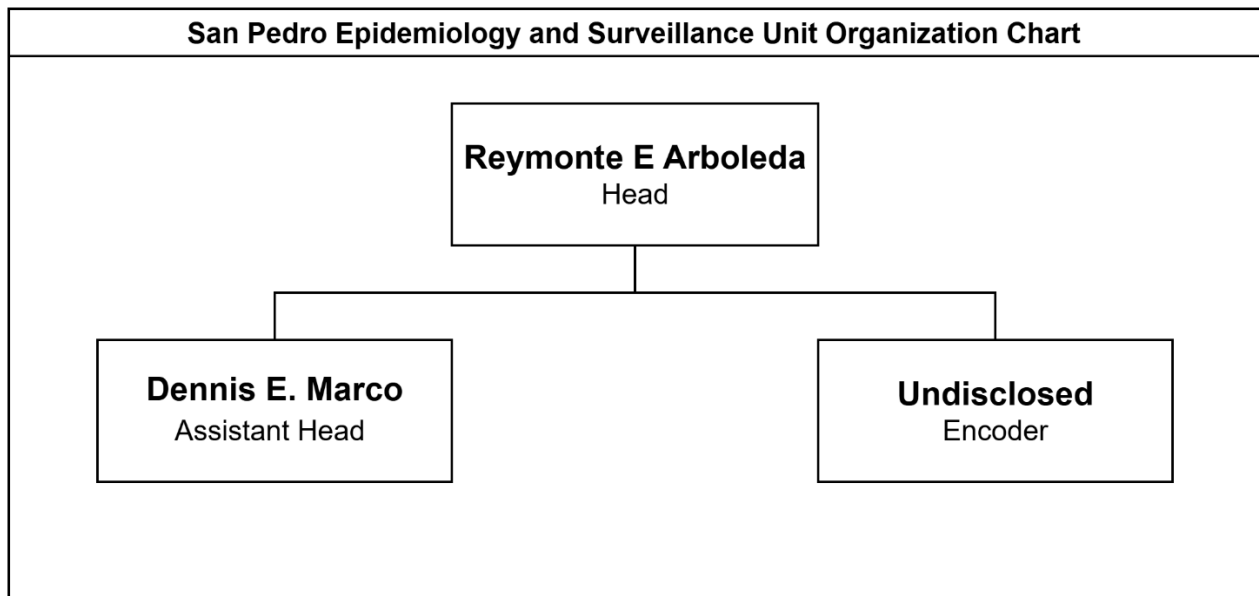
Appendix G

Interview with San Pedro ESU



Appendix H

San Pedro Epidemiology and Surveillance Unit Organization Chart



Appendix I

Sample DOH COVID-19 Data Drop Record

Field	Description
CaseCode	Random code assigned for labelling cases;
Age	does not equate to the unique case number assigned by DOH Age
AgeGroup	Five-year age group
Sex	Sex
DateSpecimen	Date when specimen was collected
DateResultRelease	Date of release of result
DateRepConf	Date publicly announced as confirmed case
DateDied	Date died
DateRecover	Date recovered
RemovalType	Type of removal (recovery or death)
Admitted	Binary variable indicating patient has been admitted to hospital
RegionRes	Region of residence
ProvRes	Province of residence
CityMunRes	City of residence

CityMuniPSGC	Philippine Standard Geographic Code of Municipality or City of residence
BarangayRes	Barangay of residence
BarangayPSGC	Philippine Standard Geographic Code of Barangay of residence
HealthStatus	Known current health status of patient
Quarantined	(asymptomatic, mild, severe, critical, died, recovered) Ever been home quarantined, not necessarily currently in home quarantine
DateOnset	Date of Onset of Symptoms
Pregnanttab	Yes/No if patient is pregnant at any point during COVID-19 condition

	A	B	C	D	E	F	G	H	I	J	K	L
1	CaseCode	Age	AgeGroup	Sex	DateSpecimen	DateResultRelease	DateRepConf	DateDied	DateRecover	RemovalType	DateRepRem	Admitted
2	C404174	38	35 to 39	FEMALE		30/01/2020	30/01/2020			RECOVERED	07/02/2020	NO
3	C462688	44	40 to 44	MALE		30/01/2020	03/02/2020	01/02/2020		DIED	02/02/2020	NO
4	C387710	60	60 to 64	FEMALE	23/01/2020	30/01/2020	05/02/2020		31/01/2020	RECOVERED	05/02/2020	YES
5	C377460	49	45 to 49	MALE			06/03/2020			RECOVERED	27/03/2020	NO
6	C498051	63	60 to 64	MALE	05/03/2020		06/03/2020	11/03/2020		DIED	12/03/2020	NO

	M	N	O	P	Q	R	S
1	RegionRes	ProvRes	CityMunRes	CityMuniPSGC	Barangay Res	Barangay PSGC	HealthStatus
2	Region VII: Central Visayas	NEGROS ORIENTAL	DUMAGUETE CITY (CAPITAL)	PH074610000			RECOVERED
3	Region VII: Central Visayas	NEGROS ORIENTAL	DUMAGUETE CITY (CAPITAL)	PH074610000			DIED
4	Region VII: Central Visayas	BOHOL	PANGLAO	PH071233000			RECOVERED
5	Region IV-A: CALABARZON	BATANGAS	SANTO TOMAS	PH041028000			RECOVERED
6	Region IV-A: CALABARZON	RIZAL	CAINTA	PH045805000			DIED

Appendix J

Hyperparameter Estimation Techniques Used in Various Studies

Hyperparameter estimation	Type of SEIR model	Study that used this technique
Simulated Annealing	Classical SEIR model	Use of simulated annealing to determine the operational parameters of the SEIR model for the coronavirus for various jurisdictions by Leslie Mulder
	SEIR-migration model	Identifying epidemic spreading dynamics of COVID-19 by pseudocoevolutionary simulated annealing optimizers by Choujun Zhan, Yufan Zheng, Zhikang Lai, Tianyong Hao & Bing Li
	Stochastic SEIR Model	Bayesian Melding Estimation of a Stochastic SEIR Model by Luis K. Hotta
Bayesian Melding	Deterministic SEIR Model	Inference for Deterministic Simulation Models: The Bayesian Melding Approach by David Poole and Adrian E. Raftery

Appendix K

Usability Testing Questionnaire

Question Statement	1	2	3	4
	Strongly Disagree	Disagree	Agree	Strongly Agree
The SEIR math model produced consistent and reliable results across multiple simulations				
The web app was stable and did not crash or experience significant issues while running				
The web app consistently produced results without discrepancies across different runs.				
The web app performed well under different data sizes and stress conditions.				
The web application was easy to learn and use				
The user interface design was intuitive and easy to navigate				
The color scheme and contrast of the UI made it easy to distinguish different elements.				
The tooltips, labels, and descriptions provided sufficient context for understanding the functionalities.				
The visualizations and graphs representing the SEIR model results were clear and easy to understand				
Adding records and generating prediction based on the SEIR math model was efficient and required minimal effort				
The layout and process flow minimized unnecessary steps in using the SEIR model				
The web app responded quickly to user interactions without noticeable delays.				
The time taken to generate results was reasonable and acceptable				
The steps involved in running the SEIR model and visualizing were logically arranged and efficient.				

If errors occurred, the error messages were helpful in resolving the issues.

The application provided clear logs or feedback for debugging errors.

The web app functioned well on different browsers and devices without compatibility issues.

The simulation and prediction results of the SEIR model provided enough information for analyzation

The web app was able to successfully visualize the SEIR prediction without issues

The generated report was sufficient and accurate to requirements

The web app correctly handled different input values and scenarios without unexpected behavior.

The web app provided sufficient options for customizing the visualization generated.

Overall, I found the web application to be a useful tool for epidemic spread visualization and decision-making.

Question Statement	1	2	3	4
	Strongly Disagree	Disagree	Agree	Strongly Agree
The methodology for optimizing the SEIR math model using Bayesian Inference and Simulated Annealing is a clearly defined and well-justified choice				
The use of Bayesian Inference for parameter estimation improves the accuracy and reliability of the SEIR math model's prediction compared to the actual observed data.				
The use of Simulated Annealing for parameter estimation improves the accuracy and reliability of the SEIR math model's prediction compared to the actual observed data.				
The Bayesian Inference process effectively balances accuracy and computational efficiency				
The Simulated Annealing process effectively balances accuracy and computational efficiency				
The metrics used to validate the optimized model using RMSE are appropriate and clearly presented				
The metrics used to validate the optimized model using MAE are appropriate and clearly presented				
The application of Simulated Annealing demonstrates a clear improvement in predictive performance compared to the classical model, as evidenced by the validation metrics				
The application of Bayesian Inference demonstrates a clear improvement in predictive performance compared to the classical model, as evidenced by the validation metrics				
Based on the results, Simulated Annealing is better than Bayesian in optimizing parameters.				
The application of Bayesian Inference in the classic model maintains predictive stability when tested with different datasets or varying initial conditions				
The application of Simulated Annealing in the classic model maintains predictive stability when tested with different datasets or varying initial conditions				
When applied to different age groups, Simulated Annealing maintains prediction stability				
When applied to different age groups, Bayesian Inference maintains prediction stability				
When applied to different training durations, Bayesian Inference maintains prediction stability				
When applied to different training durations, Simulated Annealing maintains prediction stability				

The model validation process sufficiently accounts for overfitting and ensures generalizability.

The optimized models' predictions align with historical or real-world epidemic data, supporting its practical applicability.

The optimized SEIR model is robust when applied to datasets with varying characteristics and different epidemiologic assumptions.

The parameters optimized by Bayesian Inference are within acceptable bounds

The parameters optimized by Simulated Annealing are within acceptable bounds

The application of Bayesian Inference to classical model can be used to predict the future cases

The application of Simulated Annealing to classical model can be used to predict the future cases

Question Statement	1 Strongly Disagree	2 Disagree	3 Agree	4 Strongly Agree
The developed optimized Model provides acceptable prediction of the disease spread				
The web application correctly presents useful data in assessing the spread of the disease				
The developed optimized model can easily fit well in different epidemic phenomenon				
The optimized SEIR model used provides acceptable prediction results against actual data				
The web application can accurately provide useful visualization on how likely the disease can spread				
The parameters and the model used in predicting disease spread are applicable and correctly align to requisites in epidemiology				
The optimization techniques applied can clearly present closer approximation of prediction against actual data				
The overall web application can generally be helpful in augmenting the way epidemiologists and related public health personnel identify and observe prediction spread overtime				
The developed optimized model can handle various scales of data				
The optimized SEIR model incorporated the optimization techniques effectively				
The model maintained consistent predictive performance when applied to different data				
The web application provided sufficient visualization incorporating the developed model				
The model can be applied to new emerging infectious disease				
The web application effectively integrates predictive modeling to assess disease spread.				
The model can be effectively utilized for pandemic preparedness.				

Appendix L

Transcript of Interview

Sahagun:

Nag-dedevelop po kami ng model. Then, simulate naming yung epidemic spread gamit yung model. So, yung model po is applicable para sa mga epidemic spread. Tapos, kung may data ka po ng epidemic, magkakaroon ka po ng simulation using heat map then makikita niyo po yung spread. Tapos, using yung slider, may time frame po siya na pwede i-adjust para makita yung spread throughout the time. Then, yung purpose po ng model, yung SEIR model, Susceptible, Exposed, Infectious, and Recovered, compartmental siya na hinati yung population into compartments. Susceptible yung possible na ma-infect, yung exposed yung may contact pero hindi pa infected, yung infectious yung infected, yung recovered is from infectious naka-recovered.

Sahagun:

Hinati namin yung population, then i-cocompute yung spread using that. Tapos, may chance din na simulate namin yung future na infection, pagkalat ng spread. Pine-predict ng model kung ilan yung infected in the next few days. Yung final output is may website wherein makikita niyo yung heat map, then, pwede niyo i-adjust para ma simulate yung spread, para makit ayung epidemic. Ngayon po kasi gamit namin yung COVID-19 dataset, which is open source from DOH. So, iyon yung gamit namin currently. Pero, t-try din namin siya sa ibang epidemics.

Pomperada:

If ever po na mapagana namin yung model, kailangan lang po sa dataset is yung count ng per day na infected, confirmed cases per day sa isang time range. Then, magagamit ng model para ma-predict.

Marco:

So, technically yung users niyo is mga epidemiology surveillance units? Kasi hindi nila kaya yung sa barangay-level, eh. And kami ang makakakita.

Sahagun:

So, currently po kayo po yung napili naming research locale para mag-conduct ng study kasi dito nakita kaming post within the area na may epidemiology. Nag-search kami sa iba't-ibang browsing sites, tapos San Pedro yung may mga achievements na ESU. Sa FB page niyo, kaya lumapit kami.

Sahagun:

So, ngayon po is aalamin namin yung current background ng ESU, ano ginagawa in everyday. And yung technology na gamit to mitigate the epidemic spread currently. So let's start with the first question.

Yung first question po is what is epidemiology and surveillance units?

Marco:

So, epidemiology and surveillance units kasi is yung unit na responsible for the gathering of datas, then analysis, then yung analysis para ma-prevent yung certain spread ng disease. Kami

rin ang meron na data lang, ang records namin. So, may record keeping kami para in the future pwede magamit as reference. Saamin ang bagsakat ng lahat ng data.

Marco:

Kami ang responsible sa lahat din ng reports from other reporting units, kasama ang hospitals, barangay health stations, and line-ins. Si epidemiology surveillance units lang ang may access sa lahat ng data. For example, may namatay sa isang hospital, kailangan nami nang data, walang pirma or data privacy act, because we have *R.A.11332*. So, hindi pwede na hindi ibigay yung data, lalo kung notifiable or reportable disease, kasi ikaw ang kakasuhan namin. Pwede tanggalin ang license ng hospital.

Marco:

Aware ang San Pedro doon, kami lang ang may access sa lahat. Kami ang middleman. Kahit ibang program. For example, sa vaccination, magtatanong sa hospital, hindi sila makalapit sa hospital kasi wala silang kilalang point person doon, kay ESU dadaan. “Sir baka pwede pakisabi sa hospital”. Minsan na-ooverlook naman ang surveillance, so ibang program makikisabay na din saamin. Pag may kailangan silang data, kami na kukuha. Kami kasi ay data driven. Actually, in the near future hindi lang yung notifiable or reportable disease, pwede na din saamin mapunta yung TB, HIV. Kasi yung epidemiology is not just about notifiable or reportable disease, it’s all about health data. “Ilan ba ang may diabetes sa San Pedro? Ilan ang may hypertension?”. So, in the near future, hahawakan na din namin iyon kasi lahat ng data dapat nasaamin na. pero sa ngayon, ang sa job description palang namin is notifiable palang.

Marco:

Roles and responsibilities. Meron kami tatlong, unfortunately tatlong taop lang. yung head namin si Sir Ryan. Siya nag o-oversee ng day-to-day work. Siya mag-momonitor kung may reports. Siya rin may hawa ng COVID-19. Tehcnically, dapat overseeing nalang siya, pero since tatlo lang kami sa surveillance, siya ang nahawak pa ng COVID. Pero dapat supervise lang siya. Although wala nang COVID, nag-rereport parin, kasi ganon ang surveillance, weekends or holidays, hindi titigil ang reports.

Marco:

Ako naman ang hawak ko is yung notifiable. For notifiable meron 19 sa system. Kasama na yung dengue, rabies, influenza-like illnesses kasama din lahat, plus yung MPox. Then, hawak ko din ang data since ako yung data manager. Ako nag-bibigay ng data analysis kay Sir Ryan para iyon yung pe-present niya.

Marco:

Ang last is yung encoder. Training palang siya since bago palang. Yung role niya is encoding lang talaga para sa mga systems.

Sahagun:

Paano po nag-ooperate ang ESU in everyday? Like, pag-kuha ng data to encoding, paano po ang process niyo?

Marco:

For day-to-day ang first agenda namin lagi is yung COVID-19. Everyday dapat mag-report ang barangay, hindi pwede na hindi mag report, regardless kung may case or wala. So, first is yung COVID, then rereport yung data for the day.

Marco:

So everyday may excel file kami maski zero. After noon hihintayin naman namin report ng hospital. Yung hospital kasi ay lagi may deadline ng 12 noon. So, everyday is mag rereport sila regardless kung Sabado, Linggo, Holiday. Kasi, before 5, ako naman ang mag-papasa sa provincial. Aside from that, may system kasi ang San Pedro, which is ginawa namin. Iyon yung capstone project ko nung nursing, which is “E-Marites Mo”. Mobile app, gumamit ako ng third party mobile app provider.

Marco:

Mobile app siya before, pero naging google form nalang kasi pre-subscription lang dati. Ginawa nalang amin google form kasi hindi na kaya ng mobile app. Sa ngayon, bukod doon sa hospital, report sa barangay. Pagka-report nila, titignan namin kung pasok siya sa case definition ng sakit. Magbibigay sila ng syndromes, then through that symptoms, titignan namin kung aling disease. Iyon yung daily namin.

Sahagun:

Sa, mag-rereport po ang mga barangay?

Marco:

Then consolidate namin.

Marco:

Unfortunately, maski tatlo lang kami, wala kami weekend or holiday. Kasi kailangan ang reporting is continuous. Actually, wala naman kaso kung hindi kami nag-report, may hinahabol lang kami ng awardings. Kaya diligent kami ng Sir Ryan sa par-rereport.

Sahagun:

Paano po yung from ESU na aakyat to authorities? Meron po ba process na from sainyo, mapupunta sa city ESU, provincial ESU... may process po ba?

Marco:

The EDCS-IS. Lahat ng na-encode, mapupunta doon.

Marco:

Medyo allergic kami sa EDCS kaya ayaw na namin i-memorize. Kasi lagi siya naka-down. Kasi kami gusto namin real-time, so every day lagi kami nag-bubukas ng system kasi gusto ko makita kung tatasas yung case. Yung problema, tuwing Sabado at Linggo, lagi down. Problema din, for example may maling address, hindi ma-edit. It's a waste of time kasi hindi pwede mag edit at delete.

Arboleda:

Hindi niyo ba tatanungin bakit lagi down ang EDCS-IS? Iniwan na kasi ng developer.

Marco:

Ang ginawa naman ng mga surveillance units ay kanya-kanyang database. Kunin ang data sa EDCS, ikaw mag-eedit, bubuo ka sa excel ng sariling database, then pivot for analytics.

Sahagun:

May overall count po kayo ng number of ESUs and DRUs na existing sa country?

Marco:

Lahat ng bayan meron through *R.A.11332*. Kahit saang bayan dapat may functional ESU. Kapag hindi functional, may penalties.

Arboleda:

And hindi kami binded ng data privacy act dahil sa *R.A.11332*. Pero, kung kayo ang kukuha saamin, data privacy act. Currently po, paano kayo nag-ttrack ng notifiable disease, bukod sa nag-mention sa report ng DRU, may other way pa po ba para kumuha ng data? And what ways po para ma-mitigate yung cases? May system po or tracking device?

Arboleda:

Dalawa lang, chismis at info from the community. Pag may narinig kami ng chismis, vavalidate namin iyon. Through community, friends, or kung ano, may ganitong kaso, binababa namin sa barangay health station tapos instruct namin mga barangay health emergency response team. During COVID, mandato dapat na lahat ng barangay offices ay may barangay emergency health response team.

Sahagun:

Kung madami po cases, ano po ang response niyo?

Arboleda:

Sa surveillance, mag-gagather kami ng data, ng reports, then analyze namin iyon. Gumagawa kami ng sariling excel para ma-pivot ni Sir Dennis, from there pwede niya laruin ang data. From there, makikita namin kung may cluster.

Appendix M

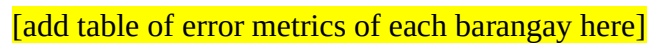
Transformed Dataset Data Dictionary

Field	Description
City	City Residence
Barangay	Barangay Residence
AgeGroup	Five-year age group
Date	Date
Count_Exposed	Number of exposed individuals

Count_Infectious	Number of infectious individuals
Count_Recovered	Number of recovered individuals
Count_exposed_9_below	Number of exposed individuals ages 9 and below
Count_exposed_10_to_17	Number of exposed individuals ages 10 to 17
Count_exposed_18_to_39	Number of exposed individuals ages 18 to 39
Count_exposed_40_to_59	Number of exposed individuals ages 40 to 59
Count_exposed_60_to_79	Number of exposed individuals ages 60 to 79
Count_exposed_80_above	Number of exposed individuals ages 80 and above
Count_exposed_unknown	Number of exposed individuals of unknown ages
Count_exposed_male	Number of male exposed individuals
Count_exposed_female	Number of female exposed individuals
Count_infectious_9_below	Number of infected individuals ages 9 and below
Count_infectious_10_to_17	Number of infected individuals ages 10 to 17
Count_infectious_18_to_39	Number of infected individuals ages 18 to 39
Count_infectious_40_to_59	Number of infected individuals ages 40 to 59
Count_infectious_60_to_79	Number of infected individuals ages 60 to 79
Count_infectious_80_above	Number of infected individuals ages 80 and above
Count_infectious_unknown	Number of infected individuals of unknown ages
Count_infectious_male	Number of male infected individuals

[illegible]

Simulated Annealing Validation on Barangay Level

306

Magsaysay	0.13	0.36	0.13
San Antonio	3.13	3.57	1.07
Maharlika	0.24	0.26	1096089825386865.5
Narra	0.56	0.58	2525159857591259.0
Pacita 1	0.93	1.26	0.86
Poblacion	0.60	0.71	0.52
Langgam	1.68	2.02	4135863226541058.5
Pacita 2	0.95	1	4274232345959057.5
Cuyab	1.73	1.86	5416156268081185.0
San Lorenzo Ruiz	0	0	0
Chrysanthemum	0.57	0.63	0.44
Laram	0	0	0
Landayan	1.99	2.34	0.99
Fatima	0.77	0.87	3452216771876333.5
San Vicente	1.67	2.13	0.80
Fatima	0.71	0.78	2939943972837004.5
San Vicente	1.67	2.13	0.80
United Better Living	0.71	0.79	2939943972837004.5
Bagong Silang	0	0	0
San Roque	0	0	0
Nueva	0	0	0
Calendola	0.82	0.90	3705878569247966.0

Sampaguita Village	0	0	0
Riverside	0	0	0
Rosario	0.43	0.47	1944427856409369.2
United Bayanihan	0	0	0
Estrella	0.47	0.52	1626548983980429.8
G.S.I.S.	0.40	0.47	0.40
Santo Niño	0	0	0

Appendix O

Screenshot of SEIR-Viz Web Application

The screenshot shows the SEIR-Viz web application interface. On the left is a sidebar with the SEIR-Viz logo and navigation links: Map, Barangay, Dataset, Reports, and Accounts (highlighted). Below the sidebar are buttons for 'Upload Data' and a user greeting 'Hello, Pomperada'. The main content area features a 'Create new account' button and a table of users.

NAME	ROLE	CONTACT NUMBER	DATE CREATED	STATUS	ACTION
Pomperada jop@email.com	Admin		Jan. 23, 2025, 4:46 p.m.	Active	
John Doe iam@email.com	Officer	09154516923	Jan. 24, 2025, 9:52 a.m.	Active	Manage User
Jon Peter Pomperada jppomperada@live.mcd.edu.ph	Officer	09154516923	Jan. 24, 2025, 9:51 a.m.	Active	Manage User
Jon Peter Pomperada jorpeterpomperada@gmail.com	Officer	09154516923	Jan. 24, 2025, 9:42 a.m.	Active	Manage User
Jon Peter Pomperada joleypomperada@gmail.com	Officer	09154516923	Jan. 24, 2025, 9:35 a.m.	Active	Manage User

The screenshot shows the 'Create New Account' form in the SEIR-Viz web application. The sidebar and navigation elements are consistent with the previous screenshot. The main content area displays the 'Accounts View > Create new account' breadcrumb and the 'Create New Account' form.

Create New Account
Admin Panel

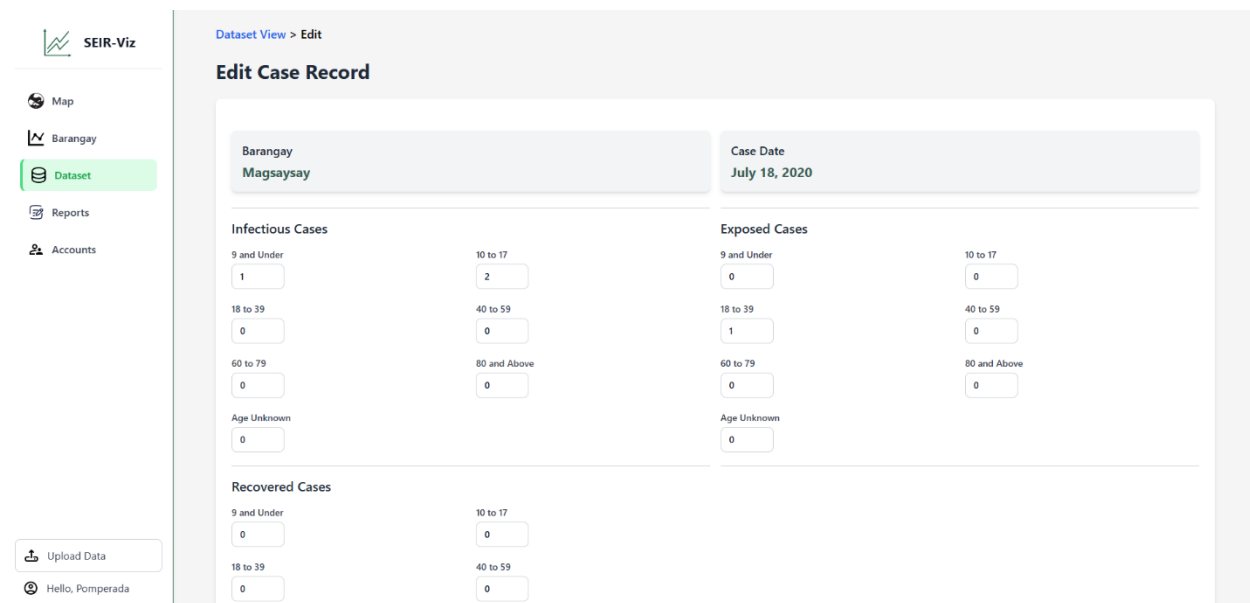
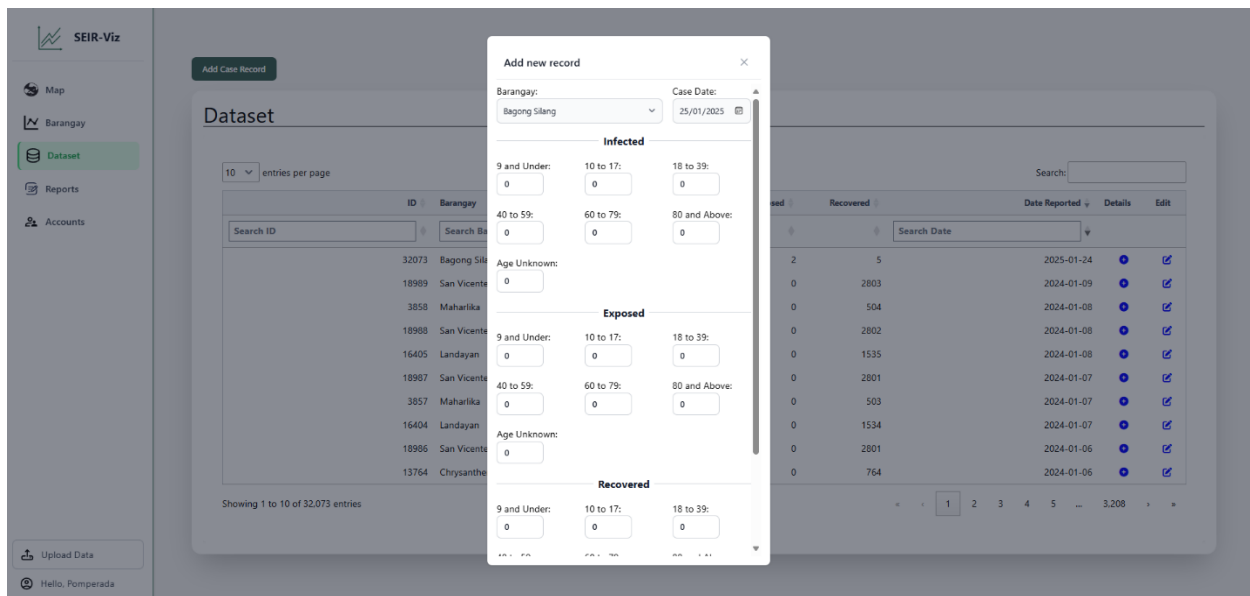
First Name: Last Name:

Email:

Password: Confirm Password:

Contact Number:

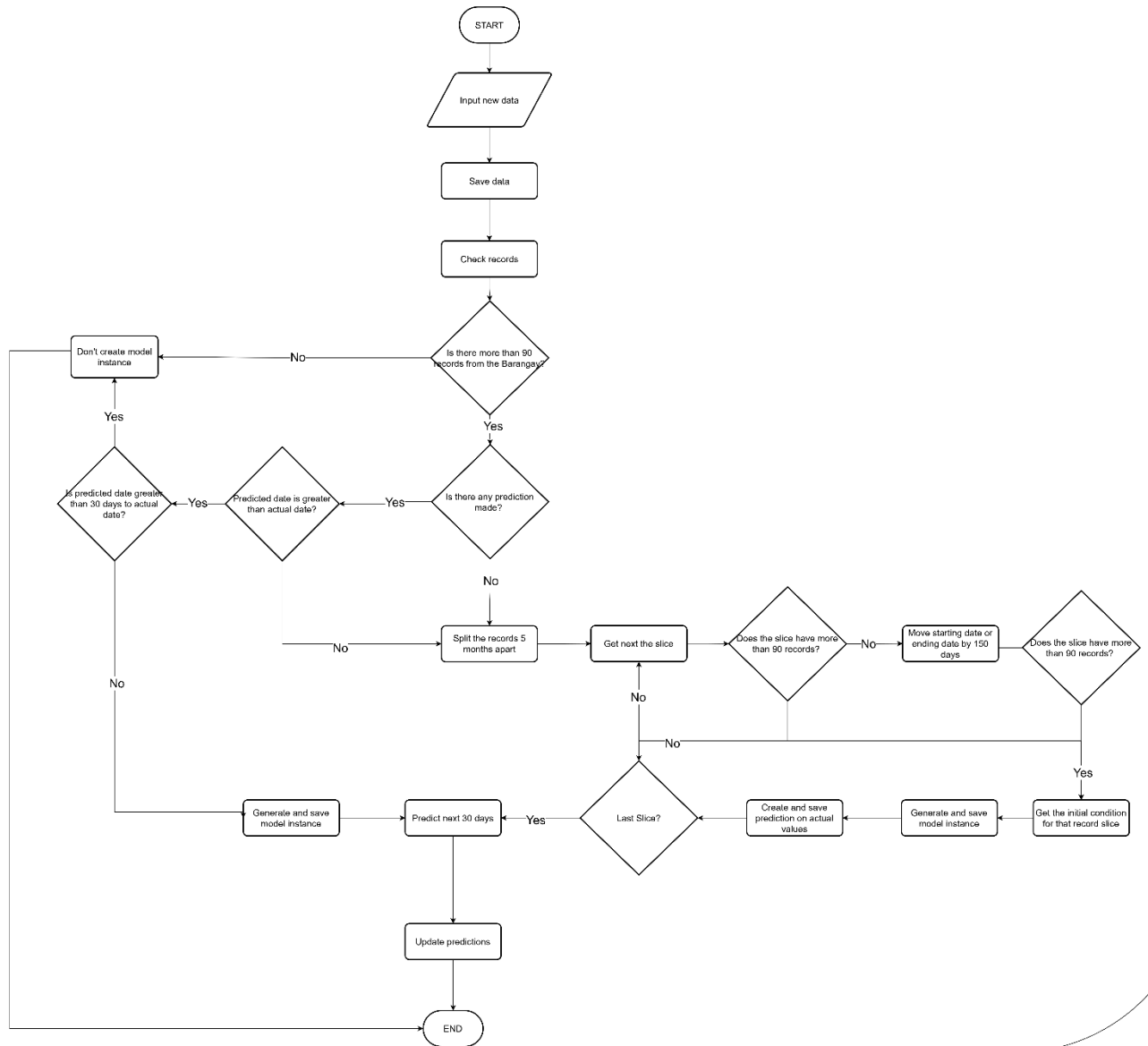
[Create Account](#)



Appendix P

Model Instance Generation Flow Chart

Model Instance Flow Chart



Appendix Q

Functional Testing Activity

TEST CASE #1-3

Test Case #1-3 System: SEIR-Viz Designed by: Jon Peter Pomperada Executed by: Jeremy Genato	Test Case Name: Map Page Data Design Date: March 10, 2025 Execution Date: March 11, 2025
--	---

Pre-Conditions

- Load Map Page fetch data functions

Step	Action	Expected System Response	Pass/Fail	Comment
1.	Test if map current and percentage data is fetched accurately	Map current and percentage infected counts are correct	Pass	
2	Test if map total data is fetched accurately	Map total infected data is correct	Pass	
3	Test if top barangay cases count is accurate	Top barangay matches map data	Pass	

Post-Conditions

- Map page data outputs are correct

TEST CASE #4-5

Test Case #4-5	Test Case Name: Test Choropleth Map
System: SEIR-Viz	Design Date: March 10, 2025
Designed by: Jon Peter Pomperada	Execution Date: March 11, 2025
Executed by: Jeremy Genato	

Pre-Conditions

- Test map page if choropleth heatmap is loading

Step	Action	Expected System Response	Pass/Fail	Comment
1.	Test if map page load shapefile loads the choropleth map	Show heatmap with infected count values	Pass	
2.	Test if create choropleth with frame data is fetched properly by moving time slider	Map changes heatmap based on date	Pass	

Post-Conditions

- Shows the choropleth heatmap of San Pedro and time slider frame change

TEST CASE #6

Test Case #6 System: SEIR-Viz Designed by: John Bernard Sahagun Executed by: Jon Peter Pomperada	Test Case Name: Barangay Exposed Count Design Date: March 10, 2025 Execution Date: March 11, 2025
---	--

Pre-Conditions

- Test Barangay Exposed Data

Step	Action	Expected System Response	Pass/Fail	Comment
1.	Test Exposed 3 Months count	Output exposed count for 3 months given test barangay	Pass	
2	Test Exposed 6 months count	Output exposed count for 3 months test barangay	Pass	
3	Test Exposed All date count	Output exposed count for all dates for test barangay	Pass	

Post-Conditions

- Output the correct data count depending on date filter

TEST CASE #7

Test Case #7 System: SEIR-Viz Designed by: John Bernard Sahagun Executed by: Jon Peter Pomperada	Test Case Name: Barangay Infected Count Design Date: March 10, 2025 Execution Date: March 11, 2025:
---	--

Pre-Conditions

- Test Barangay Infected Data

Step	Action	Expected System Response	Pass/Fail	Comment
1.	Test Infected 3 Months count	Output Infected count for 3 months given test barangay	Pass	
2	Test Infected 6 months count	Output infected count for 3 months test barangay	Pass	
3	Test Infected All date count	Output infected count for all dates for test barangay	Pass	

Post-Conditions

- Output the correct data count depending on date filter

TEST CASE #8

Test Case #8 System: SEIR-Viz Designed by: John Bernard Sahagun Executed by: Jon Peter Pomperada	Test Case Name: Barangay Recovered Count Design Date: March 10, 2025 Execution Date: March 11, 2025
---	--

Pre-Conditions

- Test Barangay Recovered Data

Step	Action	Expected System Response	Pass/Fail	Comment
1.	Test Recovered 3 Months count	Output Recovered count for 3 months given test barangay	Pass	
2	Test Recovered 6 months count	Output Recovered count for 3 months test barangay	Pass	
3	Test Recovered All date count	Output Recovered count for all dates for test barangay	Pass	

Post-Conditions

- Output the correct data count depending on date filter

TEST CASE #9

Test Case #9	Test Case Name: Barangay Age Group Count
System: SEIR-Viz	Design Date: March 10, 2025
Designed by: John Bernard Sahagun	Execution Date: March 11, 2025
Executed by: Jon Peter Pomperada	

Pre-Conditions

- Test Barangay Age Group for infected for a given time range

Step	Action	Expected System Response	Pass/Fail	Comment
1.	Test if all age groups in a barangay's infected count for the past 3 month	Output age group data of infected in a barangay for 3 months	Pass	
2	Test if all age groups in a barangay's exposed count for the past 6 month	Output age group data of exposed in a barangay for 6 months	Pass	
3	Test if all age groups in a barangay's recovered count for the past 6 month	Output age group data of recovered in a barangay for 6 months	Pass	

Post-Conditions

- Output the correct age group data count depending on the category and date filter

TEST CASE #10

Test Case #10 System: SEIR-Viz Designed by: John Bernard Sahagun Executed by: Jeremy Genato	Test Case Name: Test Barangay Page SEIR Graph Design Date: March 10, 2025 Execution Date: March 11, 2025
--	---

Pre-Conditions

- Test barangay page SEIR graph loads

Step	Action	Expected System Response	Pass/Fail	Comment
1.	Test if barangay page SEIR graph renders properly	Barangay Page Infected, Exposed, and Recovered graph renders	Pass	

Post-Conditions

- Barangay Page SEIR graph renders properly

TEST CASE #11

Test Case #11 System: SEIR-Viz Designed by: John Bernard Sahagun Executed by: Jeremy Genato	Test Case Name: Test Barangay Page Age Distribution Chart Design Date: March 10, 2025 Execution Date: March 11, 2025
--	---

Pre-Conditions

- Test barangay page age distribution graph loads

Step	Action	Expected System Response	Pass/Fail	Comment
1.	Test if barangay page age distribution chart renders properly	Barangay Page Age distribution chart is visible on the page	Pass	

Post-Conditions

- Barangay Page Age distribution chart renders properly

TEST CASE #12

Test Case #12 System: SEIR-Viz Designed by: Jon Peter Pomperada Executed by: Jeremy Genato	Test Case Name: Test Report Page Section A Design Date: March 10, 2025 Execution Date: March 11, 2025
---	--

Pre-Conditions

- Load One-pager Report

Step	Action	Expected System Response	Pass/Fail	Comment
1.	Test if Positive Surveillance Testing reports are generated	Section A of One-pager tables and figures loads value based on date range	Pass	

Post-Conditions

- One-pager Section A Reports Load Successfully

TEST CASE #13

Test Case #13 System: SEIR-Viz Designed by: Jon Peter Pomperada Executed by: Jeremy Genato	Test Case Name: Test Report Page Section B Design Date: March 10, 2025 Execution Date: March 11, 2025
---	--

Pre-Conditions

- Load One-pager Report

Step	Action	Expected System Response	Pass/Fail	Comment
1.	Test if Prediction Reports reports are generated	Section B of One-pager tables and figures loads value based on date range	Pass	

Post-Conditions

- One-pager Section B Reports Load Successfully

TEST CASE #14

Test Case #14	Test Case Name: Test Download CSV
System: SEIR-Viz	Design Date: March 10, 2025
Designed by: Jon Peter Pomperada	Execution Date: March 11, 2025
Executed by: Jeremy Genato	

Pre-Conditions

- Load Download CSV report

Step	Action	Expected System Response	Pass/Fail	Comment
1.	Test if download csv report is functional	Downloaded csv report contains value	Pass	

Post-Conditions

- Download CSV report is populated and successful

TEST CASE #6

Test Case #6 System: SEIR-Viz Designed by: John Bernard Sahagun Executed by: Jeremy Genato	Test Case Name: Test None Prediction Design Date: March 10, 2025 Execution Date: March 11, 2025
---	--

Pre-Conditions

- Test Barangay Prediction if there is no sufficient predicted value

Step	Action	Expected System Response	Pass/Fail	Comment
1.	Test if prediction result to none if barangay has no predicted values	Output none to barangay if there is no existing predicted value	Pass	

Post-Conditions

- Output the correct no data for none existing predicted value

TEST CASE #7

Test Case #7	Test Case Name: Test Prediction Data
System: SEIR-Viz	Design Date: March 10, 2025
Designed by: John Bernard Sahagun	Execution Date: March 11, 2025
Executed by: Jeremy Genato	

Pre-Conditions

- Test Barangay Prediction if there is sufficient data

Step	Action	Expected System Response	Pass/Fail	Comment
1.	Test if prediction in the next 7 days is fetched accurately	Output prediction for the next 7 days of test barangay	Pass	
2.	Test if prediction in the next month is fetched accurately	Output prediction for the next 30 days of test barangay	Pass	

Post-Conditions

- Output the correct prediction data of test barangay

TEST CASE #15-18

Test Case #15-18	Test Case Name: Model Integration Tests
System: SEIR-Viz	Design Date: March 10, 2025
Designed by: John Bernard Sahagun	Execution Date: March 11, 2025
Executed by: Jon Peter Pomperada	

Pre-Conditions

- Load dataset containing barangays with insufficient and sufficient amount of time for prediction

Step	Action	Expected System Response	Pass/Fail	Comment
1.	Test if prediction will proceed with more than 90 data	Prediction is populated on barangay	Pass	
2.	Test if prediction will not proceed with 90 or less data	No prediction is populated on barangay	Pass	
3.	Test if prediction will proceed if there are 30 days projected cases	Retain prediction for projected cases	Pass	
4.	Test if model instance is created	Model instance is added to database	Pass	

Post-Conditions

- Prediction is populated on barangay case data

TEST CASE #19

Test Case #19	Test Case Name: Database Prediction Populate
System: SEIR-Viz	Design Date: March 10, 2025
Designed by: John Bernard Sahagun	Execution Date: March 11, 2025:
Executed by: Jon Peter Pomperada	

Pre-Conditions

- Upload dataset or Add record

Step	Action	Expected System Response	Pass/Fail	Comment
1.	Test if case data on prediction will populate prediction	Case data prediction is generated	Pass	

Post-Conditions

- Update model instance table and create case data prediction instance

TEST CASE #20-21

Test Case #20-21	Test Case Name: San Pedro Records Update
System: SEIR-Viz	Design Date: March 10, 2025
Designed by: John Bernard Sahagun	Execution Date: March 11, 2025
Executed by: Jon Peter Pomperada	

Pre-Conditions

- Upload dataset or Add record

Step	Action	Expected System Response	Pass/Fail	Comment
1.	Test San Pedro records upon upload data	San Pedro records are added or updated		
2.	Test if the cases on San Pedro is correct on different dates	San Pedro on other dates are accurate		

Post-Conditions

- San Pedro records are updated

TEST CASE #22-23

Test Case #22-23 System: SEIR-Viz Designed by: Jon Peter Pomperada Executed by: Jeremy Genato	Test Case Name: Case Data Record Database Create or Update Design Date: March 10, 2025 Execution Date: March 11, 2025
--	--

Pre-Conditions

- Upload dataset or Add record

Step	Action	Expected System Response	Pass/Fail	Comment
1.	Test if exposed and infected records are correct	Exposed and Infected Case Data are accurate	Pass	
2	Test if recovered cases are accumulating	Recovered count on Case Data are accumulating	Pass	

Post-Conditions

- Case Data Exposed, Infected, and Recovered records are accurate

Appendix R

Usability Testing Answered Forms

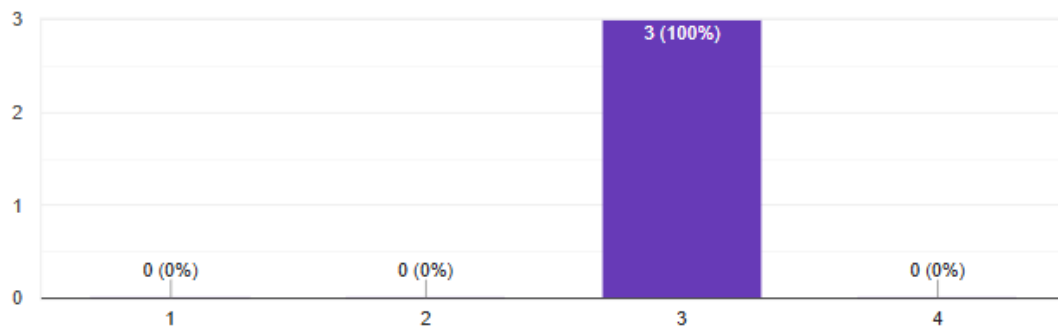
ESU Questionnaire Results



The web app consistently produced results without discrepancies across different runs.

 [Copy chart](#)

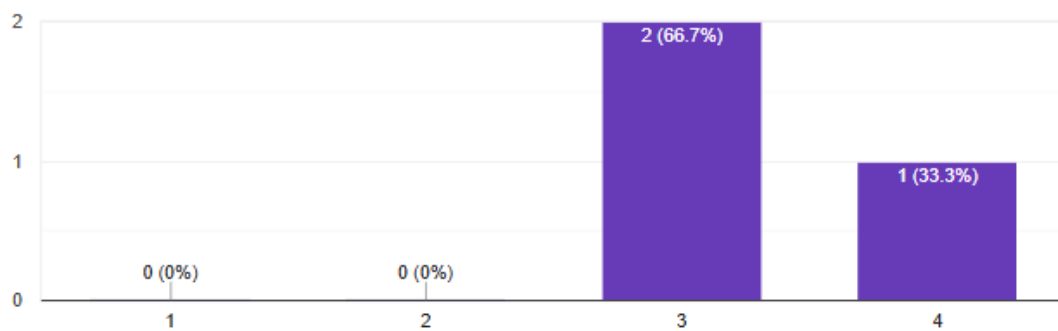
3 responses



The web app performed well under different data sizes and stress conditions.

 [Copy chart](#)

3 responses

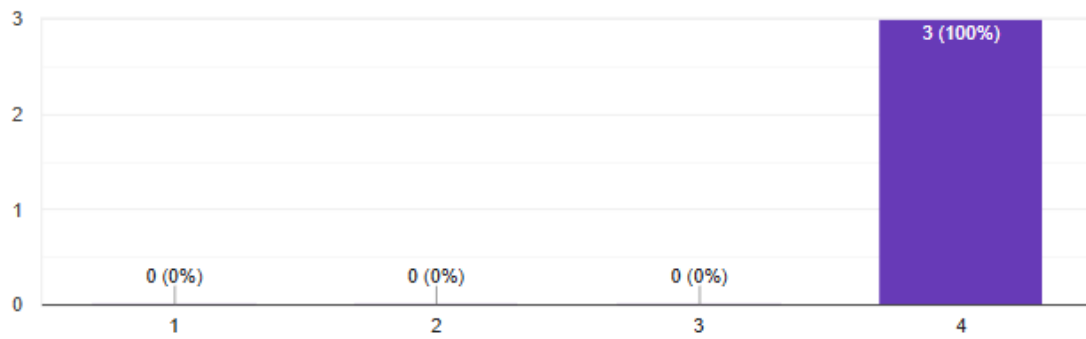


Usability

The web application was easy to learn and use

 Copy chart

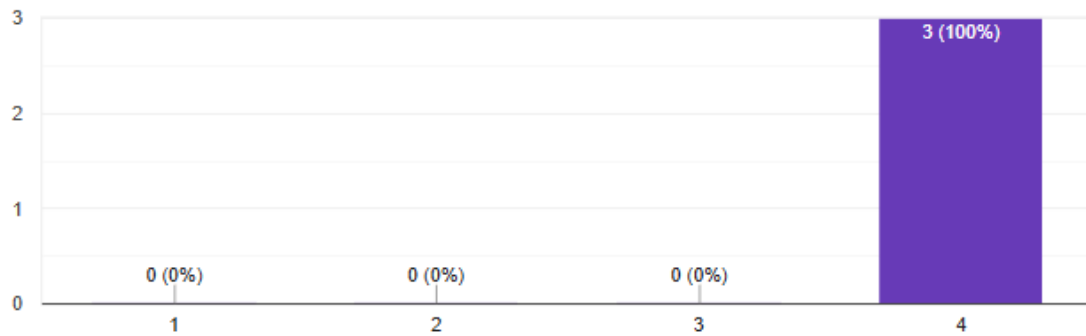
3 responses



The user interface design was intuitive and easy to navigate

 Copy chart

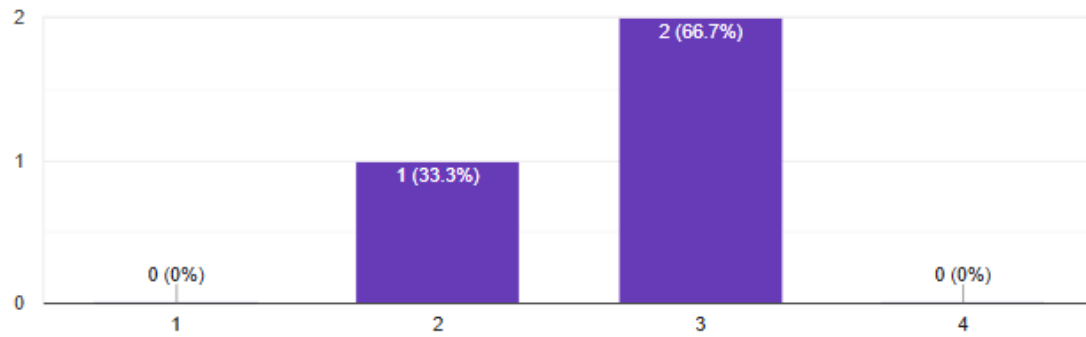
3 responses



The color scheme and contrast of the UI made it easy to distinguish different elements.

 [Copy chart](#)

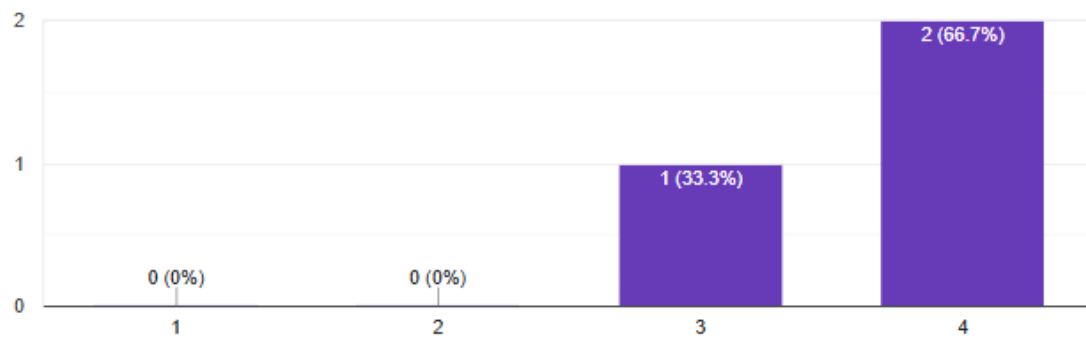
3 responses



The tooltips, labels, and descriptions provided sufficient context for understanding the functionalities.

 [Copy chart](#)

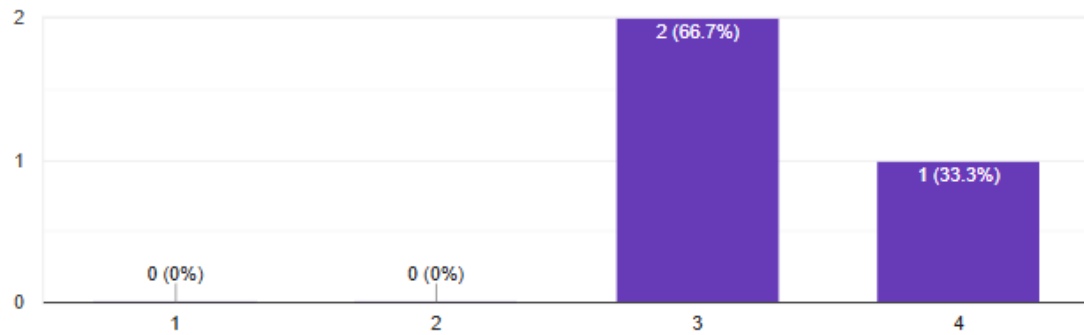
3 responses



The visualizations and graphs representing the SEIR model results were clear and easy to understand

 [Copy chart](#)

3 responses

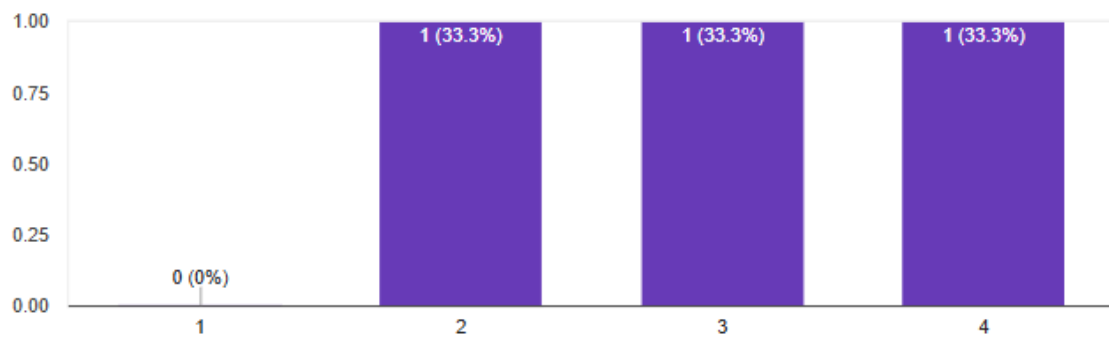


Efficiency

Adding records and generating prediction based on the SEIR math model was efficient and required minimal effort

 [Copy chart](#)

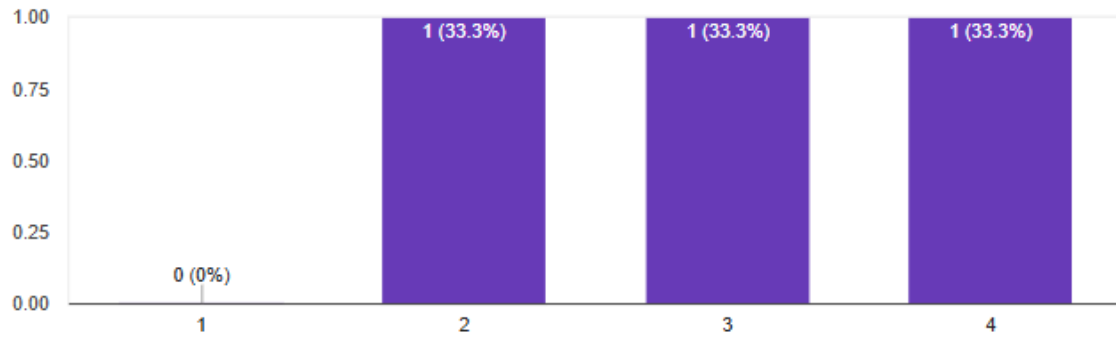
3 responses



The layout and process flow minimized unnecessary steps in using the SEIR model

 [Copy chart](#)

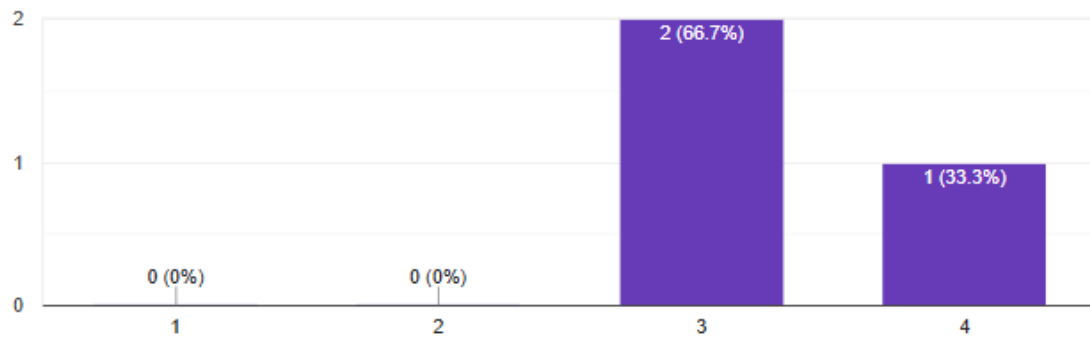
3 responses



The web app responded quickly to user interactions without noticeable delays.

 [Copy chart](#)

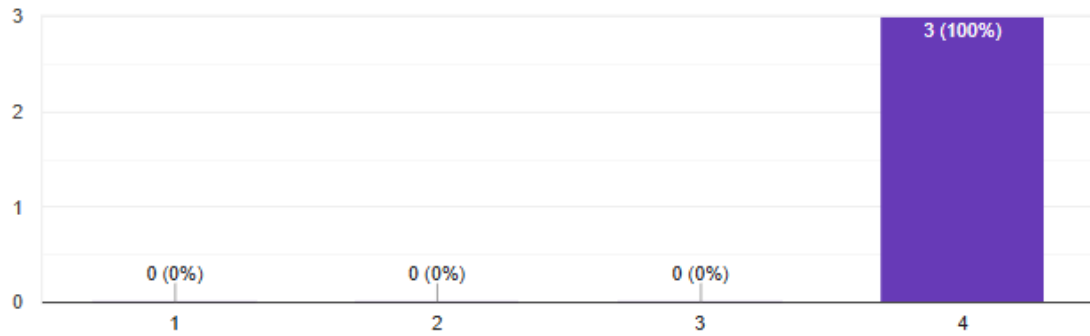
3 responses



The time taken to generate results was reasonable and acceptable

 [Copy chart](#)

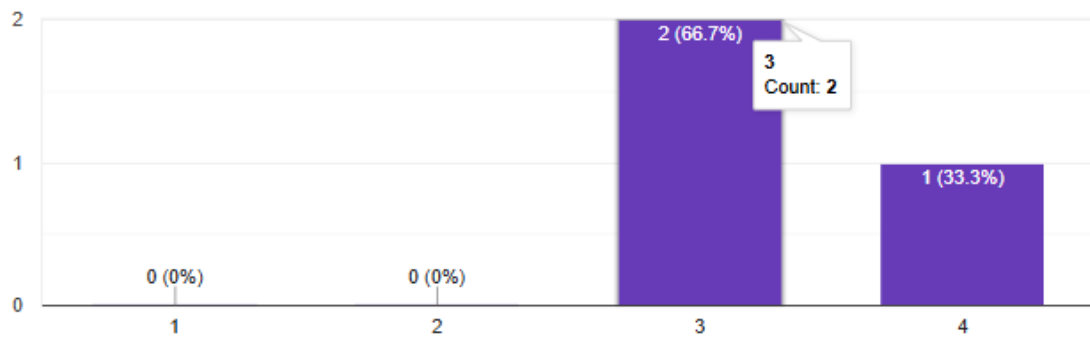
3 responses



The steps involved in running the SEIR model and visualizing were logically arranged and efficient.

 [Copy chart](#)

3 responses

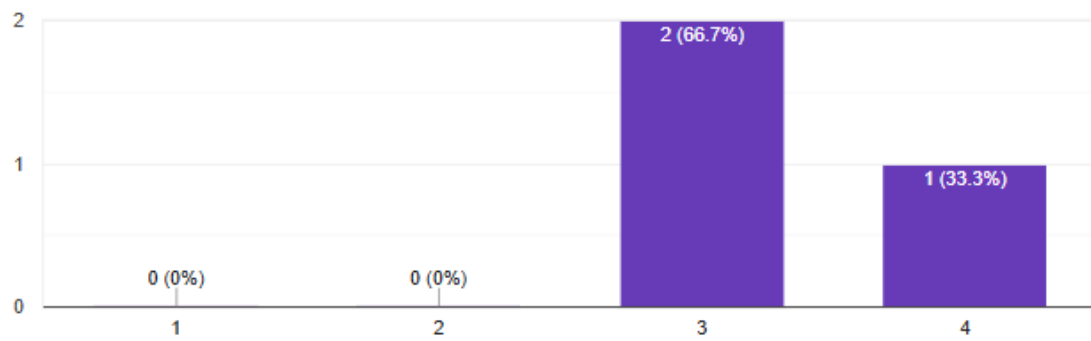


Maintainability

If errors occurred, the error messages were helpful in resolving the issues.

 [Copy chart](#)

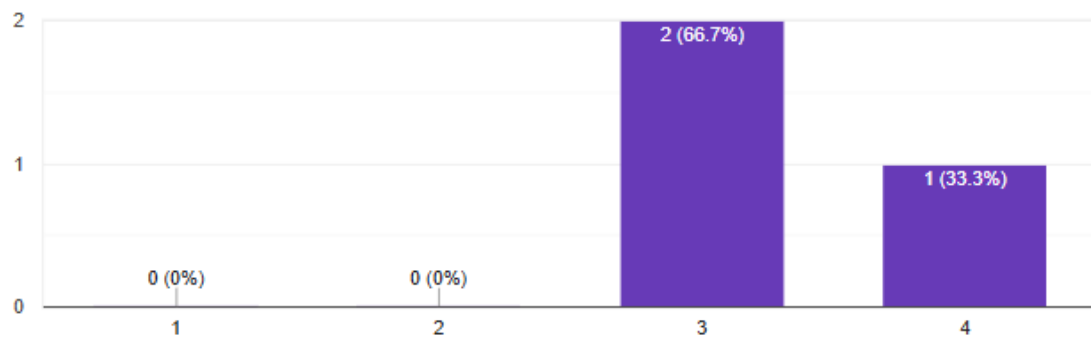
3 responses



The application provided clear logs or feedback for debugging errors.

 [Copy chart](#)

3 responses

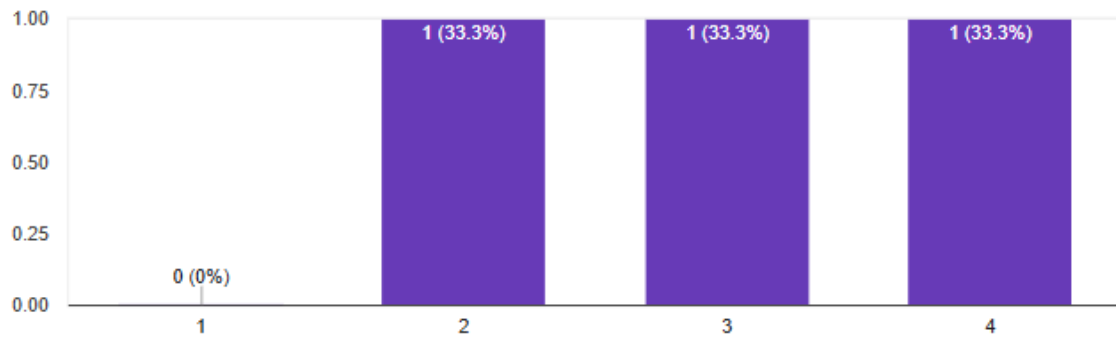


Portability

The web app functioned well on different browsers and devices without compatibility issues.

 [Copy chart](#)

3 responses

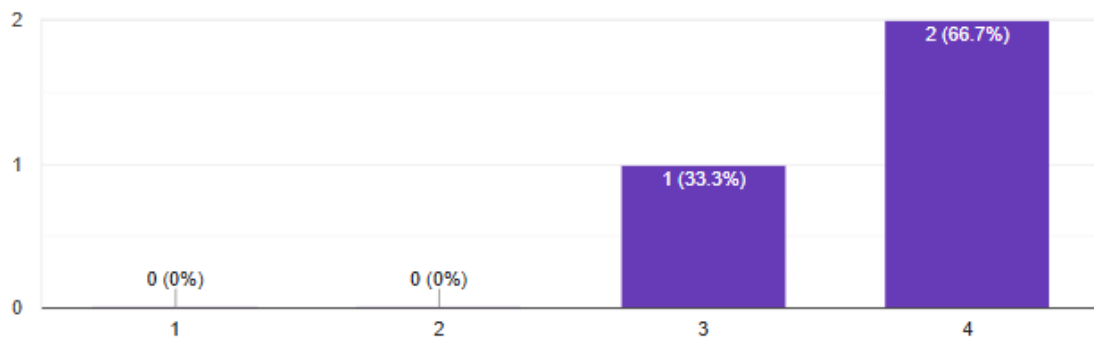


Functionality

The simulation and prediction results of the SEIR model provided enough information for analyzation

 [Copy chart](#)

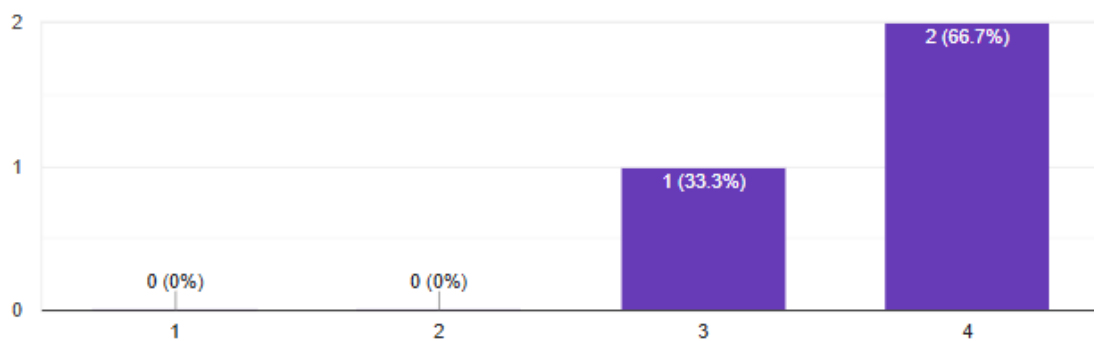
3 responses



The web app was able to successfully visualize the SEIR prediction without issues

 [Copy chart](#)

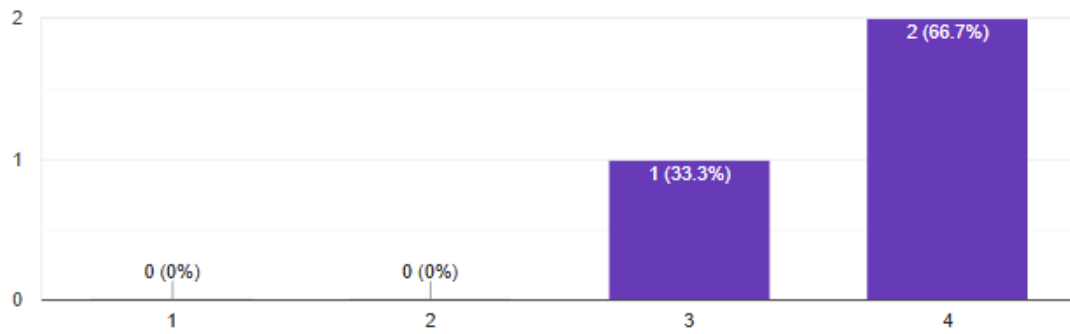
3 responses



The generated report was sufficient and accurate to requirements

 [Copy chart](#)

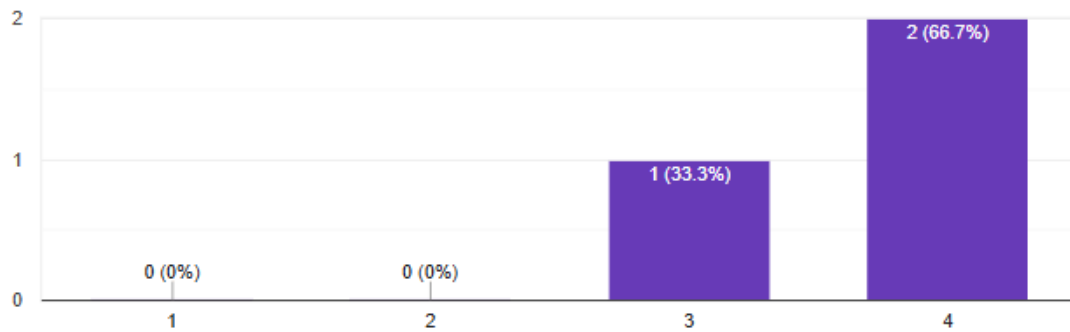
3 responses



The web app correctly handled different input values and scenarios without unexpected behavior.

 [Copy chart](#)

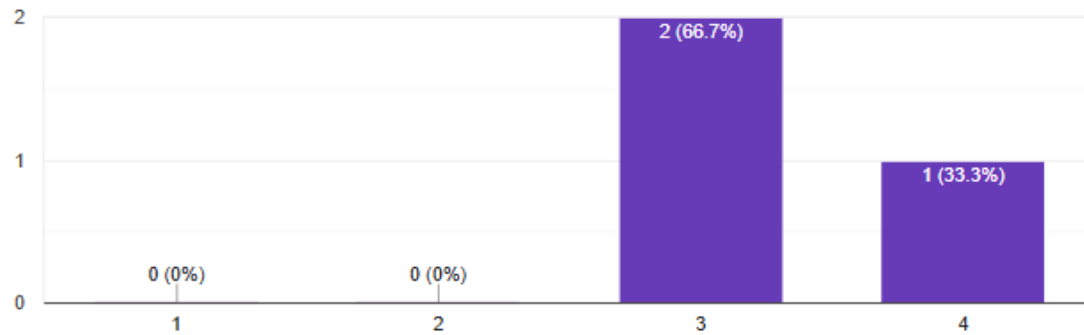
3 responses



The web app provided sufficient options for customizing the visualization generated.

 [Copy chart](#)

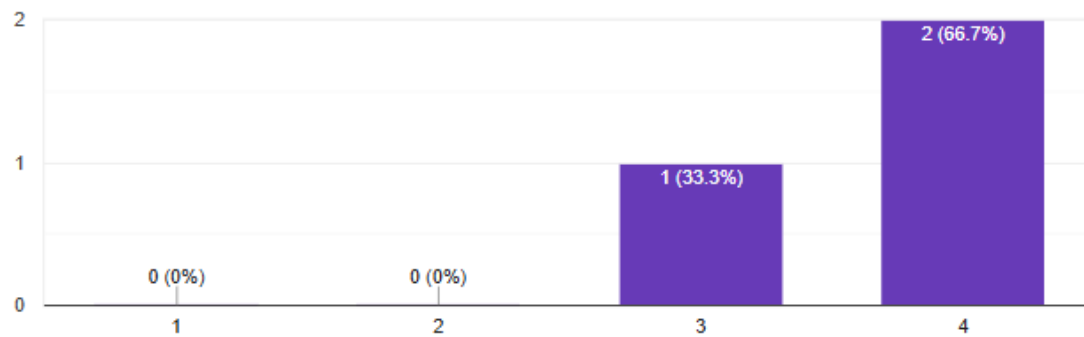
3 responses



Overall, I found the web application to be a useful tool for epidemic spread visualization and decision-making.

 [Copy chart](#)

3 responses



Expert Questionnaire Results

Expert 1

Question Statements *	Strongly Disagree	Disagree	Agree	Strongly Agree
The developed optimized Model provides acceptable prediction of the disease spread	☐	☐	☐	☒
The web application correctly presents useful data in assessing the spread of the disease	☐	☐	☐	☒
The developed optimized model can easily fit well in different epidemic phenomenon	☐	☐	☒	☐
The optimized SEIR model used provides acceptable prediction results against actual data	☐	☐	☐	☒
The web application can accurately provide useful visualization on how likely the disease can spread	☐	☐	☐	☒

The parameters and the model used in predicting disease spread are applicable and correctly align to requisites in epidemiology



The optimization techniques applied can clearly present closer approximation of prediction against actual data



The overall web application can generally be helpful in augmenting the way epidemiologists and related public health personnel identify and observe prediction spread overtime



The developed optimized model can handle various scale of data



The optimized SEIR model incorporated the optimization techniques effectively



The model maintained consistent predictive performance when applied to different data

☐☐☒☐

The web application provided sufficient visualization incorporating the developed model

☐☐☐☒

The model can be applied to new emerging infectious disease

☐☐☐☒

The web application effectively integrates predictive modeling to assess disease spread.

☐☐☐☒

The model can be effectively utilized for pandemic preparedness.

☐☐☐☒

Expert 2

Question Statements *

	Strongly Disagree	Disagree	Agree	Strongly Agree
The developed optimized Model provides acceptable prediction of the disease spread	☐	☐	☒	☐
The web application correctly presents useful data in assessing the spread of the disease	☐	☐	☐	☒
The developed optimized model can easily fit well in different epidemic phenomenon	☐	☐	☒	☐
The optimized SEIR model used provides acceptable prediction results against actual data	☐	☐	☒	☐
The web application can accurately provide useful visualization on how likely the disease can spread	☐	☐	☐	☒

The parameters and the model used in predicting disease spread are applicable and correctly align to requisites in epidemiology



The optimization techniques applied can clearly present closer approximation of prediction against actual data



The overall web application can generally be helpful in augmenting the way epidemiologists and related public health personnel identify and observe prediction spread overtime



The developed optimized model can handle various scale of data



The optimized SEIR model incorporated the optimization techniques effectively



The model maintained consistent predictive performance when applied to different data

☐☐☒☐

The web application provided sufficient visualization incorporating the developed model

☐☐☐☒

The model can be applied to new emerging infectious disease

☐☐☒☐

The web application effectively integrates predictive modeling to assess disease spread.

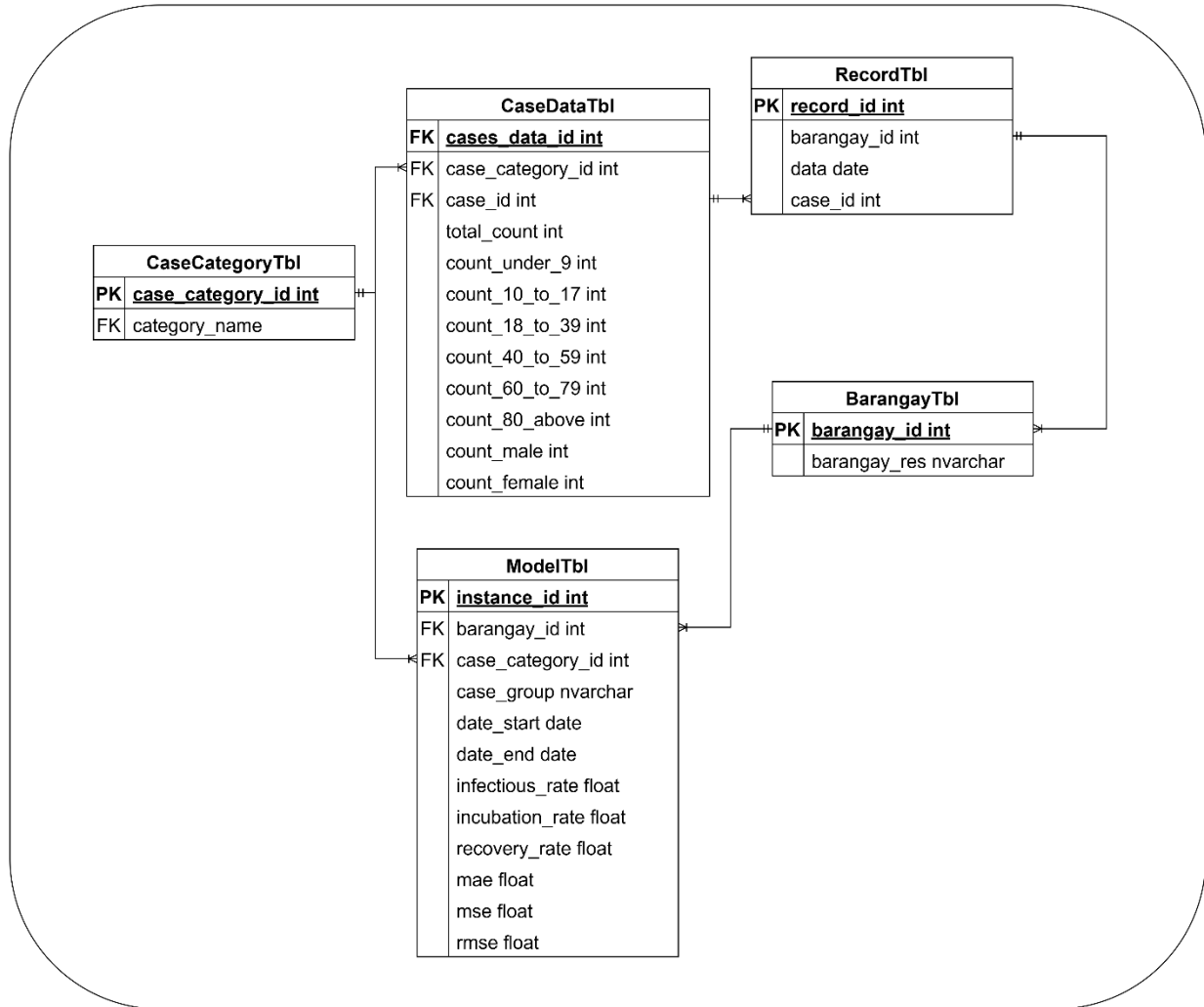
☐☐☒☐

The model can be effectively utilized for pandemic preparedness.

☐☐☒☐

Appendix S

ERD and Data Dictionary



CaseCategoryTbl Columns	Data Type	Description
Category_name	String	The types of case, can be exposed, infected, recovered and the prediction

BarangayTbl Columns	Data Type	Description
Barangay_res	String	The barangay in San Pedro where the case occurred. San Pedro is included (for total of all barangay)

RecordTbl Columns	Data Type	Description
Barangay_id	Integer	To identify what barangay did the case occurred
Date	Date	Date when the case occurred
Case_id	Integer	Key in getting the number of cases in CaseDataTbl

CaseDataTbl	Data Type	Description
Case_category_id	Integer	Identifies the type of case the counts are for
Case_id	Integer	Identifies the barangay and date the counts occurred
Total_count	Integer	The total number of cases in all age groups
Count_under_9	Integer	The total number of cases in ages 9 and below
Count_10_to_17	Integer	The total number of cases in ages 10 to 17
Count_18_to_39	Integer	The total number of cases in ages 18 to 39
Count_40_to_59	Integer	The total number of cases in ages 40 to 59
Count_60_to_79	Integer	The total number of cases in ages 60 to 79
Count_80_above	Integer	The total number of cases in ages 80 and above

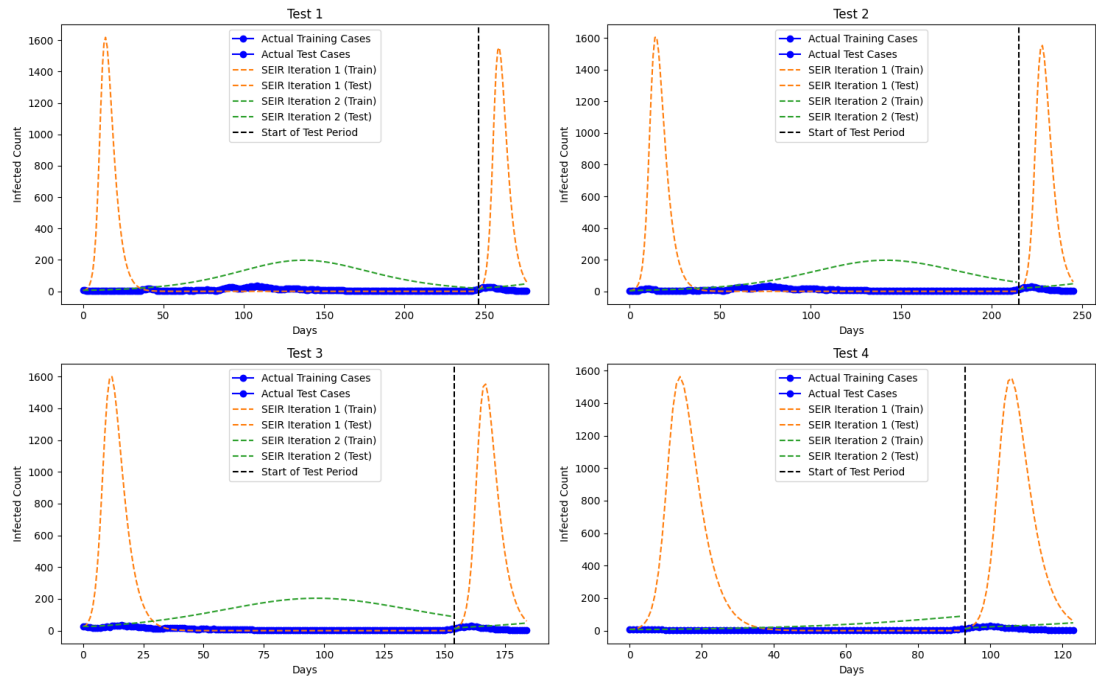
Count_unknown	Integer	The total number of cases of unknown age
Count_male	Integer	The total number of male cases
Count_female	Integer	The total number of female cases

ModelTbl Columns	Data Type	Description
Barangay_id	Integer	Barangay that the model used to train
Case_group	String	Age group that the model used to train
Date_start	Date	The start date of the data that the model used
Date_end	Date	The end date of the data that the model used
Infectious_rate	Float	The optimized infectious rate given by the model
Incubation_rate	Float	The optimized incubation rate given by the model
Recovery_rate	Float	The optimized recovery rate given by the model
mae	Float	The error rate of prediction to the actual value
rmse	Float	The error rate of prediction to the actual value
mse	Float	The error rate of prediction to the actual value

Appendix T

Barangay Level SEIR Predictions

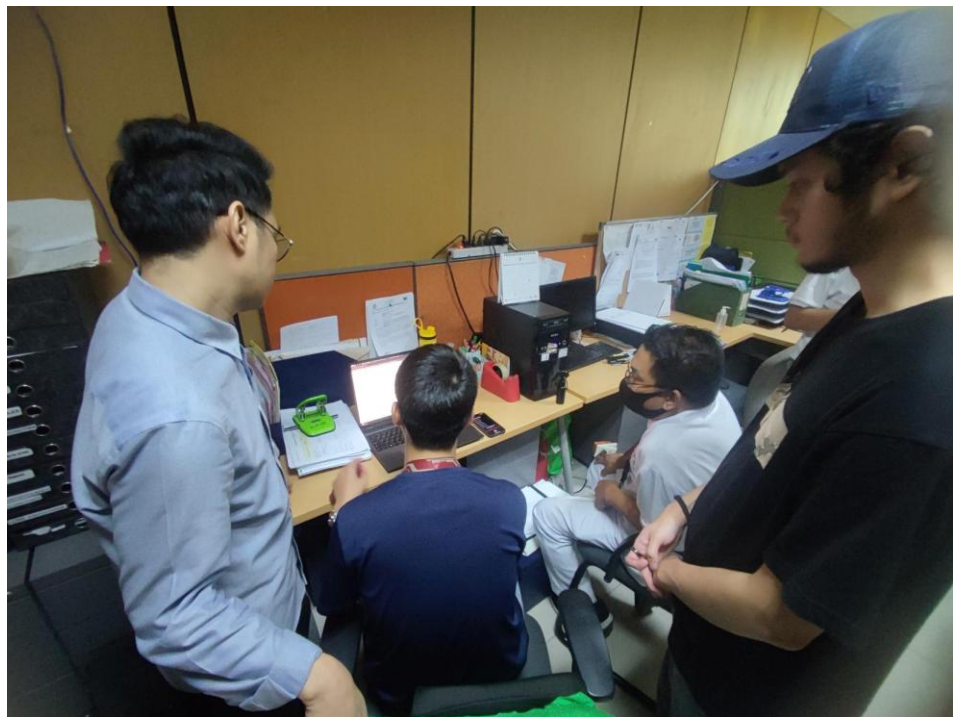
Barangay San Lorenzo Ruiz - SEIR Model Predictions



Appendix U

Usability Testing with San Pedro ESU Documentation





Appendix V

Model Comparison

Barangay	Model	RMSE	MAE	NMAE	Error Reduction	Infectious Rate	Incubation Rate	Recovery Rate
Narra	Classical SEIR	5.00	3.89	1.21	Baseline	0.20	0.30	0.14
	Simulated Annealing	2.89	2.53	0.78		1.19	0.27	0.99
	Bayesian Inference	3.32	2.99	0.93		0.46	0.28	0.37
San Lorenzo Ruiz	Classical SEIR	23.73	19.11	1.40	Baseline	0.20	0.30	0.14
	Simulated Annealing	11.54	9.08	0.67		1.24	0.20	1.00
	Bayesian Inference	14.80	12.80	0.94		0.44	0.33	0.35
San Antonio	Classical SEIR	494.29	406.50	1.91	Baseline	0.20	0.30	0.14
	Simulated Annealing	161.82	130.38	0.61		1.18	0.23	1.00
	Bayesian Inference	225.17	195.78	0.92		0.39	0.31	0.34

Appendix W

Dataset After the Cleaning Process

CaseCode	Age	Group	Sex	DateSpecimen	DateResultRelease	DateRepConf	DateDied	DateRecover	RemovalType	DateRepRem	Admitted	RegionRes	ProvRes	CityMunRes	CityMunBarangay	BarangayPSG	HealthStatus	Quarantir	DateOnset	Pregnanttab	Valid
C404174	38	35 to 39	FEMALE	1/23/2020	1/30/2020	1/30/2020			RECOVERED	2/7/2020	NO	Region VII: NEGROS C	DUMAGUETE	PH074610000			RECOVERED	NO	1/21/2020	NO	Rem
C462688	44	40 to 44	MALE	1/27/2020	1/30/2020	2/3/2020	2/1/2020		DIED	2/2/2020	NO	Region VII: NEGROS C	DUMAGUETE	PH074610000			DIED	NO	1/18/2020		
C387710	60	60 to 64	FEMALE	1/23/2020	1/30/2020	2/5/2020		1/31/2020	RECOVERED	2/5/2020	YES	Region VII: BOHOL	PANGLAO	PH071233000			RECOVERED	NO	1/21/2020	NO	Cas
C377460	49	45 to 49	MALE	2/28/2020		3/6/2020			RECOVERED	3/27/2020	NO	Region IV - BATANGAS	SANTO TOMAS	PH041028000			RECOVERED	NO			Cas
C498051	63	60 to 64	MALE	3/5/2020		3/6/2020	3/11/2020		DIED	3/12/2020	NO	Region IV - RIZAL	CAINTA	PH045805000			DIED	NO			Age
C130591	58	55 to 59	FEMALE	3/6/2020	3/7/2020	3/7/2020	3/12/2020		DIED	3/12/2020	YES	Region IV - RIZAL	CAINTA	PH045805000			DIED	NO		NO	Age
C178743	39	35 to 39	MALE	3/6/2020	3/8/2020	3/8/2020		3/21/2020	RECOVERED	3/22/2020	YES	NCR	NCR	CITY OF MAK	PH137602000		RECOVERED	YES	3/3/2020		Age
C440075	33	30 to 34	MALE	3/6/2020	3/8/2020	3/8/2020		4/5/2020	RECOVERED	4/9/2020	YES	ROF					RECOVERED	YES	3/1/2020		Age
C202135	57	55 to 59	MALE	3/6/2020	3/8/2020	3/8/2020		3/23/2020	RECOVERED	3/25/2020	YES	ROF					RECOVERED	NO			Age
C557002	86	80+	MALE	3/6/2020	3/8/2020	3/14/2020			DIED	3/15/2020	YES	NCR	NCR	CITY OF MAR	PH137402000		DIED	NO	3/1/2020		Age
C777589	51	50 to 54	FEMALE	3/2/2020	3/9/2020	3/9/2020			RECOVERED	3/25/2020	NO	NCR	NCR	CITY OF SAN	PH137405000		RECOVERED	NO		NO	Age
C985840	51	50 to 54	FEMALE	3/7/2020	3/9/2020	3/9/2020			RECOVERED	3/22/2020	NO	NCR	NCR	CITY OF SAN	PH137405000		RECOVERED	NO	3/4/2020	NO	Age
C787672	70	70 to 74	MALE	3/7/2020	3/9/2020	3/9/2020		3/27/2020	RECOVERED	3/28/2020	YES	NCR	NCR	CITY OF SAN	PH137405000		RECOVERED	YES	3/7/2020		Age
C348704	70	70 to 74	MALE	3/6/2020	3/9/2020	3/9/2020		3/19/2020	RECOVERED	3/21/2020	YES	NCR	NCR	CITY OF SAN	PH137405000		RECOVERED	YES	3/1/2020		Age
C861109	35	35 to 39	MALE	3/2/2020		3/9/2020		3/15/2020	RECOVERED	3/18/2020	YES	ROF					RECOVERED	NO			Age
C918665	48	45 to 49	MALE	3/2/2020		3/9/2020		3/13/2020	RECOVERED	3/19/2020	YES	Region IV - CAVITE	IMUS CITY	PH042109000			RECOVERED	NO			Age
C325527	59	55 to 59	FEMALE	3/2/2020		3/9/2020			RECOVERED	4/8/2020	YES	Region III: BULACAN	CITY OF SAN	PH031420000			RECOVERED	YES		NO	Rem
C261719	57	55 to 59	MALE	3/7/2020	3/9/2020	3/9/2020	3/15/2020		DIED	3/15/2020	YES	NCR	NCR	CITY OF MAN	PH133906000		DIED	NO			Age
C502781	46	45 to 49	MALE	3/8/2020	3/9/2020	3/9/2020		3/15/2020	RECOVERED	3/16/2020	YES	NCR	NCR	PASAY CITY	PH137605000		RECOVERED	NO			Age
C941779	25	25 to 29	MALE	3/2/2020		3/9/2020		3/15/2020	RECOVERED	3/18/2020	YES	NCR	NCR	CITY OF MAK	PH137602000		RECOVERED	NO	3/2/2020		Cas
C261144	68	65 to 69	FEMALE	3/3/2020	3/9/2020	3/9/2020		3/19/2020	RECOVERED	3/21/2020	YES	NCR	NCR	CITY OF SAN	PH137405000		RECOVERED	YES	3/2/2020	NO	Age
C382946	52	50 to 54	MALE	3/7/2020		3/9/2020	4/9/2020		DIED	4/9/2020	YES	NCR	NCR	QUEZON CIT	PH137404000		DIED	NO	3/3/2020		Age
C817111	46	45 to 49	FEMALE	3/9/2020		3/9/2020		3/26/2020	RECOVERED	3/28/2020	YES	NCR	NCR	CITY OF PASI	PH137403000		RECOVERED	NO	3/8/2020	NO	Cas
C557823	41	40 to 44	MALE	3/2/2020		3/9/2020		3/24/2020	RECOVERED	3/25/2020	YES	NCR	NCR	CITY OF PASI	PH137403000		RECOVERED	NO			Age
C767909	70	70 to 74	FEMALE	3/3/2020		3/10/2020		3/22/2020	RECOVERED	5/2/2020	YES	NCR	NCR	CITY OF MAR	PH137402000		RECOVERED	YES	3/5/2020	NO	Age
C870329	43	40 to 44	FEMALE	3/8/2020		3/10/2020		4/1/2020	RECOVERED	4/5/2020	YES	NCR	NCR	CITY OF MAR	PH137402000		RECOVERED	YES		NO	Age
C304126	70	70 to 74	MALE	3/9/2020		3/10/2020		3/22/2020	RECOVERED	3/26/2020	YES	NCR	NCR	CITY OF MAR	PH137402000		RECOVERED	YES			Age
C346224	60	60 to 64	MALE	3/8/2020	3/10/2020	3/10/2020		4/16/2020	RECOVERED	4/17/2020	YES	NCR	NCR	CITY OF MAK	PH137602000		RECOVERED	YES	3/3/2020		Age
C766724	82	80+	FEMALE	3/9/2020	3/10/2020	3/10/2020	3/23/2020		DIED	3/25/2020	YES	NCR	NCR	CITY OF MAR	PH137402000		DIED	YES	3/7/2020	NO	Age
C501602	29	25 to 29	FEMALE	3/8/2020		3/10/2020		3/24/2020	RECOVERED	3/25/2020	YES	NCR	NCR	CITY OF SAN	PH137405000		RECOVERED	YES	3/1/2020	NO	Age
C394725	64	60 to 64	MALE	3/3/2020		3/10/2020		3/20/2020	RECOVERED	4/22/2020	YES	NCR	NCR	CITY OF SAN	PH137405000		RECOVERED	YES			Age
C213674	31	30 to 34	MALE	3/3/2020		3/10/2020			RECOVERED	3/17/2020	NO	Region III: TARLAC	CAPAS	PH036904000			RECOVERED	NO			Rem
C202635	34	30 to 34	MALE	3/3/2020		3/10/2020		3/17/2020	RECOVERED	3/18/2020	YES	Region V: ECAMARINE	LIBMANAN	PH051718000			RECOVERED	NO			Age
C397673	72	70 to 74	MALE	3/4/2020		3/11/2020			RECOVERED	3/21/2020	NO	Region IV - BATANGAS	BATANGAS C	PH041005000			RECOVERED	NO			Age
C817385	70	70 to 74	FEMALE	3/4/2020		3/11/2020		4/1/2020	RECOVERED	4/4/2020	YES	NCR	NCR	QUEZON CIT	PH137404000		RECOVERED	YES		NO	Age
C160741	58	55 to 59	MALE	3/4/2020		3/11/2020		3/28/2020	RECOVERED	4/7/2020	YES	NCR	NCR	CITY OF SAN	PH137405000		RECOVERED	NO			Age
C233549	65	65 to 69	MALE	3/9/2020	3/11/2020	3/11/2020	4/2/2020		DIED	3/15/2020	YES	Region VII: NEGROS C	TAYASAN	PH074622000			DIED	NO	3/5/2020		Age
C393029	66	65 to 69	FEMALE	3/10/2020	3/11/2020	3/11/2020		3/24/2020	RECOVERED	4/6/2020	NO	NCR	NCR	CALOCAN C	PH137501000		RECOVERED	NO		NO	Age
C848316	50	50 to 54	MALE	3/4/2020		3/11/2020			RECOVERED	4/10/2020	NO	NCR	NCR	CITY OF MAK	PH137602000		RECOVERED	NO			Rem
C731082	51	50 to 54	MALE	3/9/2020		3/11/2020		3/26/2020	RECOVERED	3/27/2020	NO	ROF					RECOVERED	NO	3/5/2020		Age
C151323	53	50 to 54	MALE	3/9/2020	3/11/2020	3/11/2020			RECOVERED	4/8/2020	YES	NCR	NCR	CITY OF LAS I	PH137601000		RECOVERED	NO			Heal

Appendix X

Object Relational Model of Web Application

```
class Barangay(models.Model):
    barangay_id = models.AutoField(primary_key=True)
    barangay_res = models.CharField(max_length=255, unique=True)
    barangay_population = models.IntegerField()
    def __str__(self):
        return self.barangay_res

class CaseCategory(models.Model):
    case_category_id = models.AutoField(primary_key=True)
    category_name = models.CharField(max_length=255, unique=True)

    def __str__(self):
        return self.category_name

class Record(models.Model):
    record_id = models.AutoField(primary_key=True)
    barangay = models.ForeignKey(Barangay, on_delete=models.CASCADE)
    date = models.DateField()
    case_id = models.CharField(max_length=255, unique=True)

    def __str__(self):
        return f"Record {self.record_id} - {self.date}"

class CaseData(models.Model):
    case_data_id = models.ForeignKey(Record, on_delete=models.CASCADE, to_field='case_id')
    case_category = models.ForeignKey(CaseCategory, on_delete=models.CASCADE)
    total_count = models.IntegerField()
    count_under_9 = models.IntegerField()
    count_10_to_17 = models.IntegerField()
    count_18_to_39 = models.IntegerField()
    count_40_to_59 = models.IntegerField()
    count_60_to_79 = models.IntegerField()
    count_80_above = models.IntegerField()
    count_unknown = models.IntegerField()
    count_male = models.IntegerField()
    count_female = models.IntegerField()
    def __str__(self):
        return f"Case {self.case_data_id} - {self.case_category.category_name}"

class ModelInstance(models.Model):
    instance_id = models.AutoField(primary_key=True)
    barangay = models.ForeignKey(Barangay, on_delete=models.CASCADE)
    case_group = models.CharField(max_length=255)
    date_start = models.DateField(default=default_start_date)
    date_end = models.DateField(default=default_end_date)
    infectious_rate = models.FloatField()
    incubation_rate = models.FloatField()
    recovery_rate = models.FloatField()
    mae = models.FloatField()
    mse = models.FloatField(default=0.0)
    rmse = models.FloatField()

    def __str__(self):
        return f"Model Instance {self.instance_id} - {self.barangay.barangay_res} - {self.case_group}"
```


Appendix Y

Database Populating Code

```
for _, row in df.iterrows():
    barangay, _ = Barangay.objects.get_or_create(barangay_res=row['barangay_Res'])
    case_id = f'{row["barangay_Res"]}_{row["date"]}'

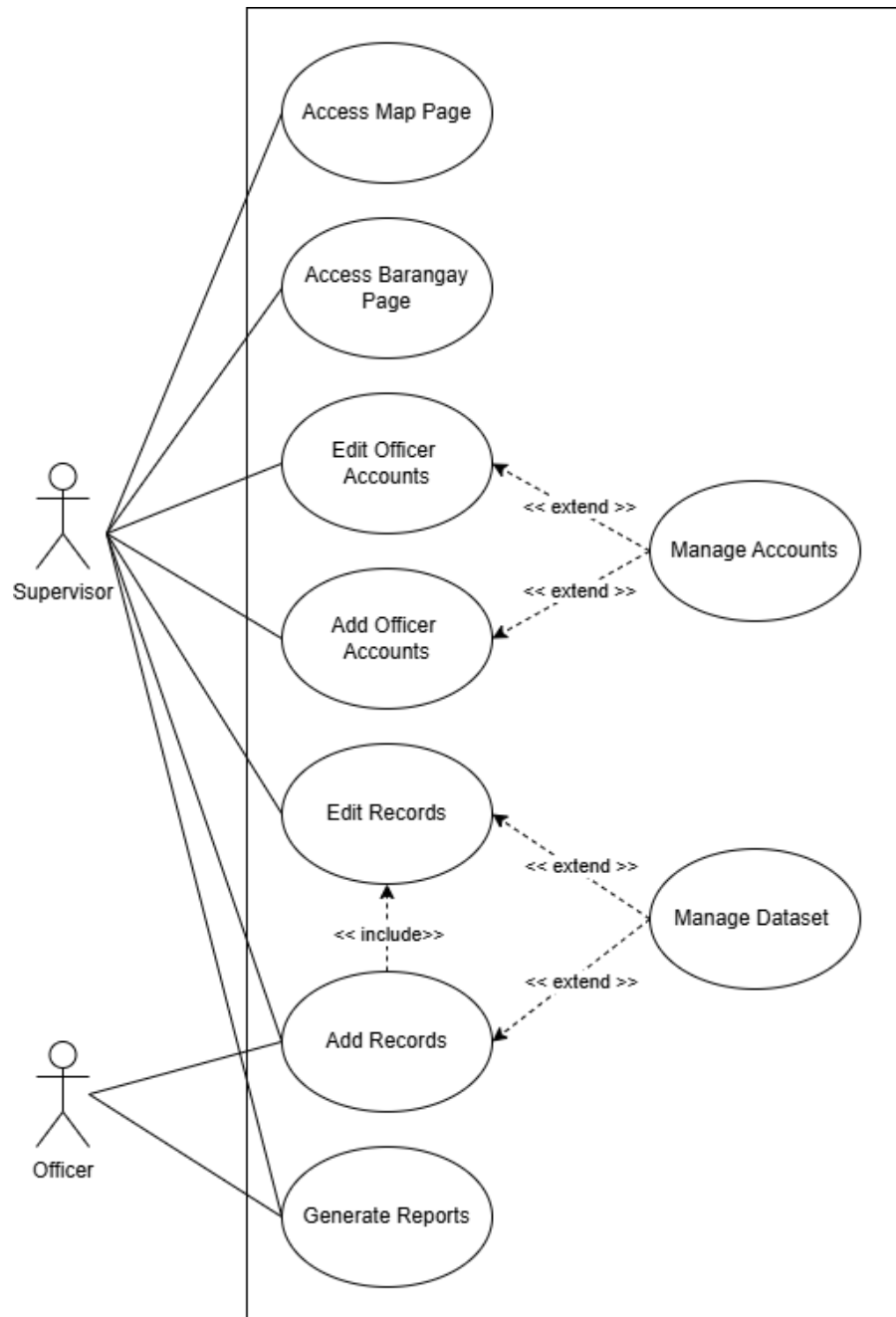
    record, _ = Record.objects.get_or_create(
        barangay=barangay,
        date=datetime.strptime(row["date"], "%Y-%m-%d").date(),
        case_id=case_id
    )

    for case_type, category in case_category_map.items():
        if case_type != 'prediction':
            if not CaseData.objects.filter(case_data_id=record, case_category=category).exists():
                CaseData.objects.create(
                    case_data_id=record,
                    case_category=category,
                    total_count=row[f'count_{case_type.capitalize()}'],
                    count_under_9=row[f'count_{case_type.capitalize()}_age_9_below'],
                    count_10_to_17=row[f'count_{case_type.capitalize()}_age_10_17'],
                    count_18_to_39=row[f'count_{case_type.capitalize()}_age_18_39'],
                    count_40_to_59=row[f'count_{case_type.capitalize()}_age_40_59'],
                    count_60_to_79=row[f'count_{case_type.capitalize()}_age_60_79'],
                    count_80_above=row[f'count_{case_type.capitalize()}_age_80_above'],
                    count_unknown=row[f'count_{case_type.capitalize()}_Unknown'],
                    count_male=row[f'count_{case_type.capitalize()}_Male'],
                    count_female=row[f'count_{case_type.capitalize()}_Female']
                )
    print(f"Successfully populated {case_id}")

self.stdout.write(self.style.SUCCESS('Successfully populated all tables with dataset records'))
```

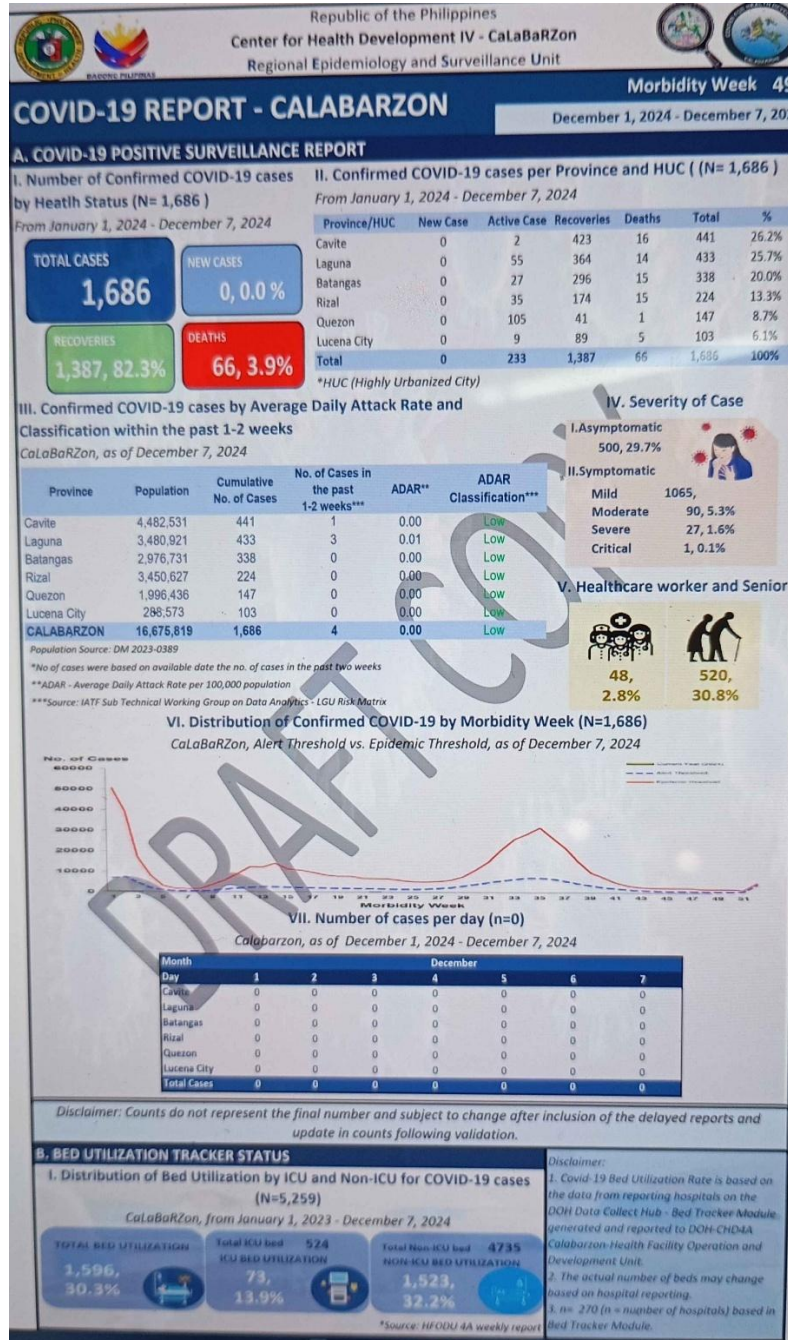

Appendix Z

Use Case Diagram



Appendix AA

Example Single Page Report



Appendix BB

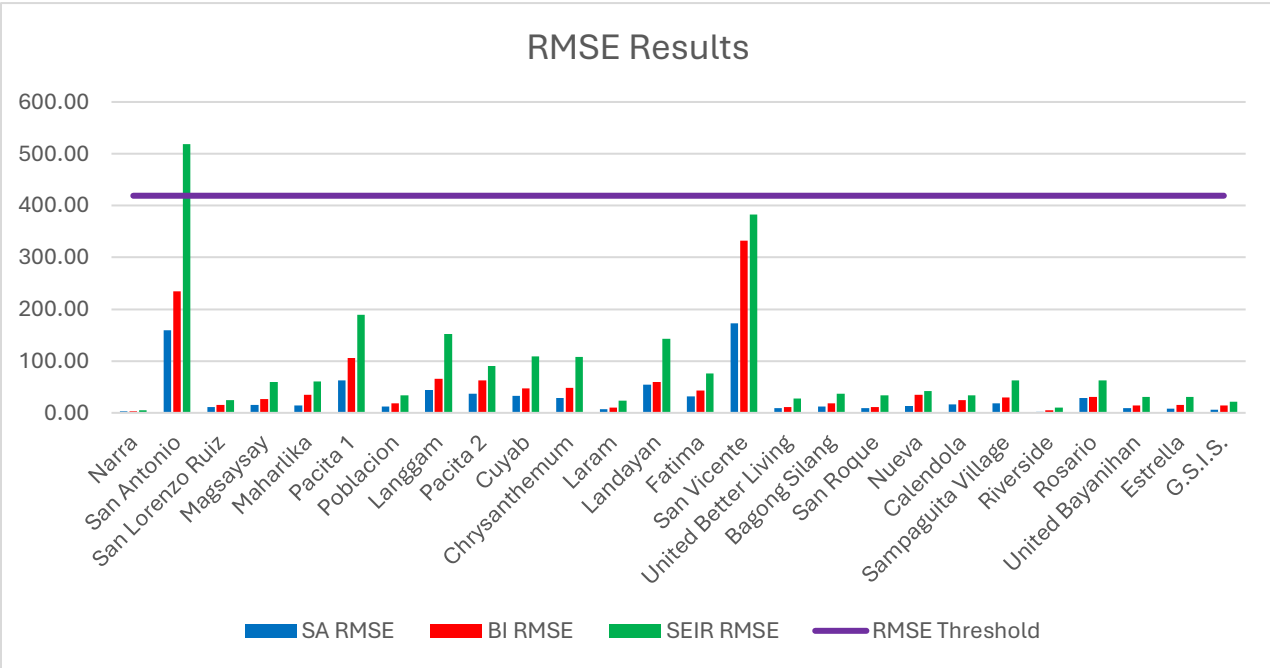
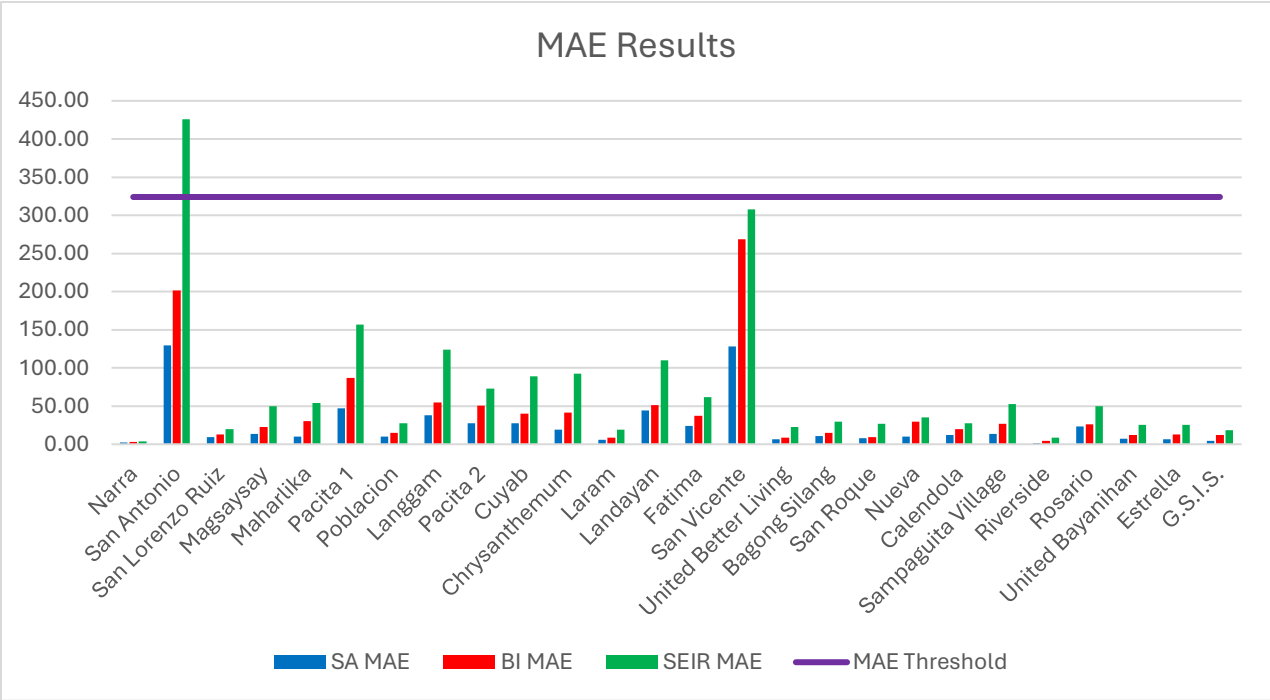
Validation Result on All Barangay of San Pedro

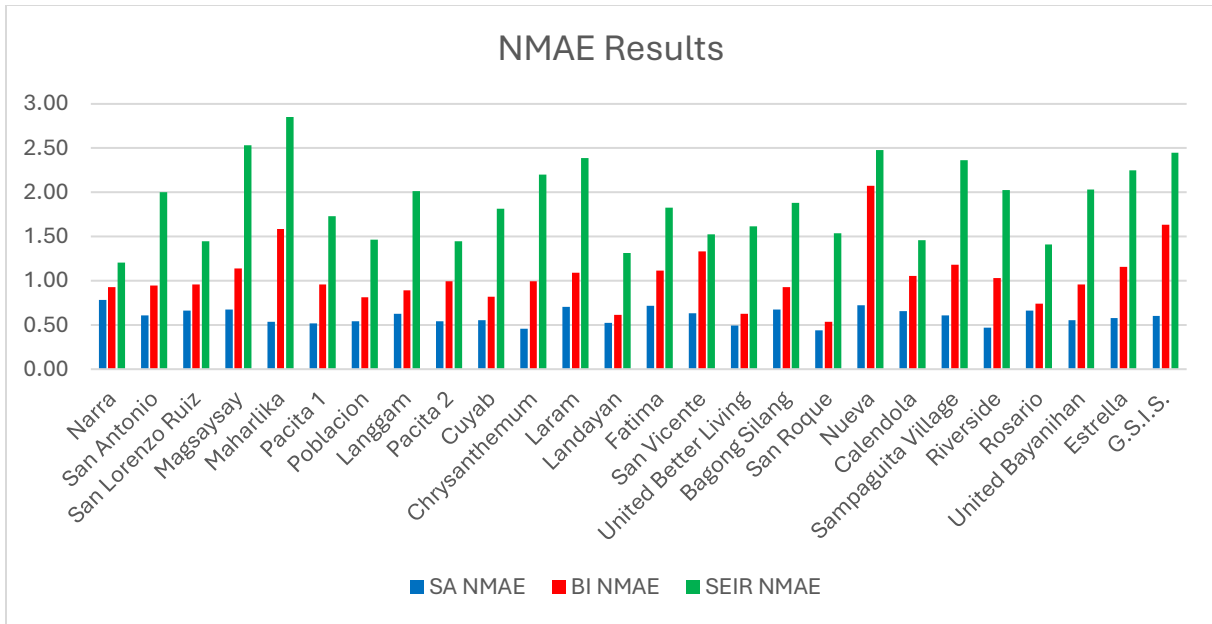
Barangay	SA MAE	SA RMSE	SA NMAE	BI MAE	BI RMSE	BI NMAE	SEIR MAE	SEIR RMSE	SEIR NMAE	SA MAE ER	BI MAE ER	SA RMSE ER	BI RMSE ER	SA NMAE ER	BI NMAE ER
Narra	2.52	2.90	0.78	2.99	3.32	0.93	3.89	5.00	1.21	35%	23%	42%	34%	35%	23%
San Antonio	129.27	159.53	0.61	201.30	234.51	0.94	426.18	518.70	2.00	70%	53%	69%	69%	70%	53%
San Lorenzo Ruiz	9.07	11.46	0.66	13.10	15.21	0.96	19.70	24.58	1.44	54%	34%	53%	53%	54%	34%
Magsaysay	13.20	15.46	0.67	22.39	26.47	1.14	49.80	60.00	2.53	73%	55%	74%	74%	73%	55%
Maharlika	10.07	13.83	0.53	29.97	35.01	1.59	53.77	60.92	2.85	81%	44%	77%	77%	81%	44%
Pacita 1	46.83	62.64	0.52	87.03	105.74	0.96	156.78	189.49	1.73	70%	44%	67%	67%	70%	44%
Poblacion	10.21	12.60	0.54	15.23	18.66	0.81	27.46	34.02	1.47	63%	45%	63%	63%	63%	45%
Langgam	38.29	44.29	0.62	54.67	65.47	0.89	123.70	152.65	2.01	69%	56%	71%	71%	69%	56%
Pacita 2	27.46	36.86	0.54	50.38	62.44	0.99	73.22	90.66	1.44	62%	31%	59%	59%	62%	31%
Cuyab	27.15	32.62	0.56	40.11	47.09	0.82	88.65	109.12	1.81	69%	55%	70%	70%	69%	55%
Chrysanthemum	19.31	28.14	0.46	41.80	47.72	1.00	92.17	107.66	2.20	79%	55%	74%	74%	79%	55%
Laram	5.72	6.60	0.70	8.89	10.59	1.09	19.40	23.46	2.39	70%	54%	72%	72%	70%	54%
Landayan	43.92	54.71	0.53	51.58	59.05	0.62	109.69	142.62	1.31	60%	53%	62%	62%	60%	53%
Fatima	24.09	32.01	0.72	37.41	43.04	1.11	61.50	75.74	1.83	61%	39%	58%	58%	61%	39%
San Vicente	128.01	172.80	0.63	268.92	331.95	1.33	307.70	382.94	1.53	58%	13%	55%	55%	58%	13%
United Better Living	6.85	8.73	0.50	8.68	10.64	0.63	22.27	27.63	1.61	69%	61%	68%	68%	69%	61%
Bagong Silang	10.62	11.93	0.68	14.58	18.04	0.93	29.52	36.98	1.88	64%	51%	68%	68%	64%	51%
San Roque	7.67	8.82	0.44	9.40	11.28	0.54	26.79	33.30	1.54	71%	65%	74%	74%	71%	65%
Nueva	10.30	13.22	0.72	29.49	34.84	2.07	35.22	41.71	2.48	71%	16%	68%	68%	71%	16%

Calendola	12.45	15.79	0.66	20.00	24.65	1.06	27.64	34.20	1.46	55%	28%	54%	54%	55%	28%
Sampaguita Village	13.56	18.01	0.61	26.51	30.06	1.18	52.89	62.14	2.36	74%	50%	71%	71%	74%	50%
Riverside	1.91	2.34	0.47	4.21	4.96	1.03	8.30	9.76	2.03	77%	49%	76%	76%	77%	49%
Rosario	23.21	29.14	0.66	26.07	30.48	0.74	49.66	62.32	1.41	53%	47%	53%	53%	53%	47%
United Bayanihan	6.95	9.06	0.55	12.10	13.93	0.96	25.60	30.81	2.03	73%	53%	71%	71%	73%	53%
Estrella	6.56	7.96	0.58	13.16	15.65	1.16	25.54	30.20	2.25	74%	48%	74%	74%	74%	48%
G.S.I.S.	4.53	5.66	0.60	12.30	14.27	1.64	18.40	21.14	2.45	75%	33%	73%	73%	75%	33%

MAE = Mean Absolute Error
RMSE = Root Mean Square Error
NMAE = Normalized Mean Absolute Error
ER = Error Reduction

SA = Simulated Annealing
BI = Bayesian Inference

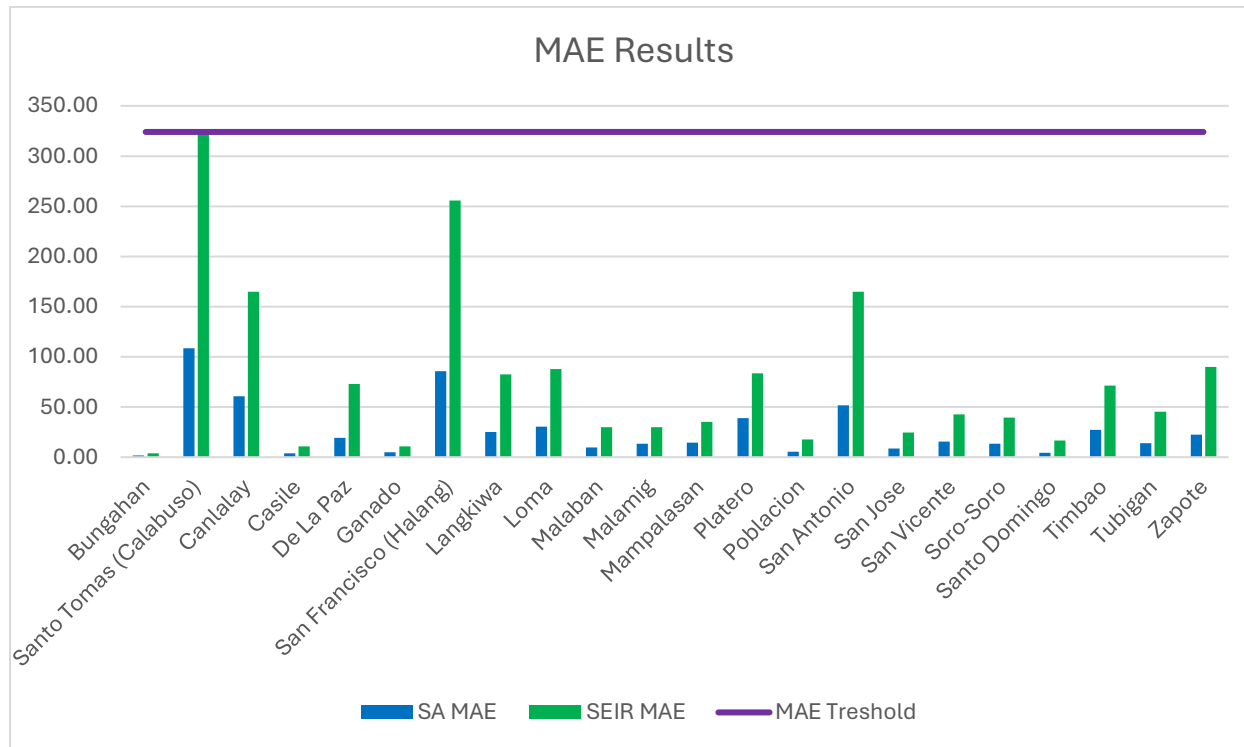


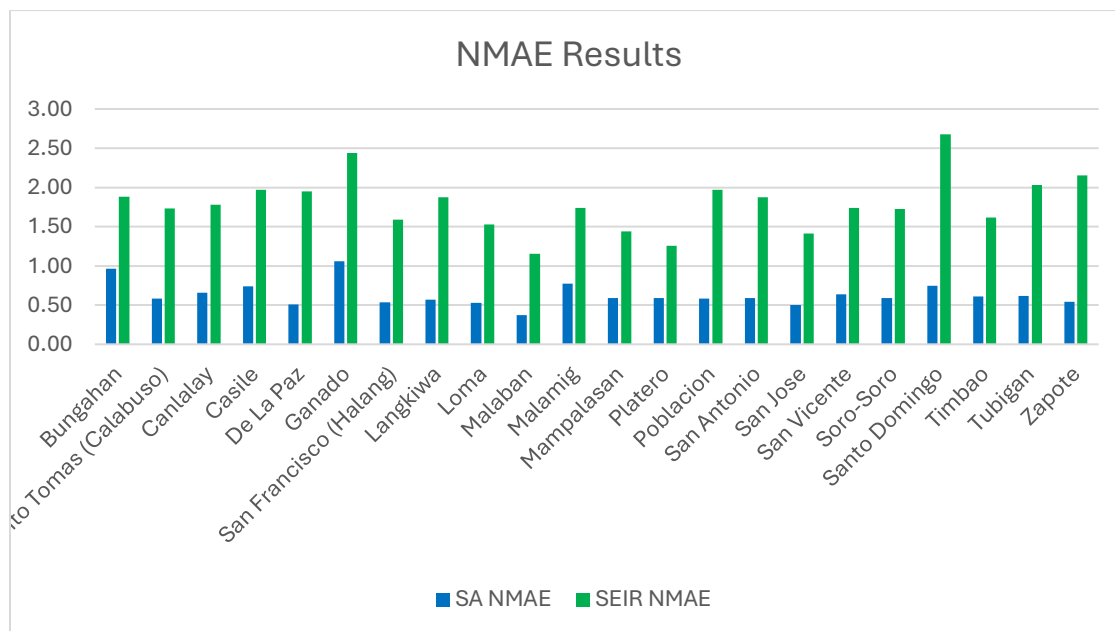
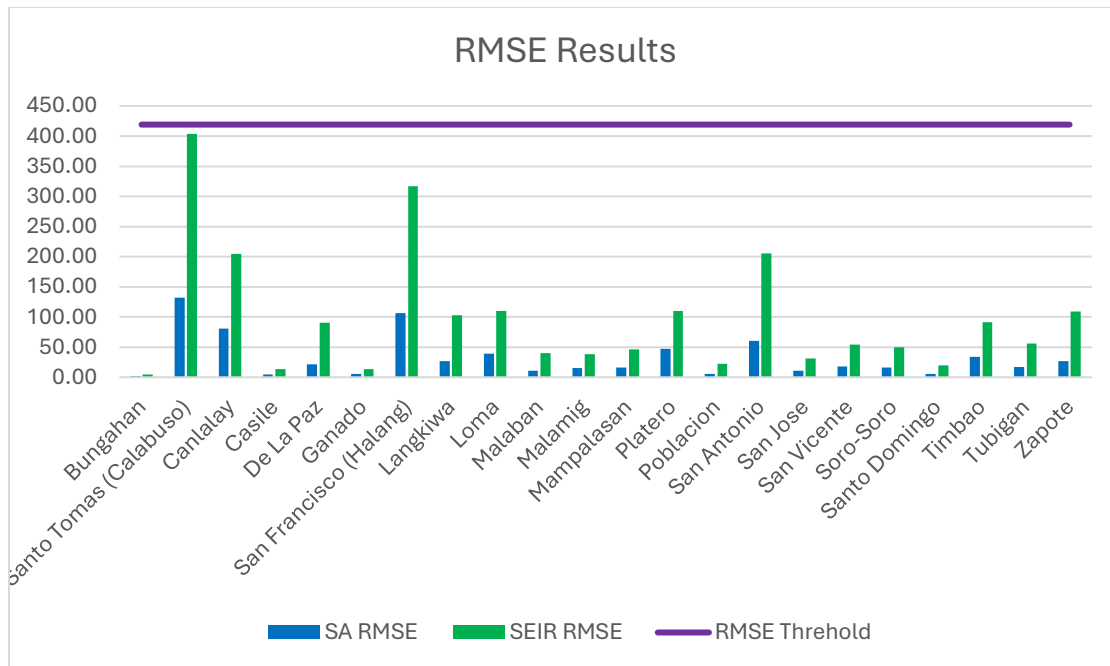


Appendix CC

Confidence Result on All Barangay of Biñan

Barangay	SA MAE	SA RMSE	SA NMAE	SEIR MAE	SEIR RMSE	SEIR NMAE	SA MAE ER	SA RMSE ER	SA NMAE ER
Bungahan	1.96	2.16	0.97	3.82	4.43	1.88	49%	51%	49%
Santo Tomas (Calabuso)	108.31	132.05	0.58	322.79	403.11	1.73	66%	67%	66%
Canlalay	60.74	80.88	0.66	164.80	204.87	1.78	63%	61%	63%
Casile	3.97	4.39	0.74	10.61	13.50	1.97	63%	68%	63%
De La Paz	19.07	21.44	0.51	72.79	90.04	1.95	74%	76%	74%
Ganado	4.74	5.23	1.06	10.95	13.58	2.44	57%	61%	57%
San Francisco (Halang)	85.92	106.05	0.53	255.72	317.20	1.59	66%	67%	66%
Langkiwa	24.85	27.10	0.57	82.26	102.88	1.88	70%	74%	70%
Loma	30.23	39.50	0.53	87.54	109.89	1.53	65%	64%	65%
Malaban	9.64	10.73	0.37	30.03	39.70	1.16	68%	73%	68%
Malamig	13.35	14.99	0.78	29.93	38.46	1.74	55%	61%	55%
Mampalasan	14.29	16.05	0.59	34.98	46.30	1.44	59%	65%	59%
Platero	39.04	46.96	0.59	83.60	109.53	1.26	53%	57%	53%
Poblacion	5.27	5.72	0.58	17.90	22.37	1.97	71%	74%	71%
San Antonio	51.77	60.53	0.59	164.74	204.99	1.87	69%	70%	69%
San Jose	8.72	10.62	0.50	24.64	31.59	1.41	65%	66%	65%
San Vicente	15.74	17.62	0.64	42.71	53.98	1.74	63%	67%	63%
Soro-Soro	13.50	16.01	0.59	39.48	50.18	1.73	66%	68%	66%
Santo Domingo	4.56	5.39	0.74	16.40	19.80	2.68	72%	73%	72%
Timbao	27.07	33.42	0.61	71.29	91.11	1.61	62%	63%	62%
Tubigan	13.69	16.85	0.61	45.35	55.53	2.03	70%	70%	70%
Zapote	22.65	26.52	0.54	90.10	109.26	2.15	75%	76%	75%

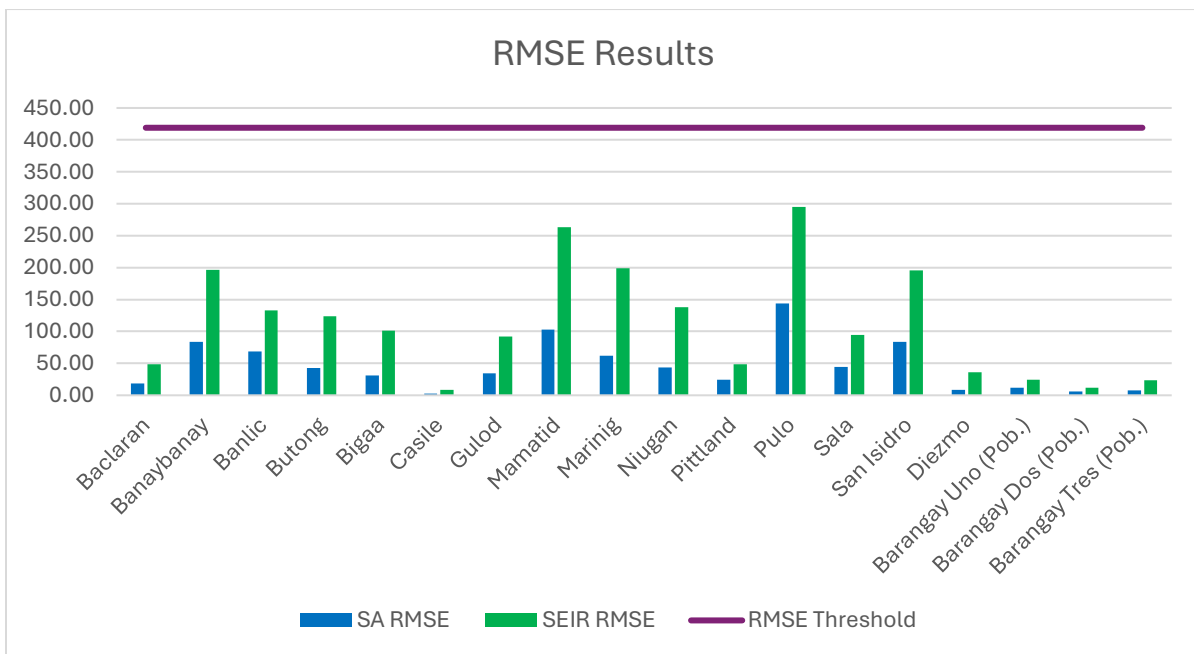
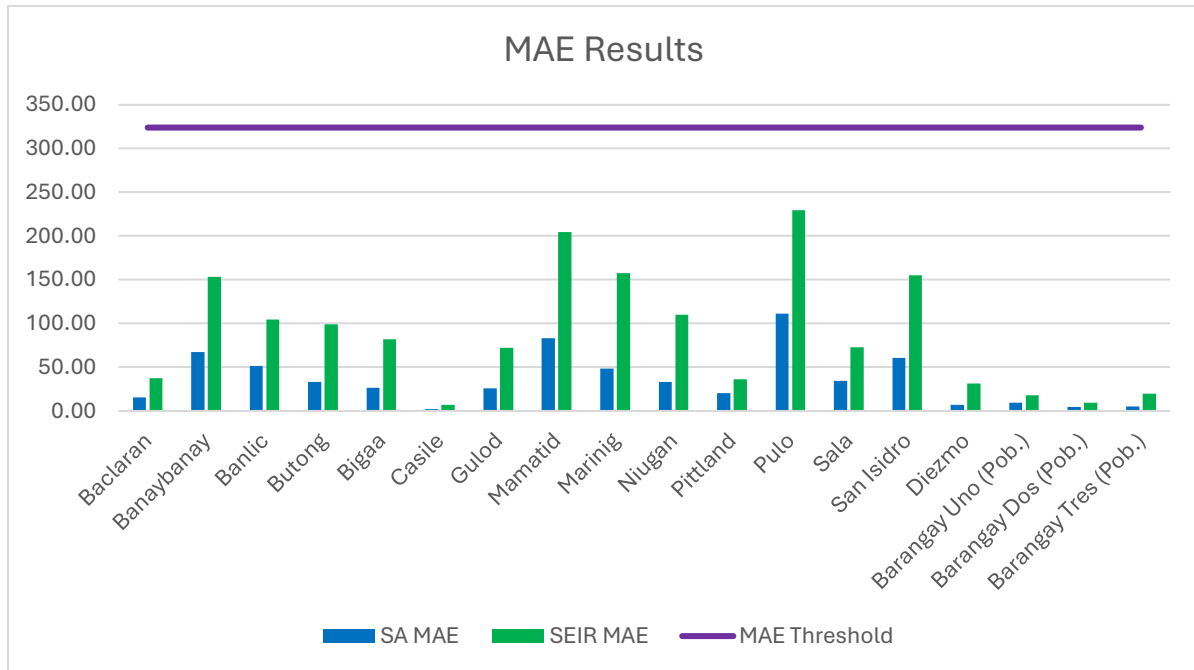


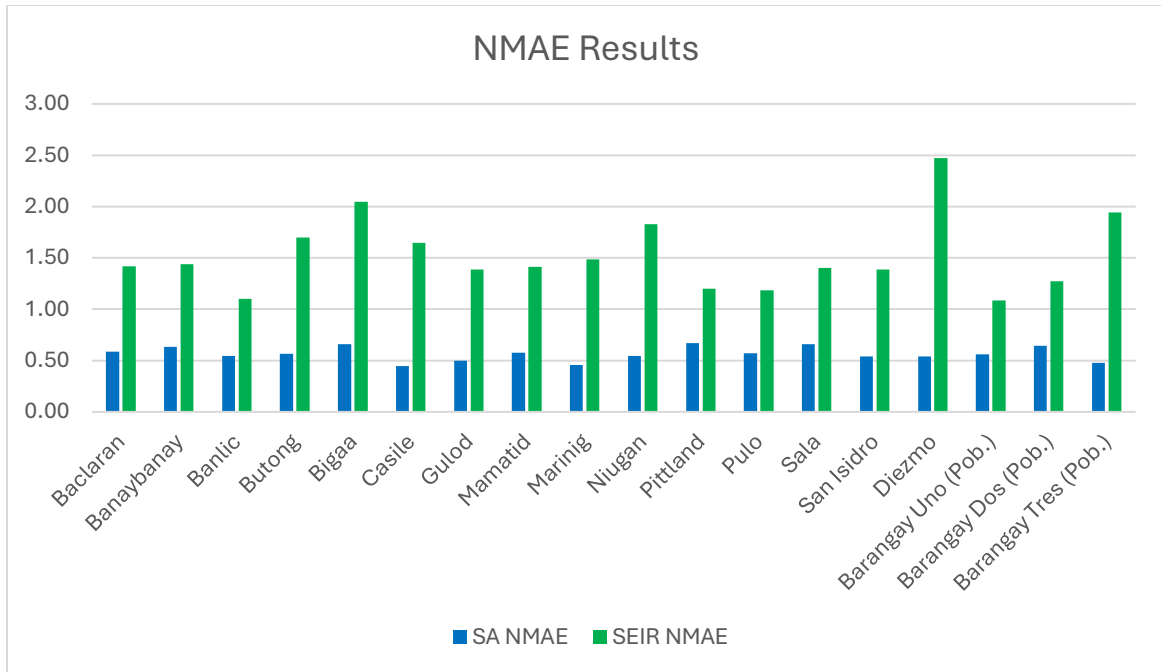


Appendix DD

Confidence Result on All Barangay of Cabuyao

Barangay	SA MAE	SA RMSE	SA NMAE	SEIR MAE	SEIR RMSE	SEIR NMAE	SA MAE ER	SA RMSE ER	SA NMAE ER
Baclaran	15.44	18.34	0.59	37.26	48.46	1.42	59%	62%	59%
Banaybanay	67.17	83.63	0.63	153.04	196.85	1.44	56%	58%	56%
Banlic	51.49	68.71	0.54	104.27	133.32	1.10	51%	48%	51%
Butong	33.21	42.95	0.57	99.20	123.77	1.70	67%	65%	67%
Bigaa	26.55	31.31	0.66	82.23	100.88	2.05	68%	69%	68%
Casile	1.89	2.32	0.45	6.95	8.64	1.64	73%	73%	73%
Gulod	25.82	34.09	0.50	72.14	92.11	1.39	64%	63%	64%
Mamatid	83.48	103.06	0.58	204.72	263.35	1.41	59%	61%	59%
Marinig	48.28	62.25	0.46	157.79	199.25	1.49	69%	69%	69%
Niugan	32.90	43.43	0.55	110.19	138.20	1.83	70%	69%	70%
Pittland	20.30	24.07	0.67	36.48	48.88	1.20	44%	51%	44%
Pulo	111.14	143.54	0.57	229.69	294.77	1.19	52%	51%	52%
Sala	34.35	44.80	0.66	72.90	94.35	1.40	53%	53%	53%
San Isidro	60.49	83.81	0.54	154.87	195.63	1.39	61%	57%	61%
Diezmo	6.82	8.30	0.54	31.28	36.26	2.47	78%	77%	78%
Barangay Uno (Pob.)	9.31	11.45	0.56	18.04	23.97	1.09	48%	52%	48%
Barangay Dos (Pob.)	4.81	5.81	0.64	9.54	11.55	1.27	50%	50%	50%
Barangay Tres (Pob.)	4.86	7.42	0.48	19.67	23.40	1.94	75%	68%	75%





Creating a New Model Using 5 Months of Data

```

class SimulatedAnnealing:
    def __init__(self, initial_conditions: list):
        self.initial_conditions = initial_conditions
        self.time_series = None
        self.S, self.E, self.I, self.R = initial_conditions
        self.cases = None
        self.beta, self.sigma, self.gamma = [0.1, 0.1, 0.1]
        # Parameter search bounds
        self.bounds = [(0.0001, 2.0), # Beta (Transmission rate)
                       (0.0001, 1.0), # Sigma (Incubation rate)
                       (0.0001, 1.0)] # Gamma (Recovery rate)
        self.solution = None

    @staticmethod
    def seir_model(y: list, t: float, beta: float, sigma: float, gamma: float):
        S, E, I, R = y
        N = S + E + I + R
        dS_dt = -beta * S * I / N
        dE_dt = beta * S * I / N - sigma * E
        dI_dt = sigma * E - gamma * I
        dR_dt = gamma * I
        return [dS_dt, dE_dt, dI_dt, dR_dt]

    def objective_function(self, params: tuple, cases: np.ndarray) -> float:
        beta, sigma, gamma = params
        # Solve SEIR model
        solution = odeint(self.seir_model, self.initial_conditions, self.time_series, args=(beta, sigma, gamma))
        I_pred = np.clip(solution[:, 2], 0, None) # Ensure no negative values
        # Compute Mean Squared Error (MSE)
        error = np.mean((I_pred - cases) ** 2)
        # Adaptive regularization to prevent extreme values
        reg_term = 0.01 * (beta**2 + sigma**2 + gamma**2)
        return error + reg_term

    def predict_trained_cases(self):
        solution = odeint(self.seir_model, self.initial_conditions, self.time_series,
                          args=(self.beta, self.sigma, self.gamma))
        self.solution = solution[:, 2]
        return solution[:, 2] # Return predicted infected cases

    def optimize_parameters(self, infectious_count: np.ndarray, time_series: np.ndarray) -> np.ndarray:
        self.time_series = time_series
        if len(infectious_count) == 0:
            raise ValueError("Infectious count data cannot be empty.")
        self.cases = infectious_count
        result = dual_annealing([self.objective_function, bounds=self.bounds,
                                args=(infectious_count,)])
        self.beta, self.sigma, self.gamma = result.x

        return result.x

    def predict_future(self, days: int, params: np.ndarray) -> np.ndarray:
        beta, sigma, gamma = params
        t_future = np.arange(days)
        solution = odeint(self.seir_model, self.initial_conditions, t_future, args=(beta, sigma, gamma))

        return solution[:, 2] # Return predicted infected cases

    def get_error_rates(self):
        from sklearn.metrics import mean_squared_error, mean_absolute_error
        mse = mean_squared_error(self.cases, self.solution)
        mae = mean_absolute_error(self.cases, self.solution)
        rmse = np.sqrt(mse)

        return (mse, mae, rmse)

```